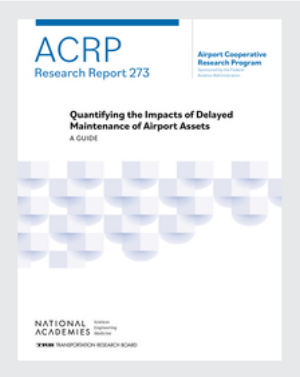
NATIONAL ACADEMIES Sciences Engineering Medicine

NATIONAL ACADEMIES PRESS Washington, DC

This PDF is available at http://nap.nationalacademies.org/27960



Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide (2024)

BUY THIS BOOK

FIND RELATED TITLES

DETAILS

146 pages | 8.5 x 11 | PAPERBACK

ISBN 978-0-309-72647-4 | DOI 10.17226/27960

CONTRIBUTORS

Pramen P. Shrestha, Scott Murrell, Brian Aho, William Robert Williams, Dawn S. Ward, Ghada G. Gad; Airport Cooperative Research Program; Transportation Research Board; National Academies of Sciences, Engineering, and Medicine

SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2024. *Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide*. Washington, DC: The National Academies Press. https://doi.org/10.17226/27960.

Visit the National Academies Press at nap.edu and login or register to get:

– Access to free PDF downloads of thousands of publications
– 10% off the price of print publications
– Email or social media notifications of new titles related to your interests
– Special offers and discounts

All downloadable National Academies titles are free to be used for personal and/or non-commercial academic use. Users may also freely post links to our titles on this website; non-commercial academic users are encouraged to link to the version on this website rather than distribute a downloaded PDF to ensure that all users are accessing the latest authoritative version of the work. All other uses require written permission. (Request Permission)

This PDF is protected by copyright and owned by the National Academy of Sciences; unless otherwise indicated, the National Academy of Sciences retains copyright to all materials in this PDF with all rights reserved.

ACRP
Research Report 273

Airport Cooperative Research Program
Sponsored by the Federal Aviation Administration

Quantifying the Impacts of Delayed Maintenance of Airport Assets

A GUIDE

NATIONAL ACADEMIES Sciences Engineering Medicine

TRB TRANSPORTATION RESEARCH BOARD

Copyright National Academy of Sciences. All rights reserved.

ACRP OVERSIGHT COMMITTEE*

CHAIR

Marianne Csaky
LanzaJet

VICE CHAIR

Kimberly A. Kenville
University of North Dakota

MEMBERS

Balram "B" Bheodari
Hartsfield-Jackson Atlanta International Airport
Darryl Daniels
Jacobsen Daniels Associates
John Dermody
Federal Aviation Administration
Rhonda Hamm-Niebruegge
St. Louis Lambert International Airport
Lance Lyttle
Seattle-Tacoma International Airport
Chad Makovsky
City of Phoenix (AZ)
Scott McMahon
Morristown Municipal Airport
Frank R. Miller
Burbank-Glendale-Pasadena Airport Authority
Ramon Ricondo
Ricondo and Associates, Inc.
Steve Sisneros
Southwest Airlines
Cathryn Stephens
Eugene Airport

EX OFFICIO MEMBERS

Justin Barkowski
American Association of Airport Executives
Sabrina Johnson
U.S. Environmental Protection Agency
Laura Rinaldi McKee
Airlines for America
Christopher J. Oswald
Airports Council International–North America
Gregory Pecoraro
National Association of State Aviation Officials
T.J. Schulz
Airport Consultants Council
Victoria Sheehan
Transportation Research Board

SECRETARY

Monique R. Evans
Transportation Research Board

* Membership as of August 2024.

TRANSPORTATION RESEARCH BOARD 2024 EXECUTIVE COMMITTEE*

OFFICERS

CHAIR: **Carol A. Lewis,** *Professor, Transportation Studies, Texas Southern University, Houston*
VICE CHAIR: **Leslie S. Richards,** *General Manager, Southeastern Pennsylvania Transportation Authority (SEPTA), Philadelphia*
EXECUTIVE DIRECTOR: **Victoria Sheehan,** *Transportation Research Board, Washington, DC*

MEMBERS

Michael F. Ableson, *CEO, Arrival Automotive–North America, Detroit, MI*
James F. Albaugh, *President and CEO, The Boeing Company (retired), Scottsdale, AZ*
Carlos M. Braceras, *Executive Director, Utah Department of Transportation, Salt Lake City*
Douglas C. Ceva, *Vice President, Customer Lead Solutions, Prologis, Inc., Jupiter, FL*
Nancy Daubenberger, *Commissioner of Transportation, Minnesota Department of Transportation, St. Paul*
Marie Therese Dominguez, *Commissioner, New York State Department of Transportation, Albany*
Garrett Eucalitto, *Commissioner, Connecticut Department of Transportation, Newington*
Chris T. Hendrickson, *Hamerschlag University Professor of Engineering Emeritus, Carnegie Mellon University, Pittsburgh, PA*
Randell Iwasaki, *President and CEO, Iwasaki Consulting Services, Walnut Creek, CA*
Ashby Johnson, *Executive Director, Capital Area Metropolitan Planning Organization (CAMPO), Austin, TX*
Joel M. Jundt, *Secretary of Transportation, South Dakota Department of Transportation, Pierre*
Hani S. Mahmassani, *W.A. Patterson Distinguished Chair in Transportation; Director, Transportation Center, Northwestern University, Evanston, IL*
Scott C. Marler, *Director, Iowa Department of Transportation, Ames*
Ricardo Martinez, *Adjunct Professor of Emergency Medicine, Emory University School of Medicine, Decatur, GA*
Michael R. McClellan, *Vice President, Strategic Planning, Norfolk Southern Corporation, Atlanta, GA*
Russell McMurry, *Commissioner, Georgia Department of Transportation, Atlanta*
Craig E. Philip, *Research Professor and Director, VECTOR, Department of Civil and Environmental Engineering, Vanderbilt University, Nashville, TN*
Steward T.A. Pickett, *Distinguished Senior Scientist, Cary Institute of Ecosystem Studies, Millbrook, NY*
Susan A. Shaheen, *Professor and Co-director, Transportation Sustainability Research Center, University of California, Berkeley*
Marc Williams, *Executive Director, Texas Department of Transportation, Austin*

EX OFFICIO MEMBERS

Michael R. Berube, *Deputy Assistant Secretary for Sustainable Transportation, U.S. Department of Energy, Washington, DC*
Shailen Bhatt, *Administrator, Federal Highway Administration, U.S. Department of Transportation, Washington, DC*
Amit Bose, *Administrator, Federal Railroad Administration, Washington, DC*
Tristan Brown, *Deputy Administrator, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, Washington, DC*
Steven Cliff, *Executive Officer, California Air Resources Board, Sacramento*
Rand Ghayad, *Senior Vice President, Association of American Railroads, Washington, DC*
LeRoy Gishi, *Chief, Division of Transportation, Bureau of Indian Affairs, U.S. Department of the Interior, Germantown, MD*
William H. Graham, Jr. (Major General, U.S. Army), *Deputy Commanding General for Civil and Emergency Operations, U.S. Army Corps of Engineers, Washington, DC*
Robert C. Hampshire, *Deputy Assistant Secretary for Research and Technology, U.S. Department of Transportation, Washington, DC*
Zahra "Niloo" Parvinashtiani, *Engineer, Mobility Consultant Solutions, Iteris Inc., Fairfax, VA, and Chair, TRB Young Members Coordinating Council*
Sophie Shulman, *Deputy Administrator, National Highway Traffic Safety Administration, Washington, DC*
Karl Simon, *Director, Transportation and Climate Division, U.S. Environmental Protection Agency, Washington, DC*
Paul P. Skoutelas, *President and CEO, American Public Transportation Association, Washington, DC*
Polly Trottenberg, *Deputy Secretary of Transportation, U.S. Department of Transportation, Washington, DC*
Jim Tymon, *Executive Director, American Association of State Highway and Transportation Officials, Washington, DC*
Veronica Vanterpool, *Acting Administrator, Federal Transit Administration, Washington, DC*
Michael Whitaker, *Administrator, Federal Aviation Administration, U.S. Department of Transportation, Washington, DC*
Vinn White, *Deputy Administrator, Federal Motor Carrier Safety Administration, Washington, DC*

* Membership as of August 2024.

Copyright National Academy of Sciences. All rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 273

Quantifying the Impacts of Delayed Maintenance of Airport Assets

A GUIDE

Pramen P. Shrestha
University of Nevada, Las Vegas
Las Vegas, NV

Scott Murrell
Applied Research Associates, Inc.
Elkridge, MD

Brian Aho
Applied Research Associates, Inc.
Elkridge, MD

William Robert Williams
Spy Pond Partners, LLC
Arlington, MA

Dawn S. Ward
Base Management, LLC
Hilo, HI

Ghada G. Gad
Independent Consultant
Pomona, CA

Subscriber Categories
Aviation • Maintenance and Preservation • Transportation, General

Research sponsored by the Federal Aviation Administration

NATIONAL ACADEMIES Sciences Engineering Medicine

TRB TRANSPORTATION RESEARCH BOARD

2024

Copyright National Academy of Sciences. All rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP RESEARCH REPORT 273

Project 09-20
ISSN 2572-3731 (Print)
ISSN 2572-374X (Online)
ISBN 978-0-309-72647-4
Library of Congress Control Number 2024946006

© 2024 by the National Academy of Sciences. National Academies of Sciences, Engineering, and Medicine and the graphical logo are trademarks of the National Academy of Sciences. All rights reserved.

COPYRIGHT INFORMATION

Authors herein are responsible for the authenticity of their materials and for obtaining written permissions from publishers or persons who own the copyright to any previously published or copyrighted material used herein.

Cooperative Research Programs (CRP) grants permission to reproduce material in this publication for classroom and not-for-profit purposes. Permission is given with the understanding that none of the material will be used to imply TRB, AASHTO, APTA, FAA, FHWA, FTA, GHSA, or NHTSA endorsement of a particular product, method, or practice. It is expected that those reproducing the material in this document for educational and not-for-profit uses will give appropriate acknowledgment of the source of any reprinted or reproduced material. For other uses of the material, request permission from CRP.

NOTICE

The research report was reviewed by the technical panel and accepted for publication according to procedures established and overseen by the Transportation Research Board and approved by the National Academies of Sciences, Engineering, and Medicine.

The opinions and conclusions expressed or implied in this report are those of the researchers who performed the research and are not necessarily those of the Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; or the program sponsors.

The Transportation Research Board does not develop, issue, or publish standards or specifications. The Transportation Research Board manages applied research projects which provide the scientific foundation that may be used by Transportation Research Board sponsors, industry associations, or other organizations as the basis for revised practices, procedures, or specifications.

The Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; and the sponsors of the Airport Cooperative Research Program do not endorse products or manufacturers. Trade or manufacturers' names or logos appear herein solely because they are considered essential to the object of the report.

Published research reports of the

AIRPORT COOPERATIVE RESEARCH PROGRAM

are available from

National Academies Press
500 Fifth Street, NW, Keck 360
Washington, DC 20001

(800) 624-6242

and can be ordered through the Internet by going to

https://nap.nationalacademies.org

Printed in the United States of America

Copyright National Academy of Sciences. All rights reserved.

The **National Academy of Sciences** was established in 1863 by an Act of Congress, signed by President Lincoln, as a private, nongovernmental institution to advise the nation on issues related to science and technology. Members are elected by their peers for outstanding contributions to research. Dr. Marcia McNutt is president.

The **National Academy of Engineering** was established in 1964 under the charter of the National Academy of Sciences to bring the practices of engineering to advising the nation. Members are elected by their peers for extraordinary contributions to engineering. Dr. John L. Anderson is president.

The **National Academy of Medicine** (formerly the Institute of Medicine) was established in 1970 under the charter of the National Academy of Sciences to advise the nation on medical and health issues. Members are elected by their peers for distinguished contributions to medicine and health. Dr. Victor J. Dzau is president.

The three Academies work together as the **National Academies of Sciences, Engineering, and Medicine** to provide independent, objective analysis and advice to the nation and conduct other activities to solve complex problems and inform public policy decisions. The National Academies also encourage education and research, recognize outstanding contributions to knowledge, and increase public understanding in matters of science, engineering, and medicine.

Learn more about the National Academies of Sciences, Engineering, and Medicine at **www.nationalacademies.org**.

The **Transportation Research Board** is one of seven major program divisions of the National Academies of Sciences, Engineering, and Medicine. The mission of the Transportation Research Board is to mobilize expertise, experience, and knowledge to anticipate and solve complex transportation-related challenges. The Board's varied activities annually engage about 8,500 engineers, scientists, and other transportation researchers and practitioners from the public and private sectors and academia, all of whom contribute their expertise in the public interest. The program is supported by state departments of transportation, federal agencies including the component administrations of the U.S. Department of Transportation, and other organizations and individuals interested in the development of transportation.

Learn more about the Transportation Research Board at **www.TRB.org**.

Copyright National Academy of Sciences. All rights reserved.

COOPERATIVE RESEARCH PROGRAMS

CRP STAFF FOR ACRP RESEARCH REPORT 273

Monique R. Evans, *Director, Cooperative Research Programs*
Waseem Dekelbab, *Deputy Director, Cooperative Research Programs*
Marci A. Greenberger, *Manager, Airport Cooperative Research Program*
Matthew J. Griffin, *Senior Program Officer*
Brittany Summerlin-Azeez, *Program Coordinator*
Natalie Barnes, *Director of Publications*
Heather DiAngelis, *Associate Director of Publications*
Alison Shapiro, *Editor*

ACRP PROJECT 09-20 PANEL
Field of Maintenance

Casey Martin, *Jacobs, Logandale, NV* (Chair)
Brigitte Codling, *Town of Haverhill, New Hampshire, North Haverhill, NH*
Scott Ira Kincaid, *Charlotte Douglas International Airport, Cramerton, NC*
Dennis Leuver, *United Airlines, Inc., Chicago, IL*
Brendalynn T. Taulelei, *Port of Seattle, Seattle, WA*
John Walewski, *Texas A&M University, College Station, TX*
Marvin Woods, *FAA Liaison*
Paul James Eubanks, *Airports Council International–North America Liaison*

Copyright National Academy of Sciences. All rights reserved.

FOREWORD

By Matthew J. Griffin
Staff Officer
Transportation Research Board

ACRP Research Report 273: Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide will help airport practitioners understand the consequences and costs of delayed maintenance of airport assets. It will also better quantify the value of a preventive maintenance plan. The responsible parties for airport assets, including airfield markings and pavement, passenger loading bridges, baggage handling systems, and airport snow removal equipment, will be able to leverage the guide and its associated tools.

Airport owners are developing more sophisticated and detailed asset management programs to help improve and extend the life cycle of their facilities and systems. A key step in asset management is a detailed life cycle and preventive maintenance plan to ensure the asset's full life is reached. This guide and the associated tools will help airports of all sizes develop a process to quantify the consequences of delayed maintenance of airport assets.

The research was led by the University of Nevada, Las Vegas with support from Applied Research Associates, Inc.; Spy Pond Partners, LLC; and Base Management, LLC. The guide and the tools provide airports with a general framework for quantifying the consequences of delayed maintenance of assets and can help them estimate the cost of airport maintenance and possible rehabilitation of specific airside and landside assets.

Supplemental to this report are a quick guide to using the delayed maintenance cost calculator spreadsheet tools, four airside asset spreadsheet tools, five landside asset spreadsheet tools, and Appendices A through D. They can be found on the National Academies Press website (nap.nationalacademies.org) by searching for *ACRP Research Report 273: Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide* and then reviewing "Resources at a Glance."

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

CONTENTS

1 **Chapter 1** Introduction
1 1.1 Background
2 1.2 Objectives and Scope
2 1.3 Overview of Research Approach
4 1.4 Organization of This Report

5 **Chapter 2** Literature Review Findings
5 2.1 Budget Prioritization and Allocation Process
6 2.2 Airport Asset Management
7 2.3 Preventive and Deferred Maintenance
10 2.4 Summary of Relevant Research Publications and Key Findings

12 **Chapter 3** Airport Interview Findings
12 3.1 Airport Budget Preparation Processes
16 3.2 Airports' Maintenance Budget Allocation Processes
18 3.3 Impacts of Delayed Preventive Maintenance of Airport Assets
20 3.4 Summary of Interview Findings

22 **Chapter 4** Framework for Quantifying the Impact of Delayed Maintenance of Airport Assets
24 4.1 Defining Airport Asset Maintenance and Preservation Policies
24 4.2 Keeping and Updating Asset Inventory Databases
25 4.3 Collecting Asset Condition Data
25 4.4 Analyzing Asset Condition Data
26 4.5 Identifying Maintenance and Replacement Activities
26 4.6 Keeping and Updating Maintenance and Replacement Cost Databases
26 4.7 Determining Maintenance and Replacement Budgets with and without Delayed Scenarios

27 **Chapter 5** Procedures to Quantify the Impact of Delayed Maintenance on Airport Airside Assets
27 5.1 Primary Runway Pavement
40 5.2 Airfield Markings
52 5.3 Runway Lighting
64 5.4 Airfield Signs

74 **Chapter 6** Procedures to Quantify the Impact of Delayed Maintenance on Airport Landside Assets
74 6.1 Airport Service Vehicles: Pickup Trucks
82 6.2 Airport Snow Removal Equipment
89 6.3 HVAC Systems
98 6.4 Baggage Handling System
105 6.5 Passenger Boarding Bridges

Copyright National Academy of Sciences. All rights reserved.

113 **Chapter 7** Procedures to Quantify the Impact of Delayed Maintenance on Other Airport Assets
113 7.1 Terminal Building Envelopes
118 7.2 Airport Hangars
121 7.3 Airport Automated People Mover Systems

126 **Chapter 8** Summary and Recommendations for Future Research
126 8.1 Key Points
127 8.2 Spreadsheet Tools to Assist Airport Asset Managers
128 8.3 Areas of Future Research

130 References

134 Appendices

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 1

Introduction

1.1 Background

The Federal Aviation Administration (FAA) provides public resources to build, maintain, and operate airports. All airport projects funded by federal agencies are regulated by the FAA. According to the Government Accountability Office (2015), most larger airports obtain more than half of their capital funding through on-site-generated income, while smaller airports, including small non-hubs and general aviation airports, tend to rely on Airport Improvement Program (AIP) grants. In Fiscal Year (FY) 2019, about $16.1 billion was spent to construct and safely operate U.S. airports. Out of this, about $9.9 billion was spent on operations, which includes airport asset maintenance. According to *ACRP Report 138: Preventive Maintenance at General Aviation Airports Volume 2: Guidebook*, airport maintenance consists of maintaining airside and landside assets. Airside assets may include runway and taxiway pavements, airfield lighting, markings, signs, ramp areas, drainage systems, hydrant fueling systems, and visual and navigational aids (NAVAIDS). Landside assets may include structures and systems, such as terminal buildings; baggage handling systems; passenger boarding bridges; heating, ventilation, and air conditioning (HVAC) systems; fuel storage infrastructure; service and access roadways; environmental works; snow removal equipment; ground and aircraft support equipment; aircraft rescue and firefighting (ARFF) equipment; and fleet vehicles. Airside assets can have a negative impact on aircraft safety and operation if not properly maintained. Maintenance of landside assets is also critical because these assets help passengers and airport equipment move safely and efficiently.

Airport maintenance can be divided into four types: operation, preventive, reactive, and predictive (Ploeger et al. 2015b). Operational maintenance is related to the removal of snow and foreign objects from pavement, and maintaining electrical/mechanical equipment, lighting systems, NAVAIDS, etc. These activities must be performed as required to keep airports safe for landing and takeoff. Preventive maintenance activities include tasks that mitigate degradation of an infrastructure system or its components. Preventive maintenance, when performed regularly as needed, can extend facility and component service life and yield optimum asset performance outputs. This results in reduced costs and improved user satisfaction. Reactive maintenance involves fixing components after they break rather than fixing the system or components when performing preventive maintenance. This type of activity incurs higher system costs compared to conducting regular preventive maintenance. Many airports are in reactive mode and spend more resources than necessary, as opposed to spending with a regular preventive maintenance approach. Reactive maintenance can have adverse effects on customer service and may reduce revenue. Predictive maintenance involves testing and monitoring equipment and facilities and predicting potential failures. This method is used in equipment and vehicles. If predictive and preventive maintenance are simultaneously implemented, it will ensure minimal impact on airport operation and safety.

Preventive maintenance is crucial to help improve and extend the life cycle of facilities, systems, and components. An asset management plan implements preventive maintenance and allows an

Copyright National Academy of Sciences. All rights reserved.

asset to reach its maximum life cycle or service life without a reactive approach. Other benefits of preventive maintenance include safety, reduction of rehabilitation cost, reduction in energy usage, legal and regulatory requirement fulfillment, reduction of environmental impacts, and improved community and customer service. To obtain the maximum benefits of asset management, airport owners must account for the asset's performance over its life cycle and decide when to implement preventive maintenance to improve the asset's life. Performing preventive maintenance before the asset condition deteriorates significantly (i.e., needs rehabilitation) can extend the life of the asset. The main objective is to spend the least amount of money by conducting preventive maintenance so that assets can last for longer periods.

As mentioned above, if the preventive maintenance of airport assets is delayed, it will have a negative impact on the life of the assets, as well as on their maintenance and rehabilitation costs. Delaying preventive maintenance of any assets is not preferred. However, due to budget shortfalls and other airport constraints, some asset maintenance must be delayed. Airports delay preventive maintenance of some assets without knowing the consequences of those delays in the total maintenance and rehabilitation costs incurred. There are also no tools or systematic methods to calculate the extra costs incurred in asset maintenance and rehabilitation when preventive maintenance is delayed by a specific number of years. Therefore, this study focuses on determining the impact of delayed maintenance on airport assets' maintenance and rehabilitation costs. It also focuses on answering the following questions related to delayed maintenance of airport assets:

1. How are the airports maintaining their assets during budget constraints?
2. Do airports consider the impact of delayed maintenance of their assets when they need to delay maintenance for a certain period?
3. Do airports have guidelines and tools available to quantify the impact of maintenance delays on maintenance and rehabilitation costs?
4. Are the airports incorporating the monetary impact caused by delayed maintenance on the overall asset life-cycle cost while preparing the maintenance budget?

1.2 Objectives and Scope

Airport owners need to ensure they have sufficient funding to conduct preventive maintenance to optimize the life of their assets. However, airport owners rarely have enough money to conduct preventive maintenance of all these assets. Airport owners should plan their airport budget strategically, taking into consideration new capital construction, as well as preventive maintenance, and the reactive maintenance of existing assets. This process needs to provide steps to efficiently apply the overall airport budget to maximize their assets' life-cycle cost. It is therefore critical to allocate airport budgets effectively so that delayed preventive maintenance will have a minimum impact on the assets' life-cycle costs, aircraft safety and operation, and user comfort. The goal of this research is to prepare a framework, process, and tools that help airports manage their budgets and assets by quantifying the impact of delayed preventive maintenance. The objectives of this research are:

- To identify the maintenance budget preparation processes used by airports.
- To develop a general framework and method to quantify the consequences of delayed maintenance of an airport's airside and landside assets.
- To develop spreadsheet tools to quantify the impact of delays on airport maintenance and rehabilitation costs for specific airside and landside assets.

1.3 Overview of Research Approach

The research for ACRP Project 09-20 builds on the qualitative and quantitative methodology to identify the practices and processes used by airports to determine the impact of delayed maintenance on their budgets. The team divided this research into two phases. In Phase I,

Copyright National Academy of Sciences. All rights reserved.

the research team interviewed the airport asset managers to identify practices in management and maintenance budget preparation, develop a framework and method to quantify the consequences of delayed maintenance, and create tools and models the airport can use to predict the impact of delayed maintenance on airport costs, operations, and users. In Phase II, the research team collected the quantitative data from case study airport projects to build the framework to quantify the impact of delayed maintenance. They also prepared spreadsheet tools to determine the extra maintenance and rehabilitation costs required when maintenance is delayed over certain periods for airside and landside assets. The research team prepared the spreadsheet tools for pavement runways, runway lighting, runway markings, and signs under airside assets. Concerning the landside assets, the research team prepared spreadsheet tools for airport service vehicles, snow removal equipment, HVAC systems, baggage handling systems, and passenger boarding bridges.

The researchers conducted a literature review and online search to collect information relevant to the current state of practices. They also compiled best practices related to the budgeting process and asset management, as well as the impact of delayed maintenance in aviation, building, highway, and transportation assets. The literature review prioritizes the budgeting and maintenance processes and the impact of delayed asset maintenance. Previous research conducted under *ACRP Report 69: Asset and Infrastructure Management for Airports—Primer and Guidebook*, *ACRP Report 138: Preventive Maintenance at General Aviation Airports Volume 2*, and *ACRP Report 139: Optimizing Airport Building Operations and Maintenance Through Retrocommissioning: A Whole-Systems Approach* was reviewed. The team also reviewed similar research conducted under *NCHRP Research Report 859: Consequences of Delayed Maintenance of Highway Assets* regarding highway assets. The information review was useful in developing a questionnaire for Phase I interviews with airport asset managers and the Phase II data collection questionnaire. It also assisted in preparing the framework and spreadsheet tools for quantifying the consequences of delayed maintenance of airport assets.

The research team then conducted virtual interviews with select airport asset managers based on the findings from the literature review and the online search. First, the research team developed the interview questionnaire and sent it to project panel members for their feedback, after which the questionnaire was updated. The research team then contacted asset managers from various airports for interviews based on airport type (large-hub, medium-hub, small-hub, and non-hub) and location. In total, 18 personal virtual interviews were conducted during this phase. The objectives of the virtual interviews were to:

- Identify reports and other documents relevant to the project that were not available through regular literature reviews and sources.
- Identify the state of practices in asset management and maintenance budget preparation.
- Identify the process for developing a framework and method to quantify the consequences caused by delayed maintenance.
- Determine whether the airports have used frameworks, tools, and models to predict the impact of delaying asset maintenance on cost, operations, and user comfort.
- Identify whether airport agencies can provide viable case study data in preparing framework and spreadsheet tools to determine the impact of delayed maintenance on airport overall budgets.
- Gather practical recommendations related to the development of frameworks and tools to determine the monetary impact delayed maintenance has on airport budgets.

The research team then collected quantitative data from six airports related to four airside (pavement, lighting, signs, and markings) and five landside assets (service vehicles, snow removal equipment, HVAC, baggage handling systems, and passenger boarding bridges). The team also conducted follow-up interviews with three airport asset managers to clarify the data provided. These data were used to develop frameworks and spreadsheet tools for the airport assets. The team also used spreadsheet tools to calculate the delayed maintenance cost using hypothetical

Copyright National Academy of Sciences. All rights reserved.

4 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

data. This analysis shows how the impact of delayed maintenance varies based on an asset's condition, age, and the number of years the maintenance is delayed.

The team sent these spreadsheet tools to six airport asset managers so that they could input information from their airports. Out of the six airports selected for validation purposes, three airports responded to the survey.

The airports that responded were satisfied overall with the spreadsheet tools developed by the research team. One airport stated that, when the asset manager used the tools to determine the delayed maintenance cost, the estimated costs were not reliable due to default maintenance costs and inflation cost factors used in the spreadsheet tools. Therefore, the research team has provided the option for airports to enter their maintenance costs.

There were some minor suggestions provided by the airports. One airport stated that the estimate would be more reliable if the assumptions provided in calculating the delayed maintenance cost could be avoided. For this, the real deterioration models of assets need to be collected. One airport stated that, due to high inflation, the maintenance cost estimate will not be reliable if the default cost inflation factor is used. All the airports stated that a user guide would be helpful to use these spreadsheet tools effectively. Based on this feedback, the team prepared a user guide the airports can reference when they have difficulties using these spreadsheet tools. In the guide, the research team also explains the process of using these spreadsheet tools when applying them to similar airport assets. Additionally, the team provides recommendations for gathering the real deterioration data needed to calculate the monetary impact of delayed maintenance. In Chapter 8 of this report, the research team has also provided topics of further research critical to the successful implementation of these products.

1.4 Organization of This Report

This report is organized into eight chapters. The introductory chapter presents background information, research objectives, the scope and approach, and the report's organization. Chapter 2 describes the literature review findings. Chapter 3 focuses on the interview's findings related to airport budget preparation processes, airport maintenance budget allocation processes, and the impact of delayed maintenance of airport assets. Chapter 4 describes a general framework for quantifying the impact of delaying maintenance on airport assets. Chapter 5 provides the procedures developed for quantifying the impact of delayed maintenance of airside assets (airport pavements, lighting, markings, and signs). This chapter also shows the findings obtained from the spreadsheet tools to calculate the delayed maintenance costs with hypothetical data for these airside assets. Chapter 6 provides the procedures developed to quantify the consequence of delayed maintenance on landside assets (airport service vehicles, snow removal equipment, HVAC, baggage handling systems, and passenger boarding bridges). This chapter also shows the findings obtained from the spreadsheet tools to calculate the delayed maintenance costs with hypothetical data for these landside assets. Chapter 7 provides the generic procedures to calculate the consequence of delayed maintenance of other airport assets not included in Chapters 5 and 6. This chapter also describes how the spreadsheet tools developed in this study can be used for other airport assets. Chapter 8 provides a summary and recommendations for future research.

Appendices A through D are supplemental to this report. They include checklists to help airport asset managers identify the maintenance activities needed for air and landside assets.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 2

Literature Review Findings

The research team focused on literature related to the budget prioritization and allocation process, preventive and deferred maintenance, and the impact of delayed maintenance of any asset type, including airports, transportation, buildings, private owners, etc. A summary of the literature reviewed on these topics follows.

2.1 Budget Prioritization and Allocation Process

Budget Allocation. Airport budgets, in the broadest sense, are composed of capital and operating costs, as well as revenue items. Capital costs consist of the construction component costs of the airport, while operating costs are expense items including interest, depreciation, or debt amortization; taxes; and maintenance and administrative costs. The revenue line item in the budget comes from various sources, including rental fees, aeronautical fees, concessions, and parking (Dempsey 2008). For example, the Minneapolis-St. Paul Metropolitan Airports Commission 2020 Report shows the airport's funds to be divided into three main funds as related to specific functions: operating funds (that include maintenance), construction funds [reflecting the Capital Improvement Program (CIP)], and debt service funds. The operating fund reserve is set by the Commission, and, as per policy, should be equivalent to 6 months of operating expenses. Transfers from the operating fund to the debt fund are made twice a year to cover debt service payments and ensure that the debt service reserve accounts are fully funded. At the end of the year, after all operating expenses and debt services are funded, any balance is typically transferred to the construction fund. The Commission follows a process for developing targets and guidelines for its yearly budget, which includes discussing critical issues to be addressed in the short term, monitoring expenses, and looking for new revenue opportunities. It also includes the approval of a seven-year CIP that (1) approves projects to start in the first year of the program, (2) identifies projects that could be reviewed further for feasibility, and (3) identifies projects in the remaining years. The CIP includes the maintenance facility upgrade projects and the ongoing maintenance programs as well (Metropolitan Airports Commission 2020).

Sources of Funds in Airports. Airports receive funding from various sources, including their cities, states, or revenue base. In terms of airport capital development funding, there are five main sources:

1. The Federal AIP, which provides federal grants to airports for planning and development, mostly for capital projects related to aircraft operations, such as runways, taxiways, and land purchase, but not for commercial revenue-producing facilities or operating costs (FAA 2019). Priority is high for AIP projects that increase capacity and safety for aircraft movements (Warren 2002).
2. Local passenger facility charges (PFCs) imposed, per federal law, on each boarding passenger. PFC revenues can be used for a broader range of projects compared to AIP funds, which could include landside projects, such as terminals and ground access improvements.

Copyright National Academy of Sciences. All rights reserved.

6 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

3. Tax-exempt bonds.
4. State and local grants.
5. State and local airport operating revenue from tenant lease and other revenue-generating activities.

Small airports are usually more dependent on AIP grants for their capital needs, compared to large- and medium-hub airports. Large airports rely more on tax-exempt bonds, or finance capital projects with the proceeds from PFCs. Different combinations of these sources (AIP funding, PFCs, tax-exempt bonds, state and local grants, and airport revenues) are used to finance projects, depending on each airport's financial situation and the project being considered (Dempsey 2008, Tang 2019). In a 2015 study, the U.S. Government Accountability Office (GAO) found that capital spending at airports was financed by airport-generated net income at 38%, AIP at about 33%, and PFCs at around 18%. Capital contributions by a sponsor, such as a state, municipality, airline, or tenant, were financed at around 6% and state grants at almost 5% (GAO 2015).

Private sector financing is a common emerging theme in airport financing given its profound advantages in transferring the risk of long-term operations and maintenance. It contractually binds a private party to ensure long-term operations and maintenance based on depicted, pre-negotiated performance requirements, with less liability to the public sector [American Society of Civil Engineers (ASCE) 2017]. Such a public-private partnership (P3) setting also ensures the developer looks at design with a life-cycle approach to minimize operations and maintenance costs. With public agencies retaining safety and regulatory oversight, it reduces the possibility of deferred maintenance, as that would have negative implications for the developer (Rapoport 2018).

Budget Prioritization Process. Airports have been successfully using existing systems to focus on managing their main assets, primarily pavements. Pavement management systems aid decision-makers in setting priorities, allocating resources, and systematically distributing funds. They also offer life-cycle cost analysis by considering different maintenance options (Moayedfar and Sajjadifard 2021). The literature review revealed that there is limited research addressing budget prioritization at airports. The studies found were mostly related to highways. For example, a study conducted on Virginia Department of Transportation bridges was focused on improving the process of allocating bridge maintenance resources in their programs and helping decision-makers effectively allocate the maintenance budget. The bridge prioritization methodologies developed utilized statistical and probabilistic risk analyses when identifying the attributes for evaluating bridges (such as average daily traffic and bridge condition). In addition to maintenance, economic and political attributes supported by sensitivity analyses were also included to prioritize bridge portfolios (Gokey et al. 2009). Such a tool could be generalized to other infrastructure, including airports.

2.2 Airport Asset Management

ACRP Report 69: Asset and Infrastructure Management for Airports—Primer and Guidebook prepared a guidebook for overall airport asset and infrastructure management (GHD Inc. 2012). This guidebook is composed of case study examples of various airports' asset management programs and their relationships with daily operations and management and longer-term capital investment planning. It also covers asset management strategies, as well as asset information and data system requirements for an efficient plan. The guidebook explains how to consider asset life-cycle costs when making an investment plan. It also proposes a performance management framework for an asset management system; performances are monitored and managed from a whole-of-airport standpoint, and at asset level.

In addition, a framework has been developed to prioritize capital asset rehabilitation and replacement decisions for transit agencies (Spy Pond Partners, LLC et al. 2012). *TCRP Report 157: State of*

Copyright National Academy of Sciences. All rights reserved.

Good Repair: Prioritizing the Rehabilitation and Replacement of Existing Capital Assets and Evaluating the Implications for Transit is a guidebook that can be used by asset managers for prioritization of new and rehabilitation projects. The guide has seven steps: (1) collect asset condition and performance data, (2) analyze data, (3) identify rehabilitation and replacement alternatives, (4) define investment scenarios, (5) prioritize projects, (6) develop an investment plan, and (7) implement the plan. Using these steps, transit agencies can select projects that best achieve their goals.

ACRP Research Report 172: Guidebook for Considering Life-Cycle Costs in Airport Asset Procurement highlights the importance of understanding, integrating, and evaluating the total cost of ownership in the procurement of assets (Fortin et al. 2017). This guidebook is helpful for procurement personnel (personnel responsible for purchasing assets) to determine how to consider life-cycle cost in addition to initial purchasing cost. Airports need to use an asset's total cost of ownership (TCO), which includes operations and management costs. The report developed an Excel-based tool that can be used by purchasing officers to determine the TCO of alternative assets during the bidding phase.

Further, ISO 55000 (2024), an international standard related to asset management principles, can also be used in airport asset management. According to ISO 55000, the objectives of any organization's asset management depend upon the nature and purpose of the organization, along with its operating context, financial constraints, and regulatory requirements, as well as the needs and expectations of its stakeholders. These factors must be considered when establishing and implementing an asset management plan. It states that asset management is based on three fundamentals: (1) asset value, (2) alignment, and (3) leadership. Every organization values its assets and needs to focus on the value the asset can provide them, rather than the asset itself. According to ISO 55000, "Good asset management translates organizational objectives into technical and financial decisions, plans, and activities." Organization leaders must value the creation, implementation, and improvement of their asset management systems. The major components of an asset management system are as follows:

- Organizational context,
- Leadership roles,
- Asset management planning,
- Organizational support,
- Asset management plan implementation,
- Asset performance evaluation, and
- Continuous improvement of an asset management plan.

2.3 Preventive and Deferred Maintenance

Public and private agencies can use various types of maintenance to keep their assets in good condition. Maintenance can be categorized into two types: proactive maintenance and reactive maintenance (Upkeep 2021). Proactive maintenance is planned maintenance, in which assets are maintained based on time used or predictive deterioration models. One of the main proactive maintenance types is preventive maintenance, which can increase asset life. In preventive maintenance, maintenance tasks are scheduled based on time used or with deterioration trends, rather than actual conditions. Sometimes, preventive maintenance is called routine maintenance. Another type of proactive maintenance is predictive maintenance, in which the maintenance tasks are scheduled based on actual asset conditions. Note there is significant variation between different industries and agencies concerning how these terms are used and how specific maintenance activities are classified.

Reactive maintenance is performed when assets need immediate maintenance that is not planned. There are two types of reactive maintenance: (1) emergency maintenance; and (2) corrective

Copyright National Academy of Sciences. All rights reserved.

8 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

maintenance. "When an asset requires immediate attention in order to keep a facility operational or safe," reactive maintenance is used (Upkeep 2021). Corrective maintenance is generally conducted when the assets need corrective actions, rather than replacement. This is also a part of emergency maintenance.

In industry, maintenance is frequently deferred due to a lack of resources, so organizations generally delay the preventive maintenance of assets which adds to backlog. This is called deferred maintenance. Many organizations are using a "run-to-fail" approach for their assets due to a lack of funds and human resources (Penny 2018). A common rule of thumb is that a run-to-fail approach can cost 3 to 10 times as much as an approach that emphasizes proactive maintenance. Therefore, research has found deferred maintenance to be very expensive for owners.

ACRP Report 138: Preventive Maintenance at General Aviation Airports Volume 1: Primer is a primer on the importance of preventive maintenance programs for airport assets (Ploeger et al. 2015a). The report includes a guide that can be used by airport asset and maintenance managers to plan, prioritize, and conduct preventive maintenance. According to the report, airport maintenance is generally divided into four types: (1) operational, (2) reactive, (3) preventive, and (4) predictive. Some maintenance activities are related to airport operations, including removing ice, fixing lights, cutting grass, sweeping pavement, maintaining escalators and elevators, and more. These operational activities are also driven by regulatory and airfield safety requirements, as provided in FAA advisory circulars. Note that a maintenance strategy that relies exclusively on reactive maintenance is equivalent to the "run-to-fail" approach described above. An HVAC system that is not regularly maintained and is used until it fails and needs replacing is one example. However, a strategy that includes preventive maintenance consists of scheduled asset maintenance activities that collectively extend asset life. Generally, in preventive maintenance, the maintenance schedule is prepared based on the number of hours an asset is used. Preventive maintenance reduces the life-cycle cost and improves user satisfaction. On the other hand, predictive maintenance requires assets' conditions to be monitored, and, based on these conditions, maintenance tasks are scheduled. For example, some airports monitor and detect the symptoms that can predict system failure and work to proactively increase the life of vehicles or equipment.

According to *ACRP Report 138: Preventive Maintenance at General Aviation Airports Volume 2: Guidebook*, airports need to prepare a preventive maintenance program that consists of the following components (Ploeger et al. 2015b):

- Prepare guiding principles of a preventive maintenance program;
- Maintain an inventory of airport assets;
- Assess asset conditions;
- Identify and schedule preventive maintenance tasks;
- Prioritize the preventive maintenance of assets based on safety, economics and operations, regulatory and FAA grant assurance requirements;
- Obtain the resources for preventive maintenance; and
- Implement a preventive maintenance program.

One of the obligations for any airport is to obtain airport operating certification from the FAA. *14 CFR Part 139* requires any airport that serves scheduled and unscheduled aircrafts with more than 30 seats, and air carrier operations in aircrafts with more than nine seats but less than 31 seats, to have a certificate (FAA 2021). This certification provides proof that the airport ensures safety in air transportation. To receive this certificate, airports must fulfill certain operational and safety standard requirements set by the FAA. To verify whether airports are meeting certificate requirements, FAA airport certification safety inspectors make inspections every year. During inspections, the review includes, but is not limited to, the following:

Copyright National Academy of Sciences. All rights reserved.

- Meeting with various airport personnel.
- Reviewing airport files and paperwork including, but not limited to, FAA Form 5010, Airport Certification Manual/Specifications, and Notices to Airmen (NOTAM).
- Checking the runway, pavement conditions, markings' lighting, signs, vehicle operations, jet or propeller blasts, and wildlife presence.
- Inspecting aircraft rescue and firefighting.
- Inspecting fueling facilities.
- Conducting a night inspection.
- Having a post-inspection briefing with airport management.

Due to *Part 139*'s requirements for any assets related to the safety and security of the airport's operations, the airport needs to maintain these assets in very good condition. Therefore, most airports keep inventory and condition assessments of assets inspected by the FAA to obtain airport certification, as well as those needed to meet the requirements for an FAA grant. All airports should have up-to-date records of these assets, as FAA inspectors can conduct inspections at any time of the year.

Airports need to have proper maintenance strategies to reduce downtime, so that customers can be served quickly, safely, and efficiently. A diligent workforce, as well as a robust computerized maintenance management system (CMMS), are needed to align the maintenance activities of airport assets (Eagle 2021). The main tasks of CMMSs are to:

- Manage asset and spare parts inventory,
- Record asset condition data,
- Plan and schedule maintenance tasks,
- Track and manage maintenance work orders,
- Track asset performance, and
- Comply with airport safety and environmental regulations.

Due to a lack of funds and human resources, an organization may have to defer maintenance of its assets. Organizations do not want to defer maintenance because it will reduce asset life, as well as increase asset life-cycle costs. Research has shown that investing $1 in pavement preventive maintenance at the appropriate time can save $4 to $10 in future costs (Agile Assets 2021). Therefore, if the pavement is not maintained, it will cost users more than its maintenance cost. One article making the case for spending more on infrastructure maintenance stated that "deferring maintenance makes fixes more expensive later" (Olson and Wessel 2017). It has been found that, due to bad pavement conditions, "the extra car repairs are equivalent to somewhere between a 50 cent to $1 tax per gallon of gasoline" (Olson and Wessel 2017). Therefore, it was suggested that all gas tax revenues be invested to repair, rehabilitate, and reconstruct existing roads and bridges. Literature has shown that deferred maintenance has negative consequences on the life of assets, which will increase assets' life-cycle costs. However, little research has been done to quantify the consequences of deferred maintenance on any type of asset. The Geaslin Group developed an inverse-square rule-of-thumb for deferred maintenance efforts (The Geaslin Group 2021). The rule states that "if a part is known to be failing but operated to failure (OTF), the resultant energy required to overcome the breakdown event to the entire organization will be the square of the cost of the primary failure part. If the breakdown event escalates, the energy required to recover from the breakdown will continue to square at each successive level of failure." An example of a car's brake shoe failure was used, with the study explaining that, if the brake shoe, costing $40, is not replaced on time, then it can cause an accident, which could cost $1,600 ($40 square) in mechanical and towing expenses. If $1,600 of mechanical and towing costs is squared, the result is $2,560,000, which is what the company needs to set aside for a personal injury lawsuit. However, relating these numbers to an actual quantification of delayed maintenance is a difficult task.

NCHRP Research Report 859: Consequences of Delayed Maintenance of Highway Assets determines the consequences of the delayed maintenance of seven highway assets, including (1) pavements,

Copyright National Academy of Sciences. All rights reserved.

10 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

(2) bridges, (3) culverts, (4) guardrails, (5) lighting, (6) pavement markings, and (7) highway signs (Chang et al. 2017). The research team identified the maintenance activities, asset performance objectives, maintenance activity decision criteria, and current asset conditions, as well as performance models for forecasting asset conditions and needs analysis to determine the consequences of delayed maintenance. Based on the asset conditions, various scenarios were created to calculate the consequences of delayed maintenance. For pavement, four delayed maintenance scenarios were created: the first is an "all needs" scenario, determined by preserving the current condition; the second is to do nothing; the third is delayed maintenance by 2 years; and the final is budget-driven with limited funds. Specific consequences considered in the research include increased agency costs, changes in the backlog of agency investment needs, changes in asset condition, increased user costs resulting from more frequent asset failures, and/or deferral of investments.

Another study conducted by Hatry and Liner (1994) developed a methodology to quantify the consequences of deferred maintenance on highway assets. The study investigated the impact of deferred maintenance of pavement treatments on agency cost, pavement performance, and cost-effectiveness. The consequence of deferred maintenance of pavement varied based on asset class and intervention types. The increase in life-cycle cost and performance loss of delayed pavement maintenance was more severe for non-interstate highway compared to interstate highways, as well as for maintenance compared to rehabilitation. Similarly, another study found that considerable cost savings could be achieved if the pavement is maintained while in good condition (Schmuck and Maerschalk 1987). The maintenance cost increases tremendously if pavements are allowed to deteriorate into poor conditions.

In summary, the literature review found that the consequences of delayed maintenance on airport assets had not been investigated by any agencies. Therefore, the main objective of this report is to fill this research gap by developing a systematic process for the quantification of delayed maintenance on various types of airport assets.

2.4 Summary of Relevant Research Publications and Key Findings

Table 2-1 shows a summary of the publications pertinent to the various topics covered in the previous sections. The research applicable to budget prioritization was limited to highway infrastructure only. Several studies related to airport asset management were found during the literature search. However, literature specifically related to quantifying the impact of deferred maintenance on airport, building, transit, water, and wastewater assets was not found during the literature

Table 2-1. Summary of relevant publications.

Topics	Sources
Funding sources	Dempsey (2008), FAA (2019), Rapoport (2018), Tang (2019), Warren (2002)
Budget allocation	Government Accountability Office (2015), Metropolitan Airports Commission (2020)
Maintenance budget prioritization	Gokey et al. (2009), Moayedfar and Sajjadifard (2021)
Airport asset management	GHD Inc. (2012), Fortin et al. (2017), ISO (2024)
Preventive and deferred maintenance	Burns et al. (2015), ASCE (2017), Penny (2018), Upkeep (2021), FAA (2021), Eagle (2021), Agile Assets (2021), Olson and Wessel (2017), The Geaslin Group (2021)
Impact of deferred maintenance	Hatry and Liner (1994), Chang et al. (2017), Schmuck and Maerschalk (1987)

Copyright National Academy of Sciences. All rights reserved.

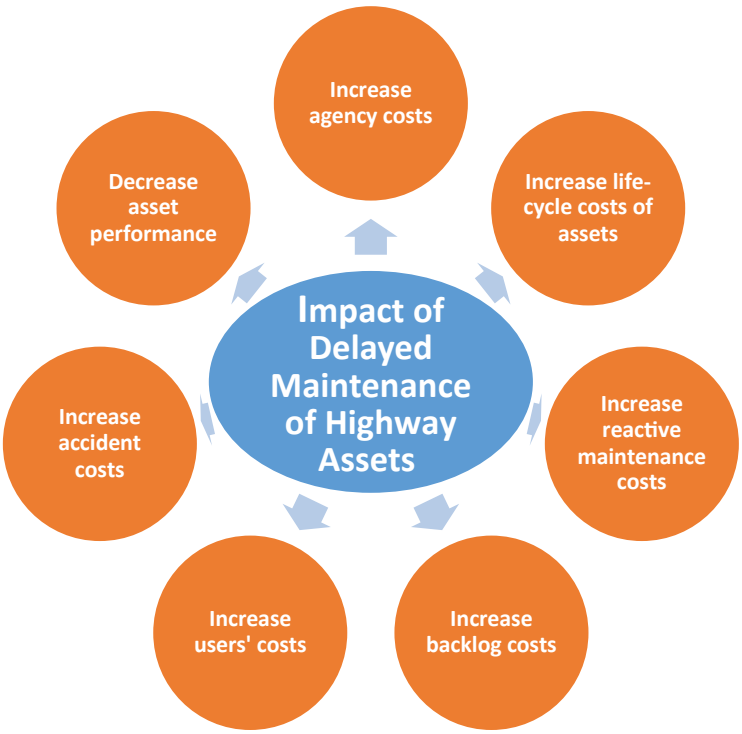


Figure 2-1. Impact of delayed maintenance of highway assets.

review. Some studies found that deferred asset maintenance has negative consequences on asset life-cycle costs. Three studies quantified the impact of delayed maintenance on highway assets. However, none of the studies have quantified the impact of delayed maintenance on airport assets. Therefore, the research on this topic for airport assets seems relevant.

Based on the literature review, the impact of delayed maintenance on highway assets is shown in Figure 2-1. *NCHRP Research Report 859: Consequences of Delayed Maintenance of Highway Assets* is a key reference because the study is comprehensive for highway assets. However, this project's focus is on airport assets that were not addressed in that report.

The key findings of the literature review are as follows:

- The major funding sources of airports are the AIP program, PFCs, tax-exempt bonds, state and local grants, and other revenue, e.g., tenant leases.
- Small-hub airports are more dependent on the AIP program, and large-hub airports' main sources of revenue are PFCs.
- For capital budgeting, airports mostly depend upon their long-term capital CIP.
- The operating budget, which includes maintenance, is set by the commission's rule and should be enough for at least 6 months.
- The pavement and bridge management system helps prioritize the maintenance budgets for pavement and bridges, respectively.
- The airport asset management studies focused more on prioritizing assets for maintenance purposes and using life-cycle costs while purchasing the assets instead of initial costs.
- The impact analysis found that deferred maintenance on highway assets increases costs associated with agency, life cycle, reactive maintenance, backlog, users, and accident costs, and decreases an asset's performance.
- The impact of delayed maintenance has been investigated primarily for highway assets.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 3

Airport Interview Findings

The research team prepared a questionnaire for individual interviews, which has three sections: (1) the airport budget preparation process, (2) airport maintenance budgeting allocation processes, and (3) the consequences of delayed preventive maintenance of airport assets. The first section has three questions related to the overall budget process for capital, operation, and maintenance. The second section consists of six questions related to maintenance budgeting processes, along with methods of allocating preventive maintenance budgets for airport assets. The third section consists of five questions related to the impact analysis of delayed asset maintenance.

The research team started contacting airport asset managers on March 10, 2021. By May 24, 2021, a total of 18 individual interviews with various types of airport asset managers were collected. The interviews were conducted in three different modes: (1) face-to-face; (2) virtual, via Webex and Zoom; and (3) telephone. Because the research team is located throughout the United States, responses were collected from all regions. Figure 3-1 shows the distribution of interviewees based on region. The highest and lowest number of responses were received from the West and Southwest, respectively.

Interviewees from all four types of airports, including small-hub, medium-hub, large-hub, and non-hub, provided interviews to the research team members. Figure 3-2 shows the percentage distribution of the types of airports whose representatives were interviewed. The highest number of interviewees were from small-hub airports, and the fewest number of interviewees were from medium-hub airports.

The team focused on interviewing the airports' asset managers. These asset managers generally work in operations, maintenance, and facilities divisions. Table 3-1 shows the number of interviewees from various airport departments. The highest number of interviewees were from operations and maintenance departments. All of the interviewees were working either in maintenance or operations and were also involved in the budgeting process of the airports' operations and maintenance. However, the names of these departments vary by airport.

The main interview findings are summarized into three sections. The first section consists of the findings related to the airport's budgeting process, including the capital, operations, and maintenance of assets. The second section includes the findings related to the budgeting processes involved in preventive maintenance of the assets, and the last section covers the findings related to the impact of delayed preventive maintenance on airport assets.

3.1 Airport Budget Preparation Processes

Airports prepare their total budgets based on fiscal year. The total budget is divided into a capital budget and an operations budget. The preventive maintenance budget is generally included in the operations budget. During the interview process, the research team found that most of the airports

Copyright National Academy of Sciences. All rights reserved.

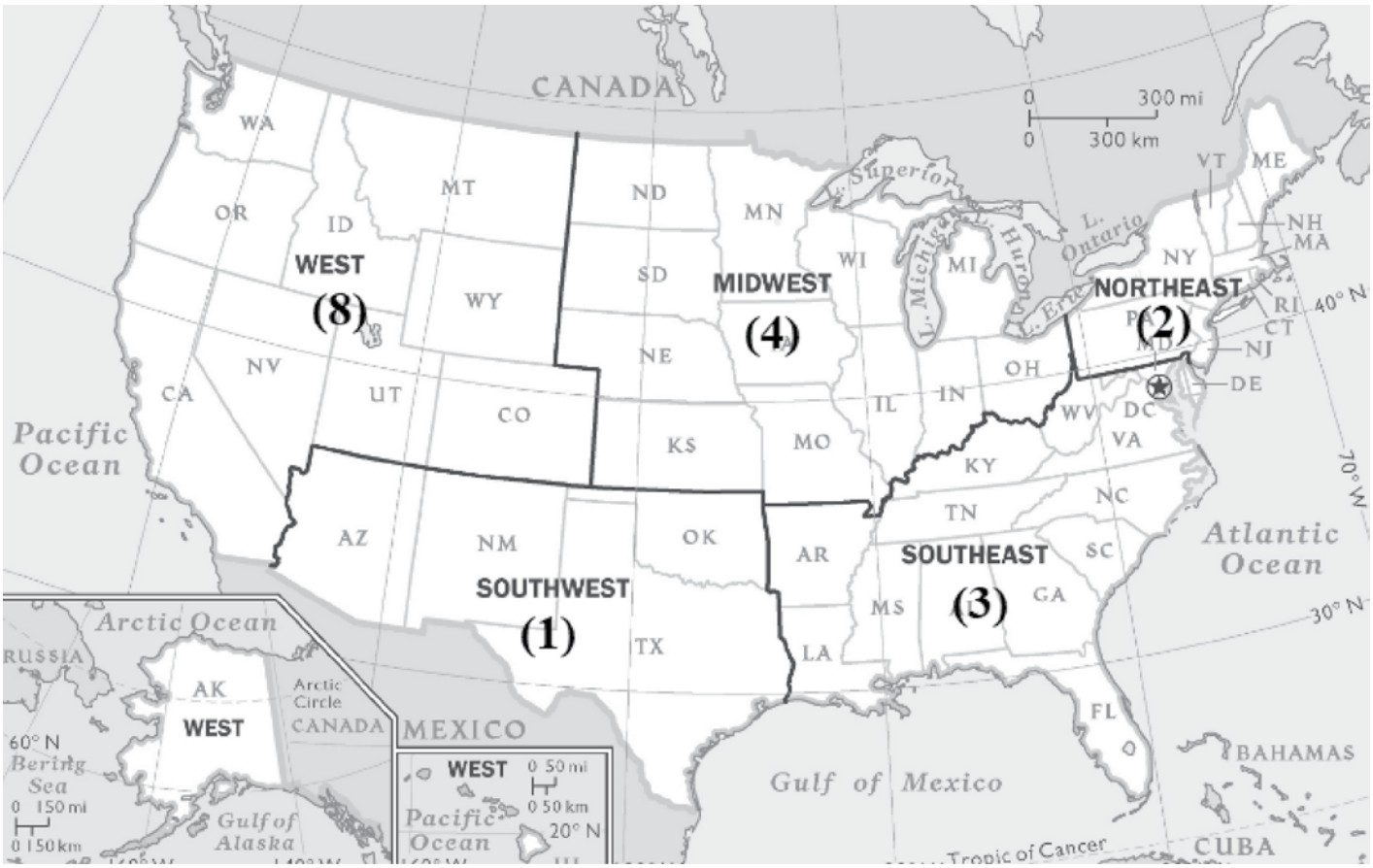


Figure 3-1. Distribution of interviewees based on U.S. regions.

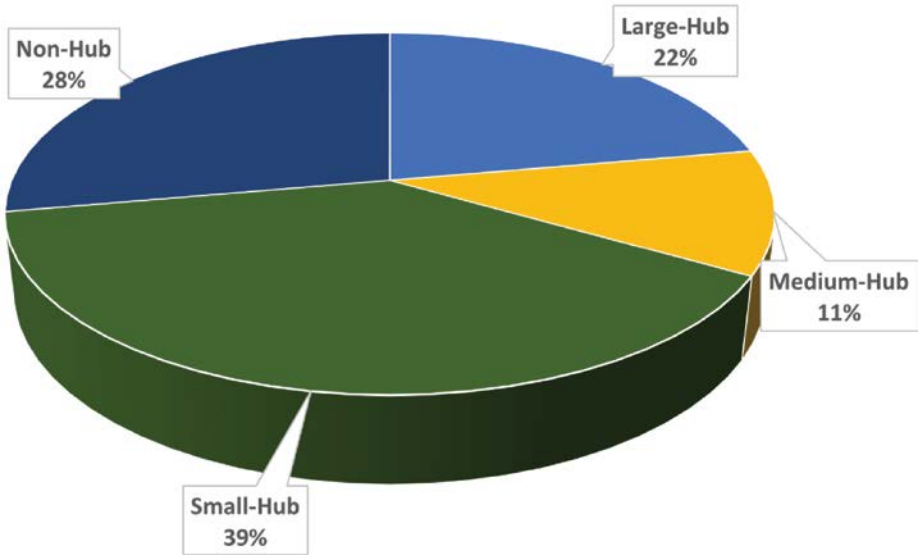


Figure 3-2. Types of airports interviewed.

Copyright National Academy of Sciences. All rights reserved.

14 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 3-1. Departments of interviewees.

Division/ Department	No. of Interviewees
Facilities Maintenance	4
Operations and Maintenance	6
Maintenance	5
Engineering, Planning, Construction, and Operations	3

(11) estimated their capital investment projects budget, and, based on that budget, they allocated their operations and preventive maintenance budgets. An airport's budget generally relies on its city or state, while some capital and operations and maintenance (O&M) budgets come from the FAA's AIP program. Some airports stated that their total budget usually depends upon the city and state legislature because they approve the budget. However, most of the airports prepared their capital investment and O&M budgets and sent them to cities or states to get approved. Some airport budgets are prepared based on the revenues collected during each fiscal year, but some general aviation airports do not generate revenues, and their budgets are provided by the state. One airport mentioned that their city used enterprise management software to help in the budgeting process.

Most of the airports agreed that their O&M budgets somehow depend upon the capital improvement budget. If there is much capital investment in their airport, then it will impact the O&M budget during that fiscal year. However, the capital and O&M budget provided by the FAA is independent of the revenues generated by airports. The budget provided to maintain assets, as per *14 CFR Part 139*, is based on whether the airports have fulfilled their grant assurance requirements. If the airport fails to meet these requirements, they become ineligible for FAA AIP funding.

Some airports have developed 5-year or 10-year master and capital improvement programs, identified all their needs, and allocated funds over those years. O&M departments also participate in CIP program development, so that the budget required for O&M can be included in that program. Some state-funded airports said that they do not receive all their requested funds because legislatures need to allocate their budgets to other areas. Some airports mentioned that their overall budgets were cut by up to 10% due to the COVID-19 pandemic. Due to budget cuts, some airports have had difficulty fulfilling the *Part 139* requirements for their assets.

When asked about strategies to allocate their budgets for capital and O&M, most airport representatives stated that the CIP is prepared based on the evaluations of asset conditions and asset life. Some airports meet about 6 months before the fiscal year ends and prepare the capital investment plan and O&M budget based on feedback from different departments; this planned budget is then sent to a higher authority for approval. Most airport representatives stated that the capital investment budget is based on the needs of new assets, and they seek input from all departments when preparing this type of budget.

One airport has developed a points-based system to allocate the replacement or capital investment budget based on an asset's importance to security, efficiency, and customer experience, as well as the asset's condition. The asset importance is ranked from 1 to 4 based on security, efficiency, and customer experience, with 4 being highly important and 1 being less important. Similarly, the assets' conditions are also ranked from 1 to 3, with 1 being very new and 3 being very old. Based on these two factors, a risk matrix is developed by multiplying the ranks of these two factors. The highest risk-value assets will get higher importance for providing a replacement, or in the capital budget, during the budgeting process. This airport uses a similar process to allocate the maintenance budgets for its assets.

The maintenance division for one of the large-hub airports surveyed developed its own 10 goals (such as customer experience and environmental and sustainability), and they create the budgets

Copyright National Academy of Sciences. All rights reserved.

accordingly to align with these goals. The budget is also developed using a bottom-up staffing model that determines the support work necessary for each group working on a maintainable asset (e.g., HVAC staff would have a full-time equivalent calculated based on that asset). The budget is mostly leveraged by historical trends, and fixed increases are added to the previous years' budget, accounting for existing facilities. Resources for new expansions of facilities or systems are additionally requested. In parallel, the maintenance division also assesses the capital development needs for the airport, which is a five-year rolling program. This assessment involves a much larger group, including an investment committee that includes operations, security, commercial management, airport dining and retail, and properties teams.

The budget, once allocated to maintenance, is used in a broad range of areas based on priority needs. Maintenance works off moving targets (not static), whether in terms of regulatory compliance or level of service provided. Even though it might be easy to prioritize the assets in terms of importance, the major challenge for this large-hub airport is when assets are of equal importance. In this scenario, criticality measures become the hardest to assess because definitions of what is critical vary.

The majority of airport representatives agreed that they do not have fixed preventive maintenance budgets for their assets. Some airports allocate preventive maintenance budgets based on departments or asset types. The airport divides asset types based on working groups or cost centers. Any assets related to electrical distribution will fall under the electrical group (lighting in roadways, interiors, ramps, airfields, etc.). The pavement maintenance management group will cover all the activities related to maintenance of the airfield's pavement (runways, taxiways, ramps, etc.). Some airports categorize assets based on cost centers (maintenance management, buildings, airfields, utilities, etc.). Other airports allocate their preventive maintenance budgets based on the condition, age, maintenance history, and work orders of those assets. When asked about written strategies developed by the airports, none could provide documentation of this process.

When asked about their budgeting processes aligning with maintenance objectives, the majority of airport representatives stated that their budgeting process aligns with their maintenance objectives. Some interviewees mentioned that their airports have developed priority lists of assets to maintain every year, and they follow that list when allocating maintenance budgets. Other interviewees mentioned that their airport's priority is to maintain the assets required by *Part 139* because the funding for those assets is provided federally. If grant assurance and *Part 139* requirements are not fulfilled, then the airport will become ineligible for the FAA AIP funding. Some interviewees stated that they provide enough maintenance funds for their assets associated with customers' safety and security. One airport has a tiered budgetary category for maintenance, e.g., an annual recurring maintenance and replacement schedule. Representatives from three airports stated that their budgeting processes do not align with maintenance objectives because their states do not provide enough funds to maintain assets. Some of the interviewees agreed that, due to COVID-19, their budgeting processing could not align with their maintenance objectives because their maintenance budgets were significantly reduced in the last fiscal year.

The airport representatives were asked for suggestions to improve their airports' budgeting processes. Out of representatives from 18 airports, three stated that they are happy with the budgeting processes they have in place now. They stated that their budgets were prepared based on all the departments' inputs, and they go through different levels before being approved. Therefore, the process is detail-oriented, and there are checks and balances in their budgeting processes. Representatives from five airports noted they need to have very good asset condition data in the CMMS, as well as the capability to analyze the data to predict the capital and maintenance budgets of their airports. One interviewee mentioned that they want to prepare a health condition index of an asset that includes various factors with weights to allocate the maintenance, capital, or replacement budget for the asset. This suggestion can be related to one airport's effort to design a risk

Copyright National Academy of Sciences. All rights reserved.

16 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

matrix when preparing its overall budget. Some interviewees mentioned that there should be better communication and working environments between different departments during the budgeting process. Every department believes that they need a higher budget than another department. Representatives from two airports mentioned that the budget approval authorities should visit the airport before approving the budget so that they are aware of the airport's real situation. Finally, three interviewees suggested that the legislature needs to be kept away from the budgeting process.

One of the airport representatives interviewed reported that they had recently added a new role in their maintenance division that functions as an "enterprise asset manager." The goal of this role is to develop a formal condition assessment and forecasting tool that can plan for renewal and replacement, as well as keep track of assets' conditions from an engineering perspective while working closely to feed maintenance with the information collected and recommendations made, which should align with the airport's overall goals. The first step in using such a tool is determining the process for standardizing the airport's inventory. Currently, the airport is using many inventory management systems that handle various individual pieces (work orders, accounting, and engineering drawings), but they do not have a comprehensive inventory system, which is the basis for any condition assessment modeling tool. The new system will be based on real data condition assessments and facilities planning, and will optimally help identify the asset condition, location, and function, among many other things, so a comprehensive assessment that can advise maintenance efforts can align the inventory to the objectives. It will become a long-term forecast and planning tool that can provide projected replacement dates and information to better inform maintenance decisions. Such a tool will analyze the data to justify the decisions recommended in advance of sponsorship-identified scope and kickoff, including financial modeling of the various options (refurbish, replace, or maintain). It will become a tool that connects various specialties from their working silos (engineers, financial analysts, etc.) and thus, improve coordination between them.

3.2 Airports' Maintenance Budget Allocation Processes

The majority of surveyed airports (14) have budget allocation processes for preventive maintenance. They allocate maintenance budgets based on asset classes. All the airports divide their assets into various classes, some based on airfield, landside, equipment, and building, while others follow the asset classes provided in the CMMS. Some airports use IBM Maximo systems to categorize asset classes. It was also found that some airports divide their assets into three main categories (high, medium, and low priority) based on their importance and regulatory *Part 139* requirements, with high-priority assets including ARFF equipment, fire alarm systems, life safety systems, safety vehicles, runways, and snow removal equipment. HVAC systems, baggage handling systems, and vehicles were categorized as medium priority based on their importance and impact on the safety and security of the airports' customers. Some airports also use a cost center for their equipment, which is based on asset function, e.g., airfield, terminal building, parking, mechanical, electrical, and information technology. Some airports divide their assets into ground, building, equipment, and miscellaneous categories, while others list assets per department, e.g., facilities, airfield, fleet, etc. Finally, some use a multiple-asset organization structure system that accommodates multiple needs from various perspectives (managerial, maintenance, etc.). Their systems encompass the Uniformat by type (which is associated with Building Information Management and Construction Specification Institute Master specifications, as well as associated with RSMeans) and Global Positioning System by location, as well as including a managerial reporting system. This does not become a single, fixed hierarchical system, but a fluid one that allows for reporting from various perspectives: department, location, and/or class.

Every year, airports develop O&M budgets based on the asset classes they have categorized. Before preparing the maintenance budget for each asset class, the asset managers, or the department

Copyright National Academy of Sciences. All rights reserved.

or group responsible for the maintenance of those assets, will prepare the budget and submit it to the airport authority. Some airports have work order management systems, and they track the required maintenance budgets from those systems. While estimating preventive maintenance budgets, the asset managers rely on their experience and asset condition data. All the surveyed airports have Pavement Condition Index (PCI) data to determine their maintenance budgets for pavement. For vehicles and equipment, the preventive maintenance budget is prepared by following the manufacturer's O&M guidelines, the equipment's life or operation hours, or the vehicle's mileage.

Most airport representatives agreed that they do not have a robust system to estimate the preventive maintenance budget of the terminal building, as compared to pavement maintenance. One airport provided the preventive maintenance budget for airfield and building maintenance based on percentage. An interviewee also mentioned that they provide budgets uniformly across all asset classes. One airport has developed a risk matrix to provide the maintenance and replacement budget for their assets. The risk matrix has two components: one is the weight provided to an asset's importance, and the other is the weight provided to the asset's life. The combination of those two components provides the quantitative value used for the budgeting process.

Representatives from two surveyed airports mentioned that they do not have any systematic approach to budget for asset maintenance. Further, the research team did not find any documentation from the airports about the maintenance budget allocation process for airport assets. However, one airport has developed a capital asset management plan that uses a risk matrix to allocate their replacement assets, as well as for maintenance purposes.

Very few airport representatives stated that their annual maintenance budgets were enough to maintain their assets. Therefore, most airports prioritize assets for maintenance based on their importance to security, regulatory requirements, and passenger comfort. The priority of maintenance is given to any assets that are covered by regulatory requirements (*14 CFR Part 139* or other state requirements). According to most of the interviewees, the highly prioritized assets for annual preventive maintenance budgets are life safety and security, fire suppression systems, electrical distribution, runway pavement, and terminal building HVAC systems. For other assets, if the budget is low, they try to defer maintenance but do not completely ignore the assets. Maintenance frequency is reduced in some assets such as jet bridges, baggage handling systems, moving walkways, service vehicles, taxiways, ramps, etc.

Some airports have a process of providing high, medium, and low prioritization to assets so that maintenance budgets can be allocated accordingly. The representatives from most surveyed airports stated that they provide the budget for those assets that the FAA requires they maintain to get federal funding for those assets. Two airports mentioned that they budget preventive maintenance based on their needs in any given year and start it from scratch. Every year they have enough funds for the maintenance of every asset, which is not true for other airports. One of the airports stated that they build a contingency fund every year so that, if any emergency maintenance or replacement scenario arises, they can use that fund to maintain or replace the assets.

The majority of the airport representatives (15) stated that they prepare the maintenance budget based on asset condition data. However, the extent of data collection and utilization for each asset varies. All airports keep PCI pavement data every 3 years and use it to prepare the preventive maintenance budgets for runways, taxiways, and ramps. Most interviewees also mentioned that they keep asset condition data and use it for budgeting purposes for assets mandated by *Part 139*, as well as those tied to federal grant assurances, but agreed that they do not have comprehensive asset condition data or software, or processes to calculate the preventive maintenance budget of every asset based on asset conditions. Some asset condition data are used to prepare maintenance budgets for airfields, vehicles, HVAC, air handlers, escalators, elevators, snow removal equipment, electrical equipment, etc.

Copyright National Academy of Sciences. All rights reserved.

18 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

For lower-level assets, the airports do not have comprehensive asset condition data. Maintenance budgets for those assets are prepared based on asset managers' experience and discretion, tracking work orders in CMMSs, and forecasting their preventive maintenance requirements. Three airport representatives stated that they do not prepare their maintenance budgets based on asset conditions. They told us that the maintenance budget should be prepared based on the asset conditions, but it is at the discretion of the asset managers and departments responsible for those assets.

When the interviewees were asked about the sufficiency of their preventive maintenance budgets for assets in fair condition, three responded negatively, and the rest stated that they had enough to operate their airports safely and efficiently. Even those who responded that their budgets were too low stated that they could operate their airports safely. Some airport representatives reported that, even though their budgets were high enough to operate the airports, they have had to delay maintenance, and there is a backlog for asset maintenance. The interviewees stated that they need to maintain the assets required by regulatory requirements, *Part 139*, and should not jeopardize the safety of customers. Most interviewees said that they maintain primary assets, e.g., pavements and life safety, in very good condition, but, for secondary assets, they might have to defer maintenance due to a lack of funds. However, one airport reported experiencing an organizational shift in how they respond to maintenance needs, moving from a reactive to a proactive approach. For example, for one of their new assets with a direct impact on their main goals, the terminal parking garage, they now have a detailed asset management system, encompassing all the various system condition ratings (structures, HVAC, etc.).

3.3 Impacts of Delayed Preventive Maintenance of Airport Assets

When the interviewees were asked whether they had delayed the preventive maintenance of their assets, 13 said that they had delayed some that were not their primary assets. The airports have not intentionally delayed asset maintenance; however, there are times, due to funding restraints, when airports must make necessary adjustments to their existing budgets and prioritize maintenance. In some cases, emergency repairs to crucial assets, such as ARFF vehicles, grass-cutting tractors, painting machines, and self-inspector vehicles, must be adjusted due to these constraints. These airports comply with the FAA's *Part 139* requirements but can face Letters of Investigation and may face civil penalties from the FAA. Therefore, assets like runways, life safety, fire alarm systems, and fire suppression, along with those required by the regulatory requirements (*Part 139*) and tied to federal funding, were not delayed. However, five airport representatives stated that they had not delayed the maintenance of any assets.

The airports that have delayed asset maintenance have not completely ignored it, but deferred maintenance frequency. One of the airports stated that they intentionally delayed the preventive maintenance of assets that will be replaced very soon. This is part of a larger plan to avoid spending their maintenance budget on assets of this type. The airport representatives also mentioned that they do not delay the preventive maintenance of critical assets that impact safety and operations. Another airport highlighted that, by using a staffing model in which each staffing unit has multiple scopes of work, they can defer the lowest-priority work and move forward with the most critical first. The risk of deferring maintenance is usually low for the assets with the least impact on customers. The interviewee from this airport also noted that there is sometimes an advantage to the deferred maintenance of some assets that would ultimately benefit from replacement, citing an instance in which a light pole was replaced with a more efficient light-emitting diode (LED).

When asked whether the airports have quantified the effect of assets' delayed preventive maintenance on costs, operation, and users, five out of 18 interviewees stated that they had not. They said

Copyright National Academy of Sciences. All rights reserved.

that their airports had never faced budget problems and had never delayed preventive maintenance of their assets. The rest of the airport representatives reported that they had assessed the damage caused by the delayed maintenance, but none followed any prescribed process, nor had any related documentation. Most interviewees who reported assessing the impact stated that they compared the risks and effects on customer safety and comfort due to delayed asset maintenance with the cost of outsourcing maintenance work. In one example, painting machines cost an interviewee three to four times more to hire a third party to complete their required airfield markings to remain compliant with the *Part 139.311* requirement. Additionally, when tractor mowers are down, one airport's alternative is to hire third-party companies to come in and complete the work, which in most cases costs the airports much more.

Some interviewees stated that they collect some data on the impact of delayed preventive maintenance, but they do not conduct a detailed analysis of those data. Two airport representatives stated that they had written white papers to the FAA to get funding for the renovation of runway pavements and had done some analysis on the economic and safety impact on airport operations. The research team reviewed the white paper provided by one airport but found that no quantitative analysis was done regarding the impact of not upgrading the existing runway. The white paper highlighted the importance of the alternative runway during natural disaster relief periods. The paper also stated that the alternative runway is utilized to separate the mix of air traffic in the airport during peak operational periods. It stated that, if the alternative runway is not renovated, its capabilities will be diminished and the taxiway capacity will be handicapped due to the increases in taxiway occupancy times, resulting in increased fuel utilization. Finally, one airport representative mentioned that most of their impact assessments are done verbally. In summary, none of the airports had a systematic process to determine the consequences of delayed maintenance.

All interviewees agreed that their airports keep asset inventory records and conduct condition assessments. However, some airports only keep asset inventory and condition assessment data for those assets that are required for the *Part 139* regulations and tied to FAA grant assurances. All the airports collect PCI data every 3 years and conduct visual inspections of runways, taxiways, and ramps. The airfield striping is also inspected at night and during the day, and readings are taken from a reflectometer to check its visibility. The airfield's lights and signs are also visually inspected, and condition data are collected. However, condition assessments of landside pavements are not conducted in as much detail as airfield pavements. Some of the airports also keep inventory and condition data of vehicles, snow removal equipment, HVAC, generators, etc.

Additionally, some airport representatives mentioned that they keep these conditional data of their assets in CMMS. Three interviewees stated that they used IBM Maximo software as a part of their CMMS systems. One stated that they use GCR's ASOCS, and another stated that they are buying Veoci software that will help maintain inventory data. Two airport representatives mentioned that they used CWorks and GIS to keep conditional data of their assets. One airport stated that they keep the work order data and condition data in ArcGIS software. The interviewee reported that they do not keep the inventory data of all assets in the system, and they are trying to upgrade the system so that every airport asset's condition and work order data can be kept. The rest of the interviewees informed us that they put the asset inventory and their condition data in spreadsheets. Finally, one mentioned that they use the asset inventory and assessment condition data to prepare a 10-year, long-term plan for vehicles.

Out of 18 interviewees, six stated that they do not establish performance objectives for their assets. Some airports did establish performance objectives for assets like airfield pavements. Some of the airports established performance objectives of assets loosely by applying a standard depreciation schedule. Some used the manufacturer's guidelines to estimate the life of the equipment. However, one airport representative mentioned that estimating the life of the equipment with manufacturer's guidelines is not a good idea. According to this interviewee, the equipment life

Copyright National Academy of Sciences. All rights reserved.

20 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

should be estimated based on the hours of operation and scheduled maintenance. Therefore, the people who are taking care of that equipment should be involved in estimating the life of the equipment based on their experience. Other airports used the experience of their staff to establish the performance objectives of their assets. Another airport representative stated that, when they determined the replacement time for their equipment, they considered not only the useful life of the equipment, but also the priority level assigned to the equipment related to safety, security, efficiency, and customer service.

One interviewee mentioned that they consider the equipment use factor to establish their performance objectives and measure the downtime of constantly running systems, like automated people movers and baggage handling systems, to determine their efficiency. This interviewee also mentioned that they conduct an alternative analysis before replacing equipment. For example, they traded in three old graders and bought two new ones based on efficiency analysis. Additionally, they sometimes rent equipment instead of buying it because it will save them money. Finally, one airport representative mentioned that their goal is to have 100% of the assets available at the time they should be used.

Some airport representatives reported that their airport vehicles for self-inspectors can put on hundreds to thousands of miles each day to meet the *Part 139* day and night airfield inspection requirement, and, in some cases, several special inspections if there are construction activities within the airfield environment. Many of these airports have back-up vehicles; however, the challenge many airports face is, when vehicles are purchased, they are purchased several at a time, or in fleets. Therefore, they age, deteriorate, and reach their useful life simultaneously. There is no strategy to purchase the vehicles in intervals so that they will not reach their useful life at the same time.

When asked about the use of FAA and other organizations' prescribed deterioration models to determine the life of assets and their preventive maintenance schedules, the majority of the airport representatives stated that they use the FAA-prescribed model for airfield pavements only. Some airports hired third-party consultants to analyze the data and prepare preventive maintenance schedules. Some non-hub airport interviewees said that they do not use the FAA deterioration model for pavement maintenance management. For other assets, they do not have any deterioration models, and instead, use their staff's experience and expertise to determine their deterioration trends. Some airports conduct a risk analysis for their fire trucks to determine their life. Some interviewees mentioned that they follow the FAA advisory circulars to determine the trends of asset deterioration and prepare their preventive maintenance budgets. One airport representative mentioned that they are in the process of buying Veoci (emergency management) software, and it might help them to predict their assets' deterioration trends.

3.4 Summary of Interview Findings

The interviews with 18 airport asset managers revealed that they have not developed systematic processes for allocating their capital and O&M budgets. Capital budgets are prepared based on the needs and conditions of assets. The airports that are dependent upon their revenues for capital investment generally plan capital projects based on their collected revenues. Most airport representatives agreed that they collect condition data for their high-priority assets, as well as assets that are associated with *Part 139* and FAA grant assurances, but they do not systematically use these data for capital and O&M budgeting purposes. Most of the airports prepare their budgets every fiscal year, and they plan the budgets about 6 to 12 months before that fiscal year. During capital and O&M budget planning, all airport departments are involved, and the budgets are reviewed by all concerned parties before being sent to the approving authorities. Therefore, there is a hierarchical budget approval process; however, there was no written documentation available from airports about preparing and managing their budgets. One surveyed airport is

Copyright National Academy of Sciences. All rights reserved.

currently in the process of developing a risk matrix to allocate its capital and O&M budgets based on asset priority level and age. In summary, there is no systematic or scientific process for preparing airport budgets, but the majority of airport representatives think that the airport capital and maintenance budgets need to be prepared based on asset conditions.

In addition, the research team could not determine any systematic approach to allocating preventive maintenance budgets. Most of the airport representatives reported that their preventive maintenance budgets need to be prepared based on their assets' conditions; however, due to a lack of funds and interdepartmental working environments, the process could not be formalized. Most interviewees stated that their preventive maintenance budgets are not enough, but they still operate their airports safely by allocating their budgets to those assets that are highly prioritized and required by *Part 139* and FAA grants. Most airport representatives stated that they keep inventories of highly prioritized assets, but do not use these data to determine their preventive maintenance budgets. Most of the airports have delayed preventive maintenance of their lower-priority assets, but they have not quantified the impact of the delayed maintenance in formal ways. The impact analyses are merely done verbally. Most surveyed airports use the FAA deterioration model for pavement; however, none of the airports have any deterioration models for other assets.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 4

Framework for Quantifying the Impact of Delayed Maintenance of Airport Assets

During the interviews with airport maintenance and operation managers in Phase I, the interviewees revealed that none of the airports have systematic methods of quantifying the regular maintenance budget and the impact of delayed maintenance on airport assets. The airport asset managers were preparing the maintenance budget based on their previous fiscal year budget. Some of the airports interviewed had a CMMS database, which they used to enter the maintenance cost and hours data; however, this database was rarely used to determine the maintenance budget of the airport assets. When asked about quantifying the impact of delayed maintenance of airport assets, none of the airports had any systematic process of determining the impacts. Therefore, airport asset managers need frameworks or tools to determine the impact of delayed maintenance on the airport's costs. This chapter addresses the development of a general framework that can help determine the maintenance budget and the impact of delayed maintenance on airport assets when existing maintenance funds are insufficient. This framework can be used for any type of asset to quantify the impact on delayed maintenance. Chapter 5 will highlight the detailed processes of determining the maintenance budget and quantifying the impact of delayed maintenance on airport budgets for specific airfield and landside assets. Chapter 6 will discuss the tools that can allow airport maintenance and operation managers to determine their budget for specific assets with and without delayed scenarios.

Every airport should have asset management goals. The main goal of any airport asset management is to keep its assets in good condition so that planes can be landed safely and travelers can comfortably use the airport terminals and other facilities to reach their destinations on time. To achieve this goal, airport asset managers track, manage, maintain, and replace assets periodically, so that they can provide safe and easy air travel through their airports. To maintain these assets in good condition, asset managers should prepare short- and long-term maintenance and replacement plans. The maintenance budget required for every fiscal year needs to be prepared based on the condition of the assets. If funds are insufficient, then the priority should be given to assets with important maintenance needs that affect the safety and comfort of air travelers. Quantification of the impact of delayed analysis should be conducted by asset managers to determine what extra funds must be spent if regular maintenance is delayed by a certain amount of time. The quantification process should be provided to the airport's higher-level authority to show them the consequence of delayed maintenance of the assets. To determine the consequences of delaying maintenance, the research team developed a framework that airport maintenance and operation managers can use. The framework shown in Figure 4-1 has seven basic steps:

1. Defining airport asset maintenance and preservation policies.
2. Keeping and updating an asset inventory database.
3. Collecting assets' condition data.
4. Analyzing assets' condition data.

Copyright National Academy of Sciences. All rights reserved.

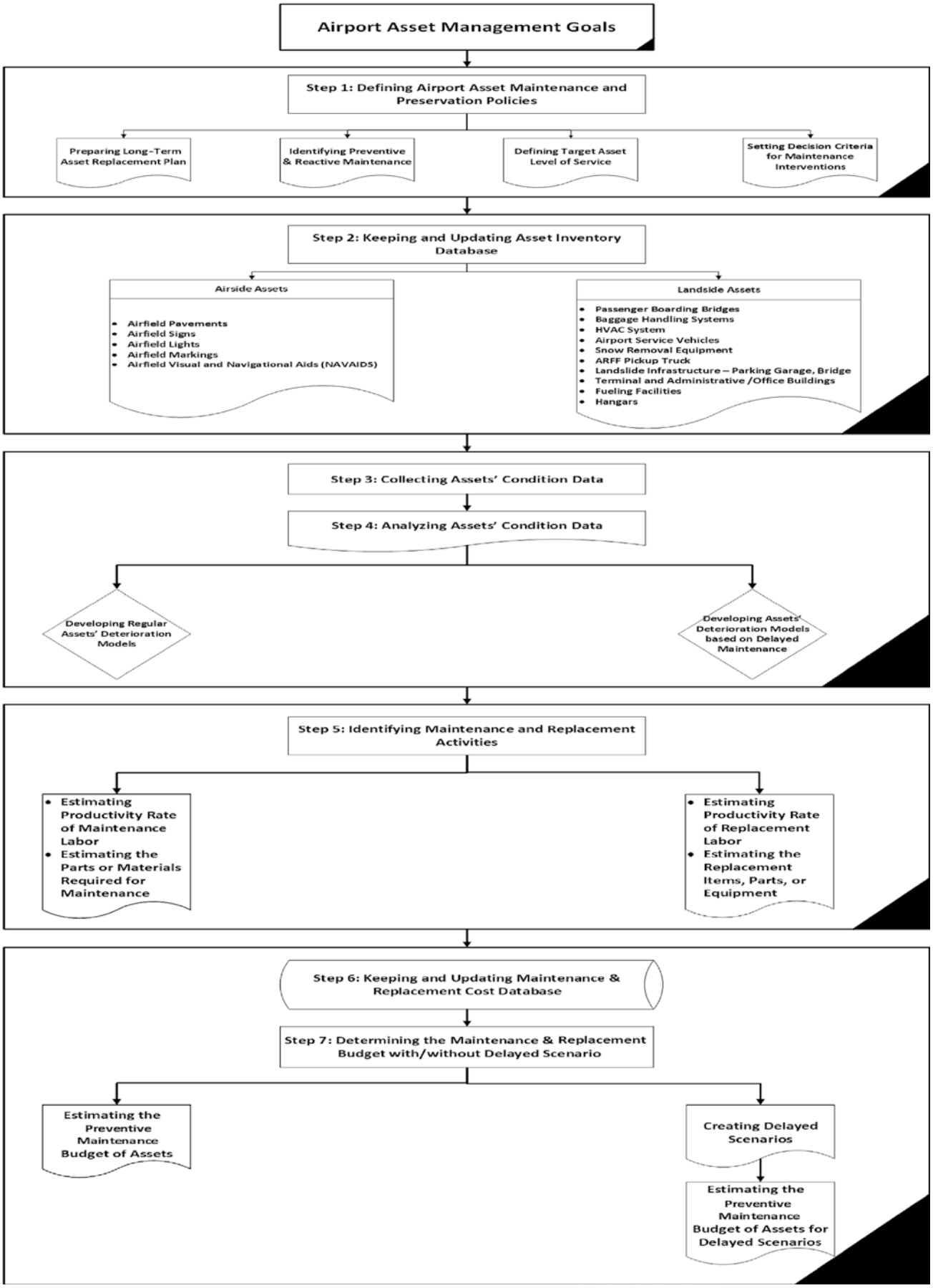


Figure 4-1. Delayed maintenance framework for airport assets.

Copyright National Academy of Sciences. All rights reserved.

24 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

5. Identifying maintenance and replacement activities.
6. Keeping and updating a maintenance and replacement cost database.
7. Determining the maintenance and rehabilitation budget with and without delayed scenarios.

Delayed maintenance in the context of airport assets is defined as deferring maintenance of assets due to a lack of funds. Notably, some airport assets are critical to plane and passenger safety, which cannot be delayed. Delaying the maintenance of these critical assets will jeopardize the safety of the plane and passengers. However, for theoretical purposes, the research team has hypothetically delayed these activities to quantify whether delaying maintenance is economical. Also, when the impact of delayed maintenance was quantified, the research team did not consider the cost of shutting down airfield pavements and airport terminals. The team only calculated the extra maintenance cost spent on assets to bring them into good operating condition. The following section describes each of the above seven steps in detail.

4.1 Defining Airport Asset Maintenance and Preservation Policies

The first step in having a good asset management plan is to define the airport asset maintenance and preservation policies. These policies should align with the asset management's overall goals. The policies related to asset maintenance should be developed to guarantee passenger safety. Similarly, the airport should build new infrastructure and maintain its existing infrastructure to ensure safety, mobility, and accessibility during air transportation. Airports should also develop and deploy innovative practices and technologies that improve the safety and performance of their airspace systems. Lastly, airports should serve with greater efficiency, effectiveness, and accountability.

Under these policies, the airport must prepare a long-term asset replacement plan. They should also identify the preventive and reactive maintenance activities to keep the assets in good condition and increase their life. The airport should define the assets' target level of service, so that an adequate maintenance budget can be spent to maintain that level. Lastly, the policy should set decision criteria for maintenance intervention. An airport must prepare these policies before acquiring new assets so that they can be maintained properly.

4.2 Keeping and Updating Asset Inventory Databases

The first step of the asset maintenance plan will be to collect the airport asset inventory data. All the assets in the airports' possession need to be kept in a database. Airports use several types of software to keep asset inventory, such as PAVEAIR, IBM Maximo, GCR's ASOCS, and Veoci. Airports generally use the term CMMS to refer to software systems like the above-mentioned proprietary software brands, or databases to keep asset inventory. This software not only holds asset information but also analyzes the asset data to assist asset managers in making informed decisions. Airports have a variety of assets, and these can be classified as airside or landside assets. The airside assets include, but are not limited to, airfield pavement, markings, signs, lighting, NAVAIDS, utilities, fencing, gates, and drainage. The landside assets include passenger boarding bridges, baggage handling systems, HVAC systems, pickup trucks, loaders, snow blowers, hangers, fueling facilities, administrative buildings, terminal buildings, overhead bridges, and parking garages. Generally, the asset inventory database consists of information related to the location, age, condition, and maintenance cost spent on the assets (labor hours and parts spent on maintenance). Some airports use radio-frequency identification (RFID) and sensors to determine the assets' exact location. If the asset inventory has the required information, it will be easy for asset managers to prepare a regular maintenance budget and long-term replacement plan, as well as quantify the impact of delayed maintenance.

Copyright National Academy of Sciences. All rights reserved.

4.3 Collecting Asset Condition Data

If the asset conditions are not assessed every year to determine their remaining life, then it will be difficult for airports to prepare maintenance budgets and develop a long-term replacement plan. During the interview in Phase I, five interviewees stated that they assess the conditions of some of their assets. All the airports interviewed have a conditional assessment of their airfield pavement. Every year the airports' in-house staff or consultants conduct the conditional assessment of the pavement and determine the PCI. The large-hub airports use PAVEAIR to store and analyze the pavement condition based on these PCI values. This value plays an important role in maintaining the airfield. Some airports also assess the condition of airport signs by rating the signs excellent, good, fair, and poor. These ratings help to determine the maintenance budget allocated to airport signs. The airports collect data on assets' conditions but do not have specific measurement metrics to determine them. For example, in airfield marking and lighting systems, airports assess their condition every year by checking the markings' reflectivity, lightings' electrical connection, and light fixtures. However, no formal process exists to record the condition of these assets in the database software. Similarly, the airports keep their inventory and check the assets' conditions but do not categorize their assets' condition levels for some landside assets, including baggage handling systems, HVAC, passenger boarding bridges, pickup trucks, loaders, and ARFF equipment. Generally, the airports depend upon their age, usage hours, or usage mileage to determine the asset conditions. For example, for pickup trucks, the airports record the mileage and determine the condition of these vehicles. Similarly, for loaders, they use the hours of operation to determine the conditions. For other landside assets, the airports interviewed use the assets' age to determine their condition. It will be easier for airports to prepare the maintenance and replacement budget as well as determine the impact of delayed maintenance if asset conditions are categorized.

4.4 Analyzing Asset Condition Data

Airport asset managers must analyze the asset condition data before preparing a maintenance budget with or without delayed maintenance. Asset condition data can be used to prepare the asset deterioration model. All the airports have their PCI data which can be used in PAVEAIR to determine the regular deterioration trend of any type of pavement (asphalt or concrete). Asphalt and concrete already have established deterioration models based on PCI values, but, for other assets like pavement markings, airfield lights, or airport signage, it is difficult to determine the empirical models like that of pavement. Most studies have used the random probability theory based on asset life to determine the cumulative failure probability of the assets. If the asset conditions can be categorized on some level (e.g., excellent, good, fair, or poor), then it will be easier to determine how the asset conditions could change each year. Also, if the asset data are collected over the course of a few years, these data can be used to determine the probability of the asset changing from one condition category to another. Categorizing the condition of the asset helps to develop deterioration models. The condition of some landside assets should be categorized as excellent, good, fair, or poor, and their condition data should be collected throughout the life of the asset. Using these data, the probability of converting these assets from one condition to another can be determined more easily. Some car and pick-up truck companies have determined the probability of failure by using the collected asset condition data. Collecting and analyzing these kinds of data requires commitment and resources that airports do not readily have access to.

Although the condition data are available, it can still be difficult for airports to develop regular deterioration models for most airport assets due to a lack of systematic data. Researchers need to depend upon theoretical models of probability to determine asset deterioration based on scenarios without maintenance. It is not possible to collect the condition data without any maintenance, because the airport cannot leave assets without preventive maintenance. Therefore,

Copyright National Academy of Sciences. All rights reserved.

the deterioration models in delayed maintenance scenarios must depend upon the researchers' assumptions and probability distribution theory. In this research, assumptions and theoretical probability distribution theory are primarily used to develop asset deterioration models that do not include preventive maintenance.

4.5 Identifying Maintenance and Replacement Activities

Every airport must identify the types of maintenance activities that need to be performed on their assets so that their lives can be prolonged. The assets cannot be maintained properly without knowing the maintenance activities and frequency, and it will be difficult to determine a budget without that information. Maintaining asphalt and concrete pavements involves different activities. Some maintenance activities performed on asphalt pavement include cold chip sealing, cold in-place recycling, cold milling, crack filling or sealing, or fog sealing. However, crack sealing, diamond grinding, diamond grooving, or dowel bar retrofitting activities can help the concrete pavement remain in good condition. Similarly, for HVAC systems, changing the airlifter regularly, cleaning the duct system, greasing the belt and motor, and maintaining the air handling system are required to keep the HVAC in good running condition. In the case of baggage handling systems, the motor of the belts needs to be greased and changed, and the conveyor belts need to be inspected and repaired to increase the life of the system. Airport asset managers need to ensure that the required maintenance activities are performed with appropriate frequency. It is also necessary to determine what activities need to be performed if the assets need replacement.

4.6 Keeping and Updating Maintenance and Replacement Cost Databases

To determine the maintenance and replacement budget, a cost database calculated using the cost of materials and labor needs to be prepared. Therefore, for all maintenance activities identified in the previous step, the airport asset managers should track the labor hours and parts required to perform these activities. If the labor productivity and parts required for maintenance are kept in a database, it will help determine the cost of maintenance. The research team found that most of the airports do not have this type of database to determine an accurate maintenance budget with and without delayed maintenance.

4.7 Determining Maintenance and Replacement Budgets with and without Delayed Scenarios

Asset deterioration models, maintenance and replacement activity types, labor productivity, and cost of labor and parts can determine a regular maintenance and replacement budget. The maintenance budget for the coming fiscal year needs to be prepared based on the predicted conditions of the assets in that year, necessitating an accurate deterioration model. Airport asset managers need to identify the types of preventive maintenance and their frequency to correctly calculate the budget. Productivity and cost data also need to be updated regularly to ascertain an exact budget. Delayed maintenance and replacement costs can be determined by using deterioration models prepared for the delayed scenarios. The accuracy of a maintenance and replacement budget for any asset will depend upon all factors mentioned in the above steps. These are estimated budgets, so there will always be variations when used during the maintenance implementation phase, but an effort must be made to increase accuracy and minimize variations between the estimated and actual budget.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 5

Procedures to Quantify the Impact of Delayed Maintenance on Airport Airside Assets

This chapter focuses on step-by-step procedures for quantifying the impact of delayed maintenance on selected airside airport assets. The FAA reports that, of the AIP grants allocated each year, about 70% are used for airfield pavement projects. In other terms, over $2 billion of the $3.3 billion awarded to sponsor airports in fiscal year 2021 funded projects specific to designing, building, and rehabilitating airport runways, taxiways, and aprons (FAA 2022a). The delay of maintenance on airport assets will have direct and indirect impacts on airport safety, operations, and passenger comfort as well as future maintenance, rehabilitation, and replacement activities. It is necessary to consider these impacts before delaying any maintenance activities. The ability to calculate the monetary impact of delayed maintenance will provide decision-makers with needed information. Because public airports' primary goal is to provide safe air travel, asset maintenance cannot be delayed if it has an unacceptable impact on safety. For example, airfield pavement maintenance cannot be delayed to the point in which airplanes can no longer safely land or take off from the airfield. Similarly, pavement markings, airport lighting, and signage maintenance cannot be delayed to the point of impacting a pilot's understanding of their aircraft's position on the airfield.

This section covers the procedures of quantifying the impact of delayed maintenance on four airport airside assets: (1) primary runway pavement, (2) airfield markings, (3) runway lighting, and (4) airfield signs. This framework details the process of quantifying the impact of delayed maintenance on these assets. As previously described, the maintenance of these critical assets should not be delayed if safety is compromised. One product of this research is an estimate of the monetary consequences of delayed maintenance scenarios that asset managers can use to understand deferral consequences and advocate for additional funding.

The research team developed spreadsheet tools to determine the delayed maintenance costs for up to 10 years for specific airside and landside assets. However, delaying the maintenance of any asset not only has an impact on maintenance and replacement costs but also user costs that are not quantified in these spreadsheet tools. The user costs are associated with airport closures that result in airports' loss of revenue and extra costs to the airlines and air passengers. Therefore, it is suggested that airports do not delay maintenance of any assets for more than 1 or 2 years so that they will not need to be closed due to failed assets.

5.1 Primary Runway Pavement

There are eight distinct steps to determine a regular maintenance budget and the consequences of delaying primary runway pavement maintenance. Figure 5-1 shows the first seven steps. The eight distinct steps are:

1. Establishing asset inventory,
2. Performing a condition assessment,
3. Identifying and updating maintenance and rehabilitation activities and their unit costs,

Copyright National Academy of Sciences. All rights reserved.

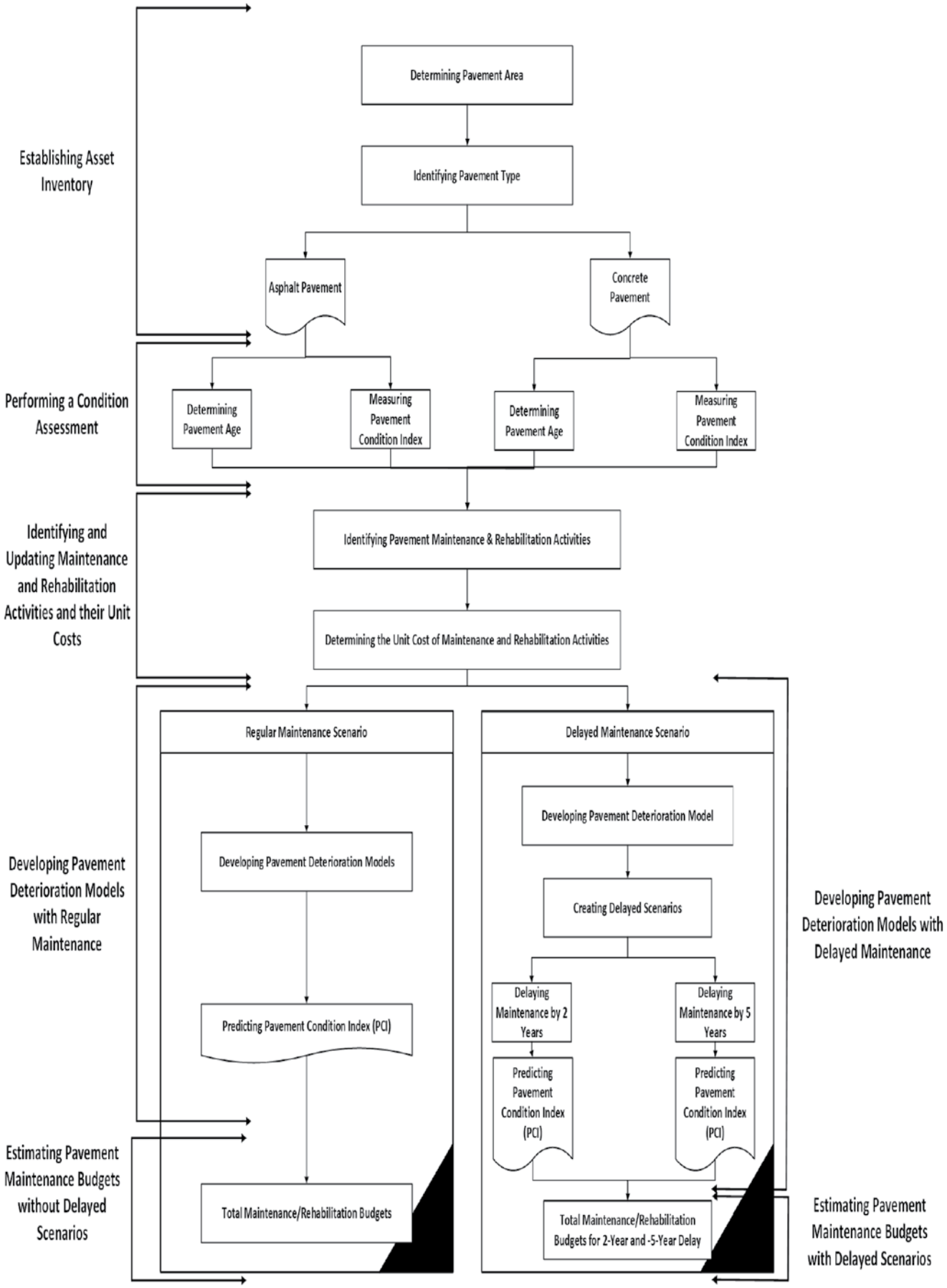


Figure 5-1. Delayed maintenance framework for runway pavements.

Copyright National Academy of Sciences. All rights reserved.

4. Developing pavement deterioration models with regular maintenance,
5. Developing pavement deterioration models with delayed maintenance,
6. Estimating maintenance budgets without delayed scenarios,
7. Estimating maintenance budgets with delayed scenarios, and
8. Reporting the impact of delayed maintenance and its consequences.

5.1.1 Establishing Asset Inventory

The airport needs to have a pavement inventory database where the dimensions and category (primary, secondary, and tertiary) of an airfield pavement can be stored. The pavement element information needed in the inventory is provided in Table 5-1 (FAA 2014b, FAA 2022b). The inventory should have the length and width of the runway. Pavement structure type, for example, concrete, asphalt, or rigid over asphalt overlay, and its surface material, are necessary. It is important to have details about pavement structure, e.g., thickness, subbase aggregate types, and the date of construction. It is also important to have the dates of minor and major rehabilitation, if applicable. The inventory should have information about surface and subsurface drainage provided during the pavement construction. The pavement design information related to the strength of the pavement and design loads of the pavement should be recorded in this database. The inventory should consist of any other information that helps airport asset managers maintain and rehabilitate the pavement, such as maintenance activities performed and frequency. These data will assist in identifying the corrective activities needed to bring the pavement to the level of service required. Most large-hub airports keep these pavement records in a pavement management system, most commonly FAA PAVEAIR. This information is key in determining the annual maintenance cost with or without delays.

5.1.2 Performing a Condition Assessment

All airports assess the conditions of their airfield pavement in accordance with FAA Advisory Circular 150/5380-7B Airport Pavement Management Program (PMP) which "is mandatory for

Table 5-1. Runway pavement element information (FAA 2014b).

Pavement element	R-1 (Original runway - North 4,000 feet 75' wide).
Pavement structure type	Rigid with asphalt overlay.
Pavement surface material	Asphalt.
Pavement structure	1. 3-inch asphalt cement concrete (ACC) (Item P-401) (1995). 2. 6-inch Portland cement concrete (PCC) (Item P-501) (1975). 3. 4-inch aggregate subbase (Item P-154) (1975). 4. 6-inch subgrade preparation (Item P-152) * 95% ASTM D698. Lean clay, plasticity index = 23–28 (1975 previously used as turf runway).
Year of construction	1990.
Year of reconstruction	2012.
Drainage features	No subsurface drainage system - Surface drainage only.
Pavement use	Heavy usage - 50-65 operations per day, average weight 45 kips dual wheel gear (DWG). Use has increased dramatically in the past 5 years.
Pavement strength	60,000 DWG.
Priority	1 (scale of 1-6, 1 highest).
Miscellaneous	North end (250') constructed near old river channel, appears to have excessive subsurface moisture.
Funding source	AIP funds used.

Copyright National Academy of Sciences. All rights reserved.

30 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

all projects funded with federal grant monies through the Airport Improvement Program (AIP) and with revenue from the Passenger Facility Charges (PFC) Program." Generally, condition assessment of pavement is done by determining the PCI of the pavement, in accordance with ASTM 5340—Standard Test Method for Airport Pavement Condition Index Surveys. PCI is "a numerical rating of the pavement condition that ranges from 0 to 100, with 0 being the worst possible condition and 100 being the best possible condition" (FAA 2022c). The PCI number shows the pavement's present condition based on the distress observed on its surface. Distress type and quantity are used to determine values deducted from 100 to calculate the PCI. Asphalt and PCC-surfaced pavements each have their own set of distress types. While some types of pavement distress (i.e., corner breaks, shattered slabs, alligator cracking, rutting) indicate a structural failure, the majority of pavement distress is caused by age or environmental factors. Therefore, most pavement maintenance is aimed at extending the functional life of the pavement as opposed to addressing a structural failure. Figure 5-2 shows the condition of the pavement based on the PCI numbers. Pavement maintenance treatments are more cost-effective when applied while the pavement is still in good/satisfactory condition. As pavement falls into the fair category, actions like a mill and inlay often become the most effective means of pavement rehabilitation. Below a PCI of 55, a mill and inlay may require additional work (thicker inlay, base repairs, additional patching) to perform as intended, and reconstruction will start to become more competitive on a life-cycle cost basis. Below a PCI of 40, complete reconstruction is typically the only feasible rehabilitation option. The PCI values are used to establish a runway's rate of deterioration and to identify future rehabilitation needs. PCI values are also used in prioritizing, funding, and executing maintenance and rehabilitation on the pavement section. Condition assessment is key in determining the health of the pavement and in predicting the rate of deterioration.

Pavement inspection is one of the most important parts of the PMP of any airport. Airports can use the pavement condition survey to determine PCI and use these values to develop pavement performance predictions. It is necessary that the PCI value is determined in accordance with the procedures contained in ASTM D5340, Standard Test Method for Airport Pavement Condition Index Surveys (FAA 2014b). Airports should also maintain runway pavements that provide surfaces with adequate friction characteristics under all weather conditions. AC 150/5320-12 provides the process and frequency of skid resistance testing. Airports should be careful about using visual observations for skid resistance determination, because the visual pavement inspections do not



Figure 5-2. Pavement condition category based on PCI Values (FAA 2022c).

Copyright National Academy of Sciences. All rights reserved.

adequately predict skid resistance. Roughness measurements can be helpful when there is evidence of roughness issues, usually in the form of frequent pilot complaints. Roughness measurements are of greater value when the pavement is in very good condition with little or no distress. They have less value if reconstruction is imminent. AC 150/5380-9, Guidelines and Procedures for Measuring Airfield Pavement Roughness, provides guidelines and procedures for measuring and evaluating runway roughness.

Various types of distress in asphalt and concrete pavement develop over time. Air traffic loading and climate/weather exposure are direct contributors to distress, leading to the need for repair and maintenance. Table 5-2 provides some of the distress seen in these two types of pavements (FAA 2014a).

5.1.3 Identifying and Updating Pavement Maintenance and Rehabilitation Activities and their Unit Costs

Identifying the major maintenance and rehabilitation activities that need to be performed on asphalt and concrete runways and taxiway pavements is important. FAA AC 150/5380-6C provides information on maintenance activities for various types of distress on asphalt and concrete pavement. Tables 5-3 and 5-4 provide the types of suggested maintenance for these two types of pavements to keep pavement in good condition, as well as to increase the pavement life. This AC also provides the types of materials used to repair asphalt distress and concrete pavements.

Once these repairs and maintenance activities are identified, the unit cost of performing them needs to be determined. These data can be updated from the contractor's unit bid price if the maintenance work is outsourced. If in-house staff performs these activities, then the labor hours

Table 5-2. Types of distress on asphalt and concrete pavement.

Asphalt Pavement	Concrete Pavement
1. Cracking • Longitudinal and transverse cracks, • Block cracks, • Reflection cracks, • Alligator or fatigue cracks, and • Slippage cracks. 2. Disintegration • Raveling, • Weathering, • Potholes, • Asphalt stripping, • Jet blast erosion, and • Patching and utility cut patch. 3. Distortion • Rotting, • Corrugations, • Shoving, • Depressions, and • Swelling. 4. Loss of skid resistance • Polish aggregate, • Contaminants, • Bleedings, and • Fuel and oil seepage.	1. Cracking • Longitudinal, transverse, diagonal cracks; • Corner breaks; • Durability "D" cracks; • Shrinkage cracks; and • Shattered slab/intersecting cracks. 2. Disintegration • Scaling, map cracking, and crazing; • Alkali-Silica Reactivity (ASR); • Joint spalling; • Corner spalling; • Blowups; • Pop-outs; and • Patching. 3. Distortions • Pumping, and • Settlement or faulting. 4. Loss of skid resistance • Polished aggregates, and • Contaminants. 5. Joint seal damage

Copyright National Academy of Sciences. All rights reserved.

32 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 5-3. Types of distress and repair activities on asphalt pavement.

Types of Distress	Repair and Maintenance Activities
Cracks	• Remove old sealer material if present, • Clean and prepare cracks, • Seal/reseal cracks, and • Use joint heating, which may be an option for longitudinal cracks when under the engineer's direction (operate heaters to avoid excessive heat on the pavement).
Alligator or fatigue cracking	• Remove and replace damaged pavement, including the base and/or subbase course if required.
Patches	• Remove/replace, and • Repair and resurface.
Surface irregularities (e.g., rutting, wash-boarding, birdbaths)	• Remove and replace damaged areas, and • Surface grinding/milling.
Loss of skid resistance	• Remove rubber/surface contamination, and • Apply surface treatment.
Bleeding	• Blot with sand and remove sand prior to resuming aircraft operations. Excessive bleeding may require removal and replacement of pavement.
Drainage	• Grade pavement shoulders, clear drainage path; and • Clean out drainage structures, e.g., edge drains, outfalls, etc.

Table 5-4. Types of distress and repair activities on concrete pavement.

Types of Distress	Repair and Maintenance Activities
Joint sealant damage	• Remove old sealant, clean joints, reseal.
Cracks	• Clean and seal cracks. • Repair/replace slab. • Evaluate adequacy of pavement structure; may require strengthening.
Corner breaks	• Remove loose material; refill with approved product; reseal. • Partial depth repair.
Slab blowup	• Replace slab in blowup area; clean and reseal joints.
Loss of skid resistance	• Remove rubber/surface contamination. • Grinding.
Drainage	• Grade pavement shoulders, clear drainage path. • Clean out drainage structures, e.g., edge drains, outfalls, etc.
Pop-outs	• Remove foreign object debris.
Patches	• Remove/replace.

Copyright National Academy of Sciences. All rights reserved.

need to be tracked during the maintenance. The cost of labor hours used to maintain the pavement can be calculated by the total hours spent multiplied by the cost of labor per hour. The materials and equipment costs can be included to determine the total maintenance cost. It is a good estimating practice to add some contingencies based on the risks associated with repair. The total cost per unit of these maintenance activities should be updated in the CMMS software so that the maintenance budget for future years can be calculated.

It is critical to determine the rehabilitation and reconstruction cost of pavement for budgeting. These costs can be determined using RSMeans cost data or the Engineering News-Record's cost data dashboard. The maintenance, rehabilitation, and reconstruction costs per unit are necessary to determine the regular maintenance budget, as well as to quantify the consequences of delayed pavement maintenance.

The research team collected extensive cost data on pavement maintenance and rehabilitation from airport projects around the United States. These data have been used to determine typical costs for the maintenance and rehabilitation of asphalt and concrete runway pavements. The team determined maintenance and rehabilitation unit cost of asphalt and concrete pavement based on PCI. These unit cost data varied based on the airport size (small-, medium-, and large-hub airports). They can be found in the supporting spreadsheet tools supplemental to this report. Airports can use these tools to estimate the regular and delayed maintenance and rehabilitation cost of the asphalt and concrete pavements.

5.1.4 Developing Pavement Deterioration Models with Regular Maintenance

FAA researchers have collected PCI values of numerous airport pavements and have developed a deterioration model to predict the PCI values based on the age of the pavement. These regression models are developed based on the pavement types and are prepared separately for runways and taxiways. These models can also differ based on airport location and size. The team collected PCI values of numerous airport pavements of various ages and developed deterioration models for all sizes of airports for asphalt and concrete pavements. Equation 5-1 shows the regression models for predicting PCI values based on the asphalt pavement's age, assuming the required maintenance work has been completed regularly in airports.

$$\text{PCI} = \left(100 - 3.8\right)\left(\text{Age of the pavement} + 0.15\right)\left(\text{Age of the pavement}\right)^2 - 0.00255\left(\text{Age of the pavement}\right)^3 \quad (5\text{-}1)$$

Figure 5-3 shows a deterioration trend based on the age of asphalt pavement of an airport. It shows that the deterioration of airfield pavement is steep in the beginning and evens out in the middle. The model shows that it generally takes 36 years to reach the PCI value of 40, assuming the implementation of regular maintenance and rehabilitation. It shows that, on average, there is a reduction of 1.7 PCI value per year.

Equation 5-2 shows the deterioration model of plain concrete runway pavement for airports. In this case, the deterioration of the runway pavement is linear.

$$\text{PCI} = \left(100 - 1.9514\right)\left(\text{Age of the pavement} + 0.0597\right)\left(\text{Age of the pavement}\right)^2 - 0.0012\left(\text{Age of the pavement}\right)^3 \quad (5\text{-}2)$$

Figure 5-4 shows the deterioration shape of a hypothetical airport's plain concrete pavement, which is similar to the asphalt pavement. Unlike the asphalt pavement, the PCI value of concrete pavement will not reach 0 at the age of 45 years. The plain concrete pavement will reach a PCI value

Copyright National Academy of Sciences. All rights reserved.

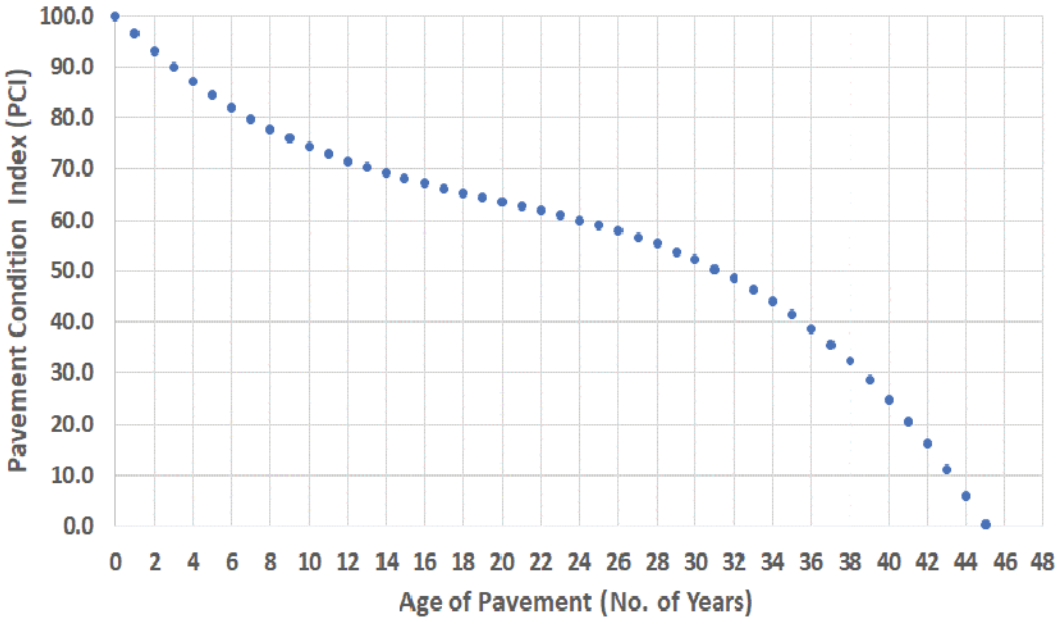


Figure 5-3. A deterioration trend based on the age of asphalt pavement of a hypothetical airport.

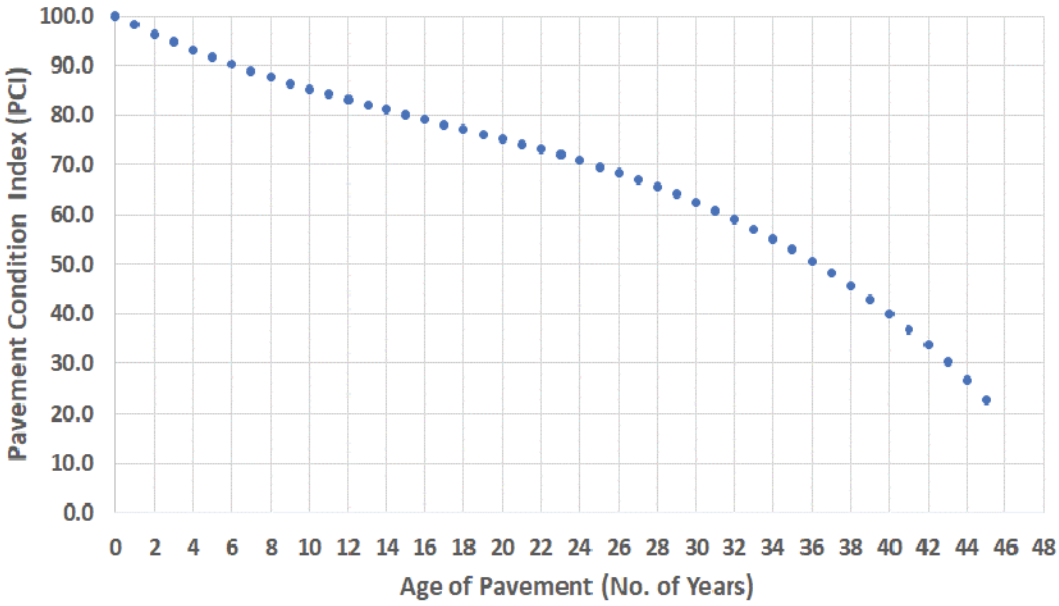


Figure 5-4. A deterioration trend based on the age of the concrete pavement of an airport.

Copyright National Academy of Sciences. All rights reserved.

of 23 at the age of 45 years. The model shows that it generally takes 40 years to reach a PCI value of 40, assuming the implementation of regular maintenance and rehabilitation. On average, the concrete pavement PCI value will be reduced by 1.5 per year, which is lower than that of asphalt pavement.

5.1.5 Developing Pavement Deterioration Models with Delayed Maintenance

When collecting PCI data from airports, the research team found that airports did not collect PCI data for runway pavements when maintenance was not performed. Therefore, it is impossible to develop the deterioration model based on delayed or no maintenance. However, to determine the deterioration trend of asphalt and concrete runway pavement for airports, the research team assumed that asphalt and concrete pavement will last for 20 and 30 years, respectively, if regular maintenance is not performed. Therefore, for asphalt runway pavement to go from a PCI value of 100 to 40, it will take 20 years. This is an average reduction of 3 PCI values each year. Similarly, for concrete runway pavement, the average reduction of PCI value per year will be about 2. Therefore, the research team reduced 3 PCI values from the PCI value of the asphalt runway pavement in a particular year. The same method was used to determine the PCI value of concrete runway pavement, with 2 PCI values reduced from its original value in a particular year. Reducing the PCI values for asphalt and concrete pavement by 3 and 2 seems reasonable because, in the regularly maintained scenarios, the average PCI value reduction for asphalt and concrete pavements are 1.7 and 1.5, respectively. Therefore, if the PCI value of the pavement of an airport in the year 2021 is 70, then delaying maintenance by one year will make the PCI values of asphalt pavement and concrete pavement 67 and 68, respectively. This reduction of PCI values is higher than the average reduction of PCI values in regularly maintained asphalt and concrete runway pavements. The following example demonstrates the process of predicting the PCI values of asphalt and concrete runway pavement with and without delayed maintenance.

For an airport with an asphalt runway pavement that is 5 years old, based on a regular maintained deterioration model, the PCI value of the pavement will be calculated using Equation 5-1.

The PCI value of the pavement this year will be 84.4. If the maintenance budget needs to be calculated for the pavement this year, a value of 84.4 will determine the maintenance budget of the pavement. However, if the maintenance of this pavement is delayed by 3 years, then the predicted PCI value based on no maintenance will be:

$$PCI = 84.4 - 3 * 3 = 75.4$$

However, if regular maintenance is done, then the pavement's PCI value after 3 years will be 77.9 based on the regular deterioration model formula shown in Equation 5-1. This demonstrates that there is a decrease of 2.5 PCI after delaying maintenance by 3 years, thus significantly increasing the cost of pavement maintenance. Similar procedures can be used to determine the difference between the PCI values of regular and delayed maintenance concrete pavements.

5.1.6 Estimating Pavement Maintenance Budgets without Delayed Scenarios

Using the deterioration models for regularly maintained asphalt and concrete runway pavement, the PCI values of these pavements can be predicted based on their age. Once the predicted PCI values are calculated for these two types of pavements for the yearly budget, the total regular maintenance budget can be calculated by multiplying the area of the runway pavement and the maintenance cost per square foot corresponding to that PCI value. To calculate the maintenance

Copyright National Academy of Sciences. All rights reserved.

36 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

cost, it is necessary to have the average maintenance cost per square foot of the runway area for every PCI value starting from 100 to 40. After PCI values reach 40, the runway pavement needs to be reconstructed. Therefore, the maintenance cost per square foot on asphalt and concrete runways needs to be updated every year for an accurate estimate of the maintenance cost. In general, the maintenance of pavement is determined based on the types and level of distress found during inspection. However, maintenance activities and PCI values are correlated. The maintenance cost spent on asphalt and concrete runway pavements can be determined for each PCI value by developing a regression model. The maintenance costs per square foot for each value of PCI can be used to estimate the maintenance cost of the asphalt and concrete runway pavement. The spreadsheet tools developed to determine the regular and delayed maintenance costs of asphalt and concrete runway pavements use this maintenance cost derived from the regression model. The maintenance cost of asphalt or concrete runway pavements used in the tool is the 2020 base cost. Inflation rates have been collected from the Bureau of Labor Statistics to determine the inflated cost (BLS 2022a). The inflation cost index of concrete pavement since 1971 shows that the inflation rate of ready-mix concrete is 4.07% per annum. Similarly, the inflation rate for asphalt was found to be 2.93% per year (BLS 2022b).

5.1.7 Estimating Pavement Maintenance Budgets with Delayed Scenarios

The total maintenance budget of asphalt or concrete runway pavements in a delayed scenario can be calculated by using the linear deterioration trends discussed in Section 5.1.4. Once the PCI value of the asphalt or concrete runway pavement is determined for the year after considering the delayed scenarios, the total maintenance cost can be estimated by multiplying the maintenance cost per square foot with the area of the runway pavements. Due to decreased PCI values, the maintenance cost of the delayed scenario runway pavements will always be higher than that of regularly maintained runway pavement. As mentioned previously, this delayed maintenance cost considers only monetary maintenance costs without considering other costs such as safety, passenger comfort, etc.

5.1.8 Reporting the Impact of Delayed Pavement Maintenance and its Consequences

Airport asset managers need to know the extra cost incurred when the maintenance of asphalt or concrete runway pavement is delayed by a number of years. The difference in the maintenance costs can be presented to higher authorities to validate that it is beneficial to spend money on regular maintenance rather than wait. The spreadsheet tool developed by the research team (found in Section 2.1 of the Quick Guide) can compute both the regular and delayed maintenance costs by selecting delayed scenarios. The user can also select the default price inflation rate or enter a new inflation rate for asphalt or concrete pavement to calculate the delayed maintenance cost. As maintenance is delayed, the maintenance cost for asphalt or concrete runway pavement will be inflated, and the updated cost needs to be considered when estimating the total maintenance cost of runway pavements. The tool also has the option of selecting a discounted rate, so that the future value of maintenance costs can be converted to present value for comparison purposes if an airport decides to use the discounted rate for its investments. Generally, private sectors use the discounted rate to determine the value of money in relation to time.

The research team has created a hypothetical airport runway with its own dimensions and pavement age to show the consequences of delaying maintenance on asphalt and concrete pavements. Two- and 5-year delayed scenarios were selected. To report the impact of delayed maintenance and its cost consequences, the team considered hypothetical data. The team hypothesized that

Copyright National Academy of Sciences. All rights reserved.

Table 5-5. Case study data of a hypothetical small-hub airport.

Name of the airport	XYZ International Airport
Type of the airport	Small-hub
Location of the airport	Southwest, United States
Runway dimensions	9,000 ft length and 150 ft width
Type of pavement	Asphalt or concrete
PCI values of the pavement	90, 80, and 70
Budget preparation year	2022
Budget to be prepared for	2023
Delayed scenarios	2 and 5 years
Price inflation factor	4.07% for concrete and 2.93% for asphalt pavement
Discount factor	0%

pavements had PCI values of 90, 80, or 70, to demonstrate the effects of aging runway pavement on the delayed maintenance cost. Table 5-5 shows the hypothetical small-hub airport case study data.

Case Study Results of the Consequences of Delaying Maintenance on Airport Runway Pavements

Results of Delaying Maintenance on the Concrete Runway Pavement of an Airport

The team used the spreadsheet tool to determine the consequences of delayed maintenance on an airport with concrete runway pavements. Three different pavements with PCI values of 90, 80, and 70 were selected. Figure 5-5 shows the total maintenance cost of this airport for concrete pavement with and without a maintenance delay. The budget was prepared in 2022 and will be used for 2023, so the maintenance cost calculated for the base scenario without delay is the cost

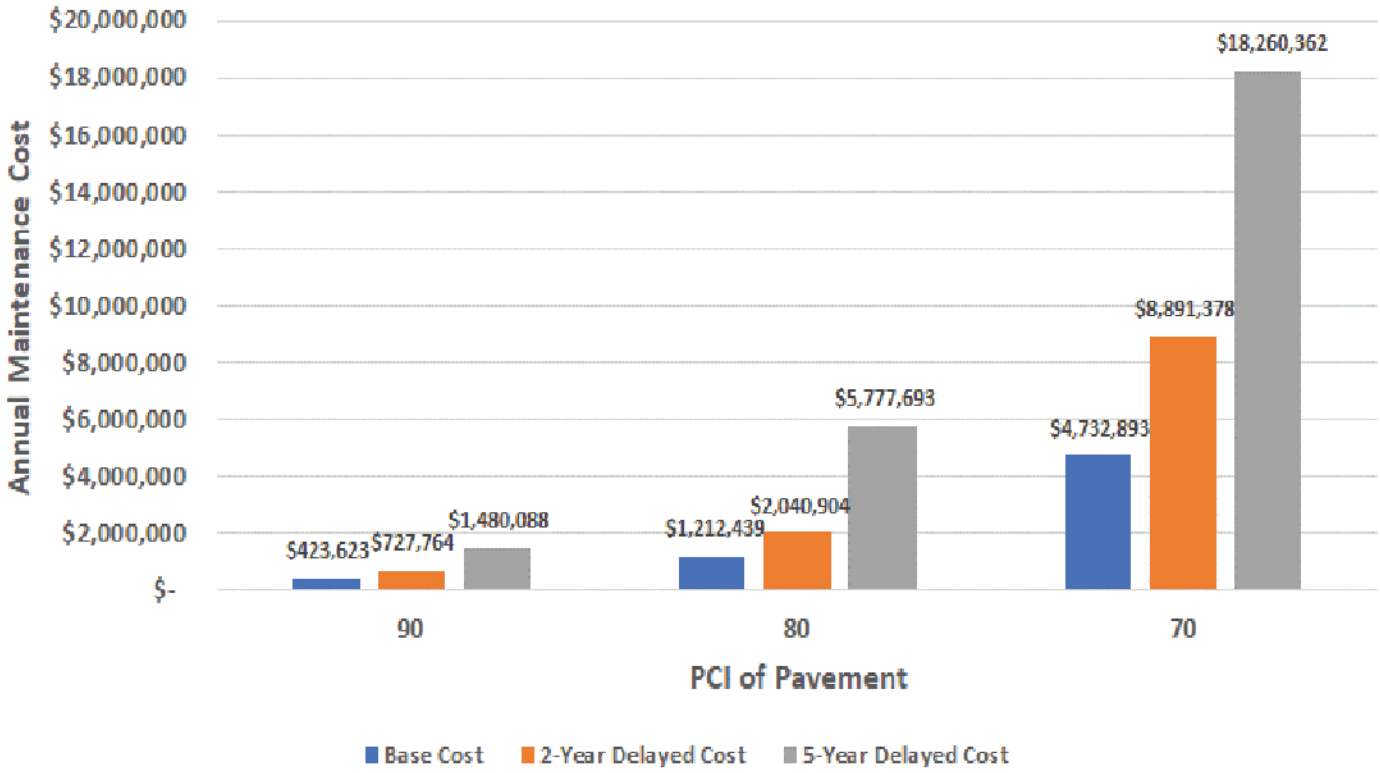


Figure 5-5. The impact of delaying maintenance on total maintenance costs of a concrete runway pavement.

Copyright National Academy of Sciences. All rights reserved.

38 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

for 2023. The delayed scenario assumes that the airport has a maintenance budget for 2023 and plans to delay for 2 and 5 years after 2023 to determine the impact on the total maintenance cost. The research team used an inflation rate of concrete derived from the historical cost index of BLS. The team also did not consider the discounted rate to convert the total maintenance cost into net present value in delayed scenarios.

The results show that the total concrete pavement maintenance cost of the airport increases significantly when the PCI values of the pavement decrease. The 5-year delayed maintenance cost of a concrete pavement with a PCI value of 90 is $1,480,088 compared to $5,777,693 for pavements with a PCI value of 80, and $18,260,362 for pavements with a PCI value of 70. Therefore, the PCI values of concrete pavement should be considered when delaying maintenance because they significantly impact the total maintenance cost.

Figure 5-6 shows an increasing trend in the maintenance cost of concrete pavements based on their PCI values. The analysis results show that the percentage the maintenance cost increases for delayed scenarios compared to the base maintenance cost is dependent on the PCI values of the pavement. For the concrete runway pavement with a PCI value of 80, the 2- and 5-year delayed maintenance cost percent increment is higher than the runway pavements with PCI values of 90 and 70. The percentage increment of the pavement with PCI value 80 is 377% when delayed for 5 years compared to its base cost. This analysis shows that there are significant consequences when an airport delays maintenance of its concrete runway pavements. For pavement with a PCI value of 70, the percentage increase of the total maintenance cost will reach 286% of its base maintenance cost when delayed by 5 years. For concrete pavements with a PCI value of 90, the total maintenance cost increases to 72% and 249%, when delayed for 2 and 5 years, respectively.

Results of Delaying Maintenance on the Asphalt Runway Pavement of an Airport

The same hypothetical case study project was considered with asphalt runway pavement. All variables were the same as the concrete runway pavement. Figure 5-7 shows the change in total maintenance cost of the asphalt runway pavement with PCI values of 70, 80, and 90 for 2- and 5-year delayed scenarios. Similar to the concrete runway pavement, the total maintenance cost of

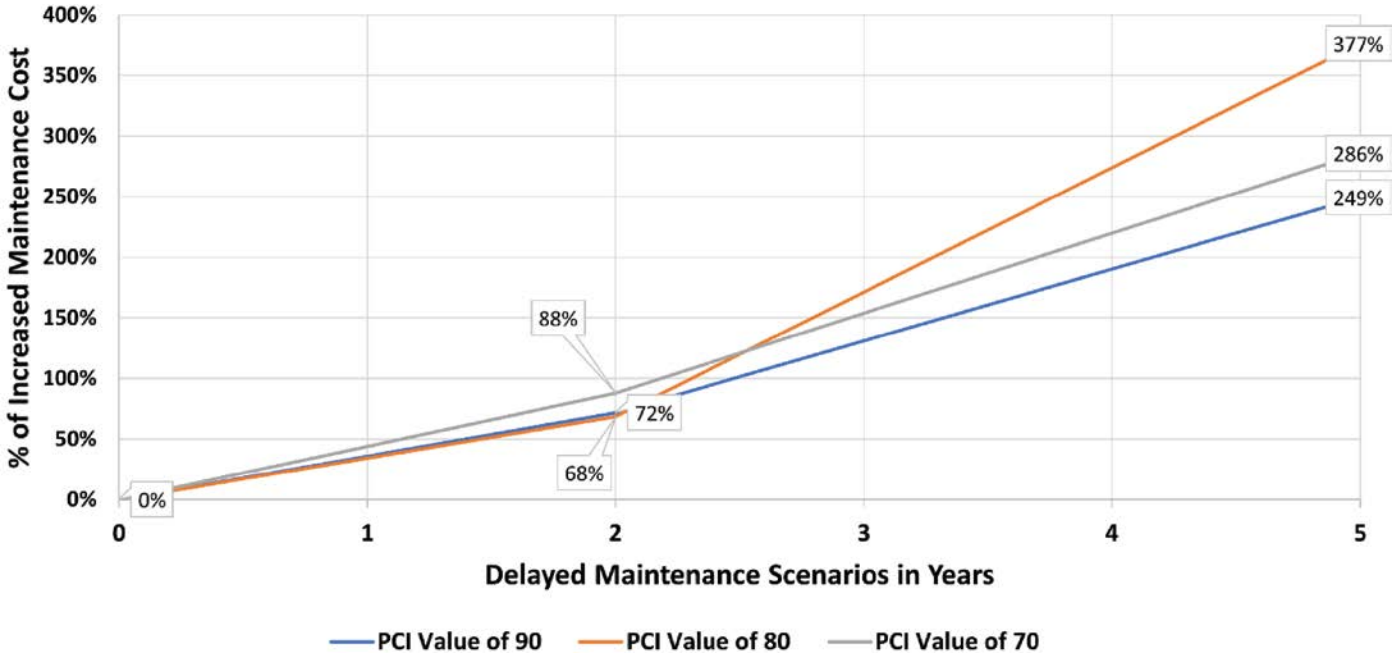


Figure 5-6. Percentage increase in the maintenance cost of a concrete runway pavement.

Copyright National Academy of Sciences. All rights reserved.

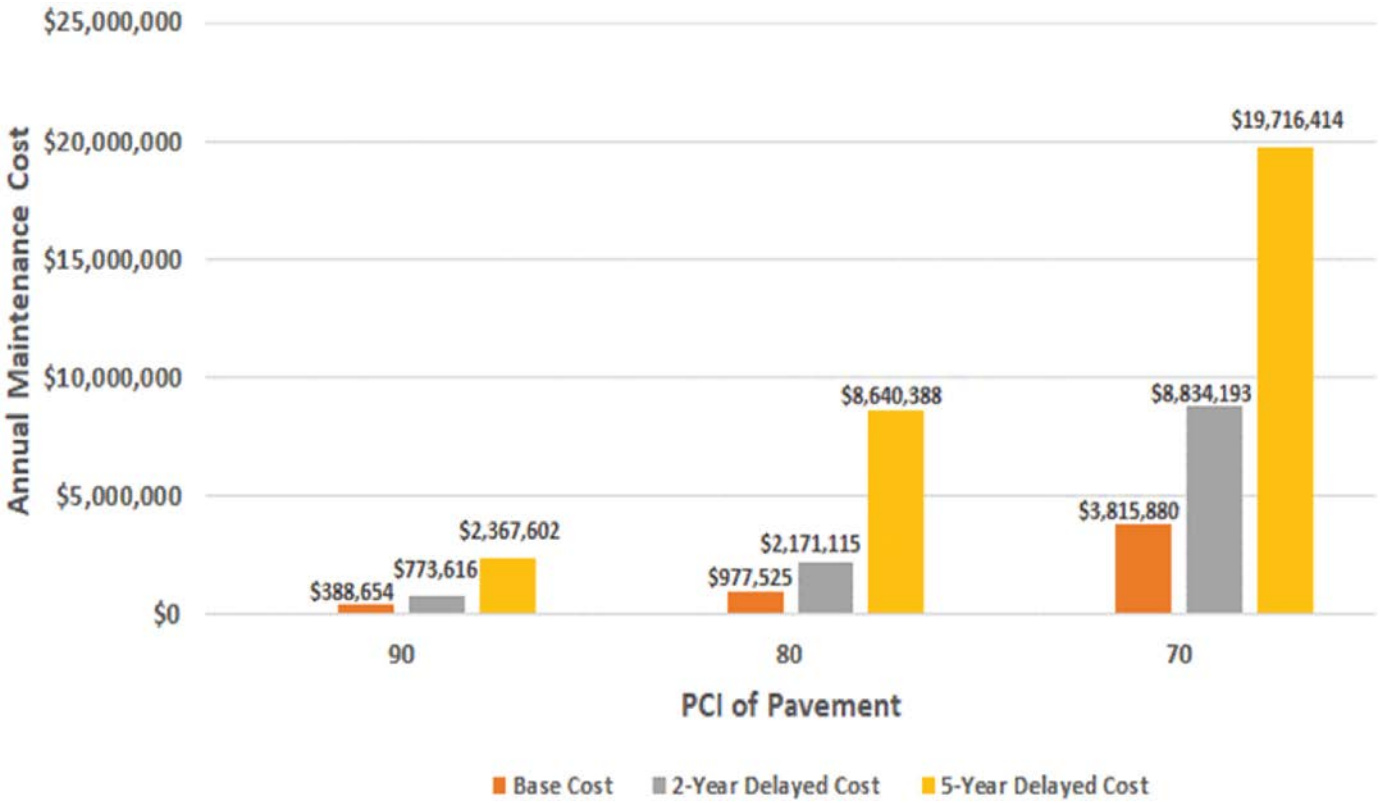


Figure 5-7. The impacts of delaying maintenance on the total maintenance cost of an asphalt runway pavement.

asphalt increases significantly when the maintenance is delayed for 2 or 5 years for pavements with PCI values of 70 and 80, compared to a PCI value of 90. For asphalt runway pavement with a PCI value of 70, the total maintenance cost increases significantly compared to its base cost. The base maintenance cost of the pavement with a PCI value of 70 is $3,815,880 but increases by 132% and 417%, respectively, when it is delayed by 2 and 5 years. These results show that the increase in total maintenance cost depends on the PCI value or age of the asphalt runway pavement.

Figure 5-8 shows the percentage increase in maintenance cost of asphalt runway pavement compared to base cost when the maintenance is delayed for 2 or 5 years. The analysis shows that the percentage increase in total maintenance cost can be seen in all three types of pavements based on their PCI values. However, when the PCI value of asphalt runway pavement is low (70), the percentage increase in maintenance cost is not significantly higher when maintenance is delayed for 2 or 5 years compared to runway pavement that has a PCI value of 90 or 80. For pavement with a PCI value of 70, the base maintenance cost without delay is already high, so the 2- and 5-year delay costs compared to base cost are not significantly different. In absolute dollar values, the 2- and 5-year delay maintenance cost of pavement with a PCI value of 70 is more than the asphalt pavement with PCI values of 90 and 80. When the PCI value of asphalt concrete is 80, the maintenance cost increases by 784% compared to its base cost when it is delayed by 5 years.

Intended Use of the Spreadsheet Tool

The spreadsheet tool (found in Section 2.1 of the Quick Guide) results showed that, when the PCI value of asphalt and concrete pavement is 70, the 2- and 5-year delay in maintenance cost increases significantly. Thus, this spreadsheet tool should generally be used for PCI values above 70 because maintenance becomes less effective, and rehabilitation is usually needed as

Copyright National Academy of Sciences. All rights reserved.

40 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

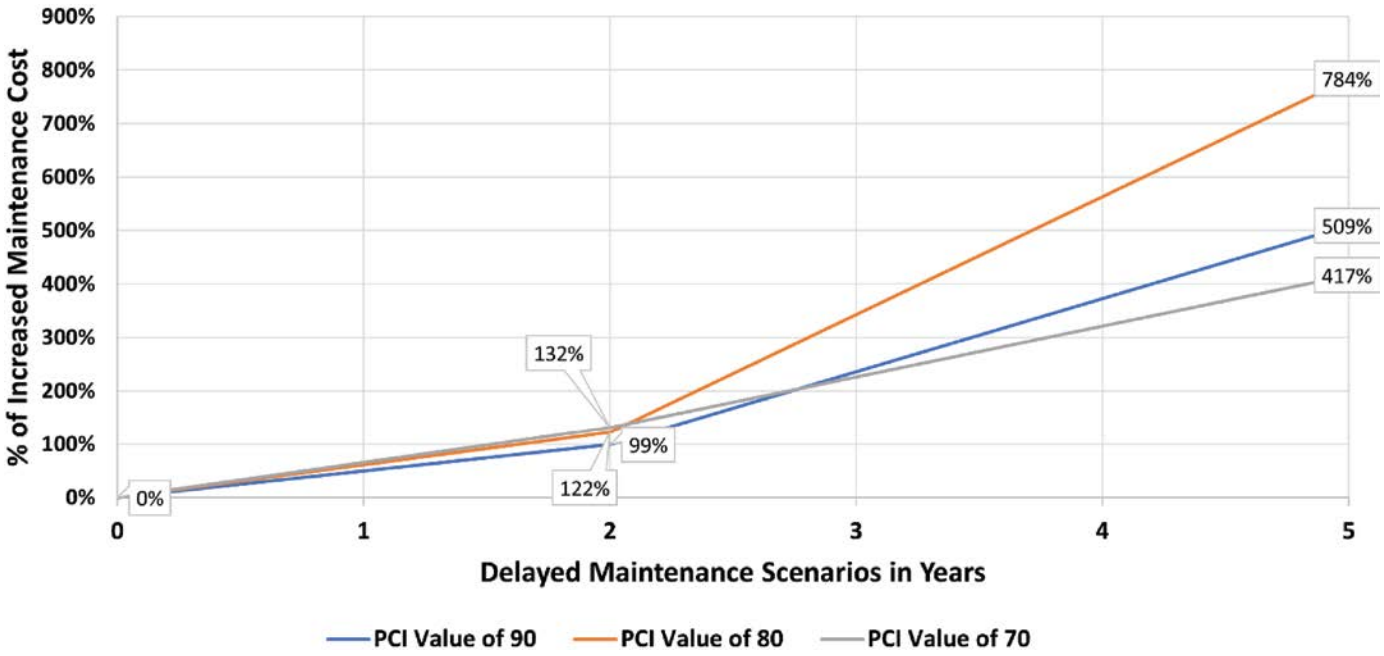


Figure 5-8. Percentage increase in maintenance cost of an asphalt runway pavement.

the pavement falls below PCI values of 65 or 70. That is why there is a big difference in delayed maintenance cost for a PCI value of 90/80 versus 70. Airports that delay maintenance when the PCI value is 70 need to pay for a mill and inlay instead of preventive maintenance.

5.2 Airfield Markings

Seven steps similar to runway pavement are used to determine the impact that delayed maintenance of airfield markings has on the total maintenance budget. Figure 5-9 shows the framework for calculating the consequence of delayed maintenance on runway markings. The steps involved in calculating the consequences of delayed maintenance are described below.

5.2.1 Establishing Asset Inventory

Airport asset managers collect pavement marking inventory related to types of runway markings, length, and width, outlining the spacing between the markings. *FAA AC 150/5340-1M—Standards for Airport Markings* discusses the various types of runway and taxiway markings and their dimensions and spacing provided in airport runways. This document describes six types of runway markings:

1. Designation markings,
2. Centerline markings,
3. Threshold markings,
4. Aiming point markings,
5. Touchdown zone markings, and
6. Edge line markings.

Figure 5-10 shows these runway markings and their locations along the runway pavement. FAA AC 150/5340-1M documents contain the number and size of the runway markings. Based on the minimum requirements of each marking, airports can determine the total area and record it in their asset inventory database.

Copyright National Academy of Sciences. All rights reserved.

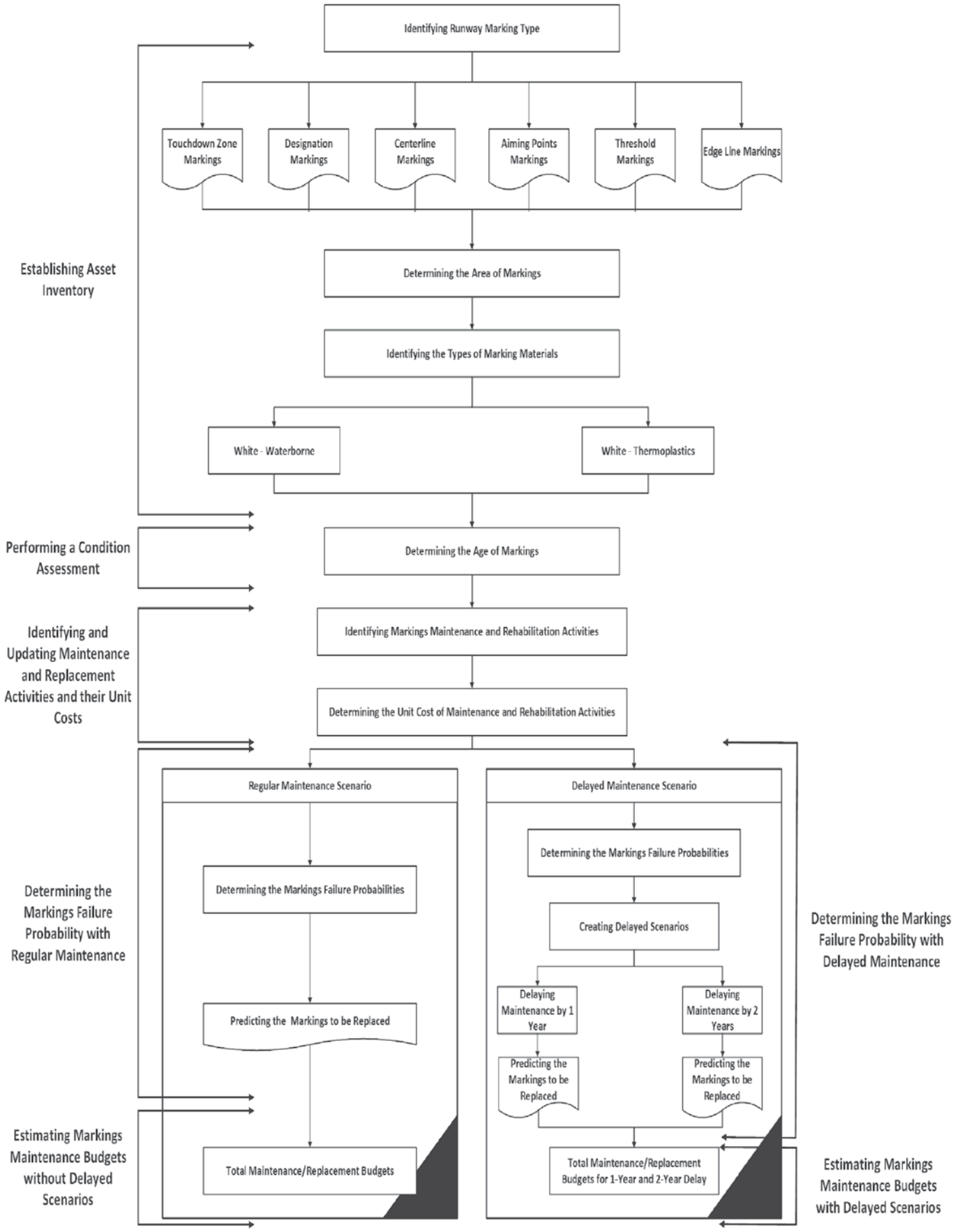


Figure 5-9. A delayed maintenance framework for runway markings.

Copyright National Academy of Sciences. All rights reserved.

42 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

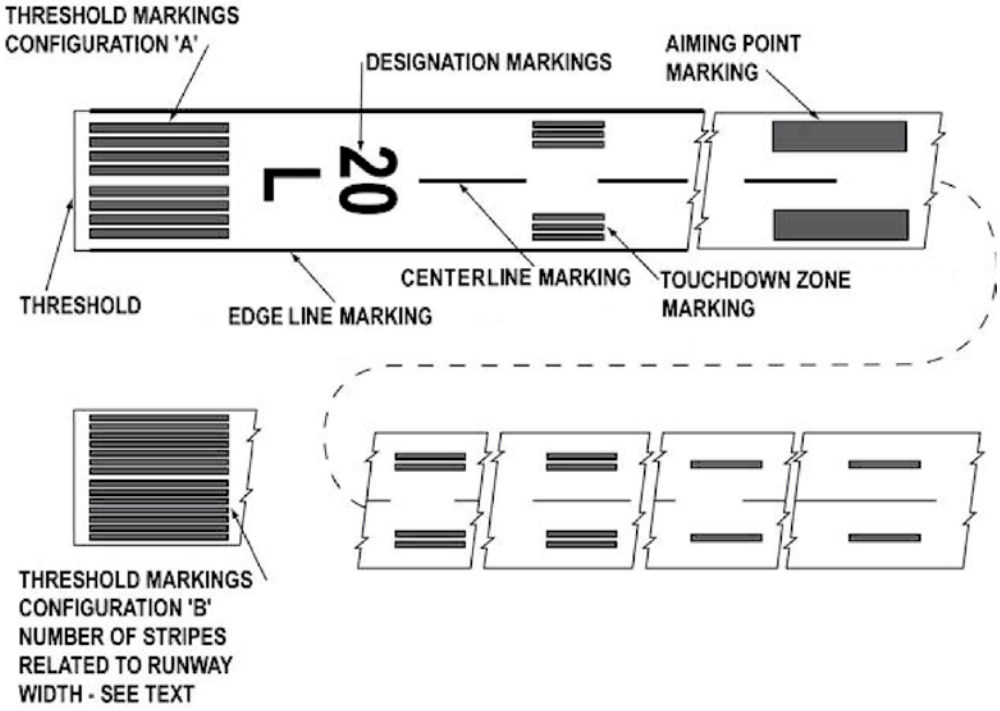


Figure 5-10. Types and locations of runway markings. Source: https://www.faa.gov/air_traffic/publications/atpubs/aim_html/chap2_section_3.html#7X7230ROBE.

Depending on the type of airport, all of these markings are not required. The FAA has prescribed the requirements of these runway markings. The minimum requirements for surface runway marking schemes are a direct function of the approach category for each runway threshold and the existence of displaced thresholds, stopways, blast pads, or extra-wide shoulders. Table 5-6 shows the complete runway surface marking schemes required for a runway based on the threshold approach category.

Table 5-6. Minimum required runway marking for paved runways (FAA AC 150/5340-1M).

Runway surface markings	Threshold approach category		
	Visual approach	Non-precision approach (And approaches with vertical guidelines not lower than ¾ -statute mile visibility)	Precision approach (Approaches with lower than ¾ -statute mile visibility)
Designation	X	X	X
Centerline	X	X	X
Threshold	Note 1	X	X
Aiming point	Note 2	Note 3	X
Touchdown zone			X
Edge line	Note 4	Note 4	X

Note 1: Required on runways serving approach categories C and D airplanes, and for runways used, or intended to be used, by international commercial air transport.
Note 2: Required on 4,200-foot (1,280 m) or longer runways serving approach categories C and D airplanes.
Note 3: Required on 4,200-foot (1,280 m) or longer instrumented runways.
Note 4: Utilized when the full runway pavement width may not be open for use.
Source: https://www.faa.gov/documentLibrary/media/Advisory_Circular/150-5340-1M-Chg-1-Airport-Markings.pdf.

Copyright National Academy of Sciences. All rights reserved.

The runway markings asset inventory should consist of the date the marking was repainted and the types of materials used. Generally, water-based or thermoplastic colors are used for runway and taxiway markings. Asset managers should record the total area of runway markings based on the types of runways. The purpose, location, color, and characteristics of these runway markings are described below.

Designation Markings

The purpose of a designation marking is to identify the runway endpoint. Runway landing designator marking(s) must be located within a certain distance from the runway threshold markings as shown in Figure 5-10. The color of this marking is generally white, and the color must be enhanced using certain techniques. A runway landing designator marking consists of a one- or two-digit number. When parallel runways exist, the number is supplemented with a letter. For more information, please refer to FAA AC 150/5340-1M.

Centerline Markings

The purpose of a runway centerline marking is to identify the physical center of the runway width and provide alignment guidelines to pilots during takeoff and landing operations. It is located along the physical center of the runway width and spaced between the runway landing designation markings as shown in Figure 5-10. The color of the runway centerline is white. A runway centerline marking consists of a line of uniformly spaced stripes and gaps of uniform width. The stripes are 120 feet in length and the gaps are 80 feet in length. The minimum width of the stripes is 36 inches for precision runways, 18 inches for non-precision runways, and 12 inches for visual runways.

Threshold Markings

A runway threshold marking, commencing 20 feet from the actual start point of the runway threshold, closely identifies the actual beginning point of the runway threshold used for landings. It starts 20 feet from the starting point of the runway threshold as shown in Figure 5-10. Its marking color is white, and it consists of a longitudinal striped pattern of uniform dimensions spaced symmetrically on the runway centerline. The runway width determines the number of longitudinal stripes and their spacing. Table 5-7 shows the number of stripes based on the width of the runway.

Aiming Point Markings

The main purpose of aiming point markings of the runway is to provide a visual aiming point for landing operations. It is located along the physical center of the runway width and spaced between the runway landing designation markings (Figure 5-10). It consists of a line of uniformly spaced stripes and gaps of uniform width. The markings are 120 feet in length, and the distance between them is 80 feet. The minimum width of the stripes is 36 inches for precision runways, 18 inches for non-precision runways, and 12 inches for visual runways. Aiming point markings are white.

Table 5-7. Number of runway threshold stripes for standard runway widths.

Standard runway width	Number of symmetrical stripes
60 feet	4
75 feet	6
100 feet	8
150 feet	12
200 feet	16

Copyright National Academy of Sciences. All rights reserved.

44 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Touchdown Zone Markings

The main purpose of touchdown zone markings is to identify the touchdown zone for landing operations along a precision runway. The touchdown zone marking is provided in 500-foot increments. It consists of symmetrically arranged pairs of rectangular bars in groups of one, two, and three along the runway centerline (Figure 5-10). It maintains a 900-foot "no-marking zone" from the midpoint of the runway toward the threshold. FAA AC 150/5340-1M states, "those pairs of surface markings that extend within 900 feet of the runway midpoint are eliminated. The intent of this painting practice is to preserve a 1,800-foot unmarked area so pilots do not confuse the surface markings during a landing with the surface markings for the other approach procedure" (FAA 2020, pp. 2–13). Touchdown zone markings are white.

Edge Line Markings

The runway edge marking provides an enhanced visual contrast between the runway edge and the surrounding terrain or runway shoulders, and delineates the suitable paved area width for runway operations. It consists of two parallel stripes, one placed along each edge of the usable runway with the outer edge of each stripe approximately on the edge of the usable paved runway. The maximum distance between the outer edges of the parallel stripes is 200 feet. It has a minimum width of 36 inches for 100 feet or wider runways and at least 18 inches on smaller runways. This marking should extend the full length of the runway (See Figure 5-10). This marking is also white.

Taxiway Markings

Various types of taxiway markings are provided in detail in the FAA AC 150/5340-1M document. Some of those taxiway markings are: (1) taxiway centerline markings, (2) taxiway edge markings, (3) surface-painted holding position signs, (4) surface-painted taxiway direction signs, (5) surface-painted taxiway location signs, (6) surface-painted gate destination signs, (7) surface-painted apron point entrance signs, (8) taxiway shoulder markings, (9) geographical position markings, and (10) apron control markings. These markings are either white or yellow.

5.2.2 Performing a Condition Assessment

The airport asset manager needs to perform a condition assessment of runway markings annually to make sure that the markings achieve the required objectives. These runway markings play an important role in assisting pilots to land their airplanes safely. It is therefore important for airport authorities to conduct condition assessments of runway and taxiway markings. Runway and taxiway markings deteriorate from wear, ultraviolet light, wind, rain, snowplowing, sweeping, and more. The FAA has prepared a guidebook for determining the effectiveness of airport pavement markings (FAA 2003). This guidebook provides quantitative criteria for inspectors and airport operators to determine the condition of airfield markings. Generally, the marking condition is assessed by visual inspections of segments of these markings, but these inspections and their validity are often subjective. The runway markings do not need to be noted every year; however, certain criteria need to be evaluated to determine whether a marking needs maintenance (Speidel et al. 2008). These criteria include:

- Faded colors or appearance.
- Poor nighttime visibility or retro-reflectivity.
- Existing markings are worn 50 percent or more.
- Existing markings are covered with contaminants.

Taking a photograph of the deteriorated markings is an effective practice to use during marking condition assessment. While performing condition assessments, the markings are evaluated for the following conditions:

Copyright National Academy of Sciences. All rights reserved.

Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

1. Layers of paint from older markings,
2. Rust discoloration,
3. Algae growth,
4. UV damage,
5. The position and dimension of existing markings,
6. The alignment of existing markings for standard compliance, and
7. The composition of existing material/coatings and compatibility with specified materials.

The FAA has developed a method for quickly and accurately evaluating paint markings. A manual method can be used to evaluate smaller surface markings. The FAA has developed an automated method for evaluating larger surface markings over a vast airport area. In addition, the method establishes a threshold pass/fail limit for white and yellow paint (Speidel et al. 2008). The runway markings can be evaluated in three ways:

1. By checking the retroreflectivity with a retrometer.
2. By checking the chromaticity (paint pigmentation) with a colorimeter or comparing it to color chips.
3. By using a transparent grid to inspect the coverage uniformity of the entire (remaining paint) markings.

Most airport asset managers inspect runway markings to check the aforementioned characteristics of the markings to maintain airport safety. However, no airport has developed a process to categorize the marking conditions (Excellent, Good, Fair, and Poor) to determine the runway markings' deterioration trend. Every airport performs visual inspections, and some use the retroreflectivity measurement instrument to check visibility. These condition data are generally kept along with asset inventory data. It is also required to record the types of paint used, either water-based or thermoplastic, to paint runway markings. The FAA has authorized the use of thermoplastic paint in taxiways. On runways, only waterborne white paints are allowed.

5.2.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

Markings need to be maintained in order to increase their life. Maintenance seeks to keep markings from discoloration due to rust, contamination, rubber deposits, and algae growth, so that the retroreflectivity of the markings will remain in excellent condition. The maintenance activities performed on airfield markings are:

1. Cleaning the markings to remove rust, contamination, rubber deposits, and algae growth.
2. Repainting the markings.
3. Maintenance activities to keep the level of retroreflectivity within a required limit.

Most airports have in-house personnel to perform marking maintenance. If major repainting or replacement needs to be done, contractors are hired. The airport needs to keep a cost database of maintenance activities to be performed on markings. If this maintenance work is done in-house, then the labor hours and materials spent need to be collected annually to estimate a maintenance budget with and without delays. Generally, waterborne airfield paints last for about 2–3 years, and thermoplastic paints last for 4–5 years if regularly maintained (Chang et al. 2017). Waterborne and thermoplastic markings do not result in a major difference in maintenance cost, however, replacement costs can vary greatly.

A case study project collected the unit cost of replacing and maintaining waterborne and thermoplastic markings. The contractor's bid cost of runway markings was used to determine the total maintenance cost of the markings. The waterborne and thermoplastic markings replacement costs were derived from the FHWA website, which calculates the cost of painting the

Copyright National Academy of Sciences. All rights reserved.

46 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

roadways (FHWA 2005). Those costs were adjusted to the 2021 base cost. Table 5-8 shows the waterborne and thermoplastic maintenance and replacement costs.

5.2.4 Determining the Marking Failure Probability with Regular Maintenance

The research team collected data on the average life of the waterborne and thermoplastic markings but was not able to collect any data on the deterioration pattern of runway and taxiway markings. The literature has shown that using Weibull distribution can help to estimate the deterioration of markings (Chang et al. 2017). The airfield marking deterioration model used in this study is based on the age of the markings, because the team could not retrieve the condition data from the airports. This age-based model estimates the remaining life of the pavement markings from historical records. The team could not use a condition-based approach to develop reliable deterioration models because it requires periodic condition assessment data. To use Weibull distribution to determine the probability of airfield marking failures, the shape and scale parameters need to be determined (Pham 2023). Equation 5-3 shows the cumulative probability of failure based on Weibull distribution.

$$f(x) = \left(\frac{\gamma}{\alpha}\right)\left(\frac{x}{\alpha}\right)^{(\gamma-1)} \exp\left[-\left(x/\alpha\right)^{\gamma}\right] \tag{5-3}$$

f represents failure, γ represents the shape parameter, and α is a scale parameter. In the above equation, the location parameter is 0. Many probability distributions are not a single distribution, but a family of distributions. The shape factor shows the types of probability distributions. These distributions are particularly useful in modeling applications, since they are flexible enough to model a variety of datasets. Depending on the different values of the shape factor, the probability distribution differs. Figure 5-11 shows the probability distribution for shape factors of 0.5, 1, 2, and 5. A shape factor of about 2 will show skewed distribution to the right. A scale parameter, sometimes called characteristic life, is related to the mean time to failure (Galar and Kumar 2017). In Weibull analysis, α is defined as the time at which 63.2% of components under analysis will have failed (Pushpa et al. 2006).

In the highway asset failure probability distribution, the shape factor is taken as 2; therefore, the team also used a shape factor of 2 to determine the failure probability of airfield markings (Chang et al. 2017). The scale factors for white waterborne and thermoplastic markings were selected as 1.3 years and 3 years because the lives of white waterborne and thermoplastic were 2–3 years and 5–6 years, respectively. Interviews with the airport asset managers during data collection confirmed that.

Table 5-9 shows the probability of failure of white waterborne runway markings derived using Weibull probability distribution. The results demonstrate that, with normal deterioration, white waterborne markings must be replaced entirely by the end of the third year.

Table 5-8. Maintenance and replacement costs used for runway markings.

Type of Cost	Cost per Unit
Maintenance cost per square feet	$0.17/ SF
Replacement cost per square feet for white waterborne markings	$2.35/SF The cost calculation is shown in the spreadsheet tool in Section 2.2 of the Quick Guide
Replacement cost per square feet for white thermoplastic markings	$3.07/SF from FHWA (2005) report The cost calculation is shown in the spreadsheet tool in Section 2.2 of the Quick Guide

Copyright National Academy of Sciences. All rights reserved.

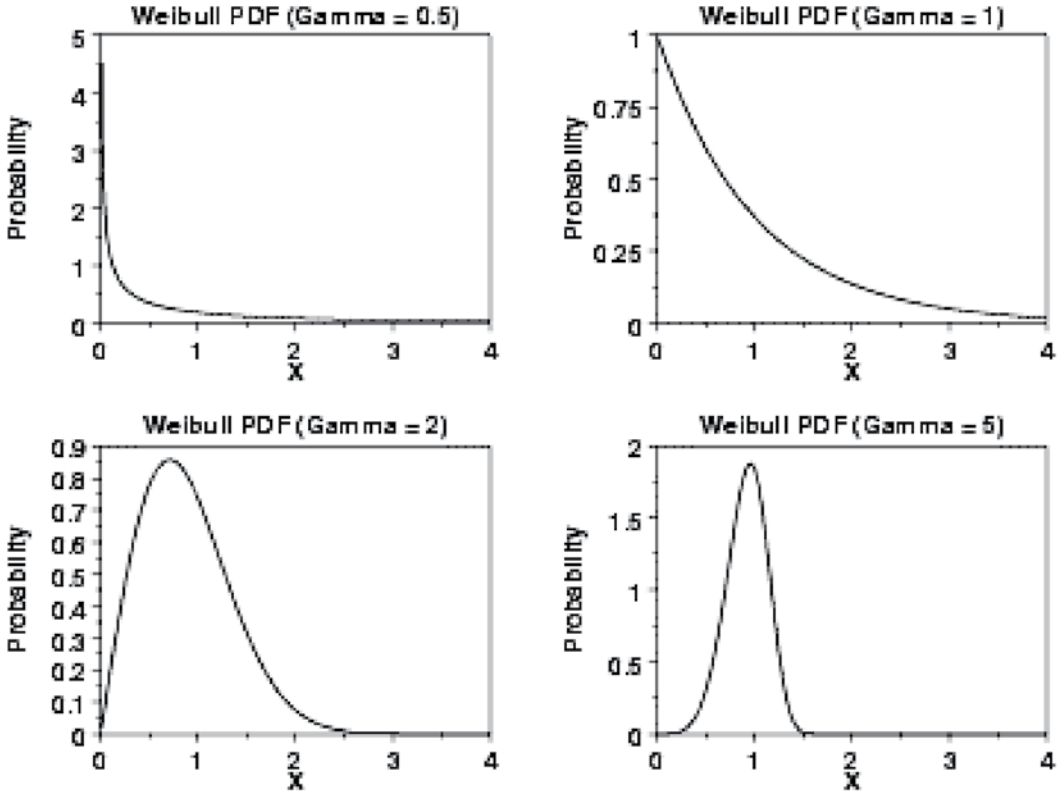


Figure 5-11. Weibull probability distribution for various shape factor values.

The research team followed a similar process to determine the failure probability of thermoplastic taxiway markings. Table 5-10 shows the probability of thermoplastic markings failure.

5.2.5 Determining the Marking Failure Probability with Delayed Maintenance

The research team also used the Weibull probability distribution model to determine the failure probability of white runway and taxiway markings with delayed scenarios. However, while calculating the failure probability in delayed scenarios, the team reduced the scale factor by 33%

Table 5-9. Weibull distribution for white waterborne runway markings.

Age of White Runway Markings	Cumulative Failure Probability
1 year	35.88%
2 years	83.10%
3 years	98.17%
4 years	100.00%

Table 5-10. Weibull distribution for thermoplastics taxiway markings.

Age of Thermoplastic Markings	Cumulative Failure Probability
1 year	10.5%
2 years	35.9%
3 years	63.2%
4 years	83.1%
5 years	93.8%
6 years	98.2%

Copyright National Academy of Sciences. All rights reserved.

48 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

due to a lack of regular maintenance. Table 5-11 shows the cumulative failure probability for white waterborne runways and thermoplastic taxiway markings assuming no maintenance was conducted. The scale factors for white waterborne and thermoplastic markings were selected as 1 and 2 years, respectively. The probability distribution results showed that the white waterborne markings need to be replaced after 2 years and the thermoplastic markings need to be replaced after 3 years due to lack of maintenance.

5.2.6 Estimating Marking Maintenance Budgets without Delayed Scenarios

After determining the types of markings used in runways or taxiways, the total area of markings to be maintained and replaced can be determined using the cumulative failure probability based on the age of the markings. The percentage of white waterborne marking areas to be replaced each year can be calculated using the values in Table 5-9. The replacement of markings will also depend upon the age of the markings. For example, if the runway marking's area is 1,000 square feet, and its age is 1 year, then the cumulative probability of waterborne white markings failure in year 2 (the next year) will be 83.10%. This means that about 831 square feet of markings need to be replaced, and the rest of the markings can be regularly maintained without replacement. Similar estimates can be performed if thermoplastics are used in taxiway markings by using the cumulative probability values provided in Table 5-10, based on the age of the markings.

Upon estimating the area of the markings to be maintained and replaced, the cost of regular maintenance can be determined by multiplying the area by the unit cost of maintenance and replacement. The research team developed a spreadsheet tool to calculate a regular maintenance budget, as well as delayed maintenance costs.

5.2.7 Estimating Marking Maintenance Budgets with Delayed Scenarios

The research team used procedures described in the previous section to estimate the maintenance and replacement cost of markings for delayed scenarios. The difference was the higher cumulative failure probability of markings due to neglected airfield marking maintenance. Delaying maintenance is not encouraged, because the team has not considered airplane safety when marking maintenance and replacement are delayed. The spreadsheet tool in Section 2.2 of the Quick Guide provides the dollar amounts that airports will need to spend when the maintenance and replacement of the airfield markings are delayed by a specific number of years. The tool uses the historical rate of inflation and provides the option to select the marking inflation factor, as the inflation of the future year is difficult to predict. The historical paint and coating manufacturing price index shows that the average inflation of the paints is about 3.23% per year (FRED 2022a). The marking maintenance and replacement cost data are from 2021. However, when the airport needs to use this spreadsheet tool in the future, the cost data will be automatically adjusted based on the budget

Table 5-11. Weibull distribution for white waterborne and thermoplastic markings without maintenance.

Age	Failure Probability (Waterborne)	Failure Probability (Thermoplastics)
1 year	63%	36%
2 years	98%	83%
3 years	100%	98%
4 years	100%	100%

Copyright National Academy of Sciences. All rights reserved.

preparation year, and the default inflation will be used to calculate the delayed maintenance cost. The airport can also enter its paint inflation rate when calculating the delayed maintenance costs.

5.2.8 Reporting the Impact of Marking Delayed Maintenance and its Consequences

The tool developed by the team was used to determine the impact of delayed maintenance on markings. A hypothetical airport case study was developed to estimate the impact of delayed maintenance of markings on total maintenance and replacement cost (Table 5-12). The delayed scenarios are 1 and 2 years, and the runway and taxiway markings are new, 1, or 2 years old.

Case Study Results of the Consequences of Delaying Maintenance on Airfield Markings

Results of Delaying Maintenance on White Waterborne Runway Markings

Figure 5-12 shows the impact of delaying maintenance on white waterborne runway markings. The maintenance cost increases as the age of the runway markings increases. When the white waterborne runway markings have been recently replaced, the delayed maintenance cost is higher compared to 3-year-old markings. The data analysis shows that the maintenance cost of newly installed runway markings will more than double when delayed by 1 year. If the maintenance is delayed for 2 years, the total maintenance cost does not change much because delaying 1 year of maintenance will require more than 98% of the markings to be replaced (Table 5-11). When the team analyzed runway white waterborne markings of ages 1 and 2 years, the impact of delayed maintenance seemed minimal, because the markings need to be replaced as soon as possible in normal deterioration conditions. Based on Table 5-11, if the runway waterborne markings are 2 years old, then the next fiscal year, the replacement percentage will be 100% based on delayed maintenance deterioration models. Delaying these failed runway markings will not have any effect on total replacement cost, but it can jeopardize airplane safety. In the case of runway markings, the effect of delaying maintenance is minimal because all airports replace their runway markings every 3 to 4 years.

Figure 5-13 shows the increase in total maintenance cost for three types of white waterborne runway markings based on the age of the markings. It clearly shows that, when the runway markings are new, the delay in maintenance of the markings has a significant impact on the total maintenance cost. However, after the runway markings start aging, the maintenance delay does not have a significant impact. This is because white waterborne markings are required to be replaced every 2 to 3 years. It is critical that airport asset managers consider airplane safety before delaying

Table 5-12. Case study data of airport runway and taxiway markings.

Name of the airport	XYZ International Airport
Type of airport	Small-hub/medium-hub/large-hub airport
Location of the airport	Southwest, United States
Runway dimensions	9,000 ft length and 150 ft width
Taxiway marking areas	30,000 square ft
Type of paint	White waterborne for all runway markings Thermoplastics for taxiway markings
Age of the markings	0, 1, or 2 years old
Budget preparation year	2022
Budget to be prepared for	2023
Delayed scenarios	1 and 2 years
Price inflation factor	3.23% for waterborne, as well as thermoplastics
Discount factor	0%

Copyright National Academy of Sciences. All rights reserved.

50 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

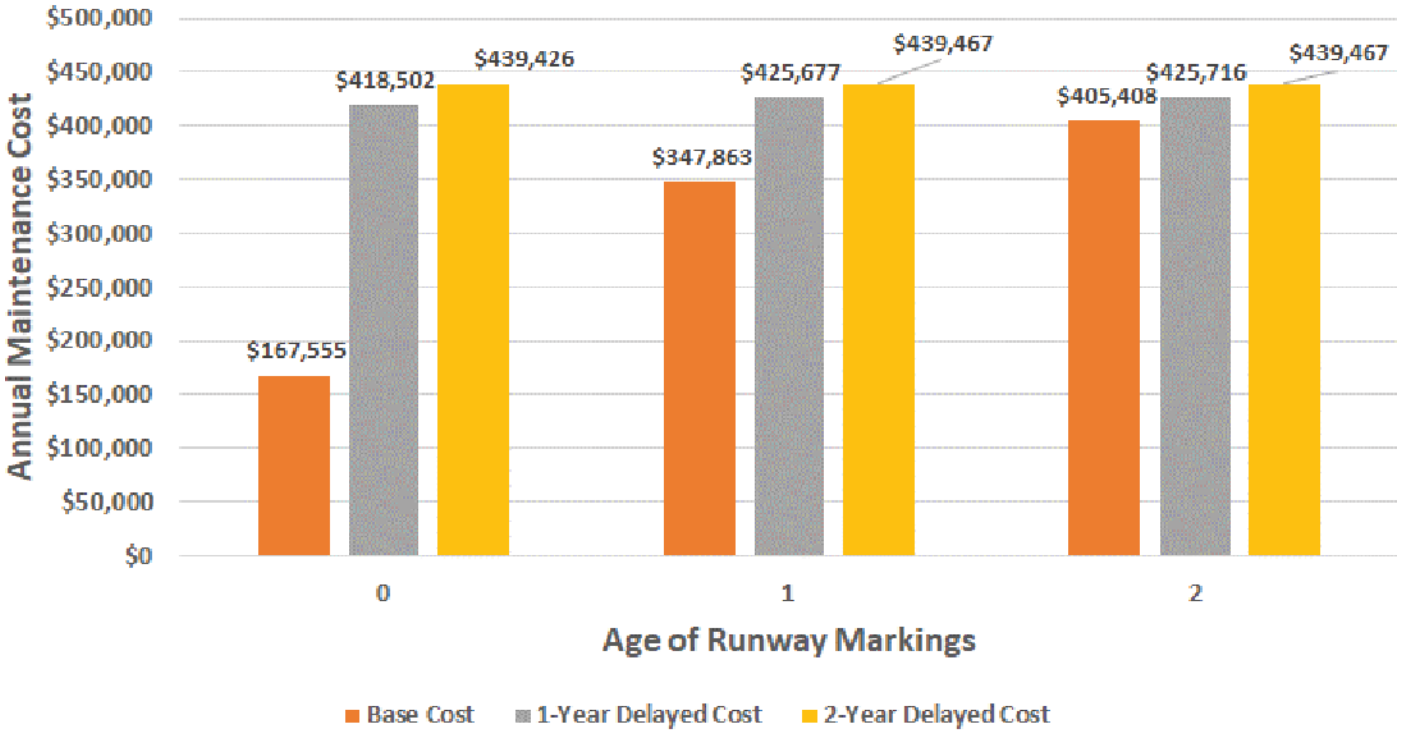


Figure 5-12. Impact of delaying maintenance on total maintenance cost of white waterborne runway markings of an airport.

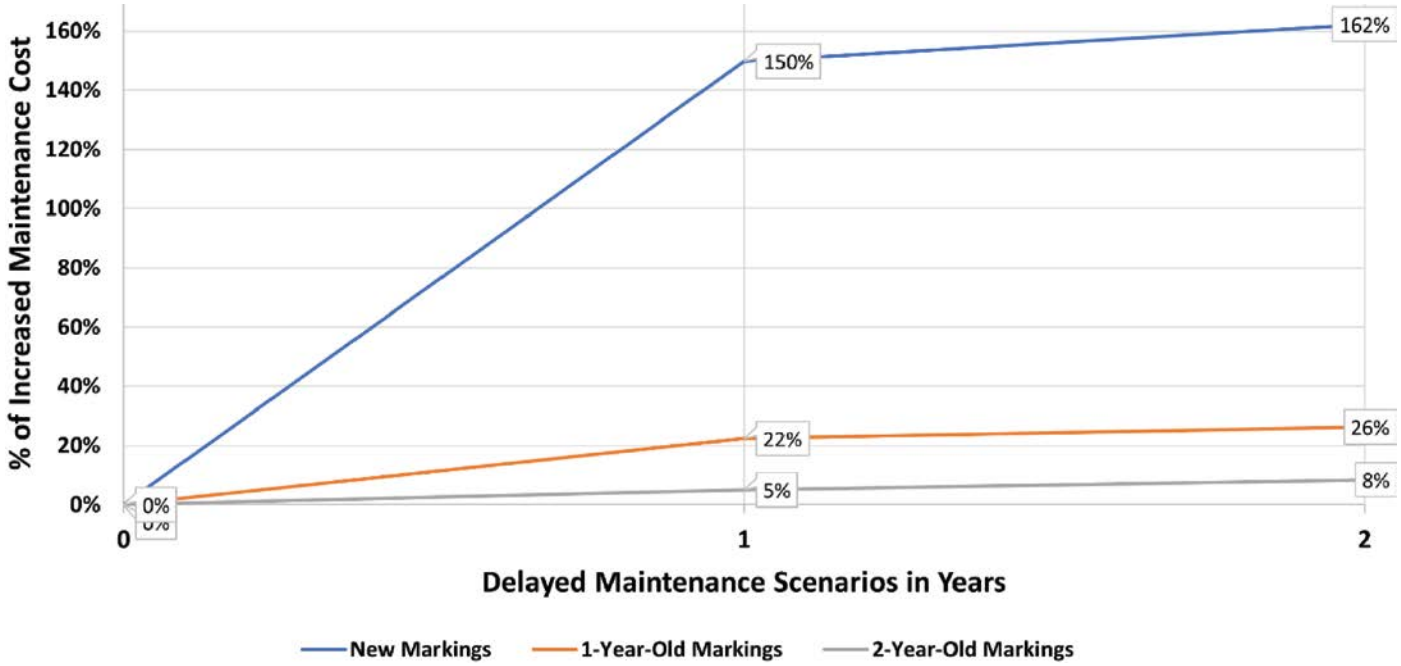


Figure 5-13. Percentage increase in maintenance cost of white waterborne runway markings.

Copyright National Academy of Sciences. All rights reserved.

maintenance or replacing runway markings. Any decision made by airport authorities related to delayed maintenance can have serious repercussions.

Results of Delaying Maintenance on Thermoplastic Taxiway Markings

The spreadsheet tool (found in Section 2.2 of the Quick Guide) developed was used to determine the impact of delayed maintenance on the total maintenance cost of thermoplastic taxiway markings. The research team used the hypothetical airport project data shown in Table 5-12. Similar to the white waterborne runway markings case study, the thermoplastic taxiway markings were assumed to be new, 1 and 2 years old. To determine the impact of delayed maintenance, the maintenance was delayed for 1 and 2 years. Figure 5-14 shows the increase in total maintenance cost of the thermoplastic taxiway markings (30,000 square feet) for new, 1-year, and 2-year-old markings, assuming the maintenance was delayed for 1 and 2 years. The base cost shows the total maintenance cost required to maintain and replace the thermoplastic taxiway markings when the markings are recently painted, 1 year, or 2 years old is $15,316, $38,798, and $64,105, respectively. The delayed scenario for 1 and 2 years shows that the total maintenance cost increases significantly compared to the base cost. As the taxiway markings' age increases, the total maintenance cost for delayed scenarios does not increase at a higher rate due to the reasons mentioned above.

Figure 5-15 shows the percentage increase in total maintenance and the replacement cost of thermoplastic taxiway markings based on the age of the markings. The graph shows that the total cost increases 456% when the markings are new and delayed by 1 year. However, delaying maintenance by another year will not have a significant impact, because it will only increase the cost by 115% compared to the 1-year delay. The reason is that the life of thermoplastic markings is about 3 to 4 years when the markings are not regularly maintained. By the end of 2 years, more than 83% of the markings must be replaced (See Table 5-11). Delaying another year will increase the marking replacement cost by 15%, which has an insignificant impact on the total maintenance and replacement cost. If the markings are 1 year old, the cost to maintain them in the 2nd year

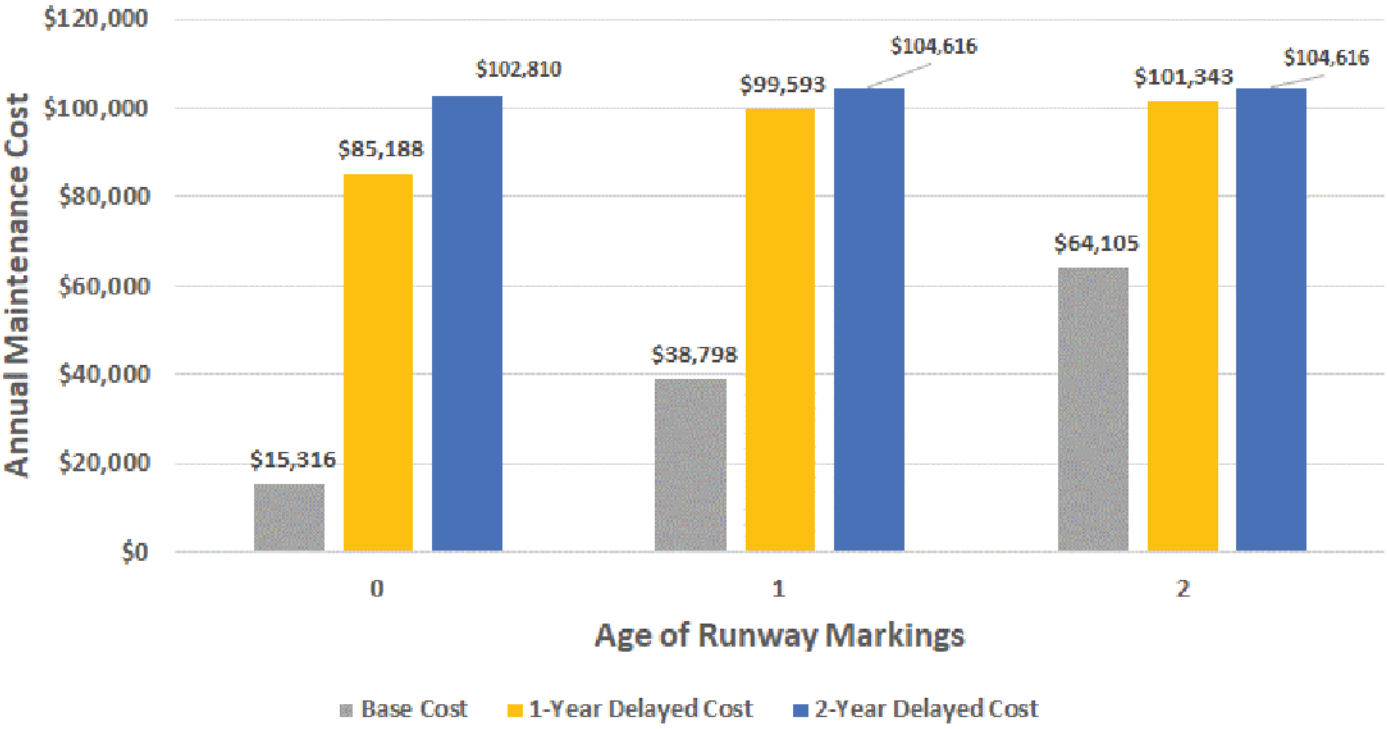


Figure 5-14. Impact of delaying maintenance on total maintenance cost of thermoplastic taxiway markings.

Copyright National Academy of Sciences. All rights reserved.

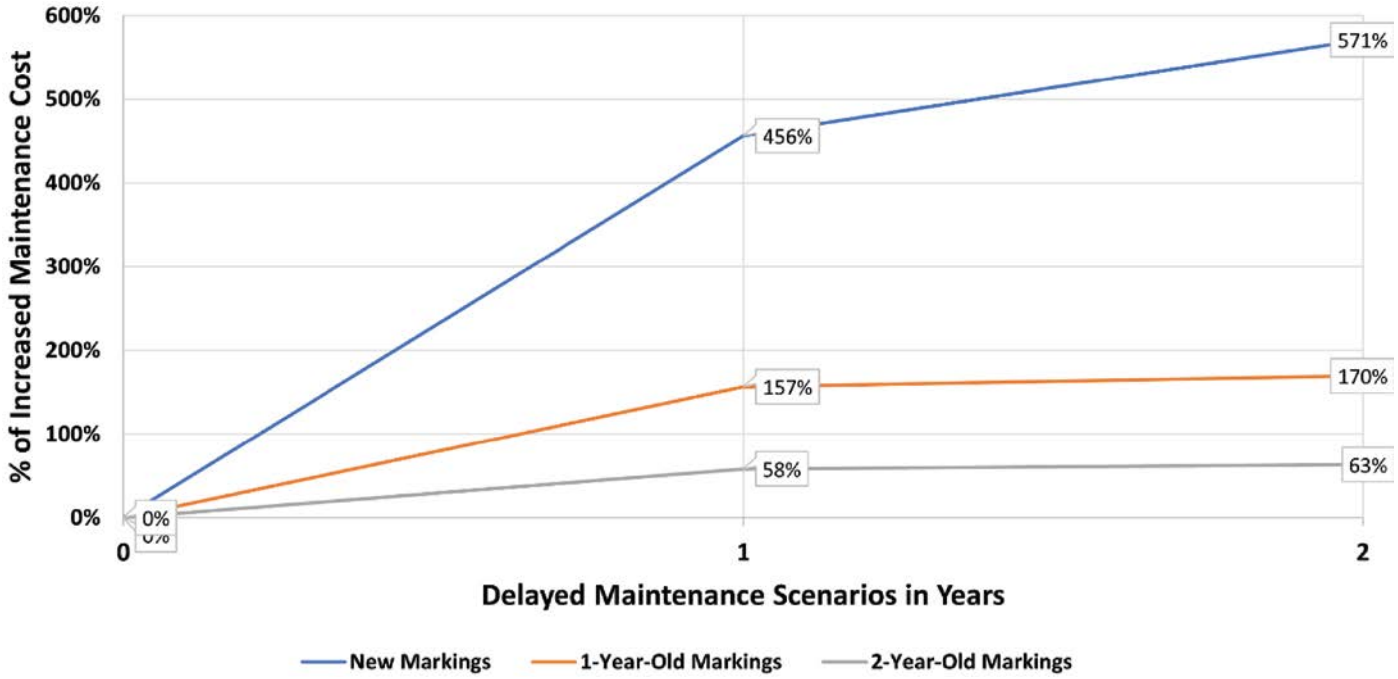


Figure 5-15. Percentage increase of maintenance cost of thermoplastic taxiway markings.

will be comparatively high. This is because the failure probability distribution shows that 35.9% of the thermoplastic markings need to be replaced in the 2nd year, which is 25.4% higher than new markings (See Table 5-10). In cases with 1- and 2-year-old markings, delaying the maintenance and replacement by 1 or 2 years will not have a significant increase in the total cost due to the reasons described above.

5.3 Runway Lighting

Runway lighting is critical infrastructure in an airport. These lights assist the pilots in landing the planes safely at night, so these lights should be maintained frequently. If the lighting maintenance is delayed, then night flights need to be canceled because planes cannot land without these lights. The framework and tools developed in this research can hypothetically determine the effects of delaying the maintenance of runway lights on the budget. Figure 5-16 shows the framework for calculating the cost impact of delayed maintenance on runway lighting. The framework is described below in detail.

5.3.1 Establishing Asset Inventory

This study focuses on lighting provided around runways. The spreadsheet tool that was developed does not cover any lighting provided in the apron or taxiways. FAA AC 150/5345-46E describes various types of runway lights. The major types are (1) runway edge lights, (2) runway centerline lights, (3) runway approach lights, (4) runway end lights, (5) runway threshold lights, and (6) touchdown zone lights. There are other lights provided around runways such as the Medium Intensity Approach Lighting System, Runway Status Lights (RWSL), Takeoff Hold Lights, and Runway Intersection Lights. Figure 5-17 shows the layout of runway lighting.

Table 5-13 shows lighting color and spacing in the runways. Asset managers need to keep inventory of these lights along with the electrical line distribution, the record of the transformer,

Copyright National Academy of Sciences. All rights reserved.

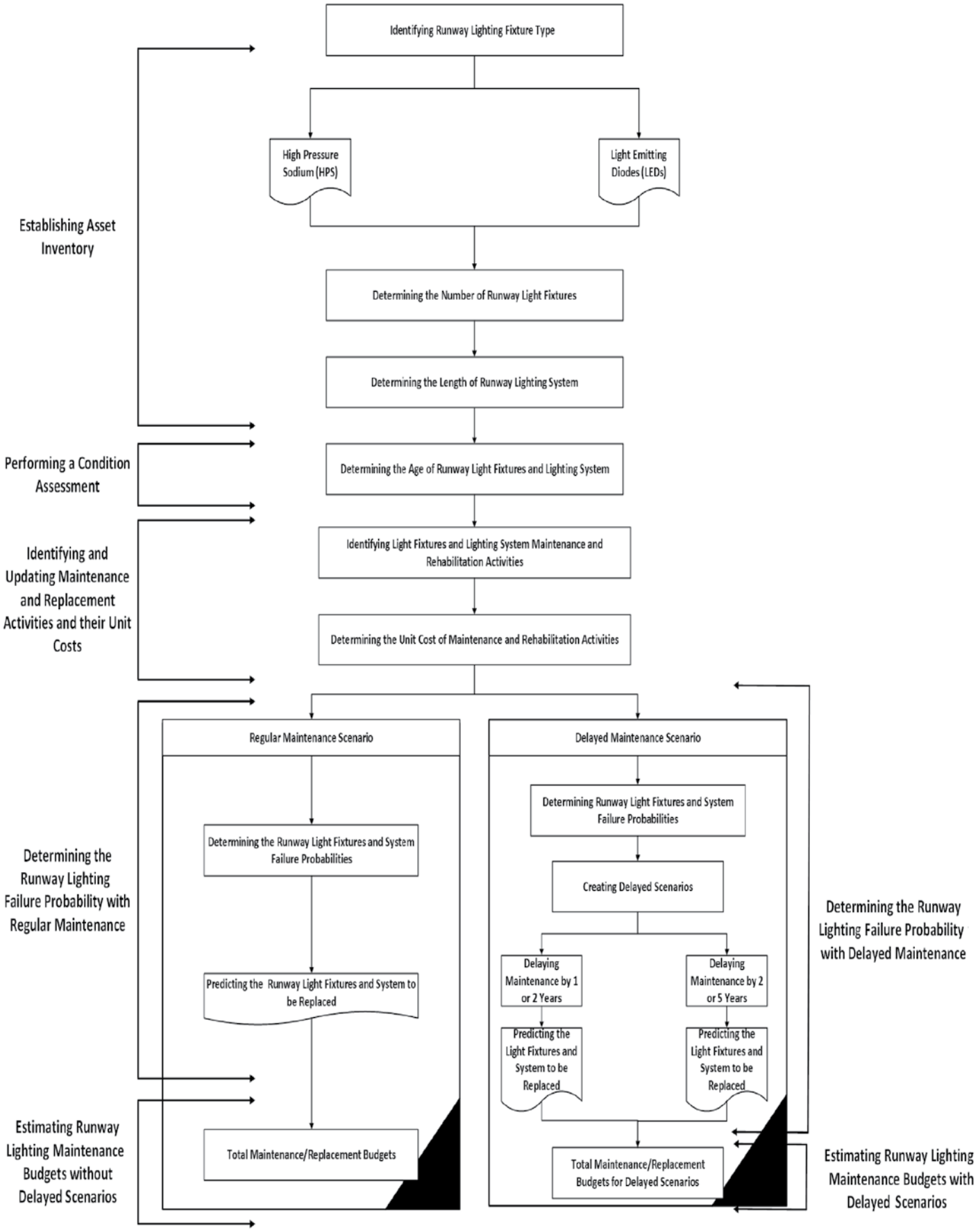


Figure 5-16. A framework for delayed maintenance of runway lighting.

Copyright National Academy of Sciences. All rights reserved.

54 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

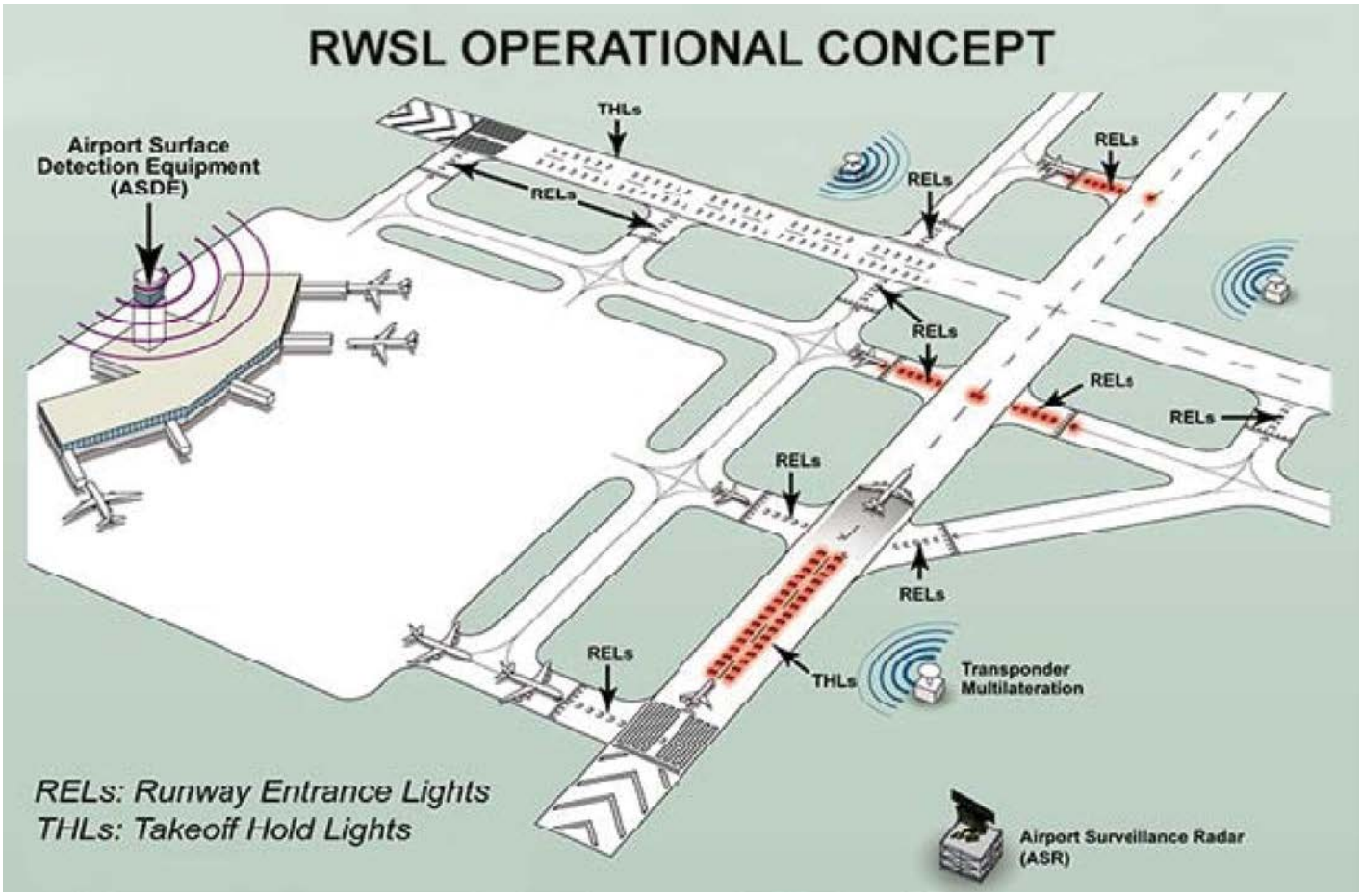


Figure 5-17. Various types of lighting systems around runways. Source: FAA, https://blog.alquilercastilloshinchables.info.

Table 5-13. Runway lighting characteristics (FAA AC 150/5345-46E and Aviation Services).

Runway Lighting Types	Light Direction and Color	Light Spacing
Runway edge lights	Bidirectional: white-white, white-yellow, white-red, yellow-red, yellow-green	200 ft
Runway centerline lights	Bidirectional: white-white, white-red Unidirectional: white, red	50 ft
Runway threshold lights	Bidirectional: green-red, red-red Unidirectional: green Unidirectional: red	10 ft
Runway end lights	Bidirectional: green-red, red-red Unidirectional: green Unidirectional: red	10 ft

Copyright National Academy of Sciences. All rights reserved.

voltage regulators, main switchboards, etc., to calculate the accurate maintenance and replacement cost. Therefore, the length of the underground wiring, locations of the transformer, voltage regulators, switch boxes, and connectors must be collected to determine the maintenance cost of these systems.

5.3.2 Performing a Condition Assessment

One of the asset manager's most important jobs is to perform condition assessments of the runway lighting systems. Asset managers need to collect the condition of light fixtures and electrical lines and ideally categorize them by level, e.g., excellent, good, fair, and poor. Most of the airports the research team contacted don't have these data because the asset managers have not collected the data and entered it into the system. However, the airports inspect the light fixtures and electrical distribution system and conduct preventive maintenance annually, if necessary. The condition of lighting systems generally deteriorates due to environmental factors, like sunlight, rain, and wind. The light fixtures also degrade naturally. This includes bulbs, electrical conduits, transformers, connectors, and switch systems. In terms of runway lighting, the airports have not collected detailed condition assessment data. To estimate the maintenance and rehabilitation cost of the runway lighting and conduits, these data play a valuable role in determining reliable deterioration models. However, most of the airports have inventory data of the number of light fixtures, types of fixtures [LED or high-pressure sodium (HPS)], condition of the transformer, switching systems, and fixture connectors. Airports had not used these data to come up with a systematic method of categorizing the condition of runway lighting systems. Generally, the light fixtures used are either LED or HPS. Airports change the lighting fixtures as a part of their maintenance program. However, the replacement of electrical conduits and distribution lines rarely occurs. The maintenance of transformers, connectors, and switching systems is regularly performed by either in-house electricians or outside electrical contractors. *ACRP Report 148: LED Airfield Lighting System Operation and Maintenance* highlights inspection processes to determine the airfield lighting system conditions. According to *ACRP Report 148*, the airfield lighting system inspection should consist of the following:

- Visual and physical inspection,
- Electrical testing,
- Photometric testing, and
- System burn-in.

The type of power feed supplying the sign and its nameplate data should be verified. The field location and orientation of all the airfield lights should be inspected in terms of whether the exterior and interior lights are clean and clear of debris; the power supply is in a sealed enclosure or is encapsulated for its protection; the light covers and frame are free of scratches, cracks, or breaks; and all bolts and tethers are properly installed and secured. The components of lighting and electrical distribution systems should be inspected regularly to avoid breaking. These include connectors, grounding, spacers, top extensions or flange rings, bolt lengths, transformers, cable lengths, and conduit ends, among others.

5.3.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

Because runway lighting systems are vital for airplane safety, all of the airports spent the required budget to maintain them. The airports also performed regular maintenance work on runway lighting systems. *ACRP Report 148: LED Airfield Lighting System Operation and Maintenance* highlighted

Copyright National Academy of Sciences. All rights reserved.

56 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

some of the maintenance activities to be conducted on the runway lighting systems to keep these lights in working condition. Some of these activities include:

- Inspect for outages and repair as necessary,
- Check the cleanliness of lenses,
- Perform photometric testing (HIRL) and check light alignment and orientation,
- Re-align lights as needed,
- Clean fixtures and sockets,
- Check light elevation,
- Check for moisture in lights,
- Inspect fixture for rust and deterioration,
- Check lamp fitting and clean contacts,
- Check gaskets, and
- Remove snow and/or vegetation from around lights.

ACRP Report 148: LED Airfield Lighting System Operation and Maintenance found that about 57% of airports surveyed used their in-house electricians to maintain their airfield lighting systems. The study also recommended inspection methods of airfield lighting systems to conduct efficient maintenance work. Along with maintenance activities, the airport must collect and update the following costs of the runway lights to calculate the required maintenance cost:

- LED light fixture cost,
- HPS light fixture cost,
- Maintenance cost (lamp replacement labor and materials),
- Refurbishment cost, and
- Maintenance cost of electrical conduit (labor and materials).

After determining the maintenance activities to be performed for the runway lighting system, the asset manager must determine the unit cost for maintenance and replacement. During the data collection phase, the research team collected the maintenance and replacement labor and parts cost data. The team used these data to determine the unit cost of maintenance and replacement. The airports tend to use HPS or LED runway light fixtures, therefore the research team collected the cost of these lights and verified using the RSMeans Cost Guide (2021). Table 5-14 shows the maintenance cost calculations using the labor hours the airports reported.

The replacement cost of the lighting systems was also collected from the airport case study and verified with the RSMeans Cost Guide. The research team gathered the replacement costs of HPS and LED light fixtures from an online search and verified them with the RSMeans Cost

Table 5-14. Unit maintenance cost calculations for airfield lighting systems.

Category	Calculation
Number of electricians involved	4
Number of electrician hours	4 x 260 days/year x 8 hours/day = 8,320 hours
Unit cost of electricians	$60/hour
Total annual labor cost	$499,200
Parts cost (15% of total cost)	$499,200 x (0.80)/(0.20) = $1,996,800
Total cost with 20% miscellaneous	$2,995,200
Total length of lighting systems	20 miles x 5,280 ft/miles = 105,600 ft
Maintenance cost per linear foot	$28

Copyright National Academy of Sciences. All rights reserved.

Table 5-15. Replacement costs for airfield lighting systems.

Type of Lighting System	Cost per Unit
Replacement cost of lighting system per foot	$490 from the case study airport, compared to the RSMeans Cost Guide, in which the cost per linear foot of theatrical black and white wires is $255
HPS light fixtures	$750 from the case study airport, compared to the RSMeans Cost Guide, in which a 150-watt HPS bulb fixture is $764
LED light fixtures	$950 from the case study airport, compared to the RSMeans Cost Guide, in which the average cost of an LED bulb fixture ranges from approximately $800 to $1,100

Guide. Table 5-15 shows the replacement cost of the lighting system per foot, HPS, and LED runway light fixtures.

5.3.4 Determining the Runway Lighting Failure Probability with Regular Maintenance

When collecting data regarding the airfield lights, the research team found that most of the airports do not rate the conditions of their airfield lighting systems as excellent, good, fair, or poor. Therefore, to determine the deterioration trend of light fixtures, the team considered the Weibull probability distribution. However, for the electrical distribution system (conduit), failure can be calculated using the reliability theory. Reliability is defined as the probability that a component or system will operate according to its specifications for a given period. Reliability is a function of time. For the failure probability distribution of runway light fixtures, the research team used the Weibull distribution. *NCHRP Research Report 859: Consequences of Delayed Maintenance of Highway Assets* used the Weibull distribution to determine the failure probability of highway LED and HPS lighting. The team used similar processes to determine the cumulative failure probability distribution of the runway LED and HPS light fixtures. Shape factors of 5 and 14 were used for HPS and LED fixtures, respectively, to determine the failure probability. Failure probability is also dependent on the number of hours the airports use these lights. Generally, the HPS fixtures last for 4 years, resulting in a total of 14,600 hours (considering 10 hours per day and 365 days in a year). LED fixtures generally last for 15 years, which is equivalent to about 54,750 hours. Considering that 63% of the HPS light fixtures will fail by 7,000 hours, the cumulative failure probability distributions were calculated in Table 5-16.

Considering that 63% of the LED light fixtures will fail by 45,000 hours, the cumulative failure probability distributions are calculated and shown in Table 5-17.

It is also necessary to determine the failure probabilities of electrical conduit systems. In electrical engineering, the system and component failure rates can be determined by using the reliability

Table 5-16. Weibull distribution for HPS light fixtures.

Age of HPS Light Fixtures	Cumulative Failure Probability
3,650 hours (1 year)	3.8%
7,300 hours (2 years)	70.9%
10,950 hours (3 years)	99.9%
14,600 hours (4 years)	100.0%

Copyright National Academy of Sciences. All rights reserved.

58 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 5-17. Weibull distribution for LED light fixtures.

Age of LED Light Fixtures	Cumulative Failure Probability
3,650 hours (1 year)	0.0%
7,300 hours (2 years)	0.0%
10,950 hours (3 years)	0.0%
14,600 hours (4 years)	0.0%
18,250 hours (5 years)	0.0%
21,900 hours (6 years)	0.0%
25,550 hours (7 years)	0.0%
29,200 hours (8 years)	0.2%
32,850 hours (9 years)	1.2%
36,500 hours (10 years)	5.2%
40,150 hours (11 years)	18.3%
43,800 hours (12 years)	49.6%
47,450 hours (13 years)	87.8%
51,100 hours (14 years)	99.7%
54,750 hours (15 years)	100.0%

theory. The TASA Group (n.d.) defines reliability as "the probability that a component, subassembly, or system will operate according to specifications for a given period of time (t)." Reliability is the function of time (t) and can be represented by $R(t)$ and calculated using Equation 5-4.

$$R\left(t\right) = e^{-\lambda t} \tag{5-4}$$

e = Euler's Number = 2.71828
λ = Failure rate
t = Time in hours

For the airport electrical system, the team used 3,650 hours per year considering 10 hours per day of usage. Using one failure per million hours (FPMH), the value of λ can be taken as 0.000001 FPMH. This reliability value is for a single component system. If the electrical system has multiple components, the reliability can be calculated using Equation 5-5.

$$Reliability = \left(R\right)^{n} \tag{5-5}$$

n = Number of components in electrical system

In airfield electrical systems, the number of components in the system is assumed to be five. Those components are transformers, switchboards, connections to airfield lights, connections to airfield signs, and voltage breakers. Equation 5-5 calculates the reliability of an electrical system for each year. The calculation results are shown in Table 5-18. The failure probability can be calculated by subtracting the reliability values from 1, and those results, up to 15 years, are also shown in Table 5-18.

Tables 5-16 to 5-18 determine the failure probability of HPS and LED light fixtures as well as electrical systems, while calculating the total maintenance cost of runway lights and electrical conduit systems. The airfield lights and electrical conduit systems deteriorate, even when regular maintenance is performed on these components. So, the deterioration models used will address the failure probability of light fixtures and electrical systems for normal conditions.

5.3.5 Determining the Runway Lighting Failure Probability with Delayed Maintenance

There are no reliability and failure probability models for electrical systems that are not regularly maintained. There are also no data available to determine the deterioration trend of light fixtures and

Copyright National Academy of Sciences. All rights reserved.

Table 5-18. Reliability and failure probability of electrical conduit systems.

Age of Electrical System	Reliability	Failure Probability
3,650 hours (1 year)	99.6%	1.8%
7,300 hours (2 years)	99.3%	3.6%
10,950 hours (3 years)	98.9%	5.3%
14,600 hours (4 years)	98.6%	7.0%
18,250 hours (5 years)	98.2%	8.7%
21,900 hours (6 years)	97.8%	10.4%
25,550 hours (7 years)	97.5%	12.0%
29,200 hours (8 years)	97.1%	13.6%
32,850 hours (9 years)	96.8%	15.1%
36,500 hours (10 years)	96.4%	16.7%
40,150 hours (11 years)	96.1%	18.2%
43,800 hours (12 years)	95.7%	19.7%
47,450 hours (13 years)	95.4%	21.1%
51,100 hours (14 years)	95.0%	22.5%
54,750 hours (15 years)	94.7%	23.9%

electrical systems. The research team assumed that the failure probability of the regularly maintained runway lights and electrical system is less than that of the systems that are not regularly maintained. The research team also assumed that the HPS and LED light fixtures' scale factors would be different than the scale factors of regularly maintained light fixtures. Assuming that the scale factors of HPS and LED light fixtures are 5,000 and 25,000 hours, respectively, the cumulative probabilities for these light fixtures are calculated in Tables 5-19 and 5-20. Table 5-19 shows that the HPS light fixtures will last only 2 years and in the 3rd year will need a complete replacement. Table 5-20 shows that LED light fixtures can last for 7 years before needing replacement.

Similar to the light fixtures, the research team assumed that the deterioration of the electrical system would also be faster for a system that is not regularly maintained, compared to a maintained system. In the case of electrical systems, the systems lacking regular maintenance deteriorate at twice the rate of a well-maintained system. In Table 5-21, the probability of failure

Table 5-19. Weibull distribution for HPS light fixtures with delayed maintenance.

Age of HPS Light Fixtures	Cumulative Failure Probability
3,650 hours (1 year)	18.7%
7,300 hours (2 years)	99.9%
10,950 hours (3 years)	100.0%
14,600 hours (4 years)	100.0%

Table 5-20. Weibull distribution for LED light fixtures with delayed maintenance.

Age of LED Light Fixtures	Cumulative Failure Probability
3,650 hours (1 year)	0%
7,300 hours (2 years)	0%
10,950 hours (3 years)	0%
14,600 hours (4 years)	0%
18,250 hours (5 years)	1%
21,900 hours (6 years)	15%
25,550 hours (7 years)	74%
29,200 hours (8 years)	100%

Copyright National Academy of Sciences. All rights reserved.

60 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 5-21. Failure probabilities of electrical systems with delayed maintenance.

Age of Electrical Systems	Cumulative Failure Probability
3,650 hours (1 year)	5.3%
7,300 hours (2 years)	7.1%
10,950 hours (3 years)	8.8%
14,600 hours (4 years)	10.5%
18,250 hours (5 years)	12.2%
21,900 hours (6 years)	13.9%
25,550 hours (7 years)	15.5%
29,200 hours (8 years)	17.1%
32,850 hours (9 years)	18.6%
36,500 hours (10 years)	20.2%
40,150 hours (11 years)	21.7%
43,800 hours (12 years)	23.2%
47,450 hours (13 years)	24.6%
51,100 hours (14 years)	26.0%
54,750 hours (15 years)	27.4%

increases by about 1.7% to 1.8% every year. Therefore, the research team has concluded that the failure probabilities of systems not regularly maintained will increase by 3.5%. Table 5-21 shows the electrical systems' failure probabilities for every year if the system is not regularly maintained.

5.3.6 Estimating Runway Lighting Maintenance Budgets without Delayed Scenarios

The total maintenance cost of the runway lighting can be calculated by determining how many light fixtures are needed and the length of the electrical system that needs to be replaced, based on their failure probabilities and the cost of maintaining the remaining light fixtures and electrical conduits. Therefore, the normal probability of failure estimated in Section 5.3.4 can be used to determine the number of runway lights and length of the electrical circuits that need to be replaced. Then, the unit cost of light fixtures and electrical conduits can be multiplied to find the total cost of replacement. The total maintenance cost can be calculated by multiplying the unit cost of maintenance by the number of light fixtures, and length of electrical conduits to be maintained every year. The accurate unit replacement cost of these light fixtures and the number of fixtures to be replaced needs to be accurately estimated to find the precise replacement cost. To determine the replacement cost of electrical conduits, the estimated length of the electrical system should be calculated accurately, and the cost per unit of length needs to be precise to find a good estimate of the total replacement budget. Similarly, the maintenance cost per unit of light fixtures and the per unit length of the electrical conduit need to be collected to calculate the accurate maintenance cost of the airfield lights. The research team collected these cost data from the airports. However, not all of the cost data were available, therefore, some cost data were gathered from the RSMeans Cost Guide as well as online databases. The inflation factor for lighting maintenance and light fixtures was derived from the cost indices of Federal Reserve Economic Data (FRED). According to this database, the average inflation per year for electrical contractors is 2.33% (FRED 2022a). For HPS and LED light fixture replacements, the average inflation is estimated at 2.72% per year (FRED 2022c).

5.3.7 Estimating Runway Lighting Maintenance Budgets with Delayed Scenarios

Similar to the process described for regular maintenance budgets without delayed scenarios, the research team used Tables 5-19 to 5-21 to determine the total maintenance and replacement

Copyright National Academy of Sciences. All rights reserved.

cost of runway lights in delayed scenarios. For this purpose, the probabilities of failure for HPS and LED light fixtures are given in Tables 5-19 and 5-20. For the electrical systems, the failure probabilities were taken from Table 5-21. Once the failure probabilities were determined, the cost of HPS and LED light fixture replacement was calculated by their unit cost. Similarly, for the electrical system, the cost of replacement was calculated by multiplying the unit cost of electrical system replacement with the length of the electrical conduit to be replaced due to delayed maintenance. Also, the maintenance cost of the surviving HPS, LED light fixtures, and electrical systems can be calculated by multiplying the unit cost of maintenance with the number of light fixtures and length of electrical conduits to be maintained. The total cost of maintenance and replacement was calculated considering an inflation rate of 2.33% for maintenance and 2.72% for replacement work. However, the users can input their own inflation rate to calculate the delayed maintenance costs.

5.3.8 Reporting the Impact of Runway Lighting Delayed Maintenance and its Consequences

The spreadsheet tool the team developed was used to determine the impact of delayed maintenance on runway lighting. A hypothetical airport case study was developed to estimate the impact of delayed maintenance of runway lighting on total maintenance and replacement costs. Table 5-22 shows the hypothetical airport data to determine the impact of delayed maintenance of runway lighting. The delayed scenario was taken as 1 and 2 years, as the HPS light fixtures last only 3 to 4 years. For LED light fixtures, the delayed scenario was considered for 2 and 5 years. The spreadsheet tool to determine the impact of delayed maintenance of runway lights has an option to choose delayed scenarios from 1 to 10 years. The age of HPS light fixtures can be new, 1, or 2 years old. For LED lights, the age can be new, 5, or 10 years old. The tool can also convert future costs into present costs using various discount rates.

Case Study Results of the Consequences of Delaying Maintenance on Runway Lighting

Results of Delaying Maintenance on HPS Runway Lighting

The research team used the spreadsheet tool to determine the impact of delayed maintenance on the total maintenance and replacement costs of HPS runway lighting. The scenarios in which the airport installed new HPS lights less than 2 years prior are the only ones that were considered

Table 5-22. Case study data of airport runway lighting.

Name of the airport	XYZ International Airport
Type of airport	Small-hub/medium-hub/large-hub
Location of the airport	Southwest, United States
Runway dimensions	9,000 ft length and 150 ft width
Type of fixtures	HPS light fixtures LED light fixtures
Age of the fixtures	0-year-old, 1-year-old, or 2-year-old for HPS light fixtures and 0-year-old, 5-year-old, or 10-year-old for LED light fixtures
Age of electrical conduit	5 years old
Budget preparation year	2022
Budget to be prepared for	2023
Delayed scenarios	1 year and 2 years for HPS light fixtures 2 years and 5 years for LED light fixtures
Price inflation factor	2.33% for maintenance and 2.72% for replacement
Discount factor	0%

Copyright National Academy of Sciences. All rights reserved.

since HPS lights do not have more than 3 years of life. The delayed scenarios were considered for 1 and 2 years. Figure 5-18 shows the total maintenance and replacement cost of runway lights in the hypothetical case study considered with no delay, a 1-year delay, and a 2-year delay, based on the age of the runway's HPS light fixtures. The total maintenance and replacement cost of runway lights will increase significantly when delayed 1 year for newly installed HPS light fixtures. However, the delay in 2 years will not have a high impact on the total maintenance and replacement cost, because the failure rate of HPS light fixtures in delayed scenarios is 99.9% and 100%, respectively (Table 5-19). Similarly, when it is assumed that the age of runway lighting with HPS light fixtures is 1 and 2 years old, the total base cost of maintenance and replacement will not be significantly different because the failure probability of HPS light fixtures will reach 70.9% and 99.9%, respectively in the following years (Table 5-16). Due to the low cost of HPS light fixtures, there is no significant difference between base cost scenarios of HPS bulb fixtures aged 0, 1, and 2 years. If the maintenance is delayed for 1 and 2 years for 1- and 2-year-old HPS light fixtures, the total maintenance and replacement cost will be similar to the newly installed HPS light fixtures.

A similar analysis was conducted to find the percentage increase in total maintenance and replacement costs of runway HPS light fixture lighting for new, 1-year, and 2-year-old fixtures in delayed scenarios. Figure 5-19 shows that the age of HPS light fixtures will not have an impact on the total and maintenance cost of the runway lighting due to the reasons mentioned above. Because HPS light fixtures need to be replaced every 2 to 3 years, delaying their maintenance does not make sense. It is suggested that the airport replace the HPS light fixtures of their runway lighting every 2 to 3 years.

Results of Delaying Maintenance on LED Runway Lighting

The research team used the spreadsheet tool to determine the impact of delayed maintenance on total LED runway lighting maintenance and replacement costs. Figure 5-20 shows the regular and delayed maintenance and replacement costs of LED light fixtures. The total maintenance and

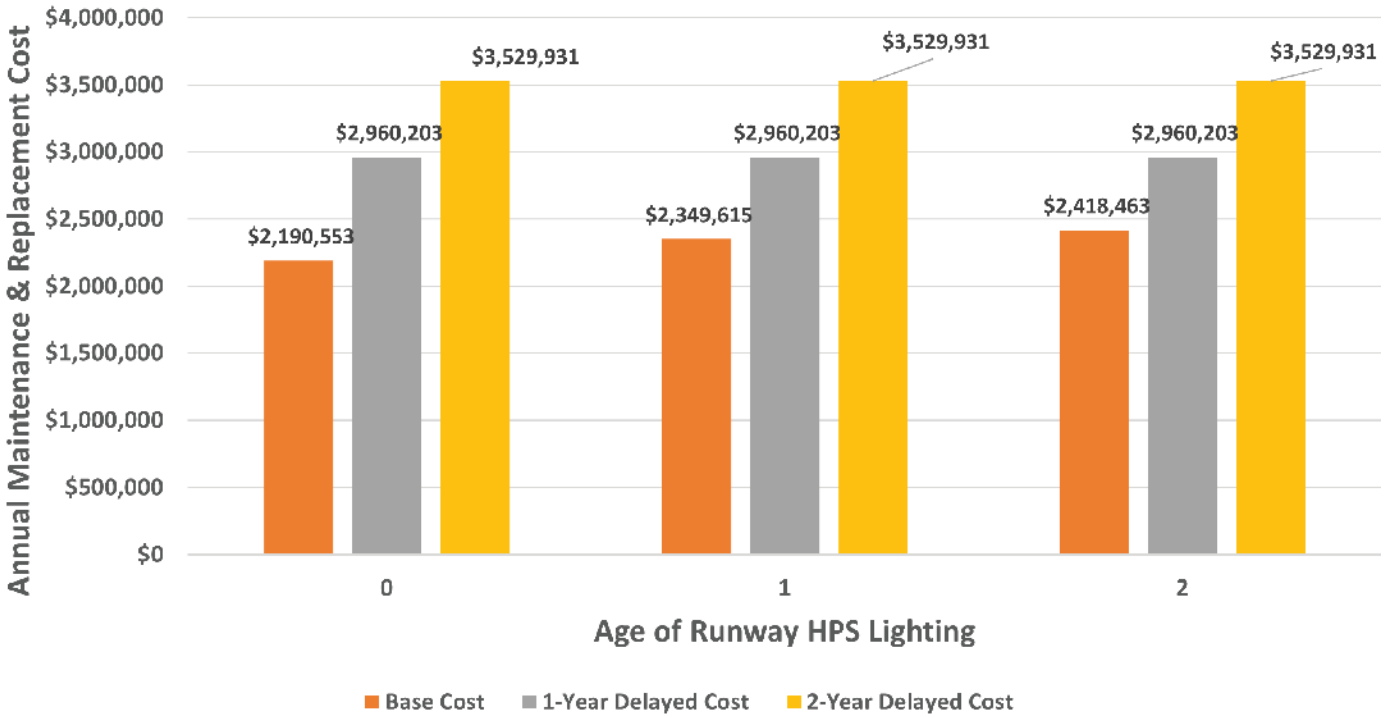


Figure 5-18. Impact of delaying maintenance on the total maintenance and replacement cost of runway HPS lighting.

Copyright National Academy of Sciences. All rights reserved.

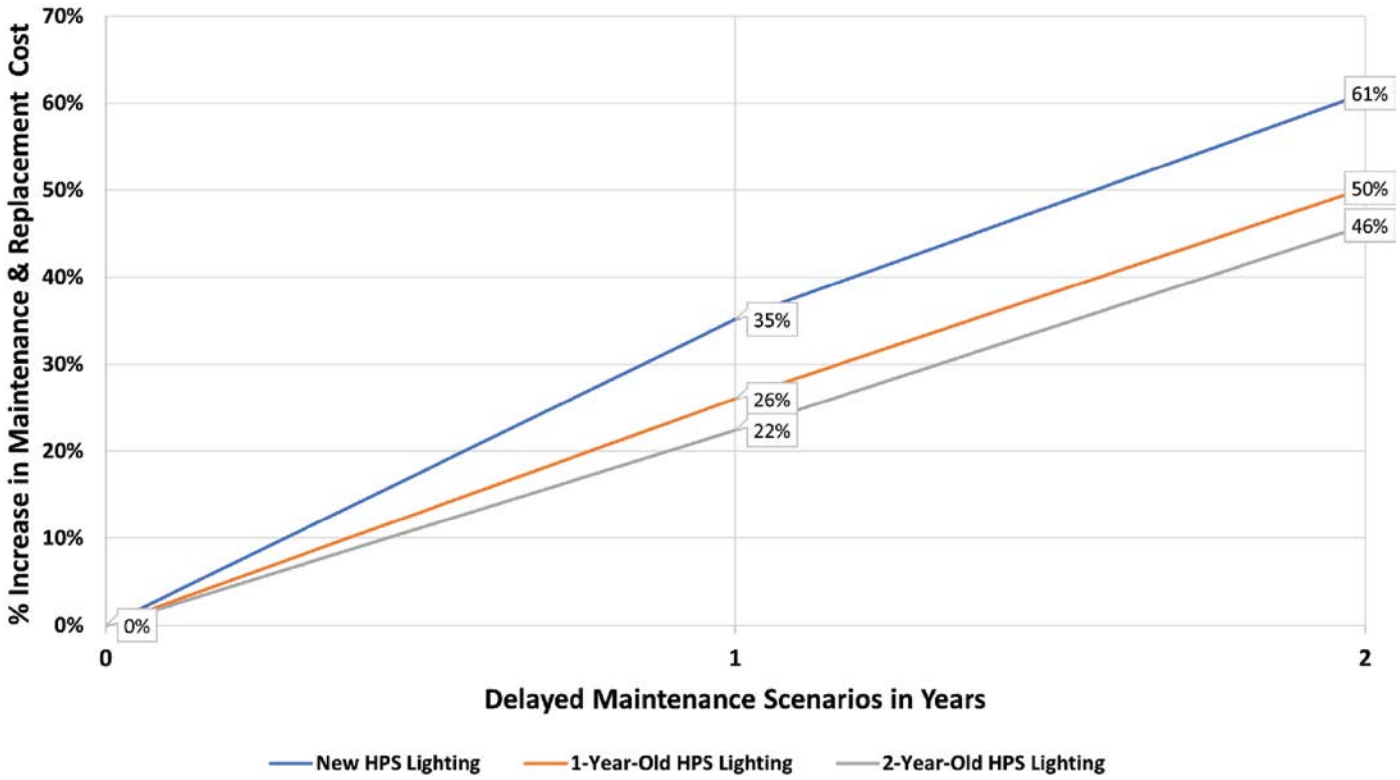


Figure 5-19. Percentage increase of HPS runway lighting maintenance and replacement cost.

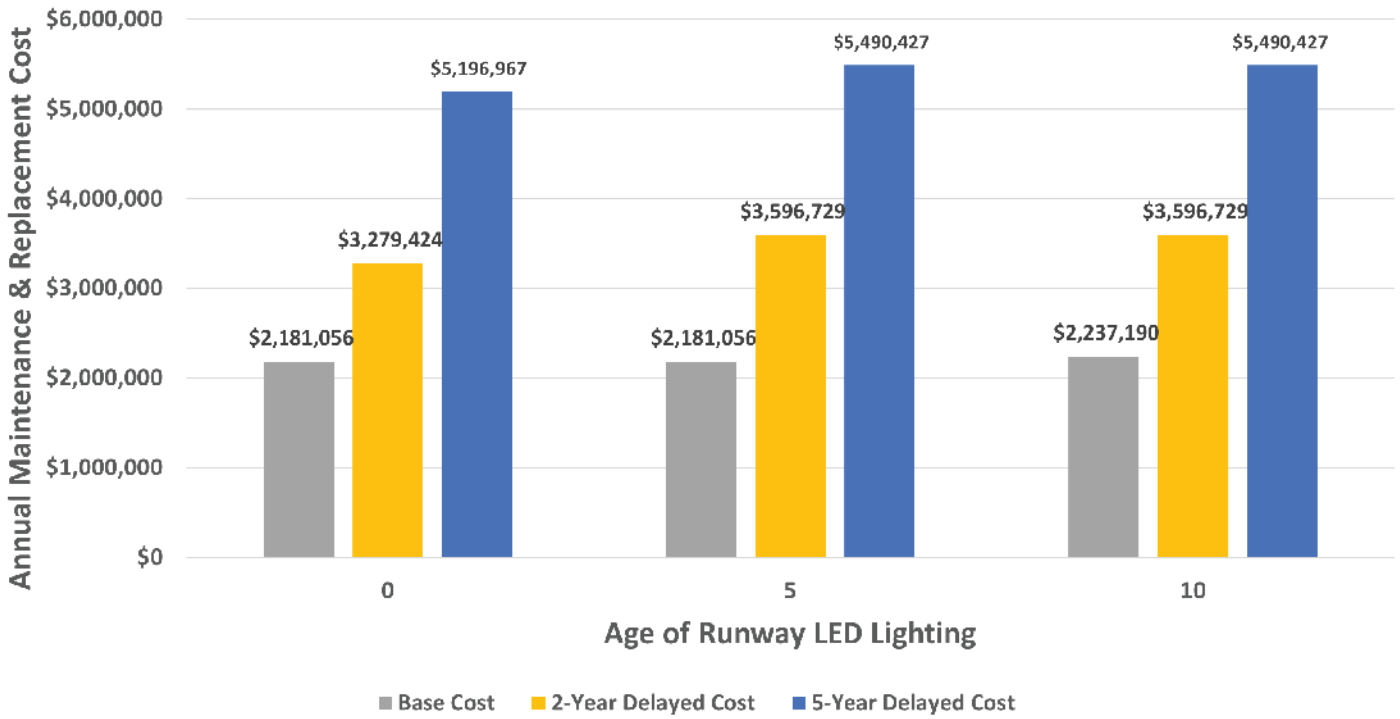


Figure 5-20. Impact of delaying maintenance on total maintenance and replacement cost of runway LED lighting.

Copyright National Academy of Sciences. All rights reserved.

64 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

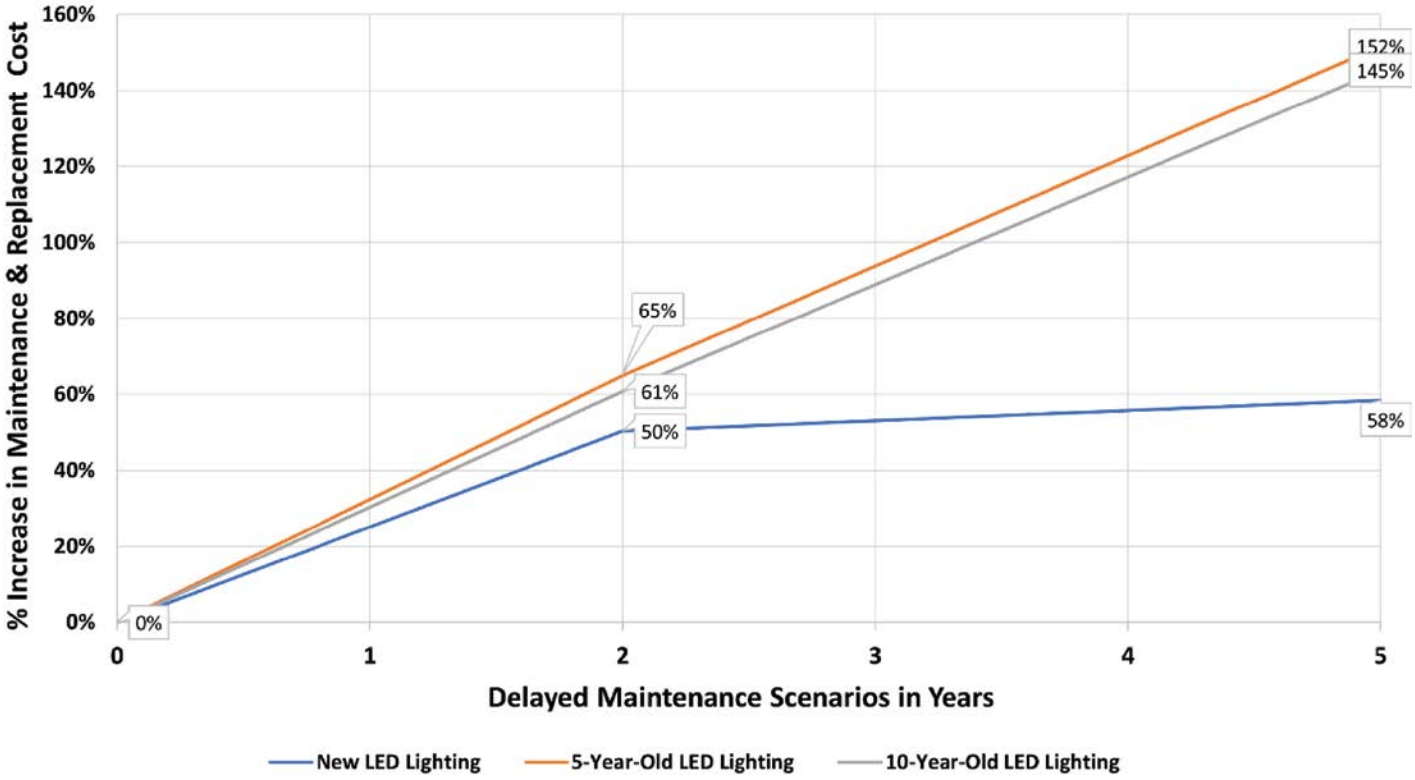


Figure 5-21. Percentage increase of LED runway lighting maintenance and replacement costs.

replacement cost for LED runway lighting increases when maintenance is delayed. However, the total maintenance and replacement costs of 5-year-old and 10-year-old LED lights do not increase based on age. When the maintenance is delayed by 5 years for these light fixtures, the probability distribution of delayed maintenance will reach 100% failure at 8 years (Figure 5-20).

A similar analysis was conducted to determine the percentage increase in total maintenance and replacement costs of runway LED light fixtures for new, 5-year, and 10-year-old fixtures in 1- and 2-year delayed scenarios. Figure 5-21 shows that the age of LED light fixtures will impact the total maintenance and replacement costs of the runway lighting due to the reasons mentioned above. Because the LED light fixtures tend to last around 8 years in a delayed maintenance scenario, the total maintenance and replacement cost will be highest when 5-year-old light fixtures are delayed up to 5 years.

5.4 Airfield Signs

There are various types of signs used in runways, taxiways, and aprons of airports. These signs play an important role in guiding pilots to navigate properly during takeoff, landing, and taxing. These signs should be maintained regularly and need to be replaced as required. Old and faded signs can confuse pilots and other airport workers, which may lead to an accident. Therefore, the important tasks of airport asset managers are to keep a record of sign inventory, track their condition, identify their maintenance activities and unit costs, and prepare a maintenance and replacement budget for the signs each year. Signs placed in runway areas are particularly crucial, and delaying their maintenance should be considered carefully to avoid jeopardizing flight safety. Signs provided in taxiways and other areas can be delayed, but the delay's impact on the safety and operation of the airplanes must first be determined. The research team prepared a

Copyright National Academy of Sciences. All rights reserved.

framework to determine the impact of delayed airport sign maintenance on total airport maintenance and replacement costs. This framework is shown in Figure 5-22. The framework consists of various phases that asset managers need to perform to determine the impact of delayed airport sign maintenance on the total maintenance and replacement costs for the airport. Furthermore, it is not logical or reasonable to convert these impacts into dollar equivalents. The following sections describe each phase.

5.4.1 Establishing Asset Inventory

All airport asset managers need to keep track of airfield signs used in their runways and taxiways. According to *FAA AC 150/5345-44K – Specification of Runway and Taxiway Signs*, the signs provided in airport runways and taxiways can be lighted or unlighted (FAA 2022d). Figure 5-23 shows the assemblies of typical stop and yield signs used in airport runways. The use of specifications in this AC is mandatory for runway and taxiway signage projects funded under the AIP or with revenue from the PFC program. Based on the specifications, there are six types of signs provided on runways and taxiways. They are:

1. Direction, destination, and boundary signs.
2. Mandatory instruction signs.
3. Taxiway location signs.
4. Runway distance remaining signs.
5. Taxiway ending marker signs.
6. One-half distance remaining signs.

The FAA AC provides advice for the sizes of sign legend panels. It discusses various styles, classes, and modes of signs. The AC provides specific guidelines related to message elements, message arrays, readability, and sign face elements, including borders, margins, edges, and arrays. It also covers the details of lighted and unlighted signs, including their mounting legs, light powers, luminance, materials, and components.

Airport asset managers need to keep a record of these signs so that the maintenance budget can be calculated accurately. Because there are various types and sizes of signs, all signs within the same category need to be collected and recorded separately. The maintenance and replacement costs of these various signs may differ, and separate records will help accurately estimate their maintenance and replacement costs. The airfield sign inventory needs to be updated every time the signs are replaced or damaged.

5.4.2 Performing a Condition Assessment

One of the important duties of asset managers in the airport is to perform condition assessments of airfield signs every year. This is done so the signs can be maintained and replaced based on their conditions. During data collection of airfield signs, it was found that some airports performed and kept the condition data based on four categories: (1) Excellent, (2) Good, (3) Fair, and (4) Poor. If the signs are in "Excellent," "Good," or "Fair" condition, these signs still need maintenance to preserve their conditions or prevent them from deteriorating rapidly. When the conditions of the signs reach the "Poor" category, they should be replaced within the following year. Condition assessment in this category will help asset managers estimate the maintenance and replacement cost of airfield signs accurately. However, it was found during data collection that some, but not all, airports put the condition data of runway and taxiway signs in the four above-mentioned categories. When the signs are inspected, their structural components and information panels need to be rated. If the signs are lighted, then their mounting legs, hardware, electrical circuits, bulbs, panels, electrical power regulator, luminance, sign materials, and

Copyright National Academy of Sciences. All rights reserved.

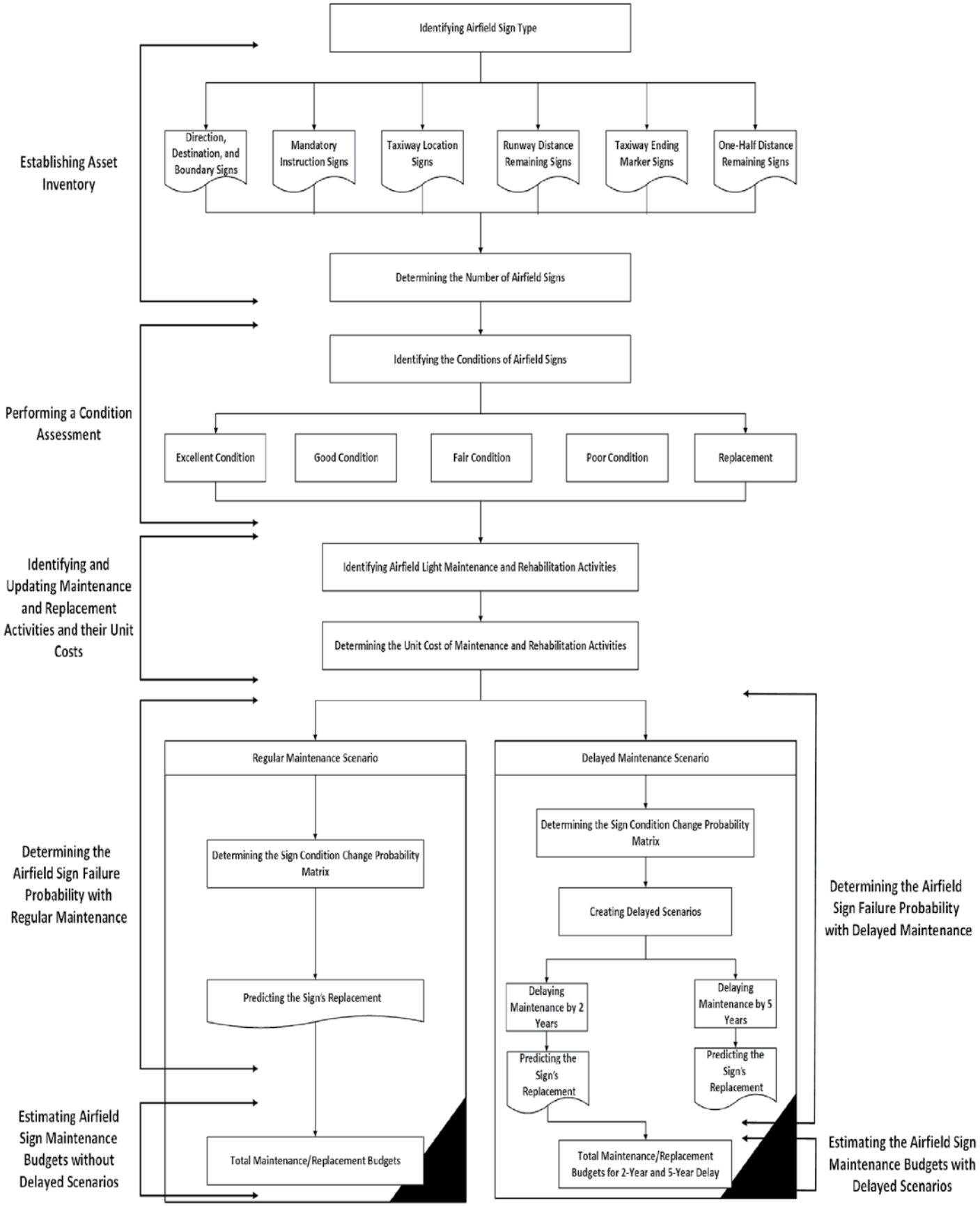


Figure 5-22. Framework for the delayed maintenance of airfield signs.

Copyright National Academy of Sciences. All rights reserved.

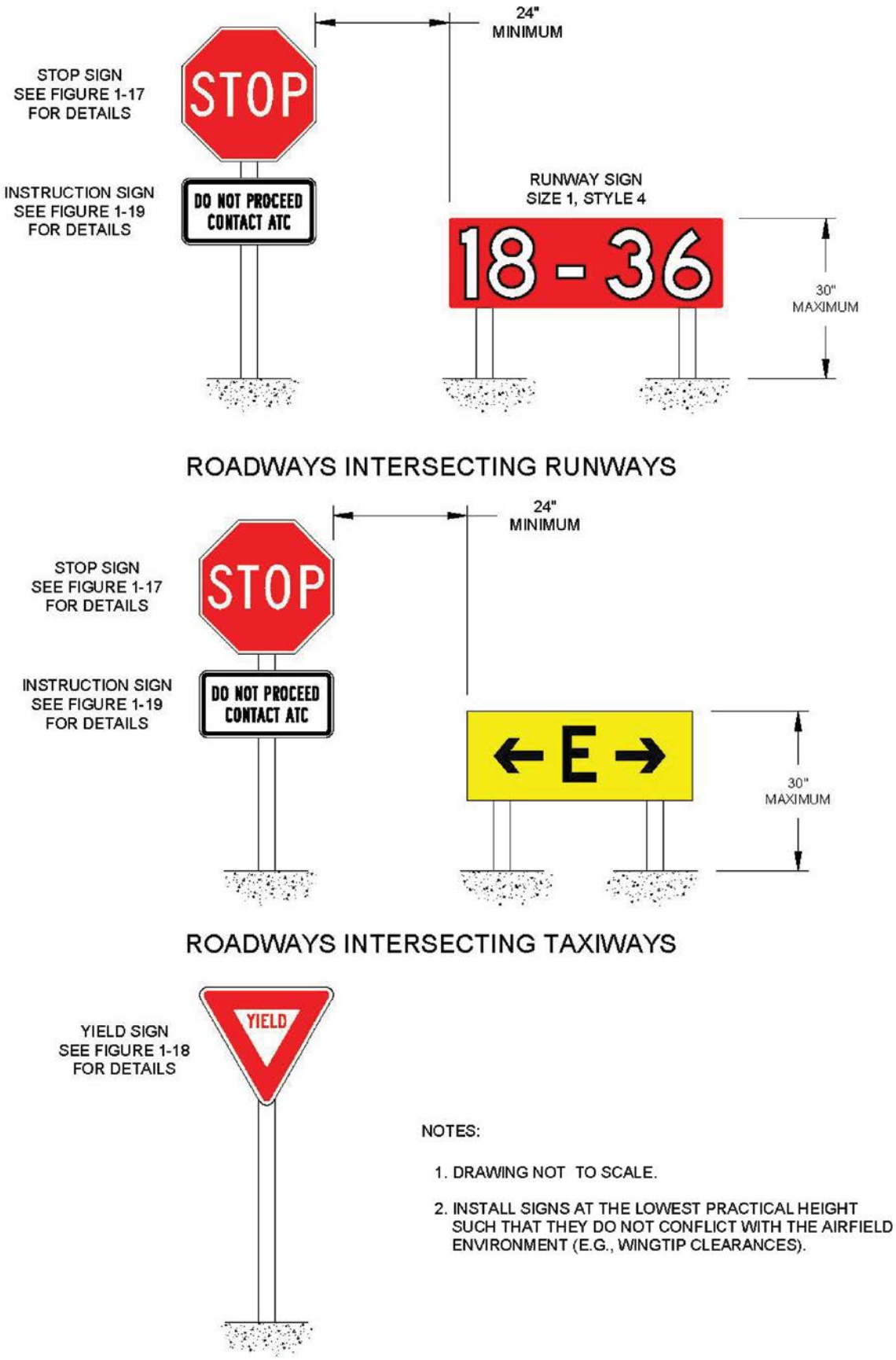


Figure 5-23. Typical stop and yield sign assemblies. Source: https://www.faa.gov /documentLibrary/media/Advisory_Circular/150-5340-18G-Chg-1-Airport-Signs.pdf.

Copyright National Academy of Sciences. All rights reserved.

68 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

components need to be inspected to make sure that the signs are working properly. FHWA has researched the inspection, maintenance, and repair of highway signs and traffic signals and developed a rating system for sign inspection. Table 5-23 shows the rating system.

5.4.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

During the inspection and condition rating of the runway and taxiway traffic signs, the asset managers need to identify what maintenance and repair activities should be performed to maintain the signs. Some of the maintenance activities to be performed for unlighted signs are (McGee 2010):

1. Sign cleaning,
2. Vegetation control,
3. Anti-theft measures, and
4. Sign support adjustments.

It is necessary to clean the sign face because high air traffic volume may cause the sign to become less visible due to a buildup of dirt, grime, mildew, or mold. During the spring and summer seasons, signs can be partially or fully blocked by vegetation or fast-growing weeds that develop quickly around them. Furthermore, signs are occasionally stolen for any number of reasons. Sign supports are also critical and need to be maintained to keep the sign working properly. Sign supports can be maintained by performing the following activities:

- Ensure the signpost is still firmly placed in the ground.
- Provide additional supports and braces, if necessary.
- Place the signs at the required distance so that an airplane or airport service vehicle may avoid hitting the sign.
- Protect sign supports from soil erosion at the base and the growth of trees around the support.

All of these maintenance and repair activities must be performed in-house or by third-party contractors. After identifying these activities and the necessary frequency, airport asset managers must estimate the labor and parts costs for performing the maintenance activities annually. This unit cost of maintenance and replacement is critical in estimating accurate maintenance costs with and without delayed maintenance.

During the data collection phase, the research team was able to collect the labor hours and money spent on all signs in the airport. Based on these cost data, the team calculated the maintenance cost per sign. The ratio of cost and parts used here is derived from the RSMeans Cost Guide 2022. The rate of the electrician used is based on average Nevada rates. The calculation of maintenance cost per sign is provided in Table 5-24.

Table 5-23. Condition rating of highway signs and signals (FHWA 2005).

Condition Rating	Description
1 (Excellent)	Element performs its intended function with a high degree of reliability
2 (Good)	Element performs its intended function with a small reduction in reliability
3 (Fair)	Element performs its intended function with a significant reduction in reliability
4 (Poor)	Element does not perform its intended function with any degree of reliability

Copyright National Academy of Sciences. All rights reserved.

Table 5-24. Unit maintenance cost calculations.

Category	Calculation
Number of electricians involved	2
Number of electrician hours	2 x 260 days/year x 8 hours/day = 4,160 hours
Unit cost of electricians	$60/hour
Total annual labor cost	$249,600
Parts cost (15% of total cost)	$249,600 x (0.15) / (0.85) = $44,047
Total cost	$293,647
Number of signs	600
Maintenance cost per sign	$490

The replacement cost of the signs was calculated based on 30% labor and 70% material costs (RSMeans Cost Guide 2021). In the case study airport, the data provided for a specific airport sign was $950. Thus, the total installation cost for this sign will be $1,371 ($960/0.7). These numbers are based on data for the year 2021; therefore, for the cost to be calculated for future years, the inflation rate should be calculated based on FRED (2022b). FRED provided the inflation cost indices from 2004 for electrical equipment and 2008 for electrical contractors. For the last 18 years, the inflation rate was found to be 2.72% annually. Similarly, for the maintenance cost of the airfield signs, an inflation rate of 2.33% derived from the last 14 years was used to estimate the maintenance cost for future years (FRED 2022b).

5.4.4 Determining the Airfield Signs' Failure Probability with Regular Maintenance

One of the important tasks for predicting the maintenance and replacement cost of airfield signs is identifying the deterioration models of the signs. Because none of the airports could provide the research team with airfield sign condition data for a few years during the data collection phase, the research team could not determine how the airfield signs were deteriorating in the airports. However, Chang et al. conducted research to determine the regular deterioration model of highway signs. The research team used the same deterioration model to determine how the airfield signs will deteriorate from Excellent to Poor condition and require replacement (Chang et al. 2017). According to the highway sign deterioration models, the percentage of Excellent signs that will be converted to Good, Fair, and Poor after successive years is shown in Table 5-25. Similarly, signs in Good, Fair, and Poor conditions that will be converted to a lower level are also shown in the deterioration transition matrix. For example, if a group of signs at year n is in condition Excellent, then there is a 52 percent probability that at year n+1, the condition will remain Excellent, a 32 percent probability that more signs will move to Good, an 8 percent probability that their condition will move to Fair, a 3 percent probability that their condition will move to Poor, and a 5 percent probability that their condition will move to Replacement.

Table 5-25. A probability matrix of airfield signs' transition after each year.

Condition	Excellent	Good	Fair	Poor	Replacement
Excellent	52%	32%	8%	3%	5%
Good		41%	34%	8%	17%
Fair			23%	42%	35%
Poor				7%	93%
Replacement					100%

Copyright National Academy of Sciences. All rights reserved.

70 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

5.4.5 Determining the Airfield Signs' Failure Probability with Delayed Maintenance

As discussed in previous sections, it is difficult to determine the deterioration trends of airport assets when the assets are not maintained regularly. Every asset life is therefore calculated assuming the assets are regularly maintained. However, research has not been done on how assets will deteriorate when regular maintenance activities are not performed. Thus, to determine the trend of airport asset deterioration without maintenance, assumptions need to be made. The rate of deterioration will be higher when maintenance is not performed on assets compared to regularly maintained assets. As described in Section 5.4.4, the probability matrix of airfield sign condition transitions was derived from highway signs and signals, which are like airport airfield signs. Both include lighted and unlighted signs. To calculate the transition probability matrix for airfield signs, it is assumed that they will deteriorate about 20% more than regularly maintained signs, and the condition of the airfield signs will move from one condition level to another each year by that increased amount. Table 5-26 shows the probability matrix of airfield sign transitions after each year under delayed maintenance scenarios. The table values show that the airfield signs in Excellent condition in the first year will result in 42% of signs remaining in Excellent condition, and the other 58% of signs move to Good, Fair, Poor, and Replacement conditions (34% to Good, 13% to Fair, 5% to Poor, and 6% to Replacement). Similarly, out of the signs that are in Good condition, 33% will remain Good, 34% will move to Fair, 13% will move to Poor, and 20% will move to Replacement the next year. Similar interpretations can be made for signs in Fair and Poor conditions.

5.4.6 Estimating Airfield Sign Maintenance Budgets without Delayed Scenarios

One of the main goals of determining the regular deterioration model for assets is to predict the conditions of assets the following year and estimate their maintenance and replacement costs. Therefore, with the help of the probability matrix in Table 5-25, asset managers can determine how many airfield signs will be in Excellent, Good, Fair, Poor, and Replacement conditions so that the budget for maintenance and replacement can be calculated. Once asset managers have performed a condition assessment of each sign in the airport and have rated them in these categories, they can estimate the number of signs to be replaced in the following year. Once the number of airfield signs to be maintained and replaced is determined, the cost can be calculated by multiplying their unit cost of maintenance and replacement.

One of the challenges during data collection was collecting the maintenance and replacement costs of airfield signs. Since there are various types of signs, the costs to maintain and replace these signs vary widely. Therefore, the research team used the number of electricians involved in maintaining airport signs and asked the asset manager to assume the ratio of labor and parts costs to maintain these signs. The research team also verified this ratio using the RSMeans Cost Guide (2021). The same process was used to determine the replacement cost of airfield signs. These cost

Table 5-26. Probability matrix of airfield signs' transition after each year under delayed maintenance scenarios.

Condition	Excellent	Good	Fair	Poor	Replacement
Excellent	42%	34%	13%	5%	6%
Good		33%	34%	13%	20%
Fair			18%	38%	44%
Poor				5%	95%
Replacement					100%

Copyright National Academy of Sciences. All rights reserved.

data were used in the spreadsheet tool developed to calculate the impact of delayed maintenance on airfield signs. However, the tool is flexible, and users can input their own airport's unit costs to calculate accurate maintenance and replacement costs with and without delayed scenarios. Table 5-24, in Section 5.4.3, shows the maintenance cost per sign calculation from one airport's case study data. Also, the replacement costs for specific airport signs are described in that section. These costs, along with the inflation rate, were used to determine regular maintenance and replacement costs with and without delayed maintenance scenarios.

5.4.7 Estimating Airfield Sign Maintenance Budgets with Delayed Scenarios

The impacts of delayed airfield sign maintenance are found by calculating the total maintenance and replacement costs and using a delayed maintenance scenario. To calculate this, the probability matrix of airfield sign condition transitions shown in Table 5-26 was used, and the cost of maintaining the airfield signs was divided into four categories (Excellent, Good, Fair, and Poor). Similarly, the number of airfield signs in the Replacement category was multiplied by the replacement cost with an inflation factor. The tool developed by the team allows the user to delay a specific number of years to find the delay's impact on the total maintenance and replacement costs of airfield signs. The user can also use different inflation factors than the ones based on historical trends. The tool can assist in converting future maintenance and replacement costs to present costs by selecting various discount rates.

5.4.8 Reporting the Impact of Airfield Signs' Delayed Maintenance and their Consequences

The research team used the tool to determine the regular maintenance and replacement costs of airfield signs when maintenance is delayed. Table 5-27 shows hypothetical airport case study data and the impact of delayed maintenance of airfield signs on total maintenance and replacement costs. To calculate the regular maintenance and replacement costs with and without delayed scenarios, it was assumed that 36, 32, 30, two, and zero airfield signs were under Excellent, Good, Fair, Poor, and Replacement conditions in the year 2022. The regular maintenance and replacement costs were calculated for the year 2023 without delayed maintenance. The maintenance and replacement costs for years 2025 and 2028 with 2- and 5-year delays were also considered to determine the impact on airport costs.

Case Study Results for the Consequence of Delaying Maintenance on Airfield Signs

Figure 5-24 shows the total maintenance and replacement cost with and without delayed maintenance of airfield direction signs for a hypothetical case study airport. The delayed scenarios

Table 5-27. Case study data of airport signs.

Name of the airport	XYZ International Airport
Type of airport	Small-hub/medium-hub/large-hub
Location of the airport	Southwest, United States
Type of signs	Direction signs
Budget preparation year	2022
Budget to be prepared for	2023
Delayed scenarios	2 and 5 years
Price inflation factor	2.33% for maintenance and 2.72% for replacement
Discount factor	0%

Copyright National Academy of Sciences. All rights reserved.

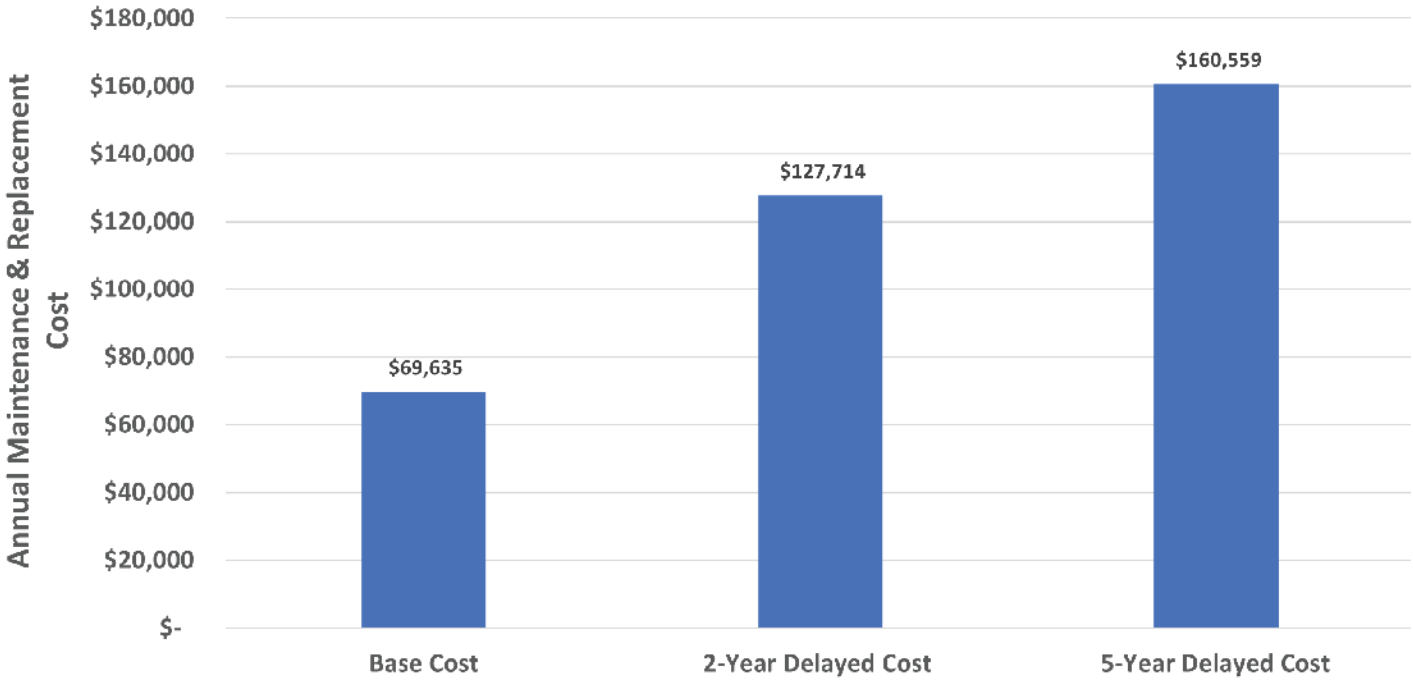


Figure 5-24. Impact of delaying maintenance on the total maintenance and replacement cost of airport direction signs.

used were 2 and 5 years. The data showed that the maintenance and replacement costs of the airfield signs increased significantly when going from no delay to a 2-year delay. However, the total maintenance and replacement costs do not increase significantly when maintenance is delayed by 5 years. This is because 75% of the total signs (76 out of 100 signs) need to be replaced during the 2-year delay. As a result, delaying by 5 years increases the replacement cost of 24 more signs. The deterioration models with delayed maintenance showed that, when the maintenance is delayed by 3 years, the number of signs to be replaced drastically reduced due to a smaller number of signs left to be maintained.

Figure 5-25 shows the percentage increase in the maintenance and replacement cost of direction signs for the case study project. It shows that the increment percentage of maintenance and replacement cost is higher for a 2-year delay than for a 5-year delay. There was an 83% maintenance and replacement cost increase when the maintenance was delayed by 2 years. However, the percentage increase was not as high when the maintenance was delayed by 5 years.

Copyright National Academy of Sciences. All rights reserved.

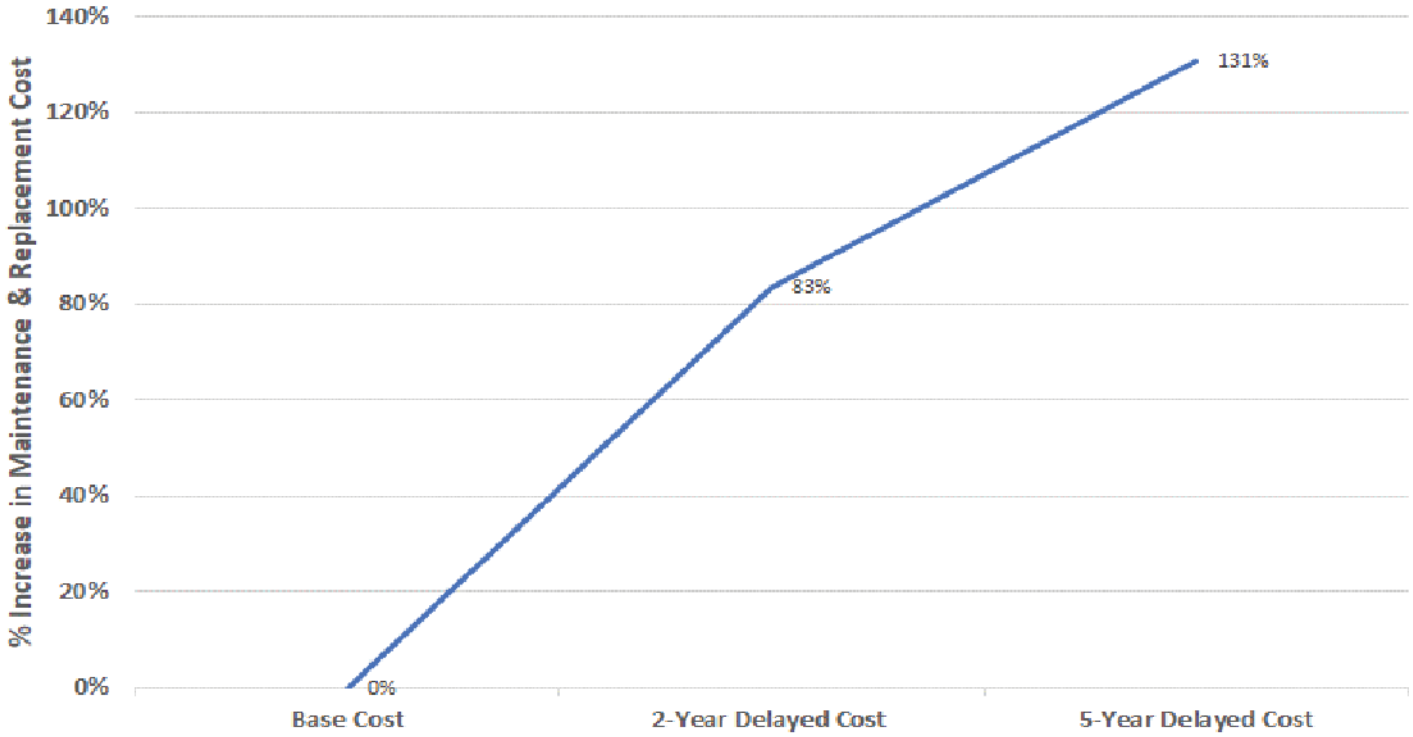


Figure 5-25. Percentage increase of maintenance and replacement cost of direction signs.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 6

Procedures to Quantify the Impact of Delayed Maintenance on Airport Landside Assets

The maintenance of landside assets, such as baggage handling systems, passenger boarding bridges, and HVAC systems, cannot be delayed to the point that these systems are inoperable, preventing air passengers from traveling on airplanes. Some landside assets, including service pickup trucks, loaders for removing snow, and maintenance of airport buildings, can be delayed for some years without affecting the comfort and safety of air travelers. In the context of airports, delayed maintenance is a critical issue and needs to be dealt with cautiously, particularly if the delay is a result of an insufficient budget. This research has prepared a framework and tools to calculate the consequences of delay for five landside assets (pickup trucks, snow removal equipment, HVAC systems, baggage handling systems, and passenger boarding bridges) in dollar amounts. The research team collected case study data from six airports to develop these tools. Follow-up interviews were also conducted with the airport asset managers to clarify the data provided by the airports.

In creating the research framework and tools, the safety, health, and comfort of air travelers were not factored in, because converting these benefits into dollar amounts is not practical. The team cannot determine the dollar amount of safety issues on airplanes, though it is possible to calculate dollar amounts for delays in flights due to a lack of maintenance. Generally, this is only in extreme situations when airports need to close, so the cost of flight delays was not factored into this model. The framework and tools calculate the extra costs associated with maintaining assets when maintenance is delayed, as the assets deteriorate at a higher rate than when they are maintained regularly. The following sections describe the procedures for quantifying the maintenance budget with and without delayed maintenance scenarios of landside airport assets.

6.1 Airport Service Vehicles: Pickup Trucks

Service vehicles like pickup trucks move maintenance workers and emergency personnel around the airport. The airport's mechanics, electricians, maintenance supervisors, inspectors, and other personnel generally use these trucks to perform their jobs. All airports own these types of vehicles and need to properly maintain them to keep the airport safe and provide comfort and services to travelers. Therefore, pickup trucks are an important part of the vehicle fleet, and airport asset managers should perform maintenance activities to keep these pickup trucks in workable condition. It is possible to delay the maintenance of these vehicles for some time, based on their condition and maintenance schedule, however, delaying maintenance for too long can break the transmission engines. The cost of fixing problems like broken transmission engines will often be higher. The research team thus developed a framework to determine the maintenance cost for pickup trucks with and without delayed maintenance. Figure 6-1 shows the framework for calculating such delayed maintenance costs. The research team also developed a spreadsheet tool, which can be used to determine the regular and delayed maintenance costs based on various scenarios.

Copyright National Academy of Sciences. All rights reserved.

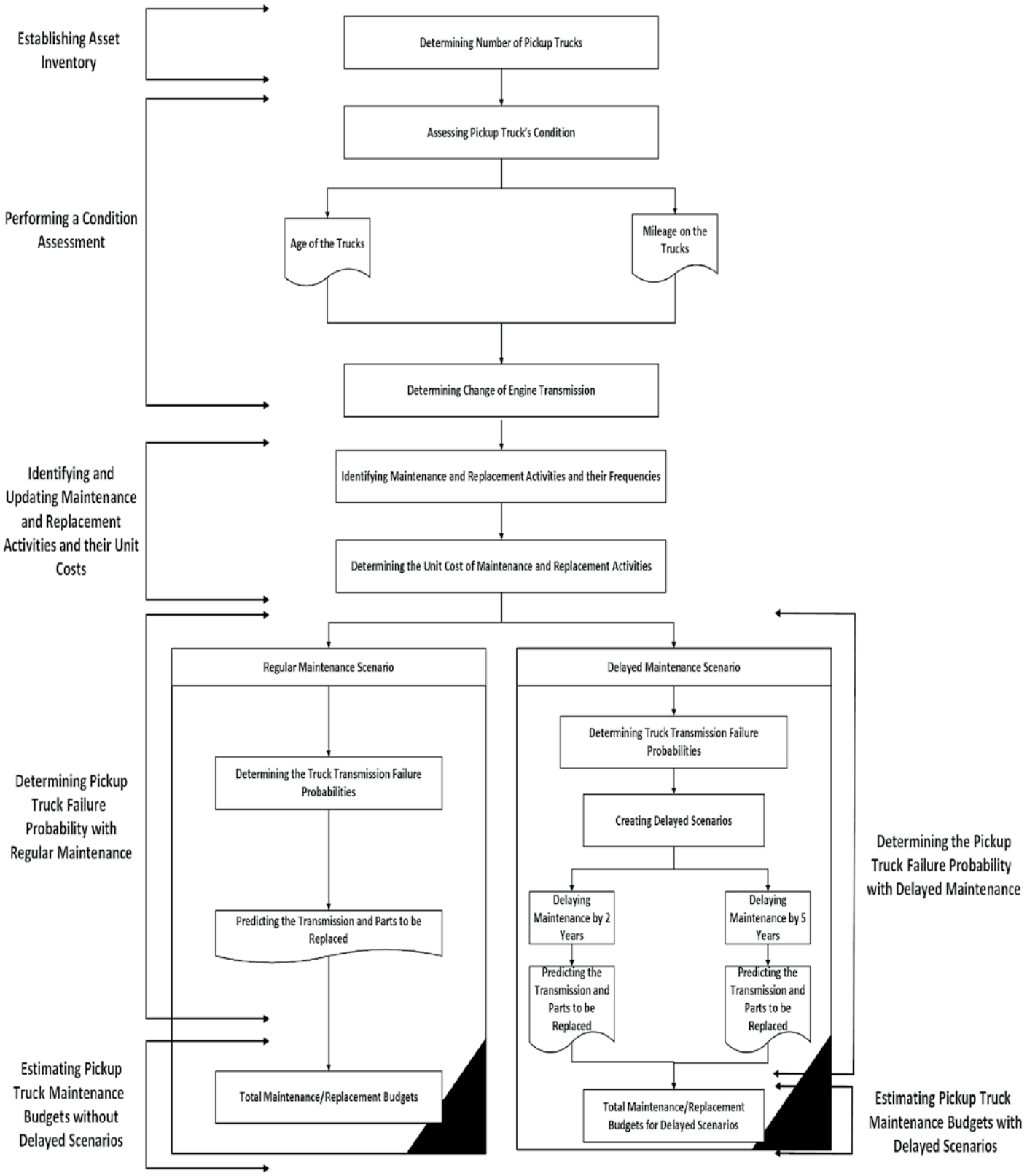


Figure 6-1. A framework of delayed maintenance of airport service vehicles (pickup trucks).

Copyright National Academy of Sciences. All rights reserved.

6.1.1 Establishing Asset Inventory

Airport asset managers must keep an inventory of service vehicles, including the make, model numbers, purchase dates, and number of vehicles in each category of vehicle. The inventory will also record the maintenance work performed each year and any major engine replacements. These records should then be stored in the airport asset inventory database system. The FAA does not have any requirements about the service vehicles' condition while driving around the runways and taxiways. However, *FAA AC 150/5210-20—Ground Vehicle Operations on Airports* provides guidelines to airport operators in developing training programs for safe ground vehicle operations and pedestrian control on the airside of an airport. This information addresses vehicle movement areas, which should be limited to those necessary for inspection and maintenance, and emergency vehicles responding to aircraft emergencies in the movement area. Requirements for vehicles will vary depending on the airport, the type of vehicle, and where the vehicle will be operating in the airport. This AC states that a vehicle shall not be permitted on the airside unless:

- It is properly marked, as outlined in *FAA AC 150/5210-5—Painting, Marking, and Lighting of Vehicles Used on an Airport.*
- It is in sound mechanical condition with unobstructed forward and side vision from the driver's seat.
- It has appropriately rated and inspected fire extinguishers (service vehicles and fuel trucks).
- It has operable headlamps and brake lights.
- Vehicles operating in the movement area shall be equipped with an amber rotating beacon or equivalent.
- All aircraft refueling vehicles and any other vehicle 8 feet or more in width shall be equipped with a flashing amber beacon and flashing front, tail, and clearance lights that are always activated when operating on the airside.

The FAA does not mention the vehicle conditions; however, the vehicles should be in good driving condition so that they do not break in the middle of runways or taxiways and obstruct the planes from landing, takeoff, or taxying. A vehicle's inventory should therefore include all of the vehicle's existing conditions, including the mileage, age, etc.

6.1.2 Performing a Condition Assessment

To assess the delayed maintenance costs, the condition of the pickup trucks needs to be assessed every year to determine the required maintenance activities. Asset managers must make sure that their vehicle department monitors and checks the conditions of the assets and keeps a record in the asset inventory database system. This condition assessment should include, but not be limited to, the make and model of the vehicles, hours or mileage used, major engine overhaul records, the frequency of preventive maintenance performed, and any type of recurring mechanical or electrical system problems. Information related to the vehicle's condition will help asset managers decide whether the vehicle needs minor maintenance, overall engine replacement, or complete vehicle replacement. Some airports purchase vehicles in fleets, so most of these vehicles' conditions will deteriorate and need replacement at the same rate. Staggering the purchase of the vehicles will help keep them in varied conditions and avoid costly fleet-wide replacements. The annual condition assessment will also assist asset managers in preparing long-term vehicle assessment plans and prioritizing vehicles that need more immediate replacement. During the data collection phase, most airports have their mechanics perform vehicle condition assessments but do not record the resulting data and do not use the data to determine a maintenance budget. Airport vehicle mechanics, however, do know the types of maintenance that keep vehicles in running condition. Various vehicle maintenance companies also keep vehicle inspection records and have developed checklists. An airport's vehicle maintenance department

Copyright National Academy of Sciences. All rights reserved.

can use these checklists to keep records of their service vehicles. A sample of a checklist is shown in Appendix A (QualityForms n.d.).

6.1.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

During annual maintenance and inspection, the mechanics determine the types of maintenance required over the following year to keep service vehicles in good condition. The preventive maintenance activities are performed based on mileage or the number of years in service. The preventive maintenance activities and their frequency based on mileage are provided in Table 6-1. Detailed maintenance tasks are provided in Appendix B (Kansas Department of Transportation 2022).

The preventive maintenance in Table 6-1 needs to be performed as scheduled to extend vehicle life. If these maintenance activities are not performed as prescribed, the vehicles risk breaking down, and repairing them can cost more than regular maintenance. To estimate a pickup truck's regular and delayed maintenance costs, the unit cost of performing these activities should be estimated. Without the unit cost for performing preventive maintenance, the accurate maintenance costs with or without delay cannot be estimated accurately. The research team contacted airports during the data collection phase to collect the unit cost data for preventive maintenance activities performed on their vehicles. None of the airports were able to provide systematic unit cost data. The team therefore conducted research to determine the cost of performing these maintenance activities on the pickup trucks. Enough maintenance cost data were available for a full-size pickup truck on the web (https://enginepatrol.com/ford-f-150-maintenance-costs-expenses/). Table 6-2 shows the cost data collected for the spreadsheet tool to determine the delayed maintenance costs (Engine Patrol 2021). When the total cost of maintenance is calculated, the team suggests adding 10% miscellaneous cost.

6.1.4 Determining the Full-Size Pickup Truck Failure Probability with Regular Maintenance

As described in the previous section, one of the critical tasks to determine the regular maintenance budget for assets is to find the deterioration rate of the asset in normally maintained conditions. In determining the regular maintenance costs of pickup trucks, it is important to find the cumulative probability distribution of engine failure in a full-size pickup truck under regularly maintained conditions. The research team contacted the case study airports to collect this failure probability but was unable to receive any data. The team searched the engine failure probability

Table 6-1. Preventive maintenance activities of vehicles and their frequencies.

Preventive Maintenance	Frequency (Miles)
Oil change	5,000
Air filter replacement	12,000
Spark plug replacement	30,000
Wiper blade replacement	25,000
Battery replacement	25,000
Brake replacement	45,000
Alternator replacement	70,000
Tire and wheel assembly rotate and balance	4,000
New transmission replacement	80,000
Engine tune-up	10,000
New tire replacement	40,000

Copyright National Academy of Sciences. All rights reserved.

78 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 6-2. Unit cost for preventive maintenance of a full-size pickup truck.

Preventive Maintenance Activities	Unit Cost (Based on 2021 data)
Oil change	$130
Air filter replacement	$45
Spark plug replacement	$250
Wiper blade replacement	$75
Battery replacement	$200
Brake replacement	$200
Alternator replacement	$550
Tire and wheel assembly rotate and balance	$85
New transmission replacement	$3,500
Engine tune-up	$450
New tire replacement	$800
New full-size pickup truck	$42,000

or reliability data of a full-size pickup truck online and found the failure probability data, which is shown in Table 6-3 (CarEdge 2022). These vehicles travel an average of 10,000 miles per year. Based on this information, 59% of the full-size pickup truck transmission engines will break at the 12-year mark.

Generally, a full-size pickup truck lasts for 15 years, which is approximately the equivalent of 150,000 miles. Because only 12 years of probability data were available, these data were used to generate a regression equation to predict the failure probability of a full-size pickup truck over 15 years. Figure 6-2 shows scatterplots with cumulative failure probability and the age of the pickup truck. Based on the line of best fit, the regression equation to predict the cumulative failure probability of the full-size pickup truck is shown in Equation 6-1. The line of best fit has an R-squared value of 0.98, which is essentially a perfect fit.

$$Cumulative\ Failure\ Probability = 0.0008 * Age\ of\ vehicle^3 - 0.0116 * Age\ of\ vehicle^2 + 0.073 * Age\ of\ vehicle - 0.0487 \quad (6\text{-}1)$$

Using Equation 6-1, the probability of the full-size pickup truck can be determined as shown in Table 6-4. The table shows that the probability of engine failure of a full-size pickup truck after 15 years will be 100%. The truck will need to be replaced entirely when it reaches 15 years or 150,000 miles. These failure probability values can be used to determine the replacement cost of transmission engines in full-size pickup trucks.

Table 6-3. Cumulative failure probability of a full-size pickup truck.

Age of the Vehicle (Years)	Vehicle Mileage	Cumulative Failure Probability
1	10,000	3%
2	20,000	5%
3	30,000	6%
4	40,000	10%
5	50,000	15%
6	60,000	17%
7	70,000	18%
8	80,000	23%
9	90,000	26%
10	100,000	31%
11	110,000	47%
12	120,000	59%

Copyright National Academy of Sciences. All rights reserved.

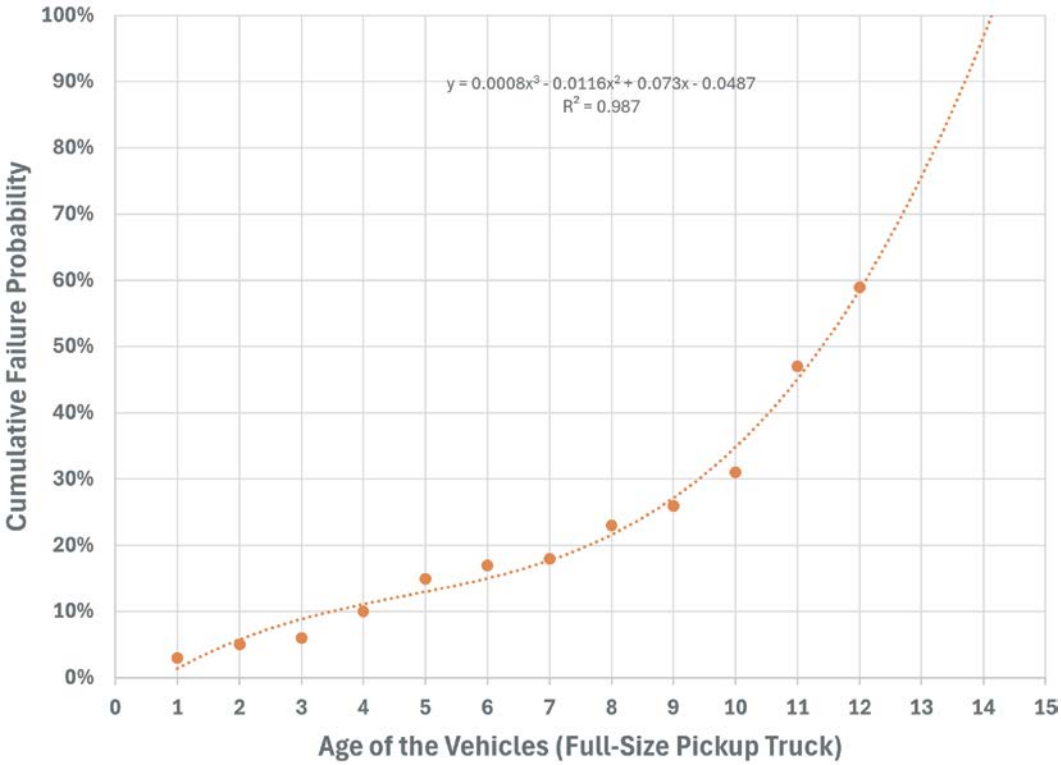


Figure 6-2. Cumulative failure probability vs. age of a full-size pickup truck.

6.1.5 Determining the Full-Size Pickup Truck Failure Probability with Delayed Maintenance

As described in Section 6.1.4, the reliability data of the full-size pickup truck was available and can be used to determine the frequency of engine transmission failure under regular maintenance scenarios. However, responsible agencies cannot keep reliable data on vehicles without maintenance. During the data collection phase, airports were asked about data that can predict the failure of the pickup trucks' transmission engine, and whether they were maintained properly;

Table 6-4. Cumulative failure probability of regularly maintained full-size pickup trucks based on regression.

Age of the Vehicle (Years)	Vehicle Mileage	Cumulative Failure Probability
1	10,000	1.4%
2	20,000	5.7%
3	30,000	8.8%
4	40,000	10.9%
5	50,000	12.6%
6	60,000	14.5%
7	70,000	16.8%
8	80,000	20.3%
9	90,000	25.2%
10	100,000	32.1%
11	110,000	41.6%
12	120,000	53.9%
13	130,000	69.8%
14	140,000	89.5%
15	150,000	100.0%

Copyright National Academy of Sciences. All rights reserved.

none of the airports were able to provide these data. They stated that the engine life of the vehicles reduces significantly when they are not maintained regularly as suggested by the manufacturers. The research team assumes that the vehicle life will reduce significantly when regular maintenance is delayed, meaning that the failure probability of the vehicle engine will likely double compared to the probability of failure with regular maintenance. Based on Table 6-4, a new cumulative failure probability for delayed maintenance of the full-size pickup truck is provided in Table 6-5. Table 6-5 shows that the failure probability increases twofold if the maintenance is delayed during that year. Based on delayed maintenance scenarios, the life of the pickup truck will be reduced to 12 years (120,000 miles) from the original 15 years (150,000 miles).

6.1.6 Estimating the Full-Size Pickup Truck Maintenance Budget without Delayed Scenarios

The maintenance cost of pickup trucks without delayed scenarios can be calculated by determining the types of maintenance that need to be conducted, their frequency every year, and the unit cost of each of these preventive maintenance activities. The research team developed a spreadsheet tool that can determine the preventive activities performed on the full-size pickup truck based on the age or mileage of the vehicles (see Table 6-4). Then, considering the activities that need to be performed, the regular maintenance cost can be calculated using the unit costs provided in Table 6-2. As described in Section 6.1.3, these unit costs are based on data from 2021. To adjust this base cost for future years, however, the research team collected the cost inflation indices of vehicle maintenance from the Bureau of Labor Statistics (2022c). Based on the cost indices available from 2008 to 2021, the average inflation rate of the repair and maintenance of motor vehicles is 2.93%. This default inflation rate can be used to calculate the total maintenance and replacement cost of the full-size pickup truck for future years. The tool also has the capability to use the recent inflation rate, if different from this default rate, to calculate future maintenance and replacement costs.

6.1.7 Estimating the Full-Size Pickup Truck Maintenance Budget with Delayed Scenarios

To determine the total maintenance and replacement costs, the research team calculated the frequency of the maintenance activities to be performed after certain periods of delay using the spreadsheet tool. Based on the frequency of maintenance, the team calculated the total maintenance and replacement costs for delayed scenarios. In the case of pickup truck delayed maintenance calculations, the cost of transmission replacement will generally increase compared to the

Table 6-5. Cumulative failure probability of full-size pickup trucks under delayed maintenance scenarios.

Age of the Vehicle (Years)	Vehicle Mileage	Cumulative Failure Probability
1	10,000	1.4%
2	20,000	10.2%
3	30,000	16.2%
4	40,000	20.5%
5	50,000	24.0%
6	60,000	27.6%
7	70,000	32.4%
8	80,000	39.2%
9	90,000	49.1%
10	100,000	63.0%
11	110,000	81.8%
12	120,000	100.0%

Copyright National Academy of Sciences. All rights reserved.

regular maintenance scenarios because the probability of transmission engine failure will be higher. Therefore, the probability of failure shown in Section 6.1.5 for delayed maintenance conditions is higher than that of regular maintenance conditions. Because the transmission replacement costs do not vary greatly, the maintenance and replacement costs in delayed scenarios will not be significantly higher than the base cost. It can also be assumed that, due to delayed maintenance, the entire pickup truck will need to be replaced, if not regularly maintained, and the replacement cost can be calculated using the purchase cost of the full-size pickup truck, which will show significant cost increases in delayed scenarios. It is difficult to predict the impact of the pickup truck engine's failure on airport maintenance and replacement costs. It is therefore suggested that delaying maintenance is avoided, and, if maintenance is delayed, it should not be delayed by more than 2 years. It is also assumed that the pickup trucks are not replaced unless their life reaches 150,000 miles under normal maintained conditions, or 120,000 miles under delayed maintenance scenarios.

6.1.8 Reporting the Impact of Full-Size Pickup Truck Delayed Maintenance and its Consequences

The research team developed a spreadsheet tool to determine the impact of delaying maintenance on full-size pickup trucks in terms of airport maintenance and replacement costs. The age or the mileage of the full-size pickup truck will have an impact on the maintenance and replacement costs of the vehicle. If the pickup truck is new, then the maintenance costs tend to be lower. However, if the age or mileage of the vehicle is high, the maintenance costs are often high. When analyzing the impact of delayed maintenance on pickup truck maintenance and replacement costs, the research team considered three types of full-size pickup trucks: one with 10,000 miles, another with 50,000 miles, and, finally, one with 70,000 miles. Two- and 5-year delayed scenarios were considered. The team calculated the trucks' regular maintenance and replacement costs for the year 2023 without delayed maintenance. Then the maintenance and replacement costs for the years 2025 and 2028 with 2- and 5-year delayed maintenance were considered to determine the impact on airport maintenance and replacement costs. It was assumed that the airport owns five full-size pickup trucks.

The spreadsheet tool developed to calculate the impact of delayed analysis of the full-size pickup truck is flexible. The tool can calculate the frequency of various maintenance activities performed within a given fiscal year. The user can also adjust the frequency of maintenance activities and enter their own values. The tool uses the unit cost of each of the activities collected during the data collection phase, and, if these unit costs are different from the ones they are using, they can enter their values. The user can also select the number of years delayed and the cost inflation factor for the maintenance and replacement activities. If the user wants to calculate the delayed maintenance costs in present value, they can select the discount rate provided in the tool.

Case Study Results of the Consequence of Delaying Maintenance on Ford Full-Size Pickup Trucks

When analyzing the total maintenance cost of full-size pickup trucks, the research team assumed that the pickup trucks would operate 10,000 miles per year. It was also assumed that, when the pickup truck has 10,000, 50,000, or 70,000 miles on it, the transmission engine has not been replaced. Figure 6-3 shows the total maintenance and replacement costs of five full-size pickup trucks with 10,000, 50,000, and 70,000 miles, with 2- and 5-year delayed scenarios. For a truck with 10,000 miles, the 2-year maintenance delay increases the total maintenance and replacement costs threefold compared to maintenance without the delay. However, when maintenance is delayed by 5 years, there is a 50% increase in the maintenance and replacement costs. The main reason for this is that, in a 5-year delay, the pickup truck mileage reaches 70,000 miles instead of 50,000 miles and the failure probability of the transmission engine increases by 8.4%; the pickup truck will also need to have its alternator changed, which increases the total maintenance and replacement costs. When the mileage in the vehicle reaches about 50,000, the cost of the delayed maintenance for

Copyright National Academy of Sciences. All rights reserved.

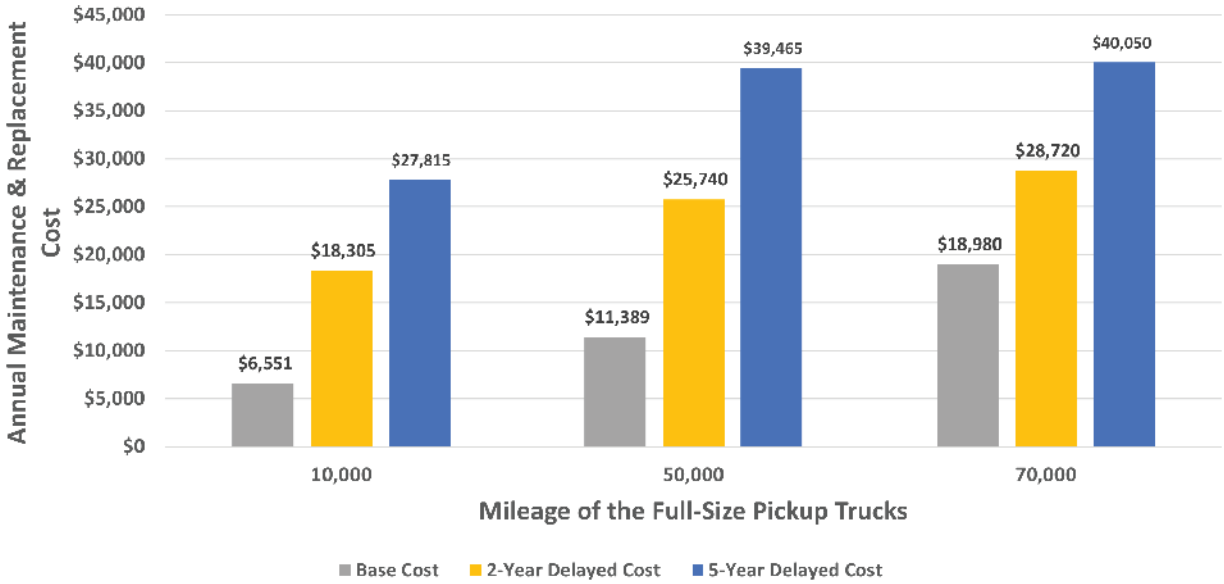


Figure 6-3. Impact of delaying maintenance on annual maintenance and replacement cost of full-size pickup trucks.

2 and 5 years will differ by a similar amount due to the previously stated reasons. The same logic applies to a pickup truck that has accumulated 70,000 miles.

6.2 Airport Snow Removal Equipment

Snow falling on runways and taxiways needs to be removed immediately so that airplanes can land and take off safely. Generally, at airports located in areas of heavy snowfall, asset managers need to keep snow removal equipment to remove snow quickly. Airports use snow blowers to remove snow when the quantity is small. If the snow is in large quantities and runways are long enough, airports generally use the wheel loader with an attached snowblower to remove the snow. Some of the airports outsource the snow removal activities to contractors. Airport asset managers must maintain the airport's snow removal equipment, e.g., blowers, loaders with blower attachments, and wheel backhoes so that snow can be removed quickly. Asset managers must keep an inventory of snow removal equipment. The inventory should consist of types of equipment, age of the equipment, hours of operation, make and model, and date purchased so that the maintenance and replacement costs can be estimated accurately. The FAA (2022d) has developed a tool to estimate the total amount of snow, and the number of front-end loaders, rotary plows, hoppers, and sweepers to remove snow from runways and taxiways based on the size of their airports. This tool is very user-friendly and can be used by asset managers to calculate the number and size of snow removal equipment needed.

FAA AC 150/5220-20A—Airport Snow and Ice Control Equipment discusses requirements for snow and ice removal control equipment (FAA 2014c). This AC provides advice to assist airport operators in the procurement of snow and ice control equipment for airport use. It emphasizes the selection process of snow removal equipment for airports. The main three steps for selection of the equipment are:

- Identifying the type of equipment for a specific task,
- Determining the number of such equipment, and
- Using the recommended equipment specifications to better ensure the equipment performs the required task.

Copyright National Academy of Sciences. All rights reserved.

Depending upon the airport and the amount of snow that needs to be removed each year, the airport can use high-speed rotary plows, snowplows, solid and liquid material spreaders, runway brooms with air blasts, and loader-mounted snow blowers. While selecting the snow removal equipment, the airport asset manager must compare the cost/benefit relationship of acquiring a multifunctional piece of equipment offering several attachments versus the acquisition of single units, each designed primarily to accomplish one task. The FAA has developed two other ACs related to snow removal: *AC 150/5200-30DAC – Airport Field Condition Assessments and Winter Operations Safety* and *AC 150/5220-18A – Buildings for Storage and Maintenance of Airport Snow and Ice Control Equipment and Materials*. The first AC provides guidelines to assist airport operators in developing a snow and ice control plan, assessing and reporting airport conditions through the utilization of the Runway Condition Assessment Matrix, and establishing snow removal and control procedures. The second AC determines the selection and design of buildings used to store and maintain airport snow and ice control equipment. However, the FAA has not developed any guidelines related to the maintenance of snow removal equipment. Therefore, the team developed a framework to determine the impact of delayed maintenance of snow removal equipment on an airport's maintenance budget. An asset manager needs to follow the framework provided in Figure 6-4 to determine the maintenance costs of snow removal equipment with and without delayed maintenance.

6.2.1 Establishing Asset Inventory

Generally, large-hub airports located in snow-prone areas will have a large fleet of snow removal equipment. Therefore, airport asset managers need to keep an inventory of all their equipment. The asset manager needs to record the amount of equipment, condition of the equipment, hours used, age, date purchased, and maintenance performed on the equipment. The make and model of the equipment, its horsepower, and whether it is a diesel or gasoline engine also need to be recorded. During the data collection phase, the airports stated that they had these records and provided them. However, the data were insufficient to develop a spreadsheet tool for quantifying the impact of delayed maintenance of snow removal equipment on an airport's maintenance budget.

6.2.2 Performing a Condition Assessment

The condition assessments of the snow removal equipment and the service pickup trucks were kept in a similar manner. None of the airports that the research team talked with have any rating of their snow removal equipment. The mechanics estimated the condition of the equipment based on their hours of operation and the age of the equipment. Although the mechanics monitor the conditions of the equipment every year, the inspection data for the equipment was not available. Also, the team was unable to retrieve any condition data for the airports' snow removal equipment. Therefore, it is necessary for airport asset managers to keep the inspection and condition data of snow removal equipment up to date, so that maintenance costs with and without delayed scenarios can be calculated more precisely for budgeting purposes. The New York State DOT has developed a maintenance inspection checklist for front-end loaders, shown in Appendix C (New York Department of Transportation 2022). This inspection checklist covers all maintenance activities performed during the loader inspection. It covers the under hood, interior, and exterior, as well as any attachments. The inspection of the loader using this kind of inspection list will help an asset manager estimate the maintenance and replacement costs for the coming year. This is important in determining the remaining life of the equipment. The inspection checklist will help identify major engines or other parts needing replacement for the coming years. Therefore, airport mechanics should use this type of inspection sheet to evaluate their snow removal equipment and rate their equipment in Excellent, Good, Fair, and Poor conditions. These data need to be entered into the conditional database so that, during the budget preparation process, they can be utilized to determine accurate maintenance and replacement costs of the snow removal equipment.

Copyright National Academy of Sciences. All rights reserved.

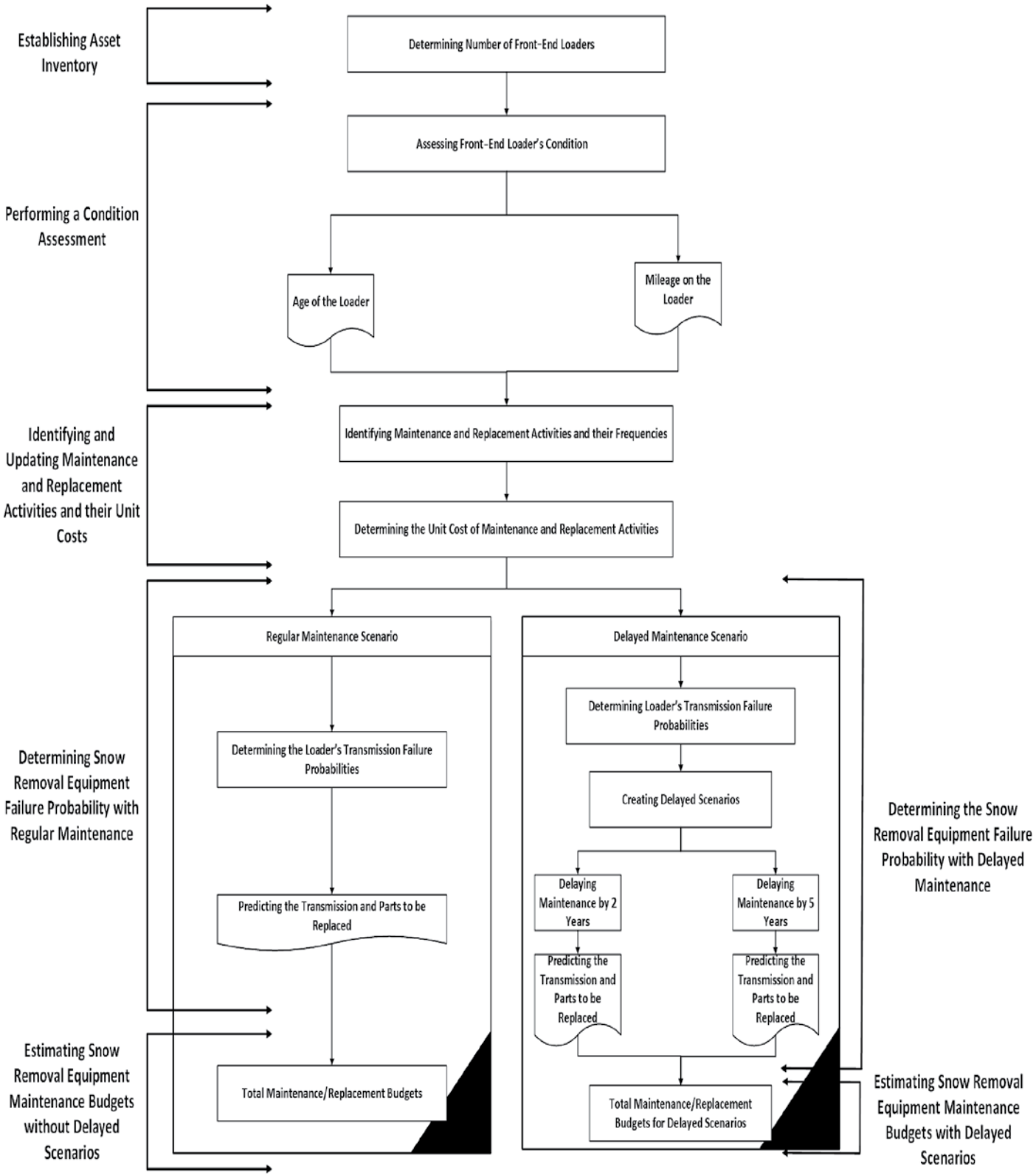


Figure 6-4. Framework for delayed maintenance of snow removal equipment.

Copyright National Academy of Sciences. All rights reserved.

6.2.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

During snow removal equipment inspection, mechanics identify the preventive maintenance activities that need to be performed to keep them in operating condition. These maintenance activities are key in determining the maintenance and replacement costs of snow removal equipment for future years. The preventive maintenance activities that need to be performed depend upon the types of equipment. However, for this research, the team gathered preventive maintenance activities for front wheel loaders with snow blower attachments (see Table 6-6) (Kramlich 2019). Table 6-6 shows the maintenance activities that need to be performed and the frequency of those activities. Generally, the life of a front-end loader with a blower attachment is 15,000 hours or 15 years, which translates to about 1,000 hours of operation every year. Other heavy equipment manufacturers like Caterpillar have developed operation and maintenance manuals for this type of equipment (Caterpillar n.d.).

After the maintenance activities of snow removal equipment are identified, asset managers must determine the unit costs to perform these tasks. Generally, to determine the unit costs of these tasks, the labor hours to perform these activities need to be estimated, and then the parts needed for these repairs should be calculated. The research team tried to collect these data from the airports; however, the airports failed to provide the data. They performed these preventive maintenance activities but did not collect labor hours and parts data. Therefore, the research team conducted an online search and found labor hours and parts data for a front wheel loader. There are no labor and parts data available for every maintenance activity to be performed on the loader; however, the total labor hours for maintenance of wheel loaders per 1,000 hours of operation and the ratio of labor and parts cost were found online (CASE Construction n.d.). Table 6-7 shows the labor hours for performing these maintenance activities on front wheel loaders for every 1,000 hours or 1-year period, as well as the hours required to replace the tires and transmission engine.

During the data collection phase, the research team could not collect the unit cost data of snow removal equipment maintenance used by the airports. Therefore, the cost available on the CASE Construction website was used to determine the unit cost of maintenance of a front-end loader and

Table 6-6. Preventive maintenance activities of front-end loaders and their frequencies.

Preventive Maintenance	Frequency
Oil change and filter	500 miles
Oil in the axles and power divider	500 miles
Service brake system	500 miles
Hydraulic oil and air filters	500 miles
New tire replacement	2,500 miles
Engine, new transmission, and axle rebuilt	8,000 miles

Table 6-7. Labor hours required for maintenance and replacement activities of front-end loaders.

Maintenance and Replacement Activities	Labor Hours
Maintenance activities per year based on 1,000 hours of work	30 hours
Replacing tires	8 hours
Replacing the transmission	24 hours

Copyright National Academy of Sciences. All rights reserved.

86 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 6-8. Unit maintenance and replacement cost calculations for front-end loaders.

Category	Calculation
Number of labor hours for maintenance	30 hours
Cost per hour for mechanic	$60/hour
Total labor cost	$1,800
Ratio of labor and parts cost	40% labor and 60% parts
Parts cost	$1,800 x (0.6) / (0.4) = $2,700
Total cost per year	$1,800+$2,700 = $4,500
Labor hours for tire replacement	8 hours
Labor cost for tire replacement	8 hours x $60/hour = $480
Cost of four tires	$26,000
Total cost of tire replacement	$26,480
Labor hours for new transmission	24 hours
Cost of new transmission	$40,000
Total cost of new transmission	$40,000 + 24 hours x $60/hour =$41,440
Cost of new loader (mid-size) with snow blower attachment	$265,000

its tire and engine replacement costs. Table 6-8 shows the cost calculation process for determining the maintenance and replacement unit costs for front wheel loaders. These costs were also verified using the RSMeans Cost Guide (2021). The numbers calculated here are based on 2021 figures.

6.2.4 Determining the Snow Removal Equipment Failure Probability with Regular Maintenance

The failure probability of snow removal equipment used by the airports is not available from asset managers because airports do not keep these data. Therefore, the research team searched the failure probability of heavy equipment online, but was unable to collect any data. Therefore, the research team used the Weibull probability distribution theory to calculate the cumulative probability of engine failure of a front-end loader. As described in the previous chapter, to calculate the cumulative probability of any electrical or mechanical system, the scale and shape factors need to be identified. For heavy equipment failure, a shape factor of 3.5 is considered, which shows the normal distribution of the failure probability of a front-end loader. Because the life of front wheel loaders is generally considered to be 15 years, assuming 1,000 hours of operation every year, the scale factor, which is 63% of the failure value, is assumed as 8,000 hours for regularly maintained loaders. Table 6-9 shows the Weibull failure probability for a front wheel loader for its total life of 15 years or 15,000 hours. This probability of failure can be used to determine the replacement costs of the regularly maintained loader. When the replacement cost is calculated, only the unit cost of installing a new transmission engine will be considered. If the age of the vehicle exceeds 15 years or 15,000 hours of operation, then the front wheel loader is considered in need of replacement.

6.2.5 Determining the Snow Removal Equipment Failure Probability with Delayed Maintenance

The research team used the probability theory to determine the transmission engine failure rate of front-end loaders in regular maintenance scenarios. For delayed maintenance scenarios, the failure rate of the transmission engine of a front-end loader was also determined using the Weibull distribution theory. In delayed maintenance scenarios, the probability distribution shape factor was similar to the one used for regularly maintained cases. However, the scale factor, which is 63% of the equipment's failure rate, was only 5 years or 5,000 hours of operation. In

Copyright National Academy of Sciences. All rights reserved.

Table 6-9. Cumulative failure probability of front-end loaders under regular maintenance.

Age of the Equipment (Years)	Hours Used	Cumulative Failure Probability
1	1,000	0.1%
2	2,000	0.8%
3	3,000	3.2%
4	4,000	8.5%
5	5,000	17.6%
6	6,000	30.6%
7	7,000	46.6%
8	8,000	63.2%
9	9,000	77.9%
10	10,000	88.7%
11	11,000	95.3%
12	12,000	98.4%
13	13,000	99.6%
14	14,000	99.9%
15	15,000	100.0%

delayed maintenance scenarios, the chances of failure of the front-end loaders' transmission will be higher. Table 6-10 shows the cumulative failure probability of transmission engines of front-end loaders in delayed maintenance scenarios. This table was used to calculate the replacement cost of the new transmission of the loader in delayed maintenance scenarios. Once the regularly maintained loader exceeds 15 years or 15,000 hours of operation, a new loader needs to be purchased. The table shows that, under delayed maintenance scenarios, the life of the front-end loader will be about 9 years.

6.2.6 Estimating Snow Removal Equipment Maintenance Budgets without Delayed Scenarios

The total maintenance cost of snow removal equipment can be calculated by determining the labor hours required for maintenance of the equipment and parts. The replacement cost of the front-end loader's transmission engine can be calculated using the failure probability shown in Table 6-9, and the labor and cost of transmission are provided in Section 6.2.3. After adding the replacement and maintenance costs, the total budget required to maintain the snow removal equipment can be estimated. Because the cost calculated and shown in Table 6-8 is based on 2021 data, these maintenance and replacement costs need to be adjusted based on the inflation factor. Inflation factors (machinery and equipment repair and maintenance) collected by the Bureau of Labor and Statistics (BLS) since 2006 were used (BLS 2022c). Based on these cost indices, the average

Table 6-10. Cumulative failure probability of front-end loader under delayed maintenance.

Age of the Equipment (Years)	Hours Used	Cumulative Failure Probability
1	1,000	0.4%
2	2,000	4.0%
3	3,000	15.4%
4	4,000	36.7%
5	5,000	63.2%
6	6,000	85.0%
7	7,000	96.1%
8	8,000	99.4%
9	9,000	100.0%

Copyright National Academy of Sciences. All rights reserved.

inflation rate of machinery and equipment repair and maintenance was found to be 2.61% annually. Using inflation cost indices from Statista (2022) for heavy equipment purchase costs since 2005, the annual inflation rate of these purchases was estimated at 2.30%. These inflation factors can be used to adjust the cost of maintenance and replacement of snow removal equipment while preparing the budget for future years.

6.2.7 Estimating Snow Removal Equipment Maintenance Budgets with Delayed Scenarios

The estimated maintenance and replacement cost of snow removal equipment with delayed scenarios was calculated using the unit cost of maintenance and replacement provided in Table 6-8 and the probability of failure of the transmission engine shown in Table 6-10. The only difference in the maintenance and replacement budget between regular and delayed maintenance equipment was that the failure probability was higher in the case of delayed maintenance compared to regular maintenance. Because these delayed scenarios are considered for future years, the unit cost of maintenance and replacement should be calculated based on the inflation rate described in Section 6.2.6. The research team developed a spreadsheet tool to estimate the maintenance and replacement costs of a front wheel loader with a snow blower attachment. For delayed maintenance scenarios, the user can enter a different inflation rate than the default value. Once the cost is adjusted based on the inflation rate, the total maintenance and replacement costs in delayed scenarios can be calculated. The user can also select the number of years the maintenance of snow removal equipment can be delayed. Additionally, the tool can convert the costs to net present value by using a discount rate.

6.2.8 Reporting the Impact of Snow Removal Equipment Delayed Maintenance and its Consequences

The research team developed a spreadsheet tool and used it to determine the impact of delayed maintenance on airport maintenance and replacement costs of snow removal equipment. The age or number of hours the front-end loader with snow blower attachment is used will have an impact on the maintenance and replacement costs of the equipment. If the loader with the attachment is new, the maintenance and replacement costs will be lower. However, if the age or hours of the equipment are high, the maintenance and replacement costs will be very high. While analyzing the impact of delayed maintenance on snow removal equipment maintenance and replacement costs, three types of front-end loaders were considered: one with 1,000 hours, another with 5,000 hours, and one with 7,000 hours of operation. Two- and 5-year delayed scenarios were used. A front-end loader with snow blower attachment's regular maintenance and replacement costs were calculated for 2023 without delayed scenarios. The team considered the maintenance and replacement costs for years 2025 and 2028 with 2- and 5-year delays to determine the impact of delayed maintenance on airport maintenance and replacement costs of their snow removal equipment. It was assumed that the equipment will be used for 1,000 hours per year.

The team developed a user-friendly spreadsheet tool to calculate the impact of delayed maintenance of snow removal equipment. This tool can calculate the labor hours required for maintenance of a front-end loader per year. The user can then use those hours instead of case study hours when investing to maintain their front-end loader. They can also use their own labor costs per hour. Similarly, if airports know the parts cost for maintenance of their wheel loader, they can use that cost in their calculations. The tool uses the unit cost of each activity (tire and transmission replacement, new loader, and snow blower attachment) from the data collection phase, and, if these unit costs are different from the user's, they can enter their own values. The user can also select the number of years delayed and the cost inflation rate for the maintenance and replacement activities

Copyright National Academy of Sciences. All rights reserved.

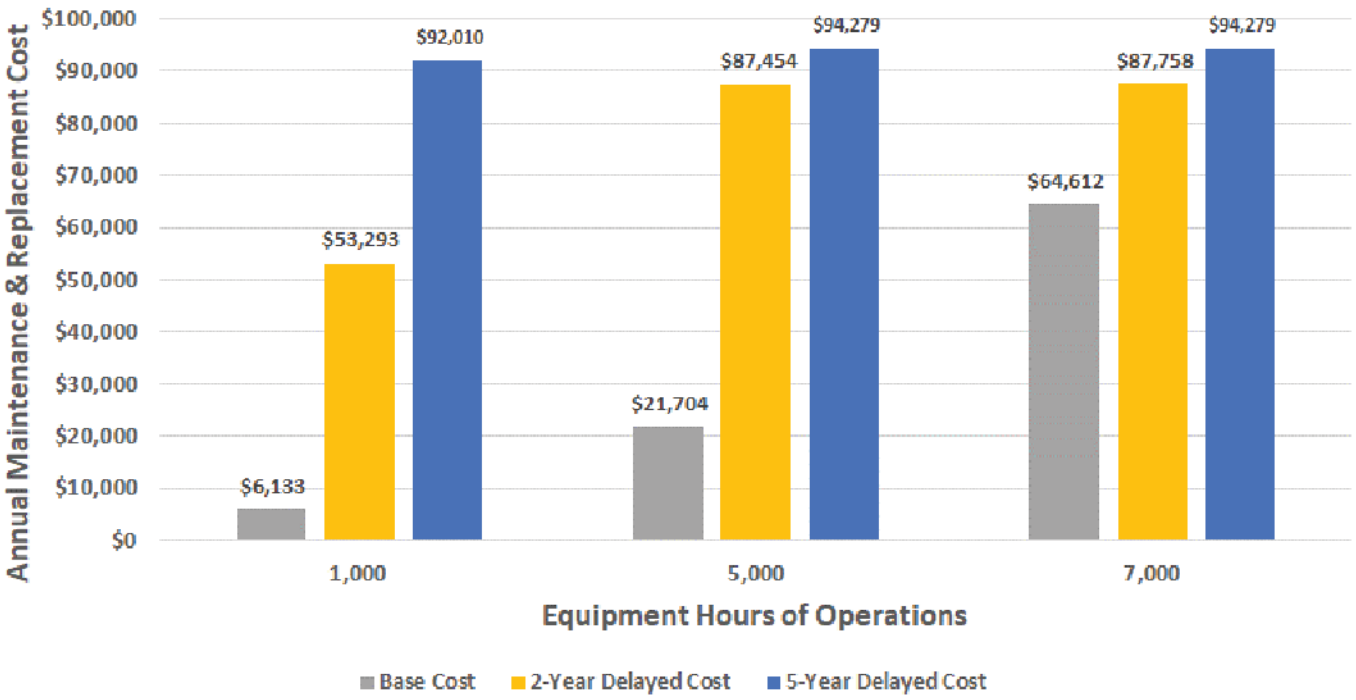


Figure 6-5. Impact of delaying maintenance on the annual maintenance and replacement cost of a front-end loader.

of snow removal equipment. If the user wants to calculate the delayed maintenance cost in present value, they can select the discount rate provided in the tool.

Case Study Results of the Consequence of Delaying Maintenance on Snow Removal Equipment

Figure 6-5 shows the total maintenance and replacement costs of a front-end loader with snow blower attachment with various hours of operations and 2- and 5-year maintenance delayed scenarios. When the snow removal equipment has just 1,000 hours of operation, the delayed maintenance cost for 2 and 5 years increases significantly. The main reason is that the probability of engine failure in delayed conditions will be significantly higher compared to non-delayed scenarios. The same logic is true for equipment with 5,000 hours of operation. However, the equipment with 7,000 hours of operation will have the same probability of engine failure as the equipment with 5,000 hours of operation in delayed scenarios, so the cost of maintenance and replacement for these types of equipment is the same.

6.3 HVAC Systems

Airports require HVAC systems to maintain the temperature in their terminals, administrative, and office buildings. Generally, the commercial types of big HVAC systems are used to cool and heat these buildings. The terminal buildings also use chillers and boilers to cool and heat the spaces. However, these chiller and boiler systems have different kinds of heating and cooling processes. They generally use pipes to send the cold and hot water to cool or heat the rooms. However, in this section, the common rooftop HVAC systems generally used in commercial buildings, like big shopping centers, office buildings, and warehouses, will be discussed. Because of their environmental exposure, rooftop units are built to withstand sunlight, rain, wind, frost, and other conditions.

Copyright National Academy of Sciences. All rights reserved.

90 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Rooftop HVAC units are also called packaged units because all the required components are packaged together as one unit (Estes Services n.d.). The roof package system consists of the heating and/or cooling system components, depending on the type of system. Apart from the rooftop units, the air conditioning unit can have split systems, which have separate equipment housed indoors and outdoors. A furnace and/or air handler is installed within the building, while an air conditioner or heat pump is installed outside the building around its perimeter.

Rooftop HVAC units are connected to the building's ductwork and heat or cool air passing through, then distribute the conditioned air throughout the interior areas. First, the air enters a rooftop unit via the air hood at its side. As air passes through, dampers operate inside to regulate airflow, and the air supply passes through the system's filters to remove contaminants. Then the air is either heated by a heat exchanger or coils or cooled by cooling coils (depending on the system configuration and operating mode). The conditioned air is circulated through the ducts of the building. All central heating and cooling systems should have an air filter. The air filter is usually located in the return air duct or blower compartment before the return air reaches the air handler. This allows the filter to clean the air from outside of the building before it enters the HVAC unit (Beauchamp and Lopez 2022).

The research team focused on rooftop package units to determine the impact of delayed maintenance on an airport's total maintenance and replacement costs. Figure 6-6 shows the framework developed to calculate the regular and delayed maintenance and replacement costs for the rooftop HVAC package units.

6.3.1 Establishing Asset Inventory

Airports with rooftop HVAC package units need to keep an inventory of the components of their system. Asset managers should keep a record of the number of HVAC units, capacity of the system in tons or British thermal units (BTU), make and model, purchase date, and Seasonal Energy Efficiency Ratio (SEER) ratings. Additionally, they should keep a record of components of HVAC units, including the condenser, compressor, heating units, air handlers, number of thermostats, number and size of air filters, number and size of ducts, and length of air duct system. These inventory data are critical in determining the maintenance and replacement costs of an HVAC system. The inventory data of these HVAC unit components will help determine the condition of each component so that airport mechanics can determine which components need to be replaced.

If airport asset managers do not have information about the capacity of the HVAC units required for their airport terminals and administrative buildings, they can calculate it using the process provided by the U.S. Environmental Protection Agency. The agency developed a tool to determine the capacity of HVAC units based on the size, number of occupants, and number of doors and windows in a building (EPA 2022). The standard also provides information on selecting HVAC units and their components. Table 6-11 provides the amount of BTU required based on the building square footage, the number of occupants, and the number of doors and windows. Using these values, airport asset managers can determine the necessary capacity of HVAC units for their buildings or terminals.

The number of vents required for the buildings can also be calculated based on the number of occupants in the building. EPA guidelines recommend providing 15 cubic feet per minute (CFM) per person of ventilation. After the number of occupants in the building is calculated, the total ventilation required in terms of CFM can be calculated by multiplying 15 CFM per person by the number of occupants. Assuming one air filter can return 2.5 CFM of air to the blower compartment, the number of return air vents can be calculated by dividing the total CFM required by 2.5 CFM per vent. Similarly, the number of air vents needed to send cool or hot air around the

Copyright National Academy of Sciences. All rights reserved.

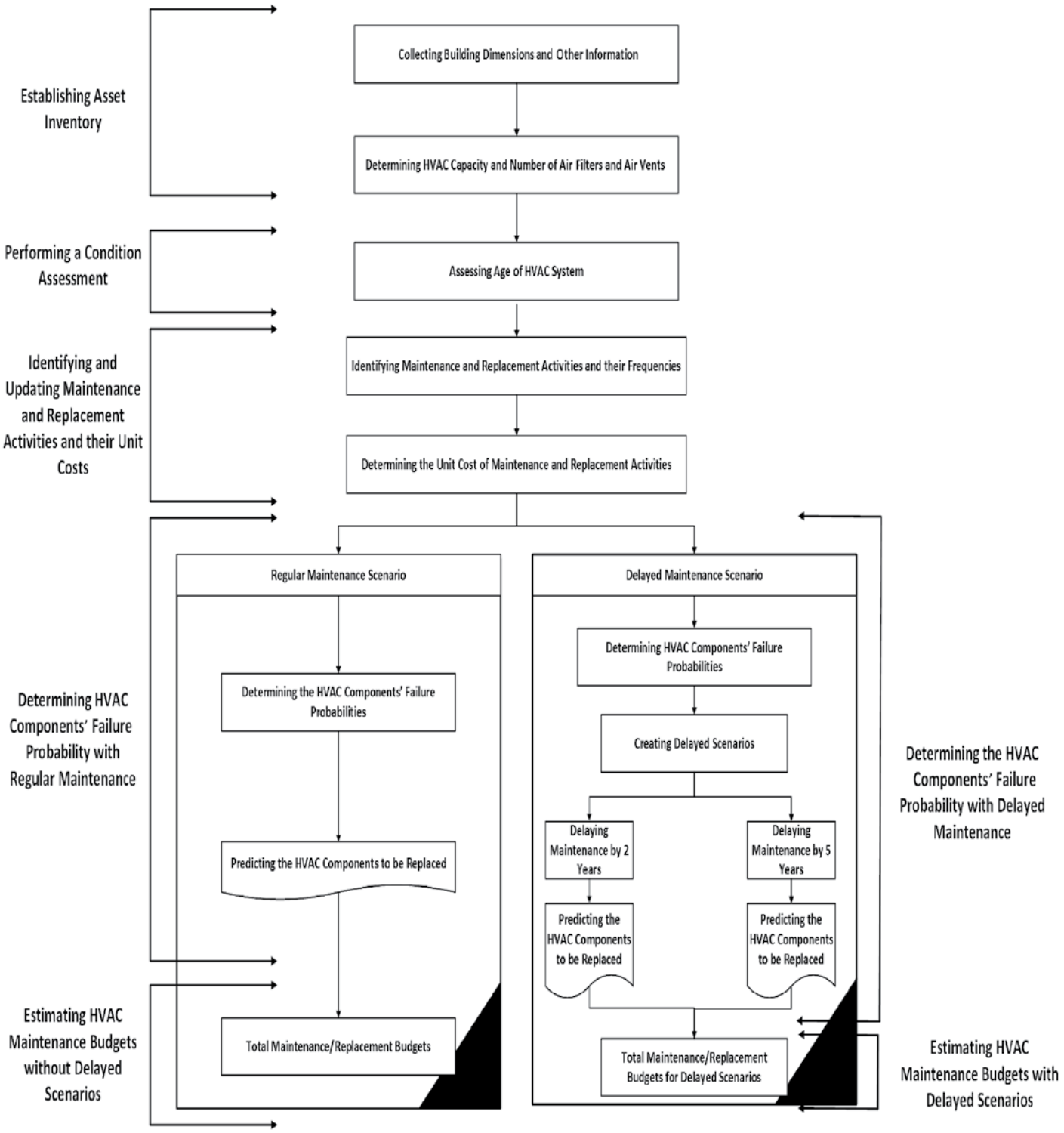


Figure 6-6. Framework for delayed maintenance of HVAC unit.

Copyright National Academy of Sciences. All rights reserved.

92 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 6-11. BTU requirements based on the building size, number of occupants, and number of doors and windows.

HVAC Load Calculator Characteristic	BTU Required
Area of building in square footage	12,000 BTU for every 500 square feet
Number of occupants	300 BTU for every occupant
Number of doors and windows	1,000 BTU for every door or window

building can be calculated by considering that one air vent is required per 100 square feet of building area. The number of thermostats can also be determined based on the number of HVAC units installed in the building. The research team developed a spreadsheet tool that can perform these calculations to determine the impact of delayed maintenance of HVACs on airport costs.

6.3.2 Performing a Condition Assessment

During the data collection phase, the research team asked asset managers to provide a condition assessment process and rating of the HVAC system used in their airports. None of the airports used systematic condition assessment procedures to rate the condition of the HVAC systems. The airport mechanics, however, assess the condition of HVAC units every year to determine what needs to be fixed to run the system without interruption. Large-hub airports have separate building departments to monitor the assets of terminals and administrative buildings. Most of the airports stated that they have plans to maintain the HVAC system and know what needs to be done. They also prepare budgets based on these requirements. However, the research team found that the airports did not have a systematic way of conducting inspections of their HVAC systems.

The research team searched the HVAC system inspection processes and procedures. There are various HVAC installations and maintenance companies that provide professional services related to HVAC systems. The EPA (2014) has also developed an HVAC inspection form. Other professional HVAC companies have suggested that the following items be inspected every year to make sure that all the components are working properly (ARISTA n.d.):

- Clean or replace filters.
- Clean condenser and evaporator coils.
- Clean drain lines for proper flow and clear clogs.
- Clear drain pans of any standing water to avoid overflows.
- Replace worn pulleys and belts.
- Inspect ducts for mold, dust, and debris.
- Observe humidity levels.
- Check refrigerant charge and for leaks.
- Test thermostats and controls to make sure temperatures and timer functions are correctly set.
- Change batteries.
- Check the electrical system and connections.
- Check the fan motor.
- Check blowers and blades for proper airflow.
- Lubricate motors, bearings, and moving parts.
- Inspect cabinet for leaks and check cabinet door for secure closure.
- Check for debris around the outside unit.

Airport asset managers need to make sure that the HVAC units are inspected yearly so that the required preventive maintenance can be performed and the airports can remain open without compromising travelers' comfort. They must also keep a record of these inspections to estimate the HVAC unit maintenance and replacement budget based on the conditions. A short HVAC unit

Copyright National Academy of Sciences. All rights reserved.

checklist developed by the professional companies that provide HVAC services is shown in Appendix D (HousecallPro 2022). During the inspection period, the mechanics can determine whether the parts need to be replaced or if regular maintenance is required. The form provided in Appendix D shows that, when the inspection is done, it is necessary to determine whether the components of the system are in good condition. If the inspection record helps determine the level of attention needed, it will be easier to prepare the maintenance budget for the HVAC system. HVACs can operate up to their full lifespan if preventive maintenance is performed on the HVAC units annually.

6.3.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

If airports perform inspections of HVAC units as mentioned in Section 6.3.2, then it will be easy to determine what major maintenance activities need to be performed. Those inspection records should focus on the necessary maintenance activities so that HVAC units can reach peak performance. There are specific maintenance activities related to each component of an HVAC unit to keep an HVAC unit in good condition. Some of the maintenance activities are shown below. For more details, please refer to the City of Lake City HVAC repair and preventative maintenance services invitation to bid (OpenGov Procurement 2021).

- Check and change filters if needed (filters shall be supplied by the contractor);
- Inspect the entire system;
- Check all controls, including thermostats, and damper operators when necessary;
- Check the operation and condition of all components;
- Adjust and check motors, starters, valves, drives, and accessories;
- Lubricate all moving parts, motors, bearings, etc., add compressor oil as needed;
- Replace or maintain compressor belts;
- Clean and check blower assembly;
- Clean ignition system; and
- Clean duct system.

After identifying all the preventive maintenance activities that need to be performed in HVAC units, the airport should record the labor hours and parts spent maintaining them. During the data collection phase, the research team found that some labor hours and money spent on parts were recorded, but not all data were available to determine how many hours were spent and on which activities. So, the team conducted an internet search to determine the labor hours required to perform these activities and their frequency. The data are shown in Table 6-12, and these labor hours were estimated using HVAC company websites. The website links are provided in the spreadsheet tool in Section 3.3 of the Quick Guide. Because other activities need to be performed besides those mentioned above, a 20% miscellaneous cost was added to the maintenance cost calculations.

Table 6-12. Maintenance activities, frequency, and labor hours required of HVAC units.

Maintenance Activities	Frequency	Labor Hours Required
Air filter replacement	Every 3 months	0.5 hours per unit
Thermostat programming	Every 1 month	0.25 hours per unit
Replacing or maintaining all compressor belts	Every 12 months	1 hour per unit
Blower assembly cleaning and checking	Every 12 months	1.6 hours per unit
Cleaning the ignition system	Every 12 months	2 hours per unit
Cleaning the duct system	Every 60 months	0.5 hours per vent

Copyright National Academy of Sciences. All rights reserved.

To calculate the maintenance cost of HVAC units, the research team first calculated the labor hours for each maintenance activity. The labor cost was then calculated using a $60/hour rate for mechanics. After the labor cost was calculated, the cost of the parts was determined by taking 70% of the labor cost and 30% of the parts cost and adding them. Then a 20% miscellaneous cost was added to the total maintenance cost of the HVAC units. Table 6-13 breaks down the component costs, labor hours required for installation, and total costs for each component for the replacement of HVACs and their major components. The cost is calculated assuming that 25 tons of commercial HVAC units are installed. These numbers are based on 2021 figures.

6.3.4 Determining the HVAC Components' Failure Probability with Regular Maintenance

To determine the replacement costs of HVAC components, it is necessary to determine their failure probability. HVAC companies have not provided these failure data; thus, the research team used the probability failure theory of machinery to determine the chances of HVAC components' failure during their life cycle. One of the probability theories the team primarily used to determine the cumulative probability of failure was the Weibull distribution theory. Researchers have used this theory successfully in various types of machines and electrical systems. The research team applied this theory to determine the failure probability of HVAC components, e.g., compressors, condensers, and air handlers, which most often break down in normal operating conditions. The shape and scale factor of these machines need to be determined to use the Weibull distribution probability theory. The shape factor is the probability distribution shape of the HVAC components. The shape of the probability distribution for these components will likely be normal, so the shape factor value used was 3.5. Because these three components are part of the HVAC system, the life of these components will be similar to the HVAC unit as a whole. The team found that the HVAC unit life is generally 15 years, so the scale factor (which denotes when 63% of the components will fail) was assumed to be 12 years in normal conditions given regular maintenance is performed annually. Table 6-14 presents the cumulative failure probabilities of these regularly maintained HVAC components using the Weibull probability theory. These cumulative failure probabilities help determine the regular replacement costs of the compressors, condensers, and air handlers.

6.3.5 Determining the HVAC Components' Failure Probability with Delayed Maintenance

As described in Section 6.3.4, the team could not find the failure probability of HVAC components tracked by any professional HVAC companies under regular maintenance conditions. Similarly, predicting the failure probabilities under delayed maintenance scenarios is difficult. Regardless, the life of HVAC components will reduce significantly if HVAC units are not regularly maintained. Therefore, the research team assumed that the Weibull distribution's scale factor would be reduced and used 6 years instead of 12. Using 6 years as a scale factor, the failure probabilities of these HVAC components were calculated. As Table 6-15 demonstrates, the life of

Table 6-13. Replacement cost of HVACs and their major components.

Replacement Components	Component Costs	Labor Hours	Total Cost
HVAC unit (25 tons)	$25,000	10 hours	$25,600
Compressor	$1,600	10 hours	$2,200
Condenser	$10,000	7 hours	$10,420
Air handler	$3,500	10 hours	$4,100

Copyright National Academy of Sciences. All rights reserved.

Table 6-14. Cumulative failure probabilities of regularly maintained HVAC components.

Age of HVAC Components	Cumulative Failure Probability
1	0.02%
2	0.2%
3	0.8%
4	2.1%
5	4.6%
6	8.5%
7	14.1%
8	21.5%
9	30.6%
10	41.0%
11	52.2%
12	63.2%
13	73.4%
14	82.0%
15	88.7%
16	100.0%

Table 6-15. Cumulative failure probabilities of HVAC components under delayed maintenance.

Age of HVAC Components	Cumulative Failure Probability
1	0.2%
2	2.1%
3	8.5%
4	21.5%
5	41.0%
6	63.2%
7	82.0%
8	93.5%
9	98.4%
10	100.0%

HVAC components will only last 10 years in delayed maintenance scenarios. These failure probabilities in delayed maintenance conditions can help calculate the HVAC component replacement costs when the system maintenance is delayed.

6.3.6 Estimating the HVAC Maintenance and Replacement Budget without Delayed Scenarios

The maintenance budget of HVAC units can be determined by calculating the labor hours required to perform the regular maintenance activities stated in Table 6-12. Based on the frequency of these activities on an annual basis, the research team calculated the total labor hours required to perform these activities for the HVAC units. Using the base cost of a mechanic per hour, the team calculated the total labor cost to maintain an HVAC unit for a year. After the total labor cost was determined, the team calculated the parts cost based on 70% labor and 30% parts distribution. Due to maintenance activities not mentioned in Table 6-12, 20% in miscellaneous costs were added to the budget to reflect the total maintenance cost in a year. The research team used the failure probabilities from Table 6-14 to determine the replacement costs of the compressor, condenser, and air handler.

Copyright National Academy of Sciences. All rights reserved.

96 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

The unit cost determined in Table 6-13 was used as a replacement unit cost for these components. If the HVAC life exceeded more than 15 years, then it was assumed that the entire HVAC would be replaced. The unit cost of maintenance and replacement was calculated based on the 2021 figures, meaning that these costs need to be adjusted to calculate future maintenance and replacement costs. To adjust the base cost, the research team used two types of cost indices. The first cost index shows the equipment cost inflation rate (BLS 2022c) and is determined as 2.95% per year. This cost inflation rate is used to adjust the cost of parts and the replacement components, e.g., compressors, condensers, air handlers, and the HVAC unit itself. The research team used the second cost index to determine the inflation rate of machinery and equipment repair and maintenance (BLS 2022c). The inflation rate for repairs and maintenance of HVAC units was calculated as 2.61% per year. The spreadsheet tool used to prepare the regular maintenance and replacement costs of HVAC units applies these two inflation factors to adjust the maintenance and replacement base costs.

6.3.7 Estimating the HVAC System Maintenance and Replacement Budget with Delayed Scenarios

The maintenance cost of an HVAC system can be calculated by multiplying the unit cost of maintenance items and the frequency of maintenance activities in delayed conditions. There is little difference in preventive maintenance costs inside or outside of delayed scenarios. However, the cost of replacing HVAC components will be higher due to higher probabilities of the failure of these components in delayed maintenance scenarios. The maintenance cost of an HVAC unit can be calculated as described in Section 6.3.6. To calculate the replacement cost of HVAC components, the research team used the failure probabilities under delayed maintenance stated in Table 6-15. Due to high failure probabilities, the total maintenance and replacement cost of HVAC systems was higher for delayed maintenance scenarios compared to regularly maintained HVAC systems. To calculate the total maintenance and replacement costs, the base cost used in the spreadsheet tool needs to be adjusted. The cost adjustment for maintenance and replacement can be completed as described in Section 6.3.6. In delayed scenarios, the user can either use the default inflation rate or enter the inflation rate they expect to occur in the future.

6.3.8 Reporting the Impact of HVAC System Delayed Maintenance and its Consequences

The research team developed a spreadsheet tool to determine the regular maintenance and replacement costs of HVAC systems. This tool can also calculate the impact of delayed maintenance on maintenance and replacement costs of airport HVAC systems. As described in previous sections, the tool uses the labor and parts cost data available from the case study and the CASE Construction website to calculate the maintenance and replacement costs with and without delayed scenarios. The tool can determine the amount of HVAC units required for the buildings. It can also calculate the number of air filters and air vents based on the area and number of occupants in the building. If an airport is already aware of these data, it can directly input the data to calculate the total maintenance and replacement costs of its HVAC systems. The same is true for the parts cost and labor hour data. The miscellaneous cost can also be changed if airports have their own data. These options provided in the tool can help airport maintenance managers calculate accurate HVAC system maintenance and replacement costs. When calculating the cost in a delayed maintenance scenario, the user can select the number of years maintenance of the HVAC system is delayed. They can also select a discounted rate to convert their future maintenance and replacement costs to net present value costs. This option helps to compare the present and future costs.

Copyright National Academy of Sciences. All rights reserved.

Table 6-16. Case study data of HVAC systems.

Name of the airport	XYZ International Airport
Type of airport	Small-hub/medium-hub/large-hub airport
Location of the airport	Southwest, United States
Area of the building	20,000 sq. ft
Number of doors and windows	10
Number of occupants	200
Capacity of HVAC unit	25 tons
Age of HVAC	2, 5, and 7 years
Budget preparation year	2022
Budget to be prepared for	2023
Delayed scenarios	2 and 5 years
Price inflation factor	Default (2.61% for repair and 2.95% for replacement parts)
Discount factor	0%

The Consequence of Delaying Maintenance on HVAC Systems Case Study Results

The research team created hypothetical airport building data to calculate the consequences of delaying maintenance of HVAC systems (Table 6-16). Based on the building dimensions, number of occupants, and number of doors and windows, the team calculated the maintenance and replacement costs with and without delayed maintenance. The team selected inflation rates for both repairs and parts replacement of 2.61% and 2.95%, respectively. The delayed scenarios selected were 2- and 5-year delays. The research team did not consider the discounted rate to convert the future values into present values.

Figure 6-7 shows the total maintenance and replacement costs of an HVAC system with and without delayed scenarios. The results show that the base maintenance and replacement costs increase with the age of the HVAC system. It also shows that the 2- and 5-year delay scenarios

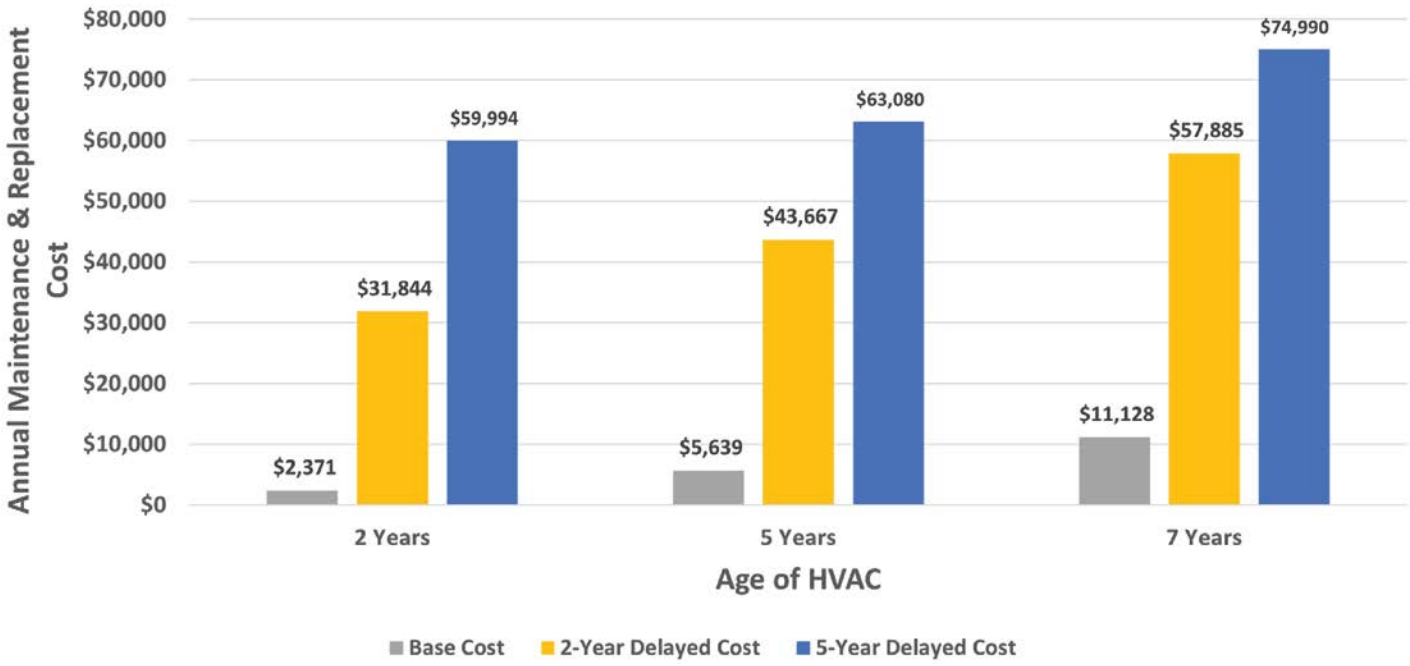


Figure 6-7. Impact of delaying maintenance on annual HVAC maintenance and replacement costs.

Copyright National Academy of Sciences. All rights reserved.

98 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

increase the cost significantly for HVAC systems of any age. The increase in the maintenance and replacement costs of a 5-year-old HVAC unit are significant when maintenance is delayed by 2 years. However, the total cost increase of 2- and 5-year-old HVAC units is not significantly different when maintenance is delayed by 5 years. When maintenance is delayed by 5 years, a 5-year-old HVAC unit will reach 11 years of age. The probability of failure in delayed scenarios reaches 100% for an 11-year-old HVAC unit (Table 6-15). Thus, delaying the maintenance of 5 years or older HVAC units by more than 5 years will not have a significant impact on the total maintenance and replacement costs.

6.4 Baggage Handling System

The airport baggage handling system is one of the most important pieces of infrastructure in the airport. The airport baggage handling system transfers baggage from ticket checkpoints to airplane loading areas, or from airplane unloading points to carousels. The FAA has not issued any AC related to baggage handling systems. Generally, the baggage handling system consists of conveyor belts that move over motors so baggage can be transported from one point to another. The baggage handling system consists of motors, belts, guard rails, frequency inverters, and photocells. The baggage handling system is continuously moving to transport baggage, and these parts function almost all hours when the airport is open. These systems are expensive and need regular maintenance and replacement of parts as they wear. If the system is not regularly maintained, it will break and need replacement entirely or new parts for the conveyor belt, which can cost thousands of dollars. Airport asset managers need to make sure that these systems are regularly inspected, and that a budget is available to maintain them. The research team collected the maintenance and replacement labor and cost data of baggage handling systems from specific airports. Based on these data, the team built a framework and a spreadsheet tool that can assist airport asset managers in calculating the regular maintenance and replacement costs of this system with and without delayed maintenance. Figure 6-8 shows the framework to determine the maintenance and replacement costs of the baggage handling system. Each step involved in this framework is described in the following sections.

6.4.1 Establishing Asset Inventory

The baggage handling system consists of various components. The conveyor belt is a major part of the system. Depending on the size of the airport, the length of the conveyor belt can range from a few 1,000 feet to a few miles. In large airports, the length of conveyor belts can be 15-20 miles long. The conveyor belt has motors that help to move the belt over the guardrails so baggage can be transported from one place to another. In addition to conveyor belts, the baggage handling system includes baggage screening points, X-ray units, pushers, power benders, power turns, merges, diverters, and nodes (loading, unloading stations, and junctions). Airport asset managers need to keep an inventory of all the components of the system so that they can track, inspect, and maintain it or replace parts on time. Airport asset managers should record the date of purchase of these components, their brand and manufacturer's name, the age of the parts, and the types of spare parts required to maintain the system. In a large airport with many terminals, these systems may be complicated. There is often a room with computers to monitor and control the movement of the conveyor belts, and these control rooms also need to be maintained efficiently. Any information related to baggage handling system components should be stored in the database.

6.4.2 Performing a Condition Assessment

Assessment of the condition of various parts of the baggage handling system is critical because the inspection and assessment of conditions will provide enough information to plan for system

Copyright National Academy of Sciences. All rights reserved.

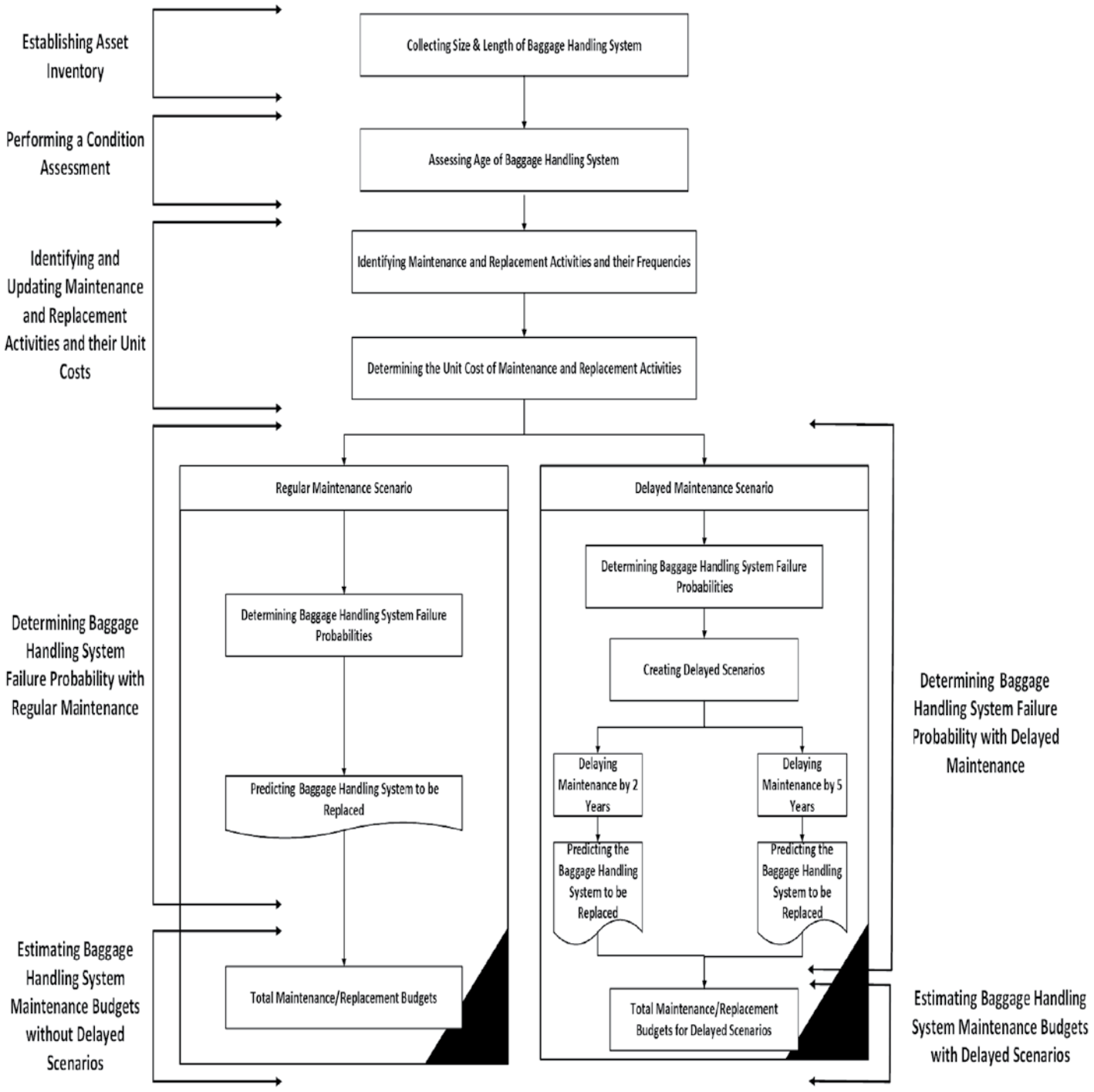


Figure 6-8. Framework to determine the impact of delayed maintenance on baggage handling systems.

maintenance and replacement. During the data collection phase, the research team did not find any specific written procedures to inspect the baggage handling system in airports. The team found that a specific department dedicated to the baggage handling system was created in large airports. Because the system consists of motors and rotary parts, the maintenance of these components is crucial. If the system breaks, it can be disruptive for air travelers. Airport mechanics monitor and regularly inspect various parts of the baggage handling system to ensure they are working properly. Some of the airports outsource baggage handling system maintenance work

Copyright National Academy of Sciences. All rights reserved.

100 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

to third parties. The following inspections should be conducted regularly to keep the baggage handling system safe and moving (Ravnholm n.d.):

- Conveyors, including check-in collection, curbside input, load and unload, and transport conveyor segments.
- Power turns, merges, high-speed diverter units, and plow diverters.
- Motor control panels and field control devices (e.g., photocells, limit switches, control stations/devices, audio/visual alarms, etc.).
- Motors, motor starters, disconnects, and push buttons.
- Baggage handling system-related computers, controls and control hardware, and control rooms.

The research team did not find an inspection checklist developed by any of the airports or other companies that provide operation and maintenance services for the baggage handling system. Because the system is complex, professionally trained people are required to conduct the system component inspection in order to execute proper maintenance.

6.4.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

As mentioned in the previous section, the baggage handling system is so complex that identifying all maintenance activities poses a challenge. Some of the regular maintenance activities are as follows:

- Cleaning all parts of conveyor belts to keep the dirt, litter, dust, spills, grime, or chemical residue away from the surface.
- Oiling and greasing the rotary parts of the conveyor belts.
- Maintaining conveyor belts including motors.
- Maintaining baggage screening areas.
- Maintaining carousels.
- Maintaining X-ray units and RFID sensors.
- Maintaining automation control hardware and software.

The frequency of these activities depends on an airport's hours of operation and the environment of the terminal given how much dirt and litter will infiltrate the system. The research team gathered the maintenance frequency of some of the key maintenance activities of the baggage handling system for the airports. These data are shown in Table 6-17. Companies like Siemens (2019) have started a condition-based maintenance plan to increase the efficiency of the baggage handling system. Various types of intelligent sensors can be installed in the components of baggage handling systems. These sensors can acquire information related to wear and tear, use, environment, and maintenance history of the system, which can be used to maintain the system

Table 6-17. Frequency of major maintenance activities of baggage handling systems.

Maintenance Activities	Frequency
Maintain conveyor belts including motors	Quarterly
Maintain baggage screening areas	Quarterly
Maintain carousels	Quarterly
Maintain X-ray units and RFID sensors	Quarterly
Maintain automation control hardware and software	Quarterly

Copyright National Academy of Sciences. All rights reserved.

components. Airport asset managers can prepare and plan the system maintenance and replacement budget using the data from these sensors.

The research team contacted several airports to collect the labor hours used in the maintenance of their baggage handling systems. None of the airports had collected these data. However, the research team found that some of the large-hub airports have a department dedicated to the operation and maintenance of baggage handling systems. The team was able to determine the number of electricians and mechanics working in this department to find how many man-hours are spent on the maintenance and operation of the baggage handling system. The cost of maintenance was calculated using a ratio of 60% for labor and 40% for parts. Then the installation cost of a new baggage handling system was retrieved from one of the large-hub airports that had recently installed the system. Table 6-18 shows the unit cost calculation for maintenance and replacement of a new baggage handling system. The research team estimated the unit costs of maintenance and replacement from the case study airport data with 20% in miscellaneous costs added to reduce variability.

6.4.4 Determining the Baggage Handling System Failure Probability with Regular Maintenance

The research team contacted the airports to determine how often the baggage handling system breaks down; however, the asset managers were not able to provide any data related to system breakdown. The team also searched online to determine whether any academic or professional organizations have developed probabilities of baggage handling system failure in normally maintained conditions but could not find any literature. The team instead used the Weibull probability function theory to determine the system failure probabilities. As described previously, two major parameters of baggage handling system failure are required to use this probability theory. These include scale factor and shape factor. Because none of the academic or industrial researchers had developed these parameters for the baggage handling system, the team assumed a normal distribution of failure probability. Therefore, the shape factor used was 3.5. During the data collection phase, the airports were asked to estimate the life of the baggage handling system. They stated that, generally, it is about 15 years. So, the team considered that 63% of system failures will occur within 10 years. A 10-year scale factor was used to calculate the cumulative failure probabilities of the baggage handling system. Table 6-19 shows the failure probabilities of the baggage handling system under regular maintenance scenarios. The cumulative probabilities show that, by year 16, the system will break down and need replacement.

Table 6-18. Unit maintenance and replacement cost calculations for the baggage handling system.

Category	Calculation
Total number of mechanics and electricians	19
Number of labor hours for maintenance	19 x 52 weeks x 40 hours/week = 39,520 hours
Total length of baggage handling system	17 miles x 5,280 ft/mile = 89,760 ft
Cost per hour of mechanic	$60/hour
Total labor cost	39,520 hr x $60/hr = $2,371,200
Labor cost per linear foot	$2,371,200 / 89,760 ft = $26.41
Ratio of labor and parts cost	60% labor and 40% parts
Parts cost per linear foot	$26.41 x (0.4) / (0.6) = $17.61
Cost per year per linear foot	$26.41+$17.61 = $44.02
Add 20% miscellaneous cost	$8.80
Total cost per year per linear foot	$53.83
Total cost of replacement system per foot	$700/ft

Copyright National Academy of Sciences. All rights reserved.

102 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 6-19. Cumulative failure probabilities of the baggage handling system under a regular maintenance scenario.

Age of the System	Cumulative Failure Probability
1	0.03%
2	0.4%
3	1.5%
4	3.9%
5	8.5%
6	15.4%
7	24.9%
8	36.7%
9	49.9%
10	63.2%
11	75.4%
12	84.9%
13	91.8%
14	96.1%
15	98.4%
16	100.0%

6.4.5 Determining the Baggage Handling System Failure Probability with Delayed Maintenance

It is difficult to collect failure probability data from airports when the baggage handling system is not maintained. This condition generally does not exist in the airport, so the research team again used the Weibull probability function to determine the failure probability of baggage handling systems under delayed maintenance scenarios. The shape factor used was 3.5, similar to a regularly maintained system. However, under delayed maintenance scenarios, the scale factor used was half that of regular maintenance scenarios because the system will only last half of its life. Hence, a scale factor value of 5 years was used. Table 6-20 provides the cumulative failure probabilities of baggage handling systems under delayed maintenance scenarios. The table shows that, if the baggage handling system is not regularly maintained, the life of the system will be about 8 years, which seems reasonable considering the life of a regularly maintained system is 15 years.

6.4.6 Estimating Baggage Handling System Maintenance and Replacement Budgets without Delayed Scenarios

The maintenance budget of a baggage handling system consists of two parts. The first is regular maintenance costs to keep the system working. The second is costs that need to be expended due to system failure. Therefore, the research team also considered that part of the system can fail even if

Table 6-20. Cumulative failure probabilities of the baggage handling system under a delayed maintenance scenario.

Age of the Equipment	Cumulative Failure Probability
1	0.36%
2	3.97%
3	15.41%
4	36.74%
5	63.21%
6	84.94%
7	96.11%
8	99.44%

Copyright National Academy of Sciences. All rights reserved.

it is regularly maintained because the failure of a mechanical or electrical system can happen even when the system is maintained. Based on the system failure probability, some funds need to be set aside to cover breakdown scenarios if the system fails. To determine the regular maintenance costs of the system, the labor hours and parts required to maintain the system need to be calculated. The research team collected the labor and parts costs based on 2021 figures, which are provided in Table 6-18. The cost should be adjusted based on the year the budget is prepared. The research team then collected the inflation cost indices from the Bureau of Labor Statistics (2022c) to determine the inflation rate per year. Since 2006, the average inflation rate of machinery and equipment repair and maintenance has been about 2.61% per year. This inflation rate was used to determine the maintenance cost of baggage handling systems for future years. The team also considered replacement costs due to the probability of failure. To determine this replacement cost, the replacement cost per unit of the system was multiplied by the total length of the baggage handling systems and the probabilities shown in Table 6-19. The unit replacement cost is based on 2021 figures and needs to be adjusted for future years. The cost indices of new equipment since 2004 were used to calculate the average inflation rate of equipment per year (FRED 2022d). The cost indices showed that the average inflation rate of equipment is about 2.95% per year. This inflation rate was used to determine the future replacement costs of the system.

6.4.7 Estimating Baggage Handling System Maintenance and Replacement Budget with Delayed Scenarios

The research team used similar procedures described in the previous sections to estimate the total maintenance and replacement cost of baggage handling systems for delayed scenarios. The difference with the baggage handling system is that the high probability of a system failure will increase the replacement cost. The system replacement cost can be calculated by multiplying the unit cost of replacement by the probabilities of failure of the system provided in Table 6-20 for delayed scenarios. The cost of replacement will be higher in delayed scenarios because the probability of failure of the system will be higher compared to regularly maintained systems. The maintenance and replacement costs also increased due to the inflation rate as described in section 6.4.6. The team developed a spreadsheet tool to determine the maintenance and replacement costs for the system. The airports can enter their own expected inflation rates in the tool for the maintenance and replacement activities when calculating the delayed costs.

6.4.8 Reporting the Impact of Baggage Handling System Delayed Maintenance and its Consequences

The research aimed to determine the impact of delaying asset maintenance on the airport's budget. This research did not address other impacts of delayed maintenance, e.g., safety, security, and the comfort of the passengers. The research team only considered the impact of higher maintenance and replacement costs due to delayed maintenance. The team used the spreadsheet tool to determine the financial impact of delayed maintenance. The team also developed a hypothetical airport case study to calculate the impact of delayed maintenance on total costs. Table 6-21 shows the hypothetical airport data used to determine the maintenance and replacement costs of baggage handling systems with and without delayed scenarios. The delayed scenarios used were 2 and 5 years. The age of baggage handling systems was assumed to be 1 year, 3 years, and 5 years. The team used the default inflation rates to calculate the impact of delayed maintenance on baggage handling system maintenance and replacement costs.

The research team developed the spreadsheet tool with a case study airport's labor hours and parts cost used to maintain the baggage handling system per year. If the airports know the labor hours used to maintain their baggage handling system, they can use those hours instead of the case study

Copyright National Academy of Sciences. All rights reserved.

104 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 6-21. Case study data of baggage handling systems.

Name of the airport	XYZ International Airport
Type of airport	Small-hub/medium-hub/large-hub airport
Location of the airport	Southwest, United States
Length of baggage handling system	2,000 ft
Age of baggage handling system	1 year, 3 years, and 5 years
Budget preparation year	2022
Budget to be prepared for	2023
Delayed scenarios	2 and 5 years
Price inflation factor	Default (2.61% for repair and 2.95% for replacement)
Discount factor	0%

hours. They can also use their own labor cost per hour. Similarly, if the airports know the amount they are spending on parts to maintain their baggage handling system, they can use that cost instead of the case study parts cost. The airports can also enter their own baggage handling system unit replacement and miscellaneous costs. If they want to calculate the delayed maintenance cost in present values, they can select the discount rate provided in the tool.

The Consequence of Delaying Maintenance on the Baggage Handling System Case Study Results

The research team used the spreadsheet tool to calculate the impact of delayed maintenance on total maintenance and replacement costs of baggage handling systems. Using hypothetical case study data available in Table 6-21, the team calculated the total maintenance and replacement costs of regular maintenance for 2023. The team then hypothesized that no funds were available for the following 2- and 5-year periods and calculated the total maintenance and replacement costs for 2025 and 2028. The budget increase reflected the impact delaying baggage handling system maintenance has on airport budgets. Figure 6-9 shows the change in the total maintenance and

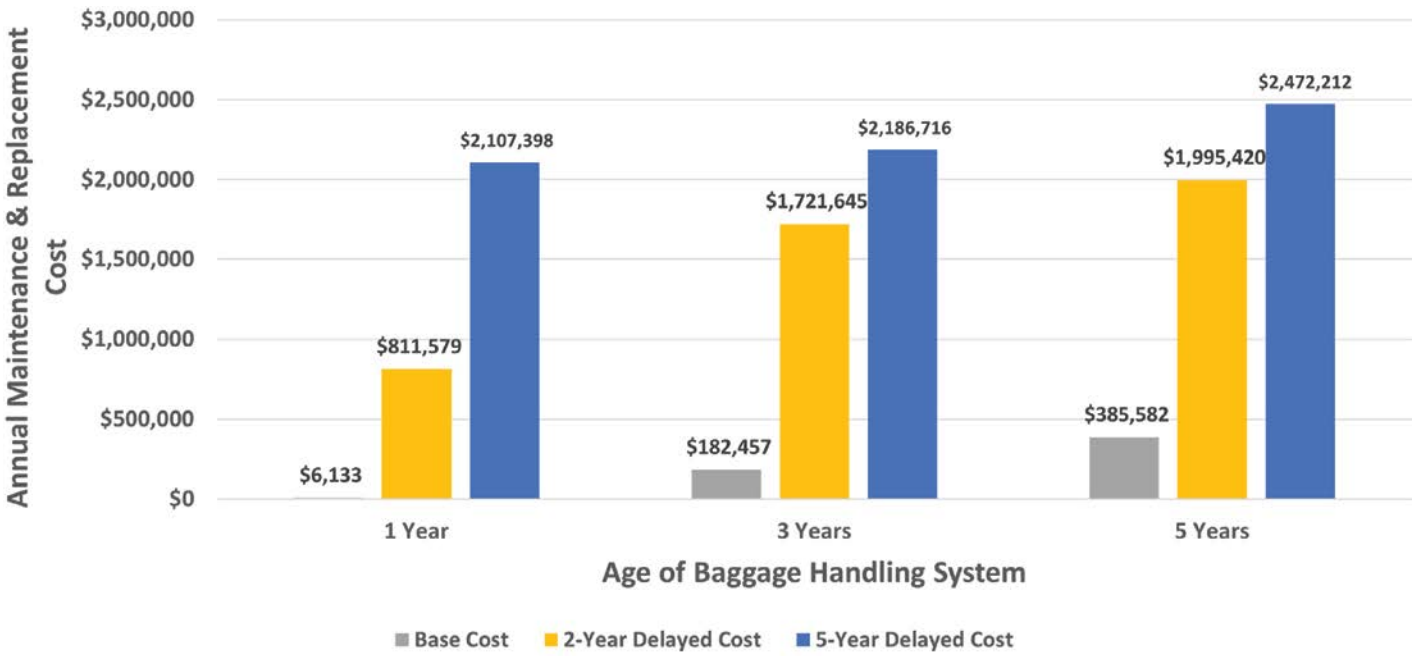


Figure 6-9. Impact of delaying maintenance on annual maintenance and replacement cost of the baggage handling system.

Copyright National Academy of Sciences. All rights reserved.

replacement costs of 1-, 3-, and 5-year-old baggage handling systems when the maintenance is delayed by 2 and 5 years. The results show that, when the system is 1 year old, the maintenance and replacement costs significantly increase at delayed maintenance scenarios of 2 and 5 years. However, for 3- and 5-year-old systems, the maintenance and replacement costs for a 2-year delay will increase significantly (but not for a 5-year delay). When the system is 7 years or older, the failure probability under delayed maintenance scenarios indicates that 96% of the system will need to be replaced (Table 6-20). This is why a 5-year delay did not significantly increase the total maintenance and replacement cost.

6.5 Passenger Boarding Bridges

The passenger boarding bridge is also called a jet bridge. The passenger boarding bridge helps passengers board the planes. Passenger boarding bridges are generally attached to terminal buildings; however, they can be extended and moved side to side to connect with the plane gates. Generally, a passenger boarding bridge has a hydraulic system, which helps the bridge extend and contract to fit into the plane doors. Passenger boarding bridges consist of five components as shown in Figure 6-10 (Aviation Learnings 2020). They include (1) the cabin unit, (2) the telescopic tunnel, (3) the elevation system, (4) the bogie, and (5) the rotunda. Manufacturers use various terms to identify the parts of passenger boarding bridges. The cabin is the part at the end of the bridge that connects with the airplane doors. The cabin is generally fitted with an operator panel, and an operator controls the movement of the bridge inside this cabin. The telescopic tunnel connects the cabin and rotunda. Some of the bridges have more than two tunnels, and intermediary tunnels are assembled between the cabin tunnel and rotunda. The elevation system supports the cabin tunnel and consists of two columns side by side. This column can collapse or elongate. The elevation system rests on the bogie, also called a traction system. This part helps the bridge to move on the ground. The rotunda is fixed over the rotunda column and anchored into the ground. The main function of the rotunda is to connect the terminal building to the mechanical bridge.

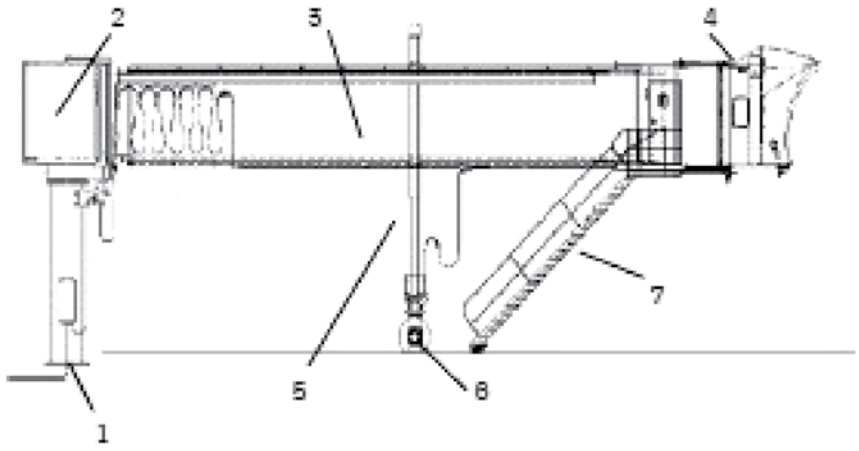


Figure 6-10. Components of the passenger boarding bridge. Source: http://alexo-llc.com/en/products/23-modeli-trapov/36-apron-drive.html.

Copyright National Academy of Sciences. All rights reserved.

106 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

FAA AC 150/5220-21C discusses the performance standards, design, manufacture, and testing of passenger boarding bridges. This AC does not review the maintenance requirements; however, it does discuss the manufacturing specifications of the bridge in detail. This research is focused on developing a framework and a tool to determine the impact of delayed maintenance on total maintenance and replacement costs for airport passenger boarding bridges. The research team only focused on the agencies' direct costs, incurred by the rapid deterioration of the bridge when regular maintenance is not conducted. Figure 6-11 shows the framework for calculating the consequences of delaying maintenance on passenger boarding bridges. The following section discusses a step-by-step process to calculate the total maintenance and replacement costs based on delayed scenarios. The team also developed a spreadsheet tool to help airport asset managers calculate the impact of delayed maintenance on passenger boarding bridges.

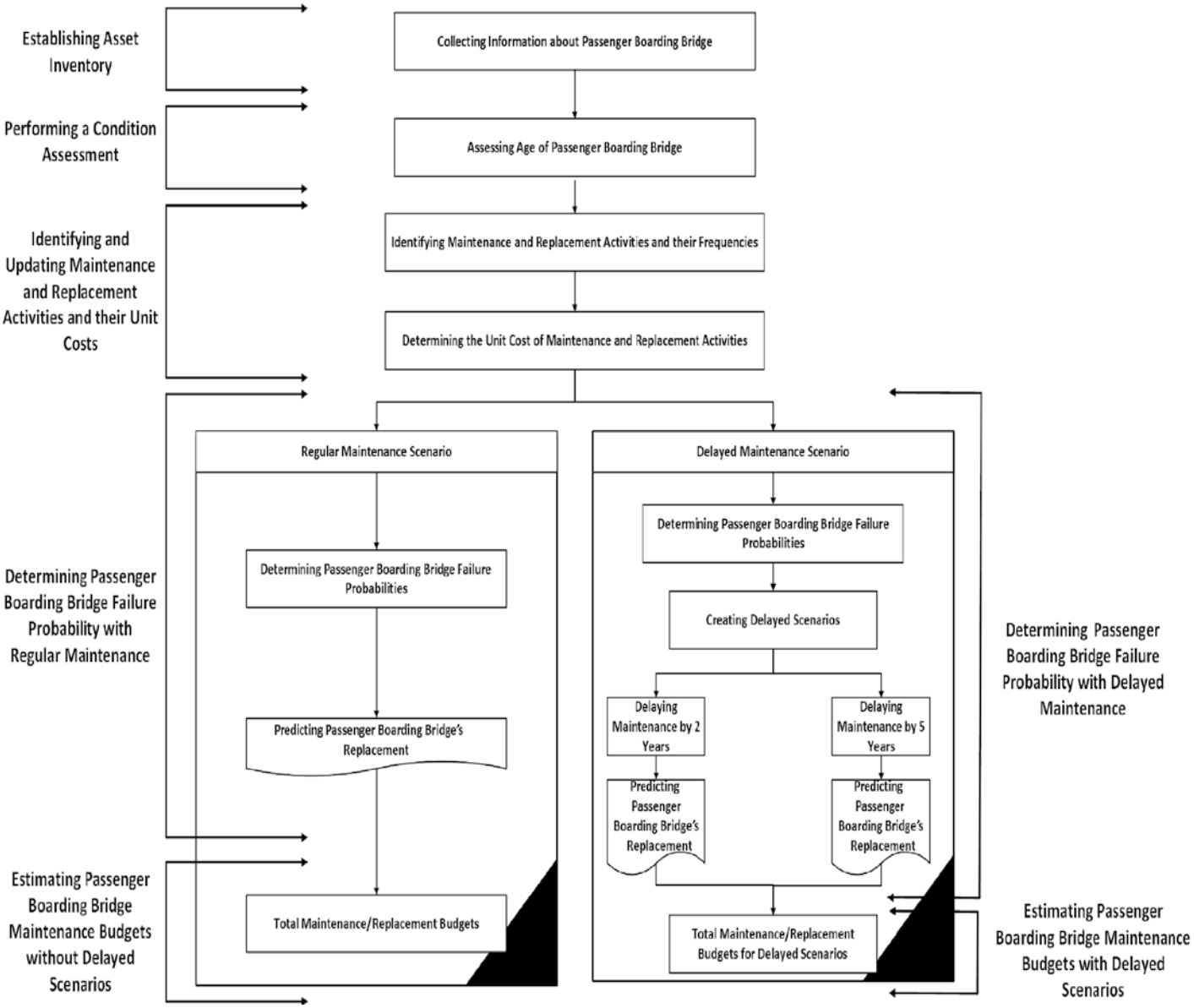


Figure 6-11. A framework for delayed maintenance of passenger boarding bridges.

Copyright National Academy of Sciences. All rights reserved.

6.5.1 Establishing Asset Inventory

As discussed in the above section, the passenger boarding bridge has various components. Asset managers should therefore not only keep records of how many passenger boarding bridges the airport owns, but also the make, model, and date of purchase of the bridges. Additionally, they need to keep an inventory of all parts of bridges purchased by the airport in order to maintain them. During the data collection phase, the research team found that some small- and medium-hub airports invited third-party service providers to maintain passenger boarding bridges. Some large-hub airports have their in-house department dedicated to the maintenance and operation of passenger boarding bridges. If the airport is performing the maintenance of the bridges in-house, then they must keep an inventory of all the components and parts required for maintenance of the bridge in their database because these components play a critical role in the maintenance and operation of the bridges. During the data collection phase, the team did not find any airport databases with inventory related to passenger boarding bridges. All the airports, however, knew the number of bridges they had, their make and model, and their purchase dates, all of which can contribute to determining the age of the bridges.

6.5.2 Performing a Condition Assessment

During the data collection phase, the team tried to collect the condition assessment data of the passenger boarding bridges. However, none of the airports had a systematic way of collecting the condition data. It was found that airport asset managers inspect the passenger boarding bridges regularly, but do not keep records of the condition data in their databases. Also, they do not use these inspection data to prepare a maintenance budget for the bridges. The budget is generally prepared based on the experience of the asset managers, with some additional costs for contingencies to cover unforeseen maintenance issues. The passenger boarding bridge is a complex system, and it can be difficult to determine the overall system conditions. The airport needs to inspect each component of the bridge to ascertain that the bridge is functioning properly. The airport mechanics need to inspect the following components of the passenger boarding bridges:

- The cabin control system;
- The canopy and its wear and tear;
- The elevation system hydraulics;
- The wheels of the bogie and other moving parts;
- The moving parts that allow the telescopic tunnels to extend and contract;
- The wear and tear of the rotunda column and rotunda;
- The tunnel electrical systems;
- The tunnel HVAC systems;
- The condition of all related mechanical, electrical, and electronic systems associated with the bridge, including cables, motors, fans, alarms, belts, indicator lights, computers, and accessories; and
- The external structure of the bridge and rust, dirt, and wear and tear of the bridge.

6.5.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

During the inspection phase, mechanics can determine the types of maintenance activities that need to be performed to keep the bridge in working condition. Identification of these maintenance activities is key to determining the maintenance and replacement costs of the passenger boarding bridges. If the maintenance activities and their cost related to labor hours and parts can be estimated, the total maintenance and replacement costs can be calculated accurately. During

Copyright National Academy of Sciences. All rights reserved.

108 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

Table 6-22. Frequency of major maintenance activities of passenger boarding bridges.

Maintenance Activity	Frequency
Cabin Unit	Quarterly
Maintaining cab unit leveler wheels, chain sprocket, curtain pad wear, passenger guardrail, and operator panel	
Telescopic Unit	Quarterly
Maintaining telescopic unit rollers, stairs, swivel casters, and movement	
Tunnel	Quarterly
Maintaining tunnel ramp hinges and inside	
Bogie	Quarterly
Lubricating pillow block bearings	
Maintaining tires	
Elevation System	Quarterly
Maintaining lift movement system	
Lubricating lift system chain, coupling breakings	
Maintaining electrical system	As required
Maintaining hydraulic system	As required
Maintaining HVAC system	As required

the data collection phase, the research team identified some of the major maintenance activities that need to be performed. The research team also searched online to identify these maintenance activities and their frequency. Table 6-22 shows the types of maintenance activities and their frequency to maintain the passenger boarding bridges (Louisville International Airport 2022).

Identifying maintenance activities helps to determine the unit cost of the activities performed every year and the total maintenance and replacement budget. However, none of the airports had determined the unit cost of these maintenance activities for the passenger boarding bridges. The research team successfully gathered the number of mechanics and electricians working in one case study airport who maintain and operate the passenger boarding bridges. Some large-hub airports have an entire unit of people responsible for maintaining passenger boarding bridges. During the interview, the asset managers were also able to estimate the percentage of the cost for labor and parts to maintain the passenger boarding bridges. The team used the case study data of this airport to determine the maintenance cost per passenger boarding bridge per year. Table 6-23 shows the unit cost calculation of the maintenance, as well as the unit cost of replacing a passenger boarding bridge. All costs are based on 2021 figures.

Table 6-23. Unit maintenance and replacement cost calculations for the passenger boarding bridge.

Category	Calculation
Total number of mechanics and electricians	27
Number of labor hours for maintenance	27 x 52 weeks x 40 hours/week = 56,160 hours
Total number of passenger boarding bridges	112
Cost per hour of mechanics	$60/hour
Total labor cost	56,160 hours x $60/hour = $3,369,600
Labor cost per linear foot	$3,369,600 / 112 = $30,086
Ratio of labor and parts cost	60% labor and 40% parts
Parts cost per bridge	$30,086 x (0.4) / (0.6) = $20,057
Cost per year per bridge	$30,086+$20,057 = $50,143
Additional 20% miscellaneous cost	$10,029
Total cost per year per bridge	$60,172
Total cost of replacement bridge	$1,000,000

Copyright National Academy of Sciences. All rights reserved.

6.5.4 Determining the Passenger Boarding Bridge Failure Probability with Regular Maintenance

It is difficult to determine the failure probability of regularly maintained passenger boarding bridges because airports do not record these failure events, and bridge manufacturers do not have publicly available data. The bridges have various components, so it is hard to predict which components will fail during operation. The research team therefore used the Weibull probability distribution to determine the failure probability of regularly maintained passenger boarding bridges. As previously described, the shape and scale factors for the bridge need to be determined in order to use this probability theory. For a regularly maintained passenger boarding bridge, the failure probability was assumed to be normally distributed, and the shape factor used was 3.5. During the data collection phase, asset managers were asked how long the passenger boarding bridges would last, and they reported that, generally, they last for 20 years. Considering this number, 63% of the passenger boarding bridges will fail after 14 years; therefore, the scale factor used was 14 years. Using the Weibull distribution theory, the cumulative failure probability for passenger boarding bridges is provided in Table 6-24. Using these parameters, the cumulative probability at 20 years is 100% bridge failure.

6.5.5 Determining the Passenger Boarding Bridge Failure Probability with Delayed Maintenance

As discussed previously for other mechanical systems, the process of determining the failure probability under delayed conditions is more difficult than regularly maintained conditions. The failure probability of a mechanical system under regularly maintained conditions can be determined by collecting the data for various years and various machines. However, most agencies do not keep their passenger boarding bridges without maintenance. Certain assumptions need to be made to calculate the failure probability of a passenger boarding bridge under delayed maintenance conditions. In this case, the team assumed that the life of the bridge would be significantly reduced if regular maintenance was not performed on the bridge. The life of the passenger boarding bridge

Table 6-24. Cumulative failure probabilities of the passenger boarding bridge under regular maintenance.

Age of the System	Cumulative Failure Probability
1	0.01%
2	0.1%
3	0.5%
4	1.2%
5	2.7%
6	5.0%
7	8.5%
8	13.2%
9	19.2%
10	26.5%
11	34.9%
12	44.2%
13	53.8%
14	63.2%
15	72.0%
16	79.7%
17	86.1%
18	91.0%
19	94.6%
20	100.0%

Copyright National Academy of Sciences. All rights reserved.

was therefore considered to be reduced by half if regular maintenance was not performed. Given this, the scale factor used was 7 years. The research team selected a shape factor value of 3.5, assuming the probability will be normally distributed like a regularly maintained bridge. Using these shape and scale factors, the Weibull probability distribution yielded the failure probabilities of passenger boarding bridges as shown in Table 6-25. The table shows that the bridge will fail 100% when it reaches 10 years of age.

6.5.6 Estimating Passenger Boarding Bridge Maintenance and Replacement Budget without Delayed Scenarios

As described above for other airport assets, the total maintenance and replacement costs of passenger boarding bridges can be calculated by the number of bridges and their unit maintenance cost, as well as the failure probability based on the age of the bridge and unit replacement costs. The maintenance cost is based on the condition of the bridge and the replacement cost is based on the probability of failure of a passenger boarding bridge. The team developed a spreadsheet tool to calculate passenger boarding bridges' total maintenance and replacement costs. These calculations were based on 2021 figures. The cost needed to be adjusted when calculated for periods after 2021. The research team gathered cost index data for machinery repair and maintenance and new equipment to adjust the cost. The team used an inflation rate of 2.61% per year to adjust the maintenance cost of the passenger boarding bridge. The team calculated this inflation rate using cost index data from 2003 to 2021 (BLS 2022d). Similarly, for new equipment purchase and installation, the team collected cost index data from 2006 to 2021 and calculated an inflation rate per year of 2.95% (FRED 2022d). These inflation rates assisted in adjusting the cost for future years.

6.5.7 Estimating Passenger Boarding Bridge Maintenance and Replacement Budget with Delayed Scenarios

The research team used procedures similar to those described in section 6.5.6 to estimate the maintenance and replacement cost of passenger boarding bridges for delayed scenarios. The difference was the higher cumulative failure probability of bridges due to neglected maintenance. During the data collection phase, the team found that most airports do not delay the maintenance of passenger boarding bridges. If budget constraints cause a delay, however, the airport should consider that an extended delay in maintenance could lead to the entire passenger boarding bridge needing replacement. The spreadsheet tool created by the research team has the flexibility to delay maintenance up to 10 years. The cost used in the tool needs to be adjusted based on future inflation rates when maintenance is delayed. The tool uses default inflation rates for maintenance and replacement, which are 2.61% and 2.95%, respectively. The tool allows the user to enter their own

Table 6-25. Cumulative failure probabilities of the passenger boarding bridge under delayed maintenance.

Age of the Equipment	Cumulative Failure Probability
1	0.1%
2	1.2%
3	5.0%
4	13.2%
5	26.5%
6	44.2%
7	63.2%
8	79.7%
9	91.0%
10	100.0%

Copyright National Academy of Sciences. All rights reserved.

inflation rate to calculate the total maintenance and replacement costs for delayed scenarios. The tool can also convert the future maintenance and replacement costs to the present value using a discounted rate.

6.5.8 Reporting the Impact of Passenger Boarding Bridge Delayed Maintenance and its Consequences

As described for other assets, this spreadsheet tool helps determine the monetary impact of delayed maintenance of passenger boarding bridges on an airport's total maintenance and replacement costs. The research team created a hypothetical airport case study to estimate the impact of delayed maintenance of passenger boarding bridges on total maintenance and replacement costs. Table 6-26 shows the hypothetical airport case study project data. In the case study scenario, there is one passenger boarding bridge. The scenario considers that the age of the bridge is 1, 3, or 5 years old. The delayed scenarios show how the total maintenance and replacement costs will change based on the age of the bridge. The delayed scenarios used were 2 and 5 years. Default inflation rates were used in this case study. The discounted rate was not used.

The spreadsheet tool developed can calculate the labor hours required to maintain the bridge per year with and without delayed scenarios. The tool uses the labor hours calculated from the case study airport. If the airports know the labor hours needed to maintain their passenger boarding bridges, they can use those hours instead of the case study hours. They can also use their own labor costs per hour. Similarly, if the airports know the parts cost to maintain and replace their bridges, they can use that cost instead of the case study cost. The tool uses the unit cost of each passenger boarding bridge replacement recorded during the data collection phase, and, if these unit costs differ, they can enter their own costs. A 20% miscellaneous cost was added to the maintenance and replacement budget to cover unforeseen costs. The airports can also use their own miscellaneous prices in this tool. They can also use their cost inflation rates for the maintenance and replacement of passenger boarding bridges. If they want to calculate the delayed maintenance costs in present value, they can select the discount rate provided in the tool.

The Consequence of Delaying Maintenance on the Passenger Boarding Bridge Case Study Results

The research team used the spreadsheet tool to calculate the total maintenance and replacement costs of a passenger boarding bridge. Using the case study data available in Table 6-26, the team calculated the total maintenance and replacement costs of a regularly maintained bridge for 2023. The team hypothesized that there would be no funds available for the following 2 and 5 years and calculated the total maintenance and replacement budget for 2025 and 2028. The budget increase demonstrated the impact of delayed maintenance of passenger boarding bridges on airport costs.

Table 6-26. Case study data for a passenger boarding bridge.

Name of the airport	XYZ International Airport
Type of airport	Small-hub/medium-hub/large-hub airport
Location of the airport	Southwest, United States
Number of passenger boarding bridges	1
Age of the bridge	1, 3, or 5 years
Budget preparation year	2022
Budget to be prepared for	2023
Delayed scenarios	2 and 5 years
Price inflation factor	Default (2.61% for repair and 2.95% for replacement)
Discount factor	0%

Copyright National Academy of Sciences. All rights reserved.

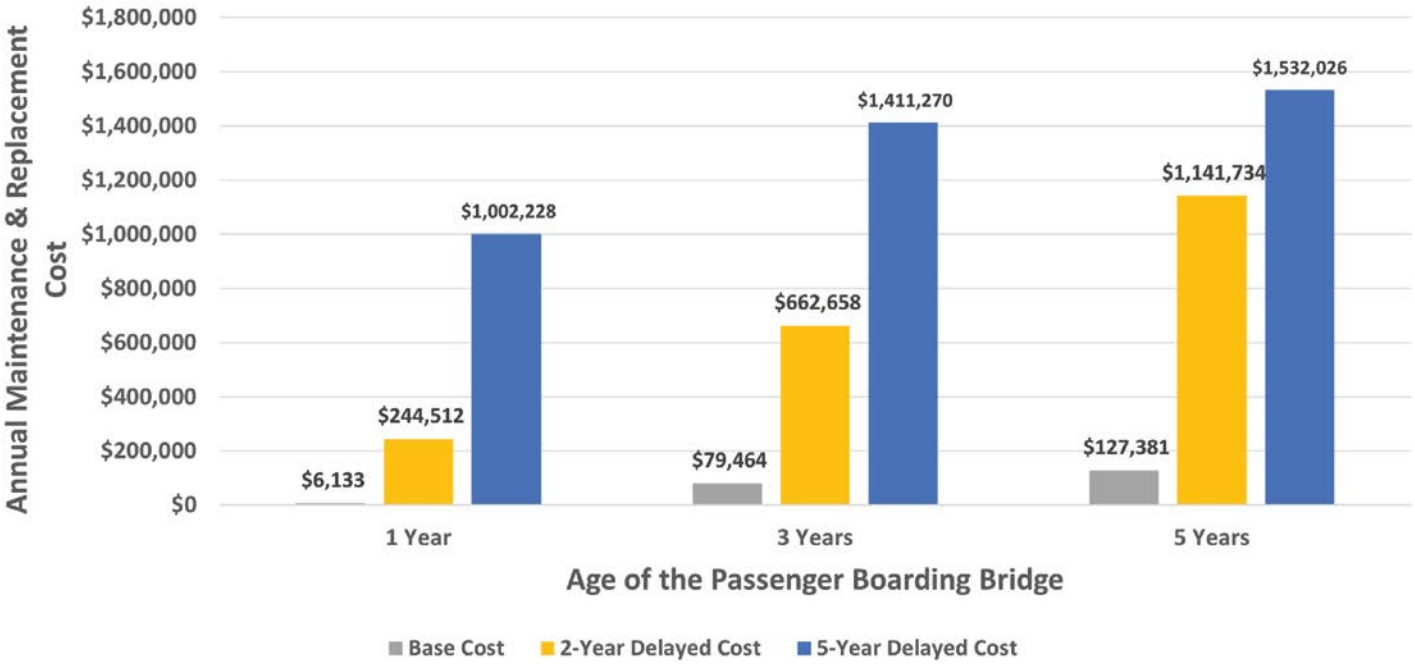


Figure 6-12. Impact of delaying maintenance on the annual maintenance and replacement cost of a passenger boarding bridge.

Figure 6-12 shows the change in the total maintenance and replacement costs of 1, 3, and 5-year-old passenger boarding bridges when the maintenance is delayed by 2 and 5 years. The results show that the impact of 2- and 5-year delays on annual maintenance and replacement costs is significantly higher when the age of the passenger boarding bridge is 1 and 3 years old. The impact of maintenance and replacement costs of passenger boarding bridges is not as high when the maintenance is delayed by 5 years for 3- and 5-year-old passenger boarding bridges. The cumulative failure probability of the passenger boarding bridges under delayed maintenance conditions at 9 years reaches 91% (Table 6-25). If the passenger boarding bridges are not maintained regularly, then they will fail after 10 years. This indicates that any maintenance delay beyond the total life of the bridge at 10 years will have a similar impact.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 7

Procedures to Quantify the Impact of Delayed Maintenance on Other Airport Assets

The research team contacted several airports to collect the case study data to prepare spreadsheet tools for critical airport assets other than those mentioned in the previous two chapters. However, due to a lack of available data, the research team could not prepare spreadsheet tools to estimate the delayed maintenance cost of other airport assets. During the interviews with airport asset managers and discussion with ACRP panel members, it was identified that the airports spent a considerable percentage of their budgets on maintaining the following airport assets:

- Terminal building envelopes including the roof,
- Hangars, and
- Automated people movers.

Therefore, airport asset managers need a step-by-step process to determine the impact of delayed maintenance of these assets on the airport's overall budget. The research team conducted an extensive literature review and referred to publicly available documents to determine some of the major issues that need to be considered when determining the impact of delayed maintenance on these assets' maintenance budgets. The process of conducting delayed maintenance cost calculation is similar to those mentioned in previous chapters. The asset managers should follow the seven basic steps mentioned in the previous chapters. They are (1) establishing an asset inventory, (2) performing condition assessments, (3) identifying and estimating maintenance activities and their unit costs, (4) determining failure probabilities of the assets in regularly maintained conditions, (5) determining failure probabilities of the assets in delayed maintenance scenarios, (6) estimating the maintenance budget without delayed scenarios, and (7) estimating the maintenance budget with delayed scenarios. The following sections provide guidelines to airport asset managers on how to conduct delayed maintenance analysis and what data and information are required to estimate the impact of delayed maintenance of the aforementioned assets.

7.1 Terminal Building Envelopes

The terminal building envelope generally consists of a foundation, walls, roofing system, glazing, doors, and other elements like chimneys, vents for stoves, restrooms, etc. Generally, the maintenance of these components of the terminal buildings is not required except when they are new. However, once these buildings age, regular maintenance is required to increase the life of the terminal buildings. To conduct the delayed maintenance cost analysis, airport asset managers need to make sure that they have up-to-date inventory data, condition assessment, and maintenance activities and their unit costs so that they can estimate the maintenance budget for terminal building envelopes. The following subsections show how to determine the maintenance and replacement costs of terminal building envelopes of airports with and without delayed scenarios.

Copyright National Academy of Sciences. All rights reserved.

114 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

7.1.1 Establishing an Asset Inventory of the Terminal Building

Asset managers should conduct an inventory survey to collect all building envelope data of their terminal buildings. The asset inventory should consist of detailed information on the building's exterior and interior parts. The asset inventory data should consist of the area of the terminal, number of levels including the basement, area and type of foundation, type of structure, area and type of exterior walls, number and type of doors and windows, area and type of roof, roof materials, and exterior drainage length and type. Asset managers should also collect the interior data of the terminal building, e.g., types of interior walls and surfaces, number of rooms, kitchens, restrooms, area and types of floors, area and types of ceiling, number and types of interior doors, hardware, and security systems. Similarly, the electrical systems including the lights, power switches, and the plumbing system should also be recorded to determine the overall maintenance costs of the terminal buildings.

The team prepared a spreadsheet tool to determine the maintenance cost of HVAC units with and without delayed scenarios. Asset managers can refer to that tool to determine the maintenance costs of their mechanical systems. If rooftop HVAC units are not used in the terminal building, they need to collect the inventory data of their heating and cooling systems. All these data are critical in estimating the maintenance cost of terminal buildings with and without delayed maintenance scenarios. An example of a terminal building inventory can be found here: https://www.cityoffrederickmd.gov/DocumentCenter/View/714/terminal-report?bidId= (City of Frederick 2005). When these data are kept and updated in the inventory, asset managers should make sure that they can be used during the maintenance budget preparation. To determine the maintenance budget with and without delayed scenarios, the airport needs to determine the unit cost of each of these components. If the airport knows the quantity of each of the components of the terminal building mentioned above, the maintenance cost can be calculated accurately.

7.1.2 Performing a Condition Assessment of the Terminal Building

Once an airport has created a database of its terminal buildings' exterior and interior components, it should conduct the condition inspection of these components at least once a year. The condition assessments will provide the required information to asset managers regarding what components of the building need to be maintained in the coming fiscal year. The building envelope regularly deteriorates as it ages; however, the main cause of the fast deterioration of building interiors and exteriors is water leakage. Therefore, airport managers should be careful in detecting water leakage in ground-level elements, below-grade building elements, roofs, walls, and windows. To conduct the water leakage inspection, an inspection checklist for water damage assessment of the terminal building such as this one can be used: https://www.hubinternational.com/-/media/HUB-International/Industries/Real-Estate-Insurance/Building-Envelope-Inspection-Checklist.pdf (HUB International 2022). An example of a building inspection report conducted in an airport terminal building can be found here: https://www.cityoffrederickmd.gov/DocumentCenter/View/714/terminal-report?bidId= (City of Frederick 2022).

The condition assessment of the entire building can require a substantial amount of time depending on the size of the terminal building. The major areas to assess conditions in the building are the exterior parts, including walls, doors, windows, roof, and drainage systems, as well as interior parts, including the kitchen, restrooms, etc., where water may leak. While performing the condition assessment, asset managers should record what needs to be fixed to improve the terminal building envelope so that accurate maintenance costs can be estimated. The identification of these maintenance activities and their quantity is key in determining the maintenance cost of the terminal building envelope with and without delayed scenarios. If the airport wants to

Copyright National Academy of Sciences. All rights reserved.

Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

determine the maintenance cost of electrical and plumbing systems of the terminal building, they need to inspect the lighting systems and the plumbing systems of their buildings.

7.1.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

During the condition assessment of the building envelope, the asset managers can also identify the types of maintenance activities that need to be performed to improve the terminal building conditions. If the building envelope components need only regular maintenance to bring in the operating conditions, then the asset managers can prepare a checklist of those maintenance activities to estimate the unit cost of those activities. However, if remodeling or a major overhaul is needed, then the replacement cost of those activities should also be calculated. To calculate the maintenance or replacement cost, those activities should be identified in detail. The building envelope maintenance activities can vary based on the components of the buildings. Some of the activities that should be performed to maintain the building's exterior wall are: (1) Prevent water filtration in the wall, (2) prevent mold growth in the exterior wall, (3) avoid loose elements falling from the walls, (4) fix the shifted or displaced elements, cracks, spalls, and stains, (5) prevent water entry into a building, (6) repair cracked masonry joints, (7) replace aged flexible sealants, and (8) clean soiled stone or concrete (Facilities Net 2022).

Another important component of the terminal building that needs to be inspected is the roof. Asset managers need to inspect and determine the maintenance activities to be performed on the terminal building's roof. The key elements of a roof to be regularly inspected are (Schultz 2016):

- The roof covering condition,
- Roof edging,
- Flashing,
- Supporting structures,
- Roof drains and scuppers,
- Roof penetrations, and
- Expansion joints.

The maintenance activities that need to be performed on the roof to keep it durable include:

- Clean the roof debris,
- Clean the drainage system of the roof,
- Protect the roof from water penetration,
- Avoid standing water or "ponding" on the roof, and
- Insulate the roof attic.

Another critical component of the terminal building is the foundation. It is crucial to maintain the foundation of the terminal building to keep the building structure stable. Some of the maintenance activities that need to be performed on the building foundations are (1) slope maintenance for proper drainage, (2) gutter and downspout inspection and maintenance, (3) proper subsurface drainage, (4) proper moisture content maintenance under the foundation, (5) maintenance of vegetation and trees that can damage the foundation, (6) plumbing leak repair, (7) foundation reinforcement steel exposure control, and (8) termite damage control (National Foundation Repair Association 2022). Airport asset managers need to maintain the interior of the terminal buildings in addition to the exterior. Some of the building interior maintenance activities and inspections are (Grainger KnowHow 2022):

- Inspect and maintain floors, ceilings, and interior walls;
- Check and maintain leaks;
- Check and maintain toilets and showers;

Copyright National Academy of Sciences. All rights reserved.

116 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

- Test and maintain smoke and carbon monoxide detectors;
- Check and maintain the fire alarm system;
- Test and maintain proper door operation;
- Inspect and fix door locks and closures;
- Clean the walkways;
- Inspect and maintain the electrical system;
- Inspect and fix all the lighting systems; and
- Inspect and maintain the plumbing system as required.

After identifying the required maintenance activities of the terminal building envelope, asset managers need to estimate the unit cost of the maintenance of these activities. The maintenance activities mentioned above have two major cost components: the labor cost and the material cost. If an airport needs equipment to perform these activities, then the equipment cost also needs to be calculated. If the airport's in-house labor performs these activities, they can determine the labor cost by calculating the total labor cost spent on these activities over a year. Then the labor cost per square foot of the terminal building can be calculated by dividing the total labor cost by the area of the terminal buildings. This unit cost can provide the average labor cost spent to maintain the building envelope per square foot. Similarly, to determine the material unit cost, the airport needs to determine the amount of money spent on material costs to maintain the terminal building per year. Dividing that amount by the total area of the terminal building will then provide the unit cost of the materials.

To calculate the total unit cost, the cost of labor and materials needs to be added, and, if equipment is used, then the cost of the equipment should also be added. It is a good practice to add 20% overhead and contingency costs to acquire the accurate total cost of maintenance. Once the average maintenance cost per square foot of the building envelope is calculated, this can be used to determine the maintenance cost of the terminal building for the following year. However, if airport asset managers would like to determine the unit cost of labor for each activity, they need to refer to the national unit cost database available from the RSMeans Cost Guide (RSMeans Cost Guide 2021). This cost guide provides the unit cost of labor, equipment, and materials required to perform these maintenance activities. Once the quantity of these activities is determined from the asset inventory database, the total cost can be calculated by multiplying the quantity by the unit cost of maintenance. An overhead or contingency cost of 20% can then be added to get the accurate maintenance cost of the terminal building. Also, these costs should be adjusted based on the year the maintenance was done using the cost escalation factors available from the RSMeans Cost Guide.

7.1.4 Determining the Terminal Building Failure Probability with Regular Maintenance

In any type of asset, there is always the probability that it can fail, even if the maintenance is conducted regularly. In the case of a terminal building, there is always the possibility that the components of the building can fail even with regular maintenance. Generally, failure can occur from natural events such as earthquakes, wind, fire, snow, or rain. In 2004, the terminal building of Charles De Gaulle Airport in Paris collapsed (Craven 2019). Despite being a newly built terminal, it collapsed due to design errors and a lack of construction quality. Although the possibility of terminal building failure is very rare, airport asset managers still need to assume that some components of the terminal building can fail, e.g., parts of the walls, roof, doors, windows, etc.

Some literature shows that the aging of a building can cause significant damage to the building components. In 2013, the University of Michigan found that 72% of all buildings in the United States were more than 20 years old (University of Michigan 2022). A commercial building has an average life of 50 to 60 years (Shingobee 2021), but the lifespan of a building depends on various

Copyright National Academy of Sciences. All rights reserved.

factors, including maintenance; external environmental factors like climate hazards and storms, harsh sunlight, heavy rainfall, and snow; internal factors like faulty plumbing, excessive heat, and improper drainage; and usage duration. Concrete decay, roof deterioration, electrical and plumbing deterioration, and door and window damage are some of the major sources of damage (Dansig 2021). Due to the above reasons, airport asset managers need to consider budgeting for emergency maintenance of their terminal buildings. It is highly suggested that these emergency funds be allocated for each component of the building separately and that the budget be based on the age of the building components. There is no accurate method to estimate this emergency budget, but generally 10% to 50% of the total maintenance budget of each component should be allotted for emergency maintenance work of the terminal building. Therefore, the asset managers need to increase the maintenance budget of the terminal building envelope, considering the probability of failure of some components of the building.

7.1.5 Determining the Terminal Building Failure Probability with Delayed Maintenance

As described in section 7.1.4, there is a probability that the terminal building can fail even when regular maintenance is conducted. If maintenance is delayed in terminal buildings, the probability of terminal building component failure significantly increases. Therefore, airport asset managers need to ensure that a significant amount of emergency funds are allocated for building envelope maintenance if the maintenance is delayed by a certain number of years. It is difficult to determine how much money will be needed for emergency maintenance, but asset managers can increase the emergency funds by twofold compared to the regularly maintained terminal buildings. There is no literature available to determine what percentage of the total maintenance budget of the terminal building should be allocated to emergency maintenance work. As described in the above section, asset managers need to allocate more money for emergency maintenance if the component maintenance within the terminal is delayed. Asset managers can also refer to the RSMeans Cost Guide to allocate contingency costs for these kinds of emergency work.

7.1.6 Estimating Terminal Building Maintenance and Replacement Budget without Delayed Scenarios

Once the quantity of each component of the terminal building and their unit maintenance costs are determined, asset managers can determine the maintenance cost per year by multiplying these two factors. Then, the asset managers can add emergency costs (contingency costs) as described in section 7.1.4 to factor into the failure probability of these building components. Depending upon the cost inflation factor, asset managers may need to increase the maintenance cost for the following year. The research team showed how to incorporate cost inflation factors to adjust the maintenance and replacement costs in the spreadsheet tools developed for other airport assets. These cost inflation factors can be found in the RSMeans Cost Guide, or asset managers can search the consumer price index data available on the Bureau of Labor Statistics website.

7.1.7 Estimating Terminal Building Maintenance and Replacement Budget with Delayed Scenarios

If an airport needs to delay the maintenance of its terminal building components due to financial constraints, it should determine which terminal building maintenance component can be delayed first, considering the lowest impact on the structural damage of the building. Once asset managers identify the building components in which maintenance must be delayed, they should predict the condition of each component after the delay. Because the deterioration models of the building with delay are not available, the airport asset manager must increase the contingency cost for emergency maintenance in the case of delayed scenarios. The emergency maintenance

Copyright National Academy of Sciences. All rights reserved.

118 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

cost due to delay needs to factor in the probability of failure as explained in section 7.1.5. Then the maintenance cost to correct that component after the delay needs to be calculated by multiplying the unit cost and the quantity of the component and adding the contingency costs. The unit cost also needs to be adjusted based on cost inflation factors. These cost inflation factors can be found in the RSMeans Cost Guide or the consumer price index data available on the Bureau of Labor Statistics website. An airport can use its own cost inflation factors for maintenance work if they are available.

7.2 Airport Hangars

The hangar is a building or structure designed to house an aircraft. Most airports own aircraft hangars. These hangars are used to keep the aircraft protected from weather and direct sunlight, and are also used for maintenance, repair, and storage. Hangars are one-story structures built of metal, wood, or concrete. Depending on the size of the airport, small or large hangars may be used. Large-size airports usually own multiple hangars to house their aircrafts for maintenance or storage. If the hangar is used for aircraft maintenance, they are generally called aircraft maintenance hangars (Tronair n.d.). The aircraft maintenance hangars are generally large, open, and have ample space for major repairs. These types of hangars also have maintenance stands, tool stands, scaffolding, and other equipment required for aircraft maintenance. The hangar should have firm, solid foundations made of concrete. The walls of the hangar can be steel columns with masonry walls, concrete, or drywall. Generally, the roof of the hangar should be made of shock-proof and fireproof materials (SAFS Steel Structure n.d.). The FAA has developed a policy on the non-aeronautical use of airport hangers. If the airport received an AIP grant to build the hangar, they should abide by these rules. The hangar should be used for aeronautical purposes, however, with the approval of the FAA, the airport can designate some areas for non-aviation use (FAA 2016b). The main purpose of an aircraft hangar is aircraft storage. If the hangar is used to store the aircraft, then storage of non-aeronautical items in the hangar does not violate FAA policy. Maintenance of the hangar is critical to keep the structure of the building intact. Like terminal buildings, airport asset managers need to maintain these hangars. Asset managers can determine the maintenance cost of the hangars with and without delayed scenarios by dividing the hangars into the following building components:

- Foundation,
- Floor,
- Wall structure, and
- Roof.

Asset managers can follow the steps provided in the following section to determine the maintenance cost of each component of the hangars with and without delayed scenarios. Then the total cost of the maintenance and replacement of the hangar can be calculated by adding these costs for each component. As mentioned for other airport assets, asset managers need to follow seven steps to determine the maintenance and replacement costs of the hangar with or without delayed scenarios.

7.2.1 Establishing the Asset Inventory of an Airport Hangar

The airport asset manager should keep the inventory of each hangar separately. If the hangars are of the same size and height, then one inventory can be used for multiple hangars. The asset manager should have the dimensions and types of materials used for each of the four major components of the hangar, keeping data related to length, width, depth, and the material used for the foundation. They should also keep the dimensions, wall size, and type of materials used on the hangar floor. The asset manager needs to collect detailed information about the

Copyright National Academy of Sciences. All rights reserved.

roofing structures, including dimensions and types of material used for the roof structure and roof shingles. In addition to this, the airport must also keep the inventory of the exterior drainage to keep the water out of the foundation. The asset manager must also keep the inventory of the door and windows related to size and material type.

7.2.2 Performing a Condition Assessment of an Airport Hangar

The aircraft hangar consists of components similar to a building. Therefore, airport asset managers can perform condition assessments of the hangar similar to the terminal buildings. However, the airports do not need a detailed assessment of the plumbing provided in the kitchen and restrooms. Asset managers can follow a process similar to the one used for terminal buildings to assess the condition of the aircraft hangars. The airport should assess the conditions of the following main components of the hangars: (1) structural system, (2) building envelope, (3) interior, (4) mechanical system, and (5) electrical system (S2 Architecture et al. 2017).

Asset managers should visually inspect the foundation of the hangar every year and determine its condition. If any structural damage issues are found in the foundation, the structural engineers should perform a detailed engineering analysis to determine the root cause of the issue. If any types of non-destructive tests need to be performed to determine the stability and loading capacity of the foundation, the structural engineering consultant should be contacted. As in the terminal building, the asset manager should also assess the condition of the building envelope every year. The building envelope consists of the exterior and the roofing of the hangar. If the exterior wall has extensive moisture-related deterioration, it should be fixed immediately. Similarly, if the asset manager finds that the roof is leaking or the roof rafters are damaged, then they should also be fixed immediately.

Airports should also assess the interior condition of their hangars. Asset managers should inspect and assess the condition of interior wall finishes, any issues related to building envelopes, water penetration affecting the hangar floor, ceiling finishes, etc. The airport should also assess the interior wood doors, frames, and plumbing fixtures. Another item to be inspected regularly is the mechanical and plumbing system in the hangar. The hangar does not typically have extensive mechanical and plumbing systems like in the terminal building. However, the plumbing required to provide a water and sewage system in the hangar must be inspected regularly to check whether it is working properly. Similarly, HVAC units in the hangar need to be inspected every year. The process of quantifying the impact of delayed maintenance of HVAC units can be found in Chapter 6. The airport also needs to assess the condition of the sewage system provided in the hangar along with the electrical system of the hangar every year. During the assessment of the electrical system, asset managers should inspect all lighting wires, light bulbs, power distribution systems, receptacles, electrical devices, and fire alarm panels. When assessing the condition of the hangar components, asset managers can divide the condition into four categories e.g., excellent, good, fair, and poor, depending on how quickly they need to be fixed.

7.2.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

The maintenance activities identified in the airport terminal building can be used to maintain the hangars. Because most of the building components in these two assets are similar, asset managers can use similar maintenance activities to those used in terminal buildings to maintain the hangar. Some specific maintenance activities that can be performed in the hangar building envelope are (S2 Architecture et al. 2017):

- Change exterior cladding, including associated flashing.
- Repair wall insulation.

Copyright National Academy of Sciences. All rights reserved.

- Repair walls, mortar joints, etc.
- Maintain or replace exterior doors and frames.
- Maintain or replace hangar doors.
- Maintain or replace hangar windows.
- Maintain or replace roofing, including cap flashings, etc.
- Maintain or replace insulation and roofing membranes on the top surface.

Airports should also assess the condition of electrical systems provided in the hangar. Some of the maintenance activities airports can perform related to electrical systems are:

- Maintain or replace existing power distribution system, including conduits and feeders.
- Maintain or replace branch circuit wiring.
- Maintain or replace lighting fixtures, including exit lights, emergency lighting, and lighting controls.
- Maintain or replace receptacles, mechanical starters, and other electrical devices.
- Maintain or replace fire alarm devices and panels.

Asset managers should determine the unit cost of these maintenance activities. As described in section 7.1.3 for terminal buildings, the airports can use their existing unit cost of maintenance, or they can determine the unit cost by using the RSMeans Cost Guide. If these maintenance activities are carried out by contractors, the airports can ask for a bid cost for maintenance budget preparation purposes. If the airports perform these maintenance activities using their in-house manpower, they can estimate the number of personnel involved in these activities per year and determine the labor cost per square foot of the hangar. Then the asset managers can determine the material costs for hangar maintenance by using the ratio of labor and materials found in the RSMeans Cost Guide. These unit costs need to be adjusted based on the budget preparation year. Cost inflation factors can be found in the RSMeans Cost Guide, or asset managers can search the consumer price index data available on the Bureau of Labor Statistics website. It is a good practice to determine the unit cost for each component of the hangar and estimate the maintenance cost of hangars with and without delay. Asset managers can also use the overall unit cost of maintenance for the hangar to determine the total maintenance cost for budgeting purposes.

7.2.4 Determining the Hangar Component Failure Probability with Regular Maintenance

The airport can use a process similar to the one mentioned in section 7.1.4 to incorporate failure probability costs in the total maintenance costs of the hangar. Because terminals and hangars have similar components, the percentage of contingency costs can be factored into the maintenance cost to incorporate the failure probability of the hangar. There is no systematic method of incorporating the hangar failure cost in the maintenance budget, but asset managers can use the contingency cost procedure to cover the failure of the hangar in normally maintained conditions.

7.2.5 Determining the Hangar Failure Probability with Delayed Maintenance

The research team could not find any literature or process that explains how to determine the failure probability of the hangar in a delayed maintenance scenario. When the maintenance is delayed, it will have an adverse effect on the condition of the hangar components. However, it is difficult to predict how fast the hangar components will deteriorate. As described for terminal buildings in section 7.1.5, asset managers can increase the contingency cost to incorporate the failure probability of hangar components in delayed scenarios. Airports can use similar percentages of contingency as described in the terminal building case.

Copyright National Academy of Sciences. All rights reserved.

7.2.6 Estimating Hangar Maintenance and Replacement Budget without Delayed Scenarios

A similar process to the one mentioned in the terminal building section can be used to determine the hangar maintenance and replacement costs without delayed scenarios. The maintenance cost includes labor, materials, and equipment (if used), and the contingency costs (added for the failure probability in regularly maintained conditions) can be added to determine the total maintenance and replacement costs of the hangar for the following year. The cost should be escalated for the following year. As described above, the cost inflation factors can be found in the RSMeans Cost Guide.

7.2.7 Estimating Hangar Maintenance and Replacement Budget with Delayed Scenarios

Airports need to determine the impact of delayed maintenance on airport assets. In the case of a hangar, the process is similar to that of the terminal building. Asset managers can add higher contingency costs with hangar maintenance costs calculated for the following year to determine the total maintenance and replacement costs in delayed scenarios. As described in section 7.1.7, the contingency cost needs to increase as the number of delay years increases. The research team could not find any literature to determine the percentage of contingency costs to be added to the maintenance cost to incorporate the failure probability of hangar components in delayed scenarios. However, the percentage of contingency costs should be greater than those without delayed scenarios, and it should increase as the number of days the maintenance is delayed increases. Asset managers also need to adjust the maintenance and replacement costs of hangars based on the future cost inflation factor. If the airport does not have the cost inflation factor data, it can use the historical cost inflation factors of the construction activities and predict the future cost inflation factors. These cost inflation factors can be found in the RSMeans Cost Guide or can be searched in the consumer price index data available on the Bureau of Labor Statistics website.

7.3 Airport Automated People Mover Systems

Large-hub airports generally have some kind of automated people mover (APM) system to rapidly move passengers across airport grounds. These APMs assist airport personnel in reducing passenger congestion and removing long walking distances for passengers. The APM system is designed and built to move passengers between airside concourses and landside from parking lots and transit lines to terminal buildings. These systems can handle high traffic volumes in large-hub airports and can be moved around the clock. In the United States, several airports, such as John F. Kennedy, LaGuardia, Oakland, Orlando, Las Vegas-Harry Reid, Philadelphia, Phoenix-Sky Harbor, Seattle-Tacoma, Tampa, San Francisco, and Washington Dulles have these APM systems. These APM systems are similar to a train system which moves over a track. The only difference between a train and an APM system is that the former is driven by a human and the latter does not have any driver and moves autonomously. A train can travel at high speeds, but an APM system can only travel at a maximum speed of 56 mph (Shen et al. 2022).

Airport asset managers who own ATM systems need to prepare the budget yearly to operate and maintain these systems. They also need to determine the consequence of delayed maintenance on the operating and maintenance costs of the systems. Asset managers who need to quantify the impact of delayed maintenance of APM in the total maintenance budget can follow the steps described below.

Copyright National Academy of Sciences. All rights reserved.

122 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

7.3.1 Establishing Asset Inventory of Airport APM Systems

Airports should keep inventory data of their APM system to determine the impact of delayed maintenance of an APM system on a maintenance budget. *ACRP Report 37: Guidebook for Planning and Implementing Automated People Mover Systems at Airports* developed a guidebook for planning and implementing APM systems at airports (Lea+Elliot et al. 2010). This report describes the requirements for APM system installments at airports. The major components of any APM system are guideway structure, stations, command, control, communication facilities, maintenance facilities, and power distribution facilities. Asset managers need to keep inventory data of the guideway structure, the number and area of stations, the number and area of command and communication systems, and the number and area of maintenance facilities. Asset managers should also keep a record of the number of passenger cars and the capacity of each passenger car. The make and model of these passenger cars and the APM system need to be recorded. These inventory data will assist asset managers in determining the maintenance and replacement cost of APM systems with or without delayed scenarios.

7.3.2 Performing a Condition Assessment of an Airport APM System

Asset managers need to assess the condition of each component of the APM system. The airport should assess the condition of its passenger cars. The asset manager should assess the interior and exterior of the passenger cars, and inspect the condition of maintenance facilities, system guideway operating areas, stations, and command and communication systems facilities. Asset managers should also assess the software system to check whether the software needs to be upgraded to run the APM systems. Asset managers should inspect the guideway structure and system frequently to make sure that the APM system can be operated smoothly and without any incidents. During the assessment, airports need to identify critical and long-term maintenance needs so that the maintenance budget can be prepared accurately. Asset managers can also categorize the condition of the components of APM systems into four categories, e.g., excellent, good, fair, and poor. These condition data should be kept in a database along with the inventory data.

7.3.3 Identifying and Updating Maintenance and Replacement Activities and their Unit Costs

An airport can identify maintenance activities during the condition assessment of APM systems components. The maintenance activities involved in the five major components of APM systems are different. Some of the maintenance to be performed on the passenger cars are as follows (Tampa International Airport 2015):

- Clean and maintain interior and exterior of the passenger cars;
- Maintain or replace the passenger car seats;
- Maintain or replace the passenger car handrail, doors, windows, and floors;
- Maintain and fix the automated door systems; and
- Maintain or replace the lighting system and light fixtures in the passenger cars.

Another important component of the APM system is the guideway structure and system. Without proper system maintenance, the APM cannot be operated efficiently. The guideway system comprises not only the super and substructures, but also special structures like tunnels, bridges, and fences along the APM line. Guideway inspection and maintenance activities are similar to those of the rail track. The American Public Transportation Association (APTA) has developed a guidebook explaining how to inspect and maintain a rail transit track. According

Copyright National Academy of Sciences. All rights reserved.

Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

to this guidebook, some of the major inspection and maintenance activities to be performed on rail tracks are (American Public Transportation Association 2017):

- Inspect and replace defective rails;
- Inspect the continuous welded rail track and fix it if found defective;
- Inspect the track geometry;
- Inspect mainline switches, track crossings, and moveable bridge lift rails and fix them if found defective;
- Inspect and replace the yard switches; and
- Inspect and fix the rail tracks in the event of a fire, flood, severe storm, or seismic activity.

If the APM guideway system has special structures like tunnels, bridges, and fences, then the following maintenance activities need to be performed annually:

- Inspect the bridges' sub- and superstructure for any defects and maintain them if found defective.
- Inspect the tunnel structure and maintain it if found defective.
- Inspect the fencing structures and maintain them if found defective.

The maintenance facilities and stations of the APM system should also be maintained. The maintenance activities involved in these facilities and stations are similar to that of the terminal buildings. If any special equipment is used, the airports also need to maintain the special equipment housed in the facilities. Command and communication facilities are critical to the APM system. In these facilities, asset managers need to perform regular maintenance, software maintenance, and upgrades. Software is key to operating the APM system effectively. Asset managers should also maintain the hardware of the APM systems.

After identifying the maintenance activities of the APM systems, airport asset managers need to determine the unit cost of these maintenance activities. Airports can refer to the RSMeans Cost Guide to determine the cost of these maintenance activities. Also, asset managers can determine the number of labor hours used to maintain their APM systems per year and convert the cost into feet or mile unit costs. Airports can determine the ratio of labor and materials spent on maintaining their APM systems and find the unit cost of materials. The total unit cost of maintaining the APM system per linear foot or mile can be calculated by adding the labor and material costs and the overhead cost. These unit maintenance costs can be used to estimate the future maintenance and replacement costs of the APM systems with or without delayed maintenance.

7.3.4 Determining APM Components' Failure Probability with Regular Maintenance

The APM system consists of physical structures as well as mechanical and electrical systems. Therefore, asset managers need to determine how to incorporate the probability of failure of these components in regularly maintained conditions. The airports can use processes similar to those in terminal buildings to determine the failure probability of structures like bridges, tunnels, guideways, stations, etc. Because there is no exact theoretical method to calculate failure probability for physical structures, the contingency cost addition method can be used to incorporate the failure costs of these structures. After calculating the maintenance and replacement costs of these structures, asset managers can add appropriate contingency costs in case of structure failures.

Asset managers can use the probability theory to determine the electrical and mechanical systems failure probability of APM systems. The Weibull distribution theory can be used to determine the failure probability of an APM system's train and guideway mechanical systems. Asset managers can determine the life of an APM system's passenger cars and guideway systems. They can estimate the duration at which 63% of the passenger cars and guideway systems will fail when

Copyright National Academy of Sciences. All rights reserved.

124 Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide

they are regularly maintained, and they can use these values in a Weibull probability distribution equation. Asset managers can use a shape factor of 3.5, considering the failure probability as normally distributed. They can refer to section 6.4.5 under the baggage handling system delayed maintenance cost calculator to determine the cumulative failure probability of passenger cars and guideway systems of APMs.

Airports can also use the Weibull probability distribution theory to determine the cumulative failure probability of APM electrical systems. Similar to the mechanical system described above, asset managers can determine the duration at which 63% of their electrical system will fail in regularly maintained scenarios. Asset managers can consider that the failure probability distribution of the electrical system of the APM will be normal and use a value of 3.5 as the shape factor. The process of determining the failure probability of electrical systems under normal conditions is shown in section 5.3.4.

7.3.5 Determining APM Components' Failure Probability with Delayed Maintenance

Asset managers can use the contingency cost estimation process to determine the impact of delayed maintenance on the physical structures of the APM system. For delayed maintenance scenarios, asset managers can double the contingency cost percentage compared to regularly maintained structures. They can increase the contingency cost based on the age of the structures as described in the previous section.

Airports can also use the Weibull distribution theory from the previous section to determine the cumulative probability of an APM's mechanical and electrical component failure in the delayed maintenance scenarios. The process is similar to the one described in section 7.2.4, however, the duration at which 63% of the components will fail when their maintenance is delayed will decrease. Asset managers can assume that this value will decrease by 50% depending upon the types of components. The cumulative probability distribution can then be determined for the mechanical and electrical components of an APM by using a shape factor of 3.5. The research team developed a spreadsheet tool for determining the delayed maintenance and replacement costs for baggage handling systems and passenger boarding bridges. Airports can refer to that sheet to determine the delayed maintenance cost of mechanical components of their APM system. They can also refer to the spreadsheet tool developed for runway lighting systems to determine the delayed maintenance costs of the electrical components of their APM system.

7.3.6 Estimating the APM Maintenance and Replacement Budget without Delayed Scenarios

Airport asset managers can use their inventory, condition assessment, unit cost of maintenance, contingency cost percentages, and the failure probability of mechanical and electrical components of the APM system to determine regular maintenance and replacement costs for the following year. They must adjust the unit cost of these maintenance activities for the subsequent year. Asset managers can use their own airport's cost inflation factors for labor and material costs or get these cost inflation factors from the RSMeans Cost Guide or the consumer price index of the Bureau of Labor Statistics. Once the unit cost of maintenance for each component of an APM system is calculated, airport asset managers can multiply this adjusted unit cost with the quantity to estimate the total maintenance and replacement costs. Airports should always consider contingency costs for the physical components of their APM system in case some components fail during operation. As described in the previous section, asset managers can estimate this contingency cost based on the age of the structures.

Copyright National Academy of Sciences. All rights reserved.

Asset managers should also add the cost related to the probability of failure of the mechanical and electrical components of the APM system. To determine this cost, they can multiply the replacement cost and the failure probability of these components. Airports should have some leverage in their maintenance and replacement budget in case some components fail abruptly during the operation of the APM system.

7.3.7 Estimating the APM Maintenance and Replacement Budget with Delayed Scenarios

Airports can use a similar process to the one described in section 7.3.6 to determine the maintenance and replacement cost of APM systems with delayed scenarios. The only difference is the failure probability cost of the APM system components. This failure cost will increase significantly depending on the number of years by which maintenance and replacement are delayed. The research team developed spreadsheet tools for other airside and landside assets for up to 10 years of delay. However, asset managers should understand the overall implication of delaying APM system maintenance on the performance of the airport. When calculating the delayed maintenance and replacement costs, the cumulative failure probability of mechanical and electrical systems will increase significantly compared to regularly maintained APM systems. Therefore, the delayed maintenance cost will increase significantly, and the incremental cost will depend on the age of the APM system. The older an APM system is, the higher the delayed maintenance and replacement cost will be because the probability of failure increases exponentially. Asset managers should use appropriate future cost inflation factors to determine this future cost. Asset managers can refer to other spreadsheet tools developed by the research team to adjust inflation factors.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 8

Summary and Recommendations for Future Research

8.1 Key Points

All types of airports, irrespective of their size, have difficulty procuring enough funds to maintain all their assets due to various constraints. Some airports' assets need to be regularly maintained due to *14 CFR Part 139* requirements, and maintenance of those assets is funded by the FAA. Airports need to create their own budget to maintain other assets. Therefore, one of the major responsibilities of the operation and maintenance division is to efficiently prepare the maintenance budget required for the following fiscal year. If the budget cannot fund maintenance of all airport assets, they need to calculate the impact of delaying maintenance on those assets. Airport asset managers can show the airport's higher authority figures the effects of delayed maintenance on the life of assets and the long-term maintenance and replacement budget.

Airport maintenance can be divided into three types: preventive, reactive, and predictive. Preventive maintenance activities include tasks that mitigate the degradation of an infrastructure system or its components. Preventive maintenance, performed regularly or as needed, can extend the service life of facilities and components and can yield the optimum performance outputs of assets. This results in reduced costs and improved user satisfaction. Reactive maintenance means fixing the system or components after they break, without performing preventive maintenance. This type of activity incurs higher system costs compared to conducting regular preventive maintenance. Reactive maintenance can have adverse effects on customer service and can reduce revenue. Predictive maintenance involves testing and monitoring equipment and facilities and predicting potential failures. This method is used for equipment and vehicles. If predictive and preventive maintenance are simultaneously implemented, it will ensure a minimal impact on an airport's operation and safety.

Airport officials and policymakers need to understand the value of preventive maintenance of their assets. Preventive maintenance of assets is critical in providing efficient service to air passengers and increasing the life of the assets. If airports cannot perform preventive maintenance of their critical assets every year as required, those assets need to be replaced before they reach the end of their economic lives. Replacing failed assets is more costly than performing regular maintenance. Some critical airfield assets include runway pavement, runway and taxiway lighting, airfield signs, airfield markings, and hangars. Landside assets include, but are not limited to, service vehicles, snow removal equipment, terminal buildings, heating and cooling systems, baggage handling systems, passenger boarding bridges, and APM systems.

The research team conducted several interviews with airport assets managers to identify the best practices for preparing maintenance budgets for airport assets and the procedure to quantify the consequence of delaying maintenance on their assets. The following are practices

Copyright National Academy of Sciences. All rights reserved.

for airports to consider while preparing their maintenance budget with and without delayed scenarios:

1. Prepare the maintenance budget based on the condition of the assets.
2. Prioritize maintenance of the assets based on their impact on the safety and operation of the planes and the comfort of air passengers.
3. Prepare the maintenance budget 6-9 months before the fiscal year begins.
4. Prepare the short-term and long-term maintenance budget so that the impact of delaying the maintenance of assets can be evaluated effectively.
5. Understand the value of short-term and long-term maintenance budgeting.
6. Involve all appropriate airport departments when preparing the long-term and short-term maintenance budgets.
7. Create working relationships with various departments while preparing maintenance budgets.
8. Implement the basic framework described in this report, and use spreadsheet tools developed to determine the consequences of delayed maintenance of some airside and landside airport assets.
9. Keep a detailed inventory of all airport assets, and update the inventory annually.
10. Perform a condition assessment of all the airport assets, and update the record in the database system.
11. Use numeric or categorical variables to describe the condition of assets. For example, PCI can be used to show the condition of pavement. Similarly, the condition categories of Excellent, Good, Fair, Poor, and Needs Replacement can be used for other airport assets.
12. Identify the preventive maintenance activities performed on assets.
13. Keep a record of labor hours spent on maintaining each asset separately.
14. Keep and update the materials used in maintaining each airport asset.
15. Use a maintenance management system, such as CMMS, however, input and update data in the system regularly.
16. Use the collected data while making decisions during maintenance budget preparations.
17. Update the labor and materials data at the end of each year.
18. Collect and record lessons learned during the preparation of the maintenance budget.
19. Train airport maintenance division personnel to follow these practices.
20. Obtain resources (funding, staffing software, tools) to maintain the database systems.

Understandably, some airports might have difficulty adopting the practices mentioned above on the first attempt. Due to levels of federal and state funding, aging airport infrastructure and other assets, and increasing air passenger volume and high-level customer satisfaction demands, airports are facing substantial pressure to be more efficient in maintaining and operating their assets. It is necessary for every airport, irrespective of its size, to be more innovative in maintaining its assets. Therefore, the airport industry must begin to apply these practices so that effective and accurate maintenance budgets can be prepared.

8.2 Spreadsheet Tools to Assist Airport Asset Managers

The research team collected data related to certain airport assets' inventory, condition assessments, maintenance unit costs, and deterioration models to develop spreadsheet tools for determining the preventive maintenance budgets of nine airport assets and quantifying the impact of delayed maintenance on overall airport budgets. The research team developed spreadsheet tools for four airside and five landside assets. The spreadsheet tools were prepared for airside assets including runway pavement, runway lighting, airfield signs, and airfield markings. In addition, the spreadsheet tools were prepared for landside assets such as service vehicles, snow removal equipment, HVAC systems, baggage handling systems, and passenger boarding bridges.

Copyright National Academy of Sciences. All rights reserved.

These spreadsheet tools are meant to be a data management resource for airport asset managers to better estimate the regular maintenance budget and to quantify the impact of delayed maintenance on these assets. These spreadsheet tools can be used as guidelines when determining the impact of delayed maintenance. These tools already have the maintenance labor hours, materials cost, cost inflation factors, theoretical deterioration models with and without delayed maintenance, and present value cost conversion process. The tools were developed to be flexible and customizable for a variety of purposes and system sizes. These spreadsheet tools have many prepopulated input cells, but the user may also easily customize those options. Airport users can use their own asset inventory, labor hours, cost, inflation factors, and deterioration models to determine the delayed maintenance costs of these nine airport assets.

The intended use of the spreadsheet tools is to show the airports that the consequence of delaying asset maintenance can be quantified monetarily. However, the spreadsheet tools generate estimates based on the input values. There are certain assumptions used to generate these delayed maintenance budgets. Therefore, if the assumptions are not valid for their airports, the number generated will not make sense. It is suggested to use these spreadsheet tools carefully when quantifying the impact of delayed maintenance on airport assets. Also, the team provided the option of delaying maintenance up to 10 years, but this may not be feasible for several airport assets due to their importance in the safety and operation of planes. These spreadsheet tools have also not considered delayed maintenance impacts on the safety of the planes and the comfort of the passengers. The estimates generated by these spreadsheet tools are merely calculated based on the impact on maintenance and replacement costs of the assets. The research team prepared an easy-to-use guide for airports so that they can effectively use these spreadsheet tools for their budgeting purposes. The guide is supplemental to this report.

8.3 Areas of Future Research

The research team collected extensive data to develop the spreadsheet tools for nine airport assets. However, due to the scope of the research, the spreadsheet tools could not be developed for other critical assets. The research team described the process of collecting and using data to develop spreadsheet tools for terminal buildings, hangars, and APMs. Therefore, future research could involve preparing spreadsheet tools for the following critical assets by collecting comprehensive data from airport case study projects:

- Terminal building envelope,
- Hangars,
- Airfield visual and navigational aids (NAVAIDS),
- Maintenance and storage buildings,
- Fueling facilities,
- Deicing facilities,
- Parking garage,
- Drainage,
- Access roads,
- APM,
- Elevators,
- Moving walkways, and
- Overhead bridges.

The research team found that most airports are using traditional ways of preparing maintenance and replacement budgets for airport assets without considering the performance of the assets. Many airports do not collect their asset conditions, except those of airfield pavements. All airports have very well-developed pavement condition data in the form of a PCI. Airports collect these data regularly to determine the condition of their runway pavements. Airports need to develop

Copyright National Academy of Sciences. All rights reserved.

similar condition assessment procedures for other airport assets. Considering this deficiency, the research team recommends that airports conduct research to develop a comprehensive plan of asset inventory and condition data collection processes for critical airport assets. State DOTs are collecting these data and using these performance measures to prepare a short-term, as well as long-term, maintenance and replacement budget for their highway assets. This research could focus on investigating the inventory data required for various types of airport assets and how to assess their conditions and store them in their database to use for budgeting purposes.

The research team used theoretical processes to determine the deterioration models for most of the airport assets, except runway pavement, to predict their condition in the future with and without delayed scenarios. Therefore, other research could focus on developing the deterioration models of airport assets based on the condition data collected from the case study airport projects. If the condition data of airport assets can be collected for multiple years, then practical models can be developed to predict the condition of these assets in the future. For example, regarding runway pavement, FAA software PAVEAIR can develop regression models to predict the PCI values based on the age and types of the pavement. If similar types of asset deterioration models can be developed for other airport assets, the accuracy of maintenance and replacement budgeting processes can be improved.

For future research, the team suggests investigating the effects of delayed maintenance on air travelers' user costs. Airports should also consider travel delay costs, airport closure costs, operating costs, and accident costs when calculating the impact of delayed maintenance of airport assets. The magnitude of the effects depends primarily on the size of the airports and the types of assets delayed. It is expected that delaying maintenance of critical airport assets will cause larger disruptions due to the cancellation of flights and increased traveler costs. This also poses the risk of asset failure and potentially increases safety issues as well as replacement costs. The spreadsheet tools should thus be used to quantify the consequence of delaying asset maintenance by considering these impacts, which the team did not include in the spreadsheet tools due to data unavailability.

Research is needed to develop guidelines for airports and address the use of performance-based management strategies in maintenance. Performance-based management approaches help owners collect the performance data of their assets and use them to develop long-term maintenance and replacement budgets. Airports prepare their maintenance and replacement budgets based on traditional methods without considering the performance of their assets. Further research could identify the process and strategies to successfully use the performance data of the assets in developing a long-term maintenance plan. It could also identify the benefits of this method and identify organization structures required to conduct performance-based budgeting.

Copyright National Academy of Sciences. All rights reserved.

References

Agile Assets. 2021. The true cost of deferred road maintenance. https://www.agileassets.com/blog/the-true-cost-of-deferred-road-maintenance/. Accessed March 22, 2021.

American Public Transportation Association. 2017. Rail Transit Track Inspection and Maintenance. https://www.apta.com/wp-content/uploads/Standards_Documents/APTA-RT-FS-S-002-02-Rev-1.pdf.

American Society of Civil Engineers. 2017. 2017 Infrastructure Report Card—A Comprehensive Assessment of America's Infrastructure. https://www.infrastructurereportcard.org/wp-content/uploads/2016/10/2017-Infrastructure-Report-Card.pdf. Accessed May 12, 2021.

ARISTA. n.d. Commercial HVAC Service. https://aristair.com/commercial-hvac-service/. Accessed February 15, 2022.

Aviation Learnings. 2020. How Jet Bridge (or Passenger Boarding Bridge) Works? – A Complete Technical Guide. https://aviationlearnings.com/passenger-boarding-bridge-jet-bridge-jetty-or-aerobridge-you-name-it/. Accessed February 23, 2022.

Beauchamp, E., and S. Lopez. 2022. How to Find the Air Filter for the HVAC System in Your Home. Today's Homeowners. https://todayshomeowner.com/how-to-find-the-air-filter-in-a-hvac-system-in-your-home/. Accessed February 20, 2022.

Bureau of Labor Statistics. 2022a. PPI Commodity Data for Nonmetallic Mineral Products-Ready-Mix Concrete, Not Seasonally Adjusted. https://data.bls.gov/timeseries/WPU1333?amp%253bdata_tool=XGtable&output_view=data&include_graphs=true. Accessed February 10, 2022.

Bureau of Labor Statistics. 2022b. PPI Data for Asphalt Pavement and Roofing Materials mfg., Not Seasonally Adjusted. https://beta.bls.gov/dataViewer/view/timeseries/PCU32412-32412-. Accessed February 10, 2022.

Bureau of Labor Statistics. 2022c. PPI Industry Data for Commercial Machinery Repair and Maintenance, Not Seasonally Adjusted. https://data.bls.gov/pdq/SurveyOutputServlet. Accessed February 20, 2022.

Burns, J., C. Dennie, S. Elshetwy, D. Lean, and J. Vigilante. 2015. *ACRP Report 148: LED Airfield Lighting System Operation and Maintenance*. Transportation Research Board, Washington, DC. https://doi.org/10.17226/22076.

CarEdge. 2022. Ford F-150 Maintenance Costs. https://caredge.com/ford/f-150/maintenance. Accessed February 20, 2022.

CASE Construction. n.d. Planned Maintenance. https://www.casece.com/en-us/northamerica/solutions/planned-maintenance. Accessed February 12, 2022.

Caterpillar. n.d. Cat® Service, Parts and Operation and Maintenance Manuals. https://www.cat.com/en_US/support/maintenance/service-manuals.html. Accessed February 20, 2022.

Chang, C.M., S. Nazahrian, M. Vavrova, M.T. Yapp, L.M. Pierce, W. Roberts, and R.E. Smith. 2017. *NCHRP Research Report 859: Consequences of Delayed Maintenance of Highway Assets*. Transportation Research Board, Washington, DC.

City of Frederick. 2005. Terminal Building Evaluation. https://www.cityoffrederickmd.gov/DocumentCenter/View/714/terminal-report?bidId=. Accessed July 6, 2022.

Craven, J. 2019. The 2004 Collapse at Airport Charles de Gaulle. ThoughtCo. https://www.thoughtco.com/charles-de-gaulle-airport-terminal-collapse-3972251. Accessed July 7, 2022.

Dansig. 2021. Structural Issues and Aging Buildings. https://dansig.com/blog/structural-issues-and-aging-buildings/. Accessed July 7, 2022.

Dempsey, P.S. 2008. *ACRP Legal Research Digest 2: Theory and Law of Airport Revenue Diversion*. Transportation Research Board, Washington, DC. https://doi.org/10.17226/23092.

Eagle. 2021. How a Proper Maintenance Strategy Reduces Your Airport Downtime. https://www.eaglecmms.com/proper-maintenance-strategy-reduces-airport-downtime/. Accessed March 25, 2021.

Engine Patrol. 2021. Ford F-150 Maintenance & Repair Costs (Complete Guide). https://enginepatrol.com/ford-f-150-maintenance-costs-expenses/. Accessed February 17, 2022.

Copyright National Academy of Sciences. All rights reserved.

Environmental Protection Agency. 2022. Heating, Ventilation and Air-Conditioning Systems, Part of Indoor Air Quality Design Tools for Schools. https://www.epa.gov/iaq-schools/heating-ventilation-and-air-conditioning-systems-part-indoor-air-quality-design-tools. Accessed February 20, 2022.

Environmental Protection Agency. 2014. HVAC Checklist - Short Form. https://www.epa.gov/sites/default/files/2014-08/documents/hvacshrt.pdf. Accessed February 21, 2022.

Estes Services. n.d. How Do Rooftop HVAC Units Work? https://www.estesair.com/blog/how-do-rooftop-hvac-units-work. Accessed February 20, 2022.

Federal Aviation Administration. 2003. Development of Methods for Determining Airport Pavement Marking Effectiveness. https://www.airporttech.tc.faa.gov/Products/Airport-Safety-Papers-Publications/Airport-Safety-Detail/development-of-methods-for-determining-airport-pavement-marking-effectiveness-1.

Federal Aviation Administration. 2014a. Advisory Circular No. 150/5380-6C. Guidelines and Procedures for Maintenance of Airport Pavements. https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-6c.pdf. Accessed February 2, 2022.

Federal Aviation Administration. 2014b. Advisory Circular No. 150/5380-7B. Airport Pavement Management Program (PMP). https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-7b.pdf. Accessed February 2, 2022.

Federal Aviation Administration. 2014c. Advisory Circular No. 150/5220-20A. Airport Snow and Ice Control Equipment. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_150_5220-20A.pdf. Accessed February 19, 2022.

Federal Aviation Administration. 2016b. Policy on the Non-Aeronautical Use of Airport Hangars. Federal Register, Vol. 81., No. 115. https://www.govinfo.gov/content/pkg/FR-2016-06-15/pdf/2016-14133.pdf. Accessed July 7, 2022.

Federal Aviation Administration. 2019. Airport Improvement Program (AIP) 2019–2021 Supplemental Appropriation. https://www.faa.gov/airports/aip/aip_supplemental_appropriation/2019. Accessed July 7, 2022.

Federal Aviation Administration. 2020. Advisory Circular No. 150/5340-1M. Standards for Airport Markings. https://www.faa.gov/documentLibrary/media/Advisory_Circular/150-5340-1M-Chg-1-Airport-Markings.pdf.

Federal Aviation Administration. 2021. What is Part 139? Part 139 Airport Certification. https://www.faa.gov/airports/airport_safety/part139_cert/what-is-part-139/. Accessed March 20, 2021.

Federal Aviation Administration. 2022a. FAA Airport Pavement Research & Development Section: 2030 Research Plan. https://www.airporttech.tc.faa.gov/Portals/0/download/File%20B_FAA%20Airport%20Pavement%20RD%20Section%202030%20Research%20Plan.pdf. Accessed February 2, 2022.

Federal Aviation Administration. 2022b. Sample Pavement Maintenance Program. https://www.faa.gov/airports/central/airport_compliance/pavement_maintenance/media/sample-maintenance-program.pdf. Accessed February 2, 2022.

Federal Aviation Administration. 2022c. FAA PAVEAIR. https://faapaveair.faa.gov/Help/PavementConditionIndexPCI.html#:~:text=The%20PCI%20value%20of%20the,during%20condition%20surveys%20(inspections). Accessed February 2, 2022.

Federal Aviation Administration. 2022d. Snow Removal Equipment Calculations. https://www.faa.gov/airports/central/airports_resources/media/sre.xls. Accessed February 19, 2022.

Federal Highway Administration. 2005. Guidelines for the Installation, Inspection, Maintenance, and Repair of Structural Supports for Highway Signs, Luminaries, and Traffic Signals. Publication No. FHWA NHI-05-036. https://www.fhwa.dot.gov/bridge/signinspection.pdf.

Fortin, J., L. Alfaqih, and P. Bell. 2017. *ACRP Research Report 172: Guidebook for Considering Life-Cycle Costs in Airport Asset Procurement*. Transportation Research Board, Washington, DC. https://doi.org/10.17226/24764.

FRED. 2022a. Producer Price Index by Industry: Paint and Coating Manufacturing. https://alfred.stlouisfed.org/series?seid=PCU3255103255101&utm_source=series_page&utm_medium=related_content&utm_term=related_resources&utm_campaign=alfred. Accessed February 10, 2022.

FRED. 2022b. Producer Price Index by Industry: Electrical Contractors, Nonresidential Building Work. https://fred.stlouisfed.org/series/PCU23821X23821X. Accessed February 10, 2022.

FRED. 2022c. Producer Price Index by Industry: Electrical Equipment Manufacturing. https://fred.stlouisfed.org/series/PCU3353133531. Accessed February 10, 2022.

FRED. 2022d. Producer Price Index by Industry: HVAC and Commercial Refrigeration Equipment. https://fred.stlouisfed.org/series/PCU3334133341. Accessed February 21, 2022.

Galar, D. and U. Kumar. 2017. *eMaintenance*: *Essential Electronic Tools for Efficiency*, Chapter 6 – Prognosis, pp. 311–370. Academic Press.

GHD Inc. 2012. *ACRP Report 69: Asset and Infrastructure Management for Airports—Primer and Guidebook*. Transportation Research Board of the National Academies, Washington, DC. https://doi.org/10.17226/22760.

Gokey, J., N. Klein, C. Mackey, J. Santos, A. Pillutla, and S. Tucker. 2009. Development of a Prioritization Methodology for Maintaining Virginia's Bridge Infrastructure Systems. 2009 Systems and Information Engineering Design Symposium, pp. 252–257. doi:10.1109/SIEDS.2009.5166190.

Copyright National Academy of Sciences. All rights reserved.

Government Accountability Office. 2015. Airport Finance: Information on Funding Sources and Planned Capital Development. Report to Congressional Committees, United States Government Accountability Office.

Grainger KnowHow. 2022. Building Preventive Maintenance Checklist. https://www.grainger.com/know-how/industry/manufacturing/kh-preventive-maintenance-checklist. Accessed July 6, 2022.

Hatry, H.P. and E.B. Liner. 1994. Issues in Deferred Maintenance. Urban Institute, U.S. Army Corps of Engineers.

HousecallPro. 2022. HVAC Inspection Checklist Template. https://www.housecallpro.com/working-in-the-trades/hvac-template/unit-inspection-checklist-template/. Accessed February 21, 2022.

HUB International. 2022. Building Envelope Inspection Checklist. https://www.hubinternational.com/-/media/HUB-International/Industries/Real-Estate-Insurance/Building-Envelope-Inspection-Checklist.pdf. Accessed July 6, 2022.

International Organization for Standardization. 2024. ISO 55000:2024. https://www.iso.org/standard/83053.html.

Kansas Department of Transportation. 2022. Preventative Maintenance Checklists. https://www.ksdot.org/Assets/wwwksdotorg/bureaus/burTransPlan/pubtrans/pdf/Vehicle%20Preventive%20Maintenance%20Checklist.pdf. Accessed February 17, 2022.

Kramlich, K. 2019. Compact Wheel Loaders: A Healthy Dose of Maintenance Goes a Long Way. *Equipment Journal*. https://www.equipmentjournal.com/on-the-job/compact-wheel-loader-maintenance/. Accessed February 20, 2022.

Lea+Elliot, Kimley-Horn and Associates, Inc., and Randolph Richardson Associates. 2010. *ACRP Report 37: Guidebook for Planning and Implementing Automated People Mover Systems at Airports*. Transportation Research Board of the National Academies, Washington, DC. https://doi.org/10.17226/22926.

Louisville International Airport. 2022. Jet Bridge Maintenance, Inspection & Repair Services at Louisville International Airport. RF-010-19. Louisville, Kentucky. https://www.bidnet.com/bneattachments?/540832374.pdf. Accessed February 10, 2022.

McGee, H.W. 2010. Maintenance of Signs and Sign Supports: A Guide for Local Highway and Street Maintenance Personnel. Federal Highway Administration, Report No. FHWA FHWA-SA-09-025. https://safety.fhwa.dot.gov/local_rural/training/fhwasa09025/.

Metropolitan Airports Commission. 2020. 2020 Operating Budget. Minneapolis-St. Paul, Minnesota.

Moayedfar, R. and A. Sajjadifard. 2021. Prioritization of Pavement Restoration and Maintenance Strategies in Airports Using APMS Technique. *International Journal of Pavement Research and Technology*, pp. 327–333. https://doi.org/10.1007/s42947-020-0244-4.

National Foundation Repair Association. 2022. Foundation Maintenance. https://foundationrepair.org/foundation-maintenance/. Accessed July 6, 2022.

New York Department of Transportation. 2022. Front End Loader Preventive Maintenance Checklist. https://www.dot.ny.gov/divisions/operating/oom/transportation-maintenance/repository/frontendloadrmaintchklst.pdf. Accessed February 19, 2022.

Olson, P. and D. Wessel. 2017. The Case for Spending More on Infrastructure Maintenance. Brookings. https://www.brookings.edu/blog/up-front/2017/01/31/the-case-for-spending-more-on-infrastructure-maintenance/. Accessed March 23, 2021.

OpenGov Procurement. 2021. HVAC Repair & Preventative Maintenance Services. City of Lake City. https://secure.procurenow.com/portal/lcfla/projects/6735?section=38876. Accessed February 21, 2022.

Penny, J. 2018. Stop Wasting Money on Deferred Maintenance. *Buildings*. https://www.buildings.com/articles/27864/stop-wasting-money-deferred-maintenance. Accessed March 15, 2021.

Pham, H. 2023. Springer Handbook of Engineering Statistics. Springer. New York, USA.

Ploeger, D., R.B. Chapman, D.G. Peshkin, and D.J. Spiedel. 2015a. *ACRP Report 138: Preventive Maintenance at General Aviation Airports Volume 1: Primer*. Transportation Research Board of the National Academies, Washington, DC. https://doi.org/10.17226/22117.

Ploeger, D., R.B. Chapman, D.G. Peshkin, and D.J. Spiedel. 2015b. *ACRP Report 138: Preventive Maintenance at General Aviation Airports Volume 2: Guidebook*. Transportation Research Board of the National Academies, Washington, DC. https://doi.org/10.17226/23435.

Pushpa, G.R., M.S. Khan, and A.H. Pasha. 2006. Empirical Analysis of the Weibull Distribution for Failure Data. *Journal of Statistics*. Vol: 13, No. 1.

QualityForms. n.d. Used Vehicle Inspection Form. https://www.qualityforms.com/used-vehicle-inspection-form.html?gclid=CjwKCAiAgbiQBhAHEiwAuQ6BkmrA5r5kYm9UK5oDdmFd14PTOB7g-z1DQCHZSkXa0XsnvzPkExLx5BoCE7EQAvD_BwE. Accessed February 17, 2022.

Rapoport, F. 2018. The Operations and Maintenance Incentive for Future Builds and Renovations of U.S. Airports. *The American City & County*. http://proxy.library.cpp.edu/login?url=https://www-proquest-com.proxy.library.cpp.edu/trade-journals/operations-maintenance-incentive-future-builds/docview/2085798053/se-2?accountid=10357. Accessed May 12, 2021.

Ravnholm, T. n.d. BHS Maintenance: How to Prepare Your Baggage Handling System for Reopening. BeumerGroup. https://www.beumergroup.com/knowledge/airport/baggage-handling-system-maintenance/. Accessed February 12, 2022.

Copyright National Academy of Sciences. All rights reserved.

RSMeans Cost Guide. 2021. RSMeans Unit Cost Data. https://login.gordian.com/GordianLogin?signin=44d32960b0385350f8c1cfc4a4c6ebdf&clientID=RsMeansOnline. Accessed December 21, 2021.

S2 Architecture, RJC Consulting Engineers, Smith + Andersen, and SMP Engineering. 2017. Condition Assessment Report. The City of Edmonton, Edmonton, CA. https://pub-edmonton.escribemeetings.com/filestream.ashx?DocumentId=11704.

SAFS Steel Structure. n.d. Aircraft Hangar Roofing System. https://www.safsteelstructure.com/aircraft-hangar-roofing-system/. Accessed July 7, 2022.

Schmuck, A. and G. Maerschalk. 1987. The Effect of Deferred Maintenance on Medium- and Long-Term Highway Maintenance Costs. Research in Road Construction and Traffic Engineering, Publisher BMDV, Berlin, Germany.

Schultz, D. 2016. Four Vital Steps of a Comprehensive Roof Maintenance Program. https://www.fibertite.com/blog/4-vital-steps-of-a-comprehensive-roof-maintenance-program. Accessed July 6, 2022.

Shen, D.L., J. Huang, and F. Zhao. 2022. Automated People Mover Applications: A Worldwide Review. http://yangdy.freeservers.com/apm/apmrev.html. Accessed July 8, 2022.

Shingobee. 2021. What is the Lifespan of a Commercial Building? – Ask the Expert. http://www.shingobee.com/About-Us/News/entryid/191/what-is-the-lifespan-of-a-commercial-building-ask-the-expert#:~:text=The%20lifespan%20of%20a%20commercial%20building%20on%20average%20ranges%20from,build%20quality%20and%20maintenance%20management. Accessed July 7, 2022.

Siemens. 2019. Airport Logistics: Siemens Increases the Efficiency and Service Life of Baggage Handling Systems with Service 4.0. https://www.siemens-logistics.com/en/news/press-releases/siemens-increases-the-efficiency-and-service-life-of-baggage-handling-systems-with-service-4-0. Accessed February 22, 2022.

Speidel, D.J. 2008. *Airfield Marking Handbook*. Innovative Pavement Research Foundation, Skokie, IL.

Spy Pond Partners, LLC, KKO & Associates, LLC, H. Cohen, and J. Barr. 2012. *TCRP Report 157: State of Good Repair: Prioritizing the Rehabilitation and Replacement of Existing Capital Assets and Evaluating the Implications for Transit.* Transportation Research Board of the National Academies, Washington, DC. https://doi.org/10.17226/22732.

Statista. 2022. U.S. Producer Price Index for Construction Machinery and Equipment from 1990 to 2020. https://www.statista.com/statistics/208541/us-producer-price-index-for-construction-equipment/. Accessed February 20, 2022.

Tampa International Airport. 2015. Maintenance Contract Terminal Automated People Mover Systems. https://www.tampaairport.com/sites/default/master/files/Maintenance%20Contract%20-%20Terminal%20Automated%20People%20Mover%20Systems%20-%20Bombardier.pdf.

Tang, R.Y. 2019. *Financing Airport Improvements*. Congressional Research Service, Washington, DC.

The Geaslin Group. 2021. Defeating Deferred Maintenance. http://www.geaslin.com/invers-square_rule.htm. Accessed March 24, 2021.

The TASA Group. n.d. Electrical Equipment Failures Cause and Liability. https://www.tasanet.com/Knowledge-Center/Articles/ArtMID/477/ArticleID/146/Electrical-Equipment-Failures-Cause-and-Liability#:~:text=Many%20insurance%20claims%20and%20lawsuits%20are%20the%20result%20of%20electrical%20system%20failures.&text=Failures%20of%20electrical%20systems%20are,service%2C%20mishandling%20and%20other%20causes. Accessed February 7, 2022.

Tronair. n.d. Aircraft Maintenance Hangar Checklist. https://www.tronair.com/resources/aircraft-maintenance-hangar-checklist/#:~:text=A%20maintenance%20hangar%20is%20different,and%20other%20equipment%20are%20housed. Accessed July 7, 2022.

University of Michigan. 2022. Commercial Buildings Factsheet. https://css.umich.edu/publications/factsheets/built-environment/commercial-buildings-factsheet. Accessed July 7, 2022.

Upkeep. 2021. Types of Maintenance. https://www.onupkeep.com/learning/maintenance-types/. Accessed March 15, 2021.

Warren, L.F. 2002. *Designing, Constructing, Maintaining, and Financing Today's Airport Projects. Proceedings of the Twenty-Seventh International Air Transport Conference.* Airport Development in a Volatile Environment. Transportation Research Board of the National Academies, Washington, DC.

Copyright National Academy of Sciences. All rights reserved.

Appendices

Appendices A through D are supplemental to this report and can be found on the National Academies Press website (nap.nationalacademies.org) by searching for *ACRP Research Report 273: Quantifying the Impacts of Delayed Maintenance of Airport Assets: A Guide*.

Appendix A Used Vehicle Inspection Report
Appendix B Preventive Maintenance Checklist
Appendix C Front-End Loader Preventive Maintenance Checklist
Appendix D HVAC Inspection Checklist

Copyright National Academy of Sciences. All rights reserved.

Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

Copyright National Academy of Sciences. All rights reserved.

Transportation Research Board
500 Fifth Street, NW
Washington, DC 20001

NATIONAL ACADEMIES Sciences Engineering Medicine

The National Academies provide independent, trustworthy advice that advances solutions to society's most complex challenges.
www.nationalacademies.org

ISBN 978-0-309-72647-4
90000
9 780309 726474

Copyright National Academy of Sciences. All rights reserved.